Group D: Late progression

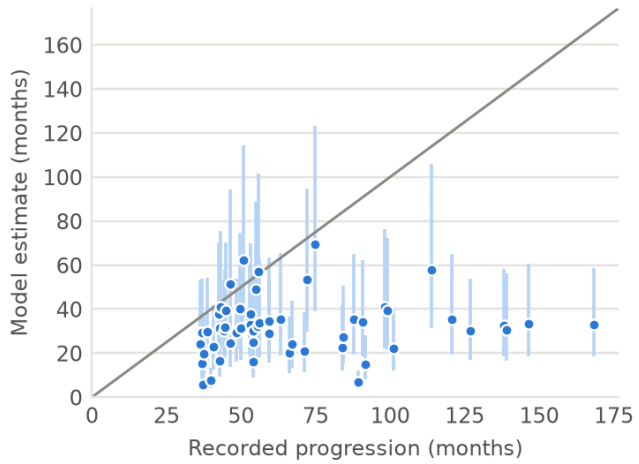
PrimeKG Patient Reasoning · Precision Oncology Research

Group D · 56 patients

Report date 14 Oct 2026

Group D · Late progression recorded progression-free survival: > 36 months

Patients	Median model risk percentile	Recorded progression	Model window agreement
56	**51**	**54.5 mo**	**25%**
5.2% of the cohort	cohort-wide risk ranking, out-of-fold	median; range 36.2–168.2 mo	estimate in this window · median |error| 23 mo



Most frequent path end-points in this group	Patients
SUMOylation of DNA damage response and repair proteins	6
Estrogen-dependent gene expression	5
Synthesis of PA	4
Extra-nuclear estrogen signaling	4
Post-translational protein phosphorylation	4
Interaction between L1 and Ankyrins	4
NRAGE signals death through JNK	3
Pregnenolone biosynthesis	3

Rules firing most often in this group	HR	Patients
F5-R1: M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	0.32	11
F5-R2: M stage = M0 AND 26.5 ≤ Age < 83.5	0.35	11
F2-R8: 40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	0.47	9
F3-R7: M stage = M0 AND 40.5 ≤ Age < 86.5	0.37	9
F3-R2: N stage = N0	0.37	6
F2-R7: M stage = M0 AND 40.5 ≤ Age < 83.5	0.39	6

Most common driver alterations

PIK3CA mutated (29%), TP53 mutated (27%), CCND1 amplified (20%), MYC amplified (18%), GATA3 mutated (12%), FGFR1 amplified (12%), CDH1 mutated (12%), ERBB2 amplified (11%), MAP2K4 mutated (7%), CBFB mutated (7%)

Patient index

Patient	Age	Stage	Subtype	Recorded	Model estimate	Window	Risk pct	Report file
TCGA-B6-A0RI	44	III	LumA	36.2 mo	29 mo	✗	71	D_progression_after_36m/TCGA-B6-A0RI.pdf
TCGA-B6-A0X7	62	X	LumA	36.3 mo	24 mo	✗	92	D_progression_after_36m/TCGA-B6-A0X7.pdf
TCGA-B6-A0I1	73	II	Basal	36.8 mo	15 mo	✗	50	D_progression_after_36m/TCGA-B6-A0I1.pdf
TCGA-BH-A0C1	61	III	LumA	36.9 mo	29 mo	✗	77	D_progression_after_36m/TCGA-BH-A0C1.pdf
TCGA-A8-A08J	52	IV	Her2	37.1 mo	6 mo	✗	100	D_progression_after_36m/TCGA-A8-A08J.pdf
TCGA-AR-A1AH	51	II	Basal	37.3 mo	20 mo	✗	64	D_progression_after_36m/TCGA-AR-A1AH.pdf
TCGA-BH-A203	78	II	Her2	38.6 mo	30 mo	✗	66	D_progression_after_36m/TCGA-BH-A203.pdf
TCGA-E2-A15O	89	I	LumA	39.8 mo	8 mo	✗	1	D_progression_after_36m/TCGA-E2-A15O.pdf
TCGA-A2-A0SX	48	I	Basal	40.7 mo	23 mo	✗	24	D_progression_after_36m/TCGA-A2-A0SX.pdf
TCGA-A2-A25B	39	II	LumB	42.4 mo	38 mo	✓	13	D_progression_after_36m/TCGA-A2-A25B.pdf
TCGA-E2-A1LL	73	III	Basal	42.8 mo	17 mo	✗	93	D_progression_after_36m/TCGA-E2-A1LL.pdf
TCGA-AR-A2LK	62	III	LumB	42.9 mo	31 mo	✗	86	D_progression_after_36m/TCGA-AR-A2LK.pdf
TCGA-AR-A0TZ	43	III	LumA	43.2 mo	41 mo	✓	98	D_progression_after_36m/TCGA-AR-A0TZ.pdf
TCGA-B6-A0X5	61	II	LumB	44.2 mo	30 mo	✗	41	D_progression_after_36m/TCGA-B6-A0X5.pdf
TCGA-A2-A0EW	53	III	LumA	44.5 mo	32 mo	✗	77	D_progression_after_36m/TCGA-A2-A0EW.pdf
TCGA-A2-A0CK	60	III	LumA	44.8 mo	39 mo	✓	75	D_progression_after_36m/TCGA-A2-A0CK.pdf
TCGA-B6-A0IN	45	X	LumB	46.4 mo	24 mo	✗	91	D_progression_after_36m/TCGA-B6-A0IN.pdf
TCGA-AR-A24L	26	II	LumA	46.5 mo	51 mo	✓	38	D_progression_after_36m/TCGA-AR-A24L.pdf
TCGA-A2-A04V	39	II	LumA	48.3 mo	29 mo	✗	7	D_progression_after_36m/TCGA-A2-A04V.pdf
TCGA-BH-A1EX	67	II	LumA	49.6 mo	40 mo	✓	46	D_progression_after_36m/TCGA-BH-A1EX.pdf
TCGA-B6-A402	47	I	missing	49.8 mo	31 mo	✗	26	D_progression_after_36m/TCGA-B6-A402.pdf
TCGA-HN-A2OB	45	II	LumA	50.7 mo	62 mo	✓	22	D_progression_after_36m/TCGA-HN-A2OB.pdf
TCGA-AR-A0U2	47	III	LumB	52.9 mo	34 mo	✗	82	D_progression_after_36m/TCGA-AR-A0U2.pdf
TCGA-AO-A03U	31	I	Normal	53.1 mo	33 mo	✗	4	D_progression_after_36m/TCGA-AO-A03U.pdf
TCGA-E2-A1L7	40	III	LumB	53.2 mo	38 mo	✓	78	D_progression_after_36m/TCGA-E2-A1L7.pdf
TCGA-AR-A256	45	II	Basal	53.9 mo	16 mo	✗	63	D_progression_after_36m/TCGA-AR-A256.pdf
TCGA-BH-A1FR	73	III	LumA	54.0 mo	25 mo	✗	72	D_progression_after_36m/TCGA-BH-A1FR.pdf
TCGA-AO-A03N	59	II	LumB	54.1 mo	30 mo	✗	35	D_progression_after_36m/TCGA-AO-A03N.pdf
TCGA-BH-A1FL	69	II	LumA	55.0 mo	49 mo	✓	44	D_progression_after_36m/TCGA-BH-A1FL.pdf
TCGA-BH-A1FU	44	I	Normal	55.5 mo	32 mo	✗	6	D_progression_after_36m/TCGA-BH-A1FU.pdf
TCGA-BH-A1EW	38	II	LumA	55.7 mo	57 mo	✓	10	D_progression_after_36m/TCGA-BH-A1EW.pdf
TCGA-BH-A0BQ	39	I	LumA	56.2 mo	34 mo	✗	1	D_progression_after_36m/TCGA-BH-A0BQ.pdf
TCGA-3C-AAAU	55	X	LumA	59.4 mo	29 mo	✗	69	D_progression_after_36m/TCGA-3C-AAAU.pdf
TCGA-E2-A1IE	61	II	LumA	59.5 mo	35 mo	✗	21	D_progression_after_36m/TCGA-E2-A1IE.pdf
TCGA-AO-A03P	54	II	LumB	63.2 mo	35 mo	✗	29	D_progression_after_36m/TCGA-AO-A03P.pdf
TCGA-B6-A0WY	40	III	LumA	66.1 mo	20 mo	✗	77	D_progression_after_36m/TCGA-B6-A0WY.pdf
TCGA-B6-A0RS	38	II	Her2	67.1 mo	24 mo	✗	64	D_progression_after_36m/TCGA-B6-A0RS.pdf
TCGA-B6-A0RH	51	II	Her2	71.2 mo	21 mo	✗	71	D_progression_after_36m/TCGA-B6-A0RH.pdf
TCGA-BH-A1FN	34	II	LumB	72.1 mo	53 mo	✓	36	D_progression_after_36m/TCGA-BH-A1FN.pdf
TCGA-BH-A1FE	31	II	LumA	74.7 mo	69 mo	✓	10	D_progression_after_36m/TCGA-BH-A1FE.pdf
TCGA-A8-A08T	64	IV	LumA	84.1 mo	22 mo	✗	99	D_progression_after_36m/TCGA-A8-A08T.pdf
TCGA-AR-A24O	43	III	LumA	84.2 mo	27 mo	✗	82	D_progression_after_36m/TCGA-AR-A24O.pdf
TCGA-BH-A0C3	47	I	LumB	87.9 mo	36 mo	✗	29	D_progression_after_36m/TCGA-BH-A0C3.pdf
TCGA-B6-A0IB	64	IV	LumB	89.3 mo	7 mo	✗	99	D_progression_after_36m/TCGA-B6-A0IB.pdf
TCGA-BH-A18K	46	I	LumA	90.8 mo	34 mo	✗	2	D_progression_after_36m/TCGA-BH-A18K.pdf
TCGA-B6-A0RP	73	III	LumA	91.6 mo	15 mo	✗	90	D_progression_after_36m/TCGA-B6-A0RP.pdf
TCGA-A2-A25E	34	III	LumB	98.3 mo	41 mo	✓	76	D_progression_after_36m/TCGA-A2-A25E.pdf
TCGA-GM-A2DA	46	II	LumA	99.1 mo	39 mo	✓	15	D_progression_after_36m/TCGA-GM-A2DA.pdf
TCGA-B6-A0RU	40	I	Basal	101.1 mo	22 mo	✗	52	D_progression_after_36m/TCGA-B6-A0RU.pdf

Patient	Age	Stage	Subtype	Recorded	Model estimate	Window	Risk pct	Report file
TCGA-BH-A1ES	35	II	missing	113.8 mo	58 mo	✓	63	D_progression_after_36m/TCGA-BH-A1ES.pdf
TCGA-BH-A1FB	60	II	missing	120.6 mo	35 mo	✗	60	D_progression_after_36m/TCGA-BH-A1FB.pdf
TCGA-B6-A0WT	61	II	LumA	127.0 mo	30 mo	✗	17	D_progression_after_36m/TCGA-B6-A0WT.pdf
TCGA-AR-A2LE	69	I	LumA	138.2 mo	33 mo	✗	3	D_progression_after_36m/TCGA-AR-A2LE.pdf
TCGA-B6-A2IU	62	II	LumA	138.9 mo	31 mo	✗	7	D_progression_after_36m/TCGA-B6-A2IU.pdf
TCGA-B6-A0IG	50	II	LumA	146.5 mo	33 mo	✗	32	D_progression_after_36m/TCGA-B6-A0IG.pdf
TCGA-B6-A0WZ	50	II	LumA	168.2 mo	33 mo	✗	14	D_progression_after_36m/TCGA-B6-A0WZ.pdf

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**44**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**EUR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0RI-01** · Aneuploidy score: **3** · TMB (nonsyn., /Mb): **0.67** · MSIsensor score: **0.02** · AJCC edition: **3RD**

2 · Progression assessment

Model risk (cohort percentile)

71st

High risk tier · fused neuro-symbolic model



Probability of progression

12% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 24%

Model-estimated progression time

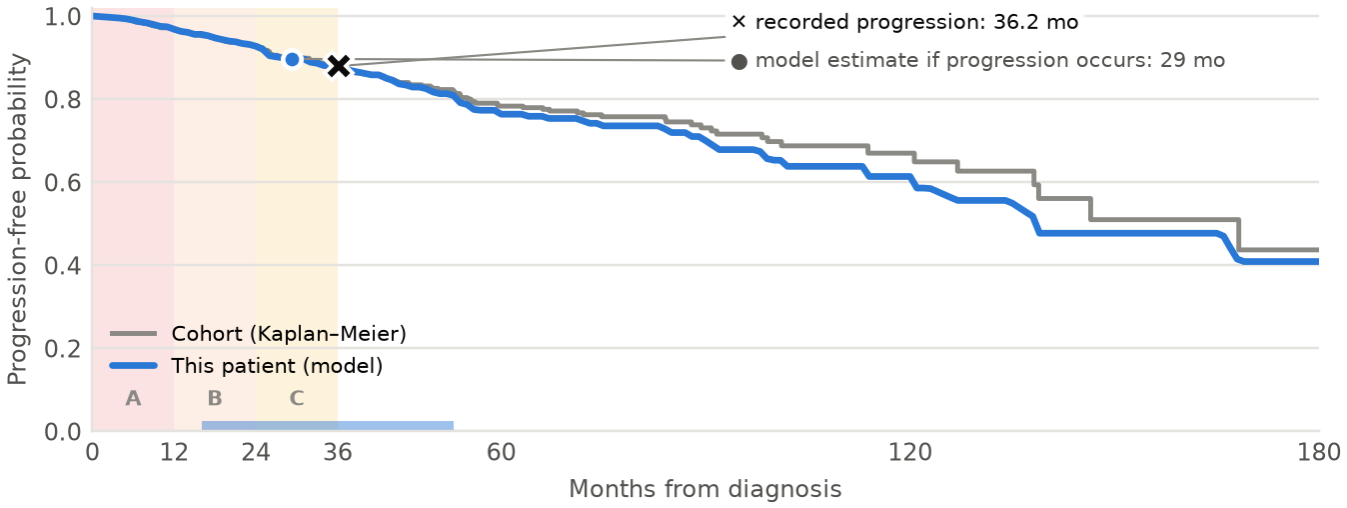
≈ 29 months

Window 24–36 months · IQR 16–53 mo
if progression occurs

Recorded progression

36.2 months

Group D · Late progression
✗ recorded window differs · |error| 7 mo



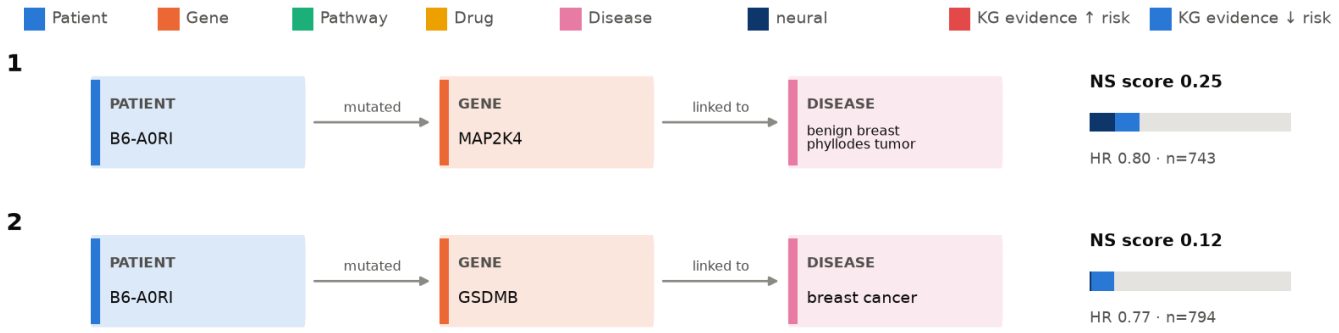
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 44-year-old patient has a LumA tumour, AJCC stage III (T4, N1, M0). The fused neuro-symbolic model places them in the **71st risk percentile** (high tier). The most significant rule covering them (*Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **MAP2K4** to **benign breast phyllodes tumor**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.80). Progression was recorded at 36.2 months; the model's estimate given progression was 29 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

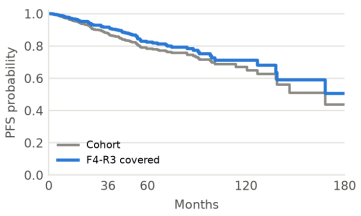


None of the 16 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-B6-A0RI** mutated → **MAP2K4** linked to → **benign breast phyllodes tumor**	21	↓ **slower progression** HR 0.80, z -0.9, n = 743 carriers of “benign breast phyllodes tumor”	**0.25**
2	**TCGA-B6-A0RI** mutated → **GSDMB** linked to → **breast cancer**	1	↓ **slower progression** HR 0.77, z -0.8, n = 794 carriers of “breast cancer”	**0.12**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

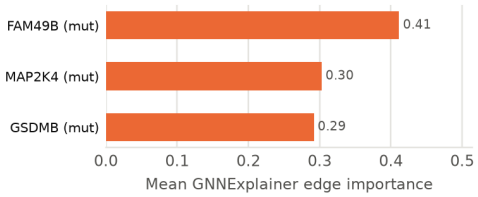
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R3 ↓ **lower risk**	Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	662 pts 77 events	**0.36** p 2.3e-07	not reached
F4-R4 ↓ **lower risk**	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo
F4-R5 ↓ **lower risk**	M stage = M0 AND Age ≥ 34.5	713 pts 89 events	**0.47** p 1.0e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MAP2K4 · driver	Mutation	p.C266Hfs*8	12	0.30
FAM49B	Mutation	p.L4R	0	0.41
GSDMB	Mutation	p.Q85*, p.S107L	4	0.29



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A1-A0SJ	39	III	LumA	0.82	no progression, followed 13.7 mo
TCGA-AR-A1AU	39	III	LumA	0.71	no progression, followed 94.3 mo
TCGA-BH-A1FR	73	III	LumA	0.70	progressed at 54.0 mo (group D)
TCGA-BH-A0AZ	47	III	LumA	0.70	no progression, followed 63.1 mo
TCGA-B6-A0RO	71	III	LumA	0.68	no progression, followed 162.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE X**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0X7-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **0.53** · MSIsensor score: **0.16** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

92nd

High risk tier · fused neuro-symbolic model



Probability of progression

17% by 36 mo

12 mo 5% · 24 mo 11% · 60 mo 33%

Model-estimated progression time

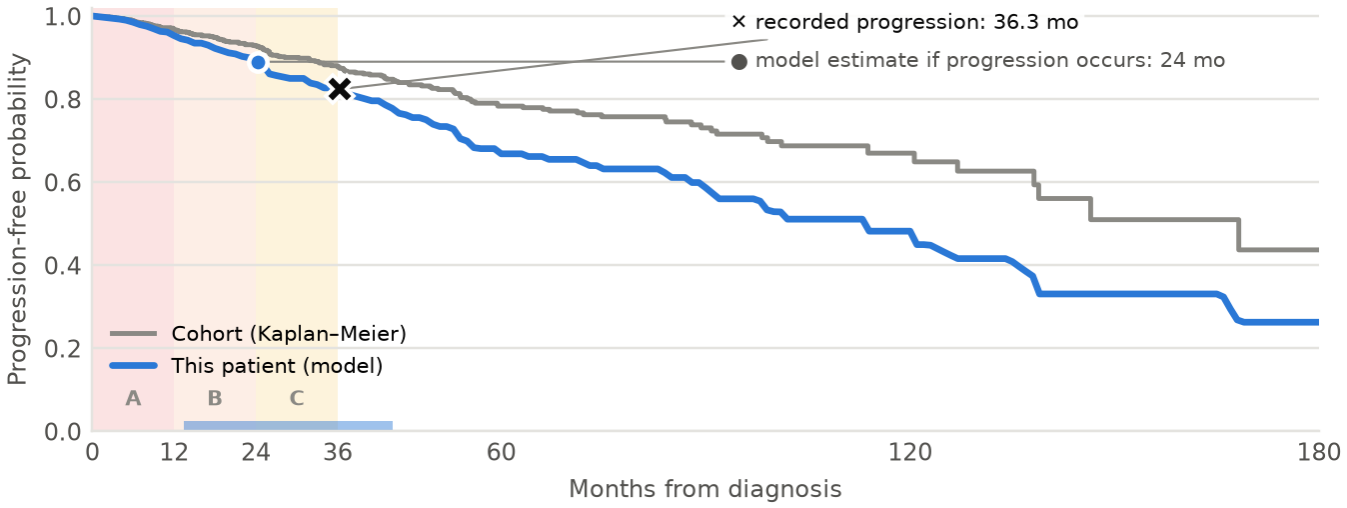
≈ 24 months

Window 24–36 months · IQR 13–44 mo
if progression occurs

Recorded progression

36.3 months

Group D · Late progression
✗ recorded window differs · |error| 12 mo



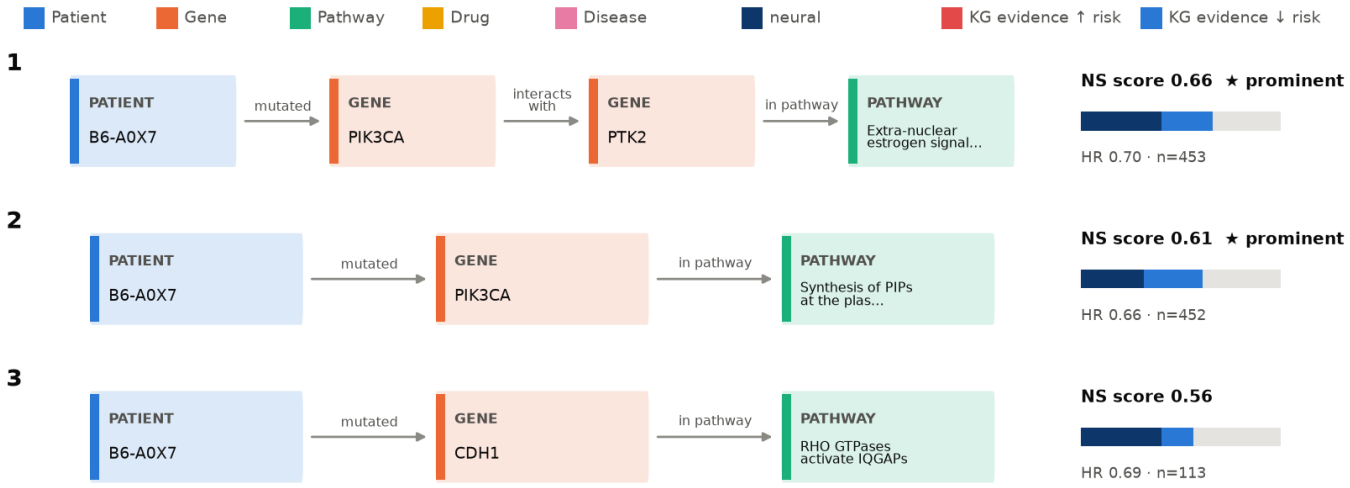
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a LumA tumour, AJCC stage X (T1, missing, MX). The fused neuro-symbolic model places them in the **92nd risk percentile** (high tier). The most significant rule covering them (*26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PIK3CA** to **Extra-nuclear estrogen signaling**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.70). Progression was recorded at 36.3 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

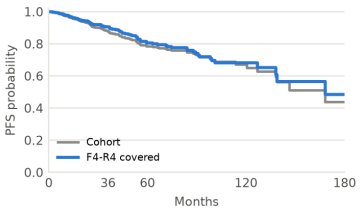


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-B6-A0X7** mutated → **PIK3CA** interacts with → **PTK2** in pathway → **Extra-nuclear estrogen signaling**	67	↓ **slower progression** HR 0.70, z -1.9, n = 453 carriers of “Extra-nuclear estrogen signaling”	**0.66**
2 ★	**TCGA-B6-A0X7** mutated → **PIK3CA** in pathway → **Synthesis of PIPs at the plasma membrane**	52	↓ **slower progression** HR 0.66, z -2.2, n = 452 carriers of “Synthesis of PIPs at the plasma membrane”	**0.61**
3	**TCGA-B6-A0X7** mutated → **CDH1** in pathway → **RHO GTPases activate IQGAPs**	67	↓ **slower progression** HR 0.69, z -1.2, n = 113 carriers of “RHO GTPases activate IQGAPs”	**0.56**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

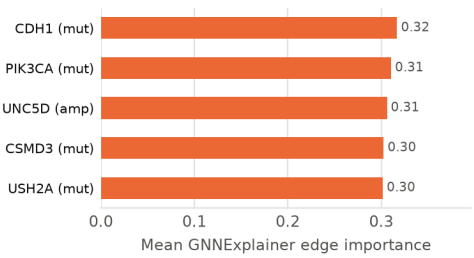
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R4 ↓ **lower risk**	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.K675Nfs*2	17	0.32
PIK3CA · driver	Mutation	p.H1047R	70	0.31
UNC5D	High-level amplification	GISTIC ±2	1	0.31
CSMD3	Mutation	p.R615=	0	0.30
USH2A	Mutation	p.T4888M	0	0.30



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A0IC	90	X	LumA	0.72	no progression, followed 50.7 mo
TCGA-XX-A899	46	III	LumA	0.65	no progression, followed 15.4 mo
TCGA-E2-A576	69	I	LumA	0.60	no progression, followed 34.3 mo
TCGA-B6-A0WS	58	X	LumA	0.60	no progression, followed 97.5 mo
TCGA-EW-A1J2	50	II	LumA	0.59	no progression, followed 13.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**73**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

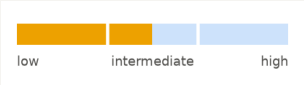
Sample: **TCGA-B6-A0I1-01** · Aneuploidy score: **9** · TMB (nonsyn., /Mb): **1.7** · MSIsensor score: **0.82** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

50th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 18%

Model-estimated progression time

≈ 15 months

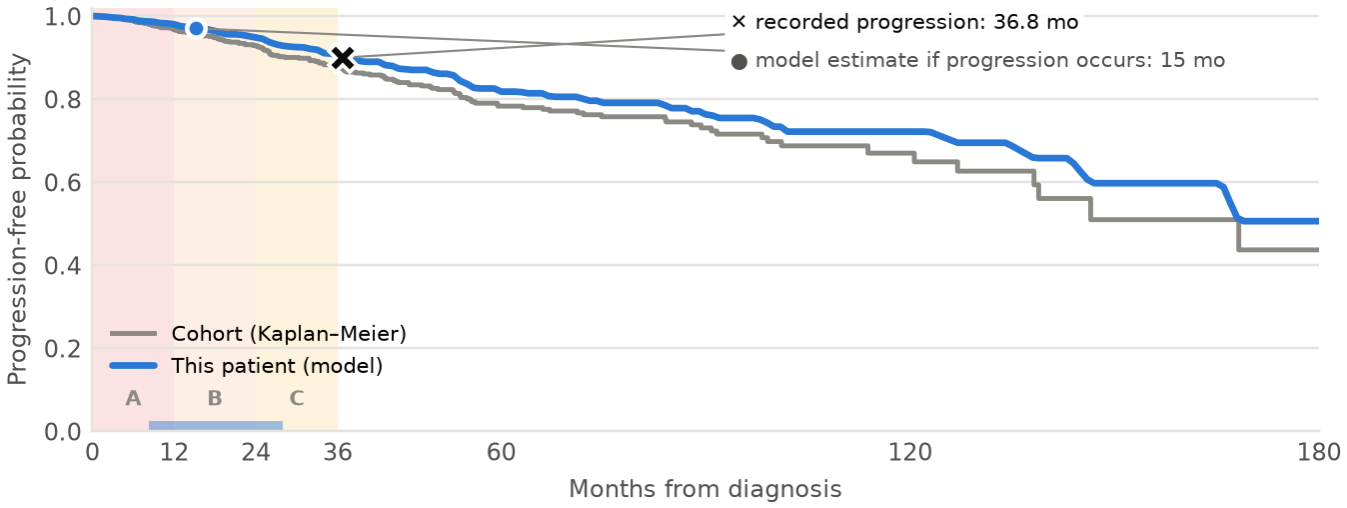
Window 12–24 months · IQR 8–28 mo if progression occurs

Recorded progression

36.8 months

Group D · Late progression

✗ recorded window differs · |error| 22 mo



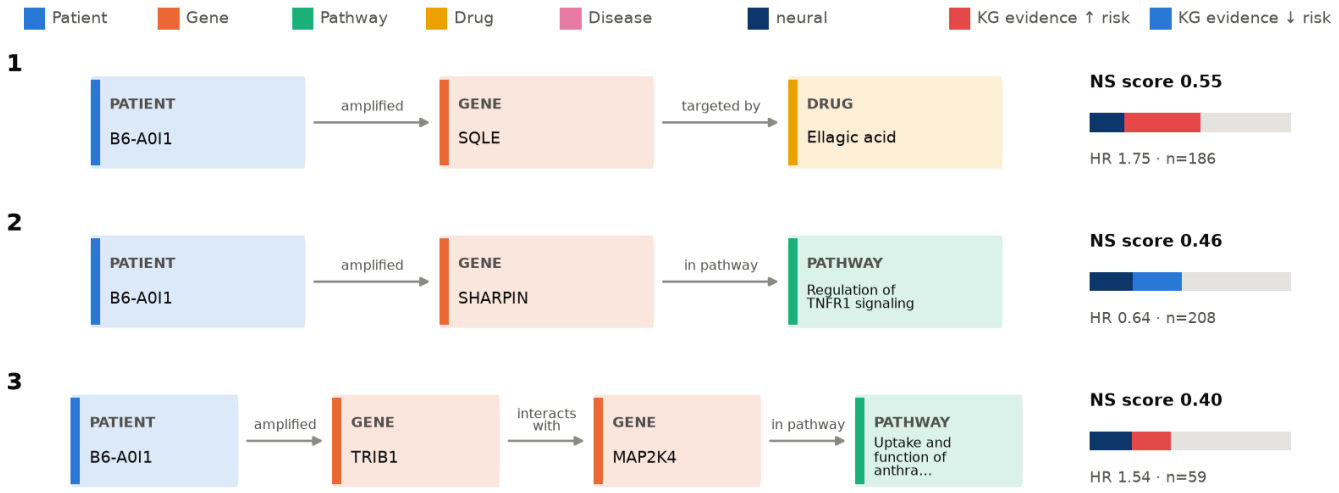
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 73-year-old patient has a Basal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **50th risk percentile** (intermediate tier). The most significant rule covering them (*N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **SQLE** to **Ellagic acid**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.75). Progression was recorded at 36.8 months; the model's estimate given progression was 15 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

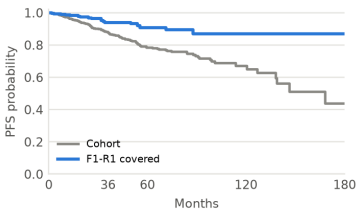


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-B6-A0I1** amplified → **SQLE** targeted by → **Ellagic acid**	29	↑ **faster progression** HR 1.75, z +2.8, n = 186 carriers of “Ellagic acid”	**0.55**
2	**TCGA-B6-A0I1** amplified → **SHARPIN** in pathway → **Regulation of TNFR1 signaling**	35	↓ **slower progression** HR 0.64, z -1.8, n = 208 carriers of “Regulation of TNFR1 signaling”	**0.46**
3	**TCGA-B6-A0I1** amplified → **TRIB1** interacts with → **MAP2K4** in pathway → **Uptake and function of anthrax toxins**	35	↑ **faster progression** HR 1.54, z +1.4, n = 59 carriers of “Uptake and function of anthrax toxins”	**0.40**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

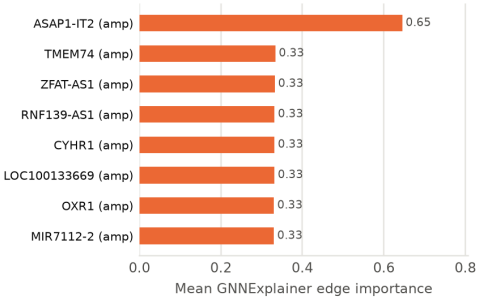
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R1 ↓ **lower risk**	N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45	365 pts 19 events	**0.26** p 4.0e-09	not reached
F1-R7 ↓ **lower risk**	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo
F1-R2 ↓ **lower risk**	N stage = N0 AND Age ≥ 31.5	410 pts 29 events	**0.36** p 4.5e-07	not reached
F1-R8 ↓ **lower risk**	AJCC stage = II AND Age ≥ 39.5 AND TMB (nonsynonymous) < 141.95	452 pts 39 events	**0.47** p 7.9e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
MYC · driver	Mutation	p.P233T	20	0.32
TP53 · driver	Mutation	p.R248Q	57	0.31
ASAP1-IT2	High-level amplification	GISTIC ±2	0	0.65
TMEM74	High-level amplification	GISTIC ±2	0	0.33
ZFAT-AS1	High-level amplification	GISTIC ±2	0	0.33
RNF139-AS1	High-level amplification	GISTIC ±2	0	0.33
CYHR1	High-level amplification	GISTIC ±2	0	0.33
LOC100133669	High-level amplification	GISTIC ±2	0	0.33
OXR1	High-level amplification	GISTIC ±2	0	0.33
MIR7112-2	High-level amplification	GISTIC ±2	0	0.33
TATDN1	High-level amplification	GISTIC ±2	0	0.33



405 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487) … and 140 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A04T	62	II	Basal	0.92	no progression, followed 73.8 mo
TCGA-E2-A14R	62	II	Basal	0.91	progressed at 31.9 mo (group C)
TCGA-A7-A0DA	62	II	Basal	0.90	no progression, followed 35.7 mo
TCGA-D8-A1XQ	69	II	Basal	0.90	no progression, followed 16.4 mo
TCGA-AR-A0TP	43	II	Basal	0.89	no progression, followed 140.5 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**61**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-BH-A0C1-01** · Aneuploidy score: **19** · TMB (nonsyn., /Mb): **1.07** · MSIsensor score: **0.08** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

77th

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 25%

Model-estimated progression time

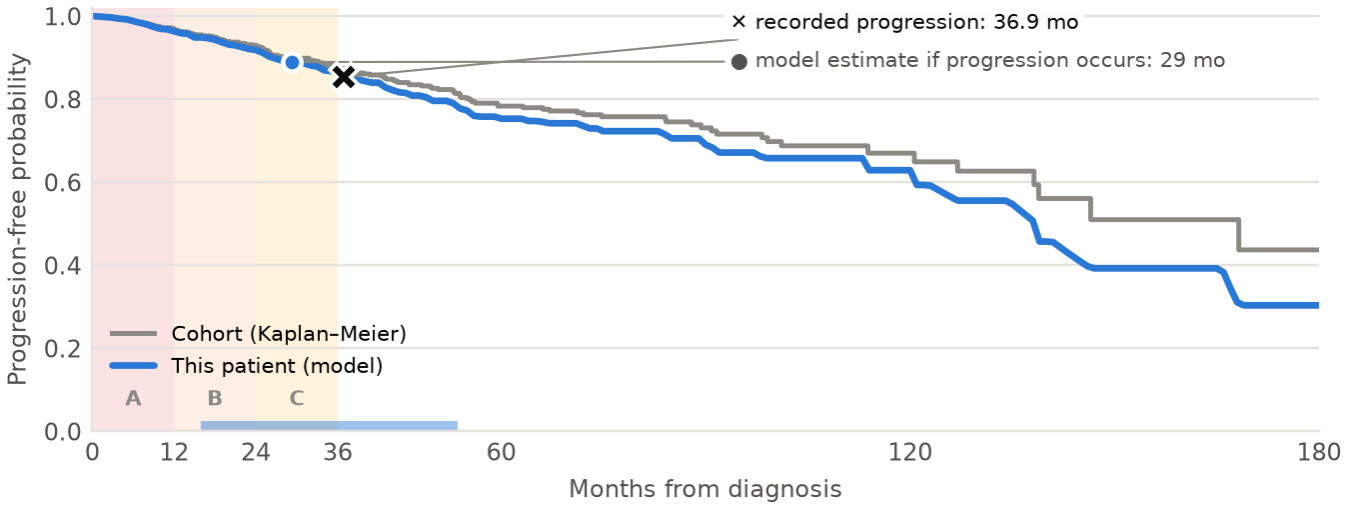
≈ 29 months

Window 24–36 months · IQR 16–54 mo
if progression occurs

Recorded progression

36.9 months

Group D · Late progression
✗ recorded window differs · |error| 8 mo



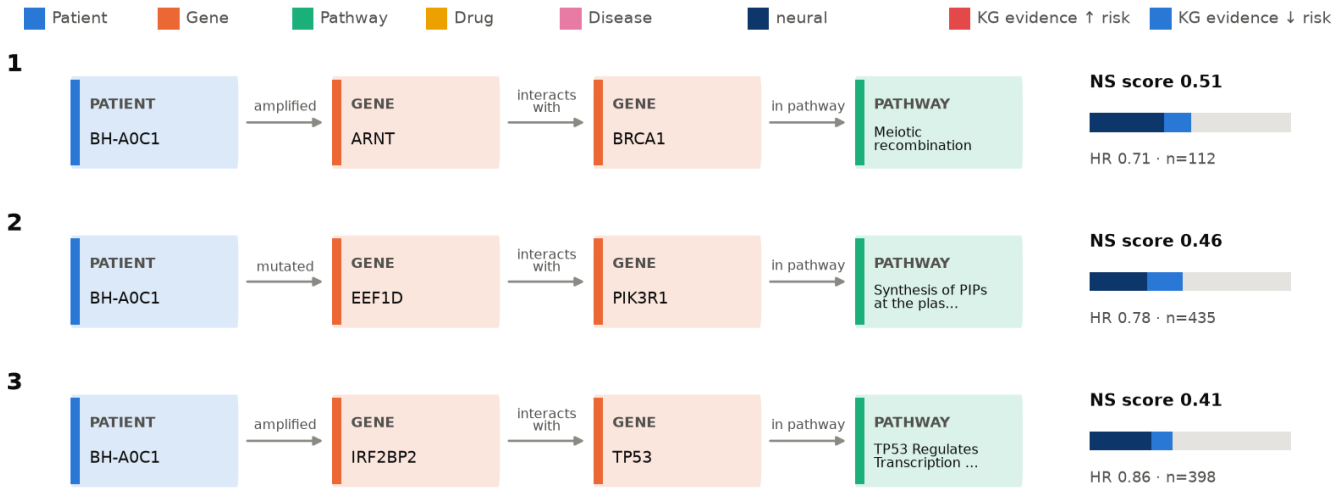
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 61-year-old patient has a LumA tumour, AJCC stage III (T3, N2, M0). The fused neuro-symbolic model places them in the **77th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ARNT** to **Meiotic recombination**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.71). Progression was recorded at 36.9 months; the model's estimate given progression was 29 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

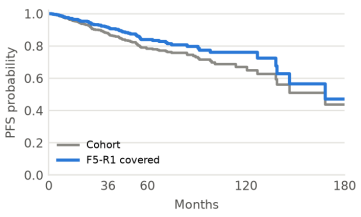


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-BH-A0C1** amplified → **ARNT** interacts with → **BRCA1** in pathway → **Meiotic recombination**	61	↓ **slower progression** HR 0.71, z -1.0, n = 112 carriers of “Meiotic recombination”	**0.51**
2	**TCGA-BH-A0C1** mutated → **EEF1D** interacts with → **PIK3R1** in pathway → **Synthesis of PIPs at the plasma membrane**	48	↓ **slower progression** HR 0.78, z -1.3, n = 435 carriers of “Synthesis of PIPs at the plasma membrane”	**0.46**
3	**TCGA-BH-A0C1** amplified → **IRF2BP2** interacts with → **TP53** in pathway → **TP53 Regulates Transcription of DNA Repair Genes**	51	↓ **slower progression** HR 0.86, z -0.8, n = 398 carriers of “TP53 Regulates Transcription of DNA Repair Genes”	**0.41**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

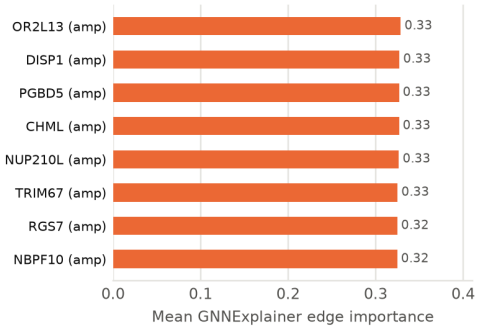
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ **lower risk**	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ **lower risk**	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R4 ↓ **lower risk**	TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5	567 pts 64 events	**0.56** p 2.1e-03	168 mo
F5-R5 ↓ **lower risk**	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.P159Lfs*56	17	0.32
RUNX1 · driver	Mutation	p.G135D	18	0.32
ERBB2 · driver	Mutation	p.V777L	55	0.31
OR2L13	High-level amplification	GISTIC ±2	1	0.33
DISP1	High-level amplification	GISTIC ±2	0	0.33
PGBD5	High-level amplification	GISTIC ±2	0	0.33
CHML	High-level amplification	GISTIC ±2	2	0.33
NUP210L	High-level amplification	GISTIC ±2	0	0.33
TRIM67	High-level amplification	GISTIC ±2	0	0.33
RGS7	High-level amplification	GISTIC ±2	2	0.32
NBPF10	High-level amplification	GISTIC ±2	0	0.32
FMN2	High-level amplification	GISTIC ±2	0	0.32



114 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 62 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A27I	58	III	LumA	0.88	no progression, followed 14.4 mo
TCGA-A8-A0AB	54	II	LumA	0.88	no progression, followed 17.0 mo
TCGA-OL-A66N	59	III	LumA	0.88	no progression, followed 26.0 mo
TCGA-BH-A0B5	40	III	LumA	0.85	no progression, followed 70.2 mo
TCGA-BH-A0BA	51	III	LumA	0.84	no progression, followed 37.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**52**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**—**	PAM50 subtype	**Her2**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N3**
Radiation therapy	**No**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8541/3**

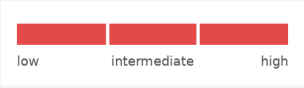
Sample: **TCGA-A8-A08J-01** · Aneuploidy score: **16** · TMB (nonsyn., /Mb): **1.2** · MSIsensor score: **0.49** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

100th

High risk tier · fused neuro-symbolic model



Probability of progression

84% by 36 mo

12 mo 38% · 24 mo 67% · 60 mo 97%

Model-estimated progression time

≈ 6 months

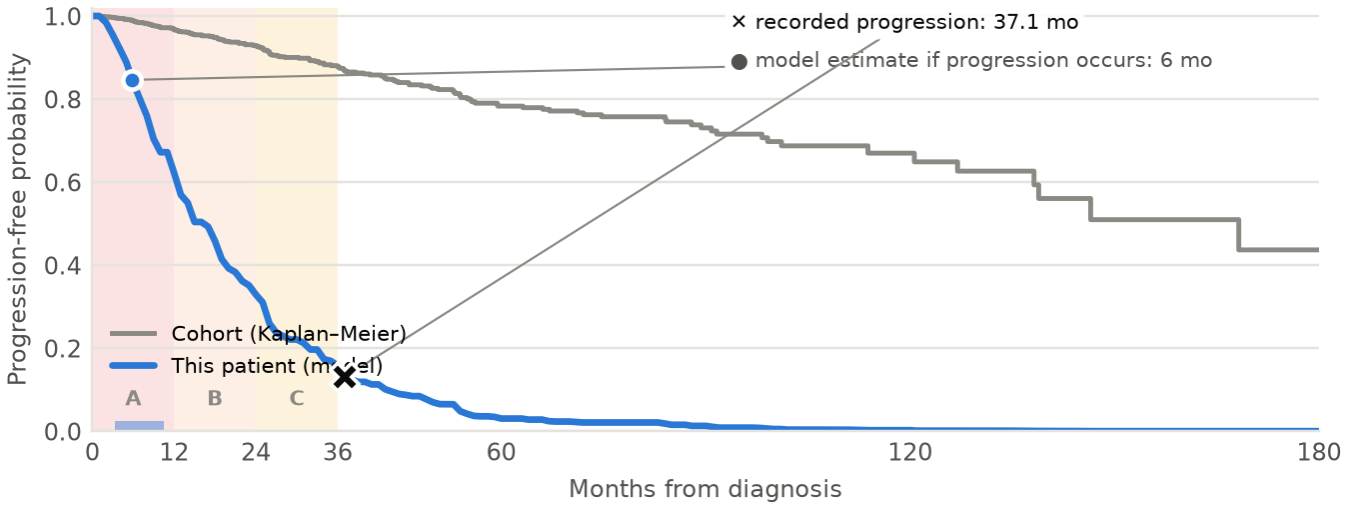
Window ≤ 12 months · IQR 3–10 mo if progression occurs

Recorded progression

37.1 months

Group D · Late progression

✗ **recorded window differs** · |error| 31 mo



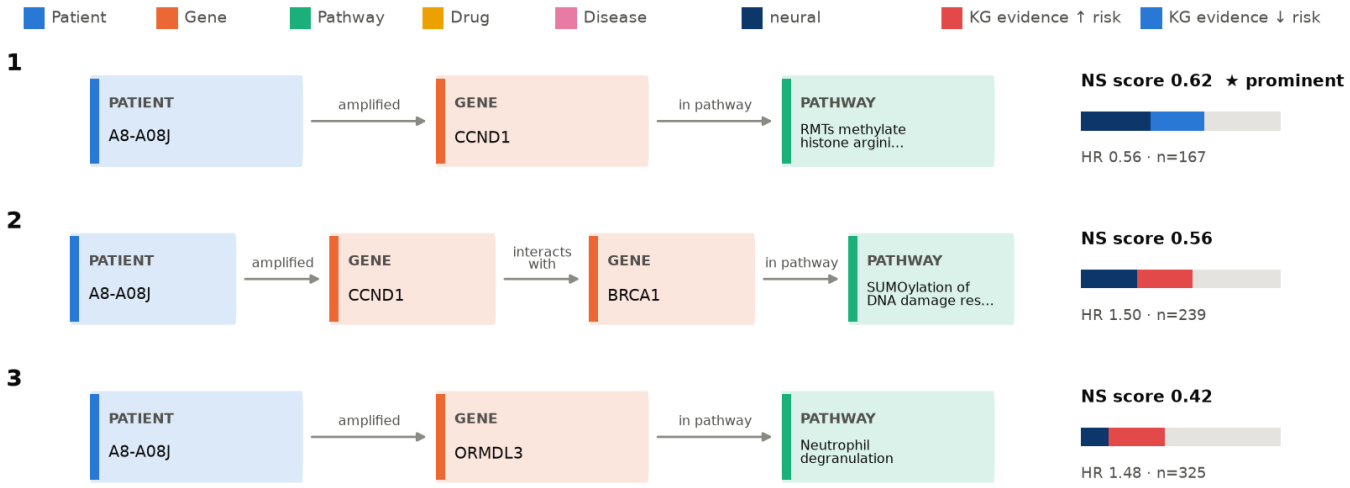
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 52-year-old patient has a Her2 tumour, AJCC stage IV (T4, N3, M1). The fused neuro-symbolic model places them in the **100th risk percentile** (high tier). The most significant rule covering them (*N stage = N3 AND Age < 72.0*) marks a group with a progression hazard 4.0× that of the remaining patients. The highest-scoring knowledge-graph path runs from **CCND1** to **RMTs methylate histone arginines**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.56). Progression was recorded at 37.1 months; the model's estimate given progression was 6 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

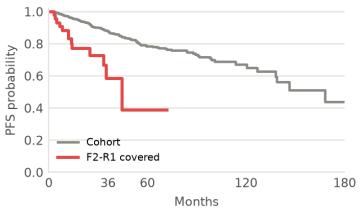


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-A8-A08J** amplified → **CCND1** in pathway → **RMTs methylate histone arginines**	58	↓ **slower progression** HR 0.56, z -2.0, n = 167 carriers of “RMTs methylate histone arginines”	**0.62**
2	**TCGA-A8-A08J** amplified → **CCND1** interacts with → **BRCA1** in pathway → **SUMOylation of DNA damage response and repair proteins**	47	↑ **faster progression** HR 1.50, z +2.1, n = 239 carriers of “SUMOylation of DNA damage response and repair proteins”	**0.56**
3	**TCGA-A8-A08J** amplified → **ORMDL3** in pathway → **Neutrophil degranulation**	23	↑ **faster progression** HR 1.48, z +2.1, n = 325 carriers of “Neutrophil degranulation”	**0.42**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

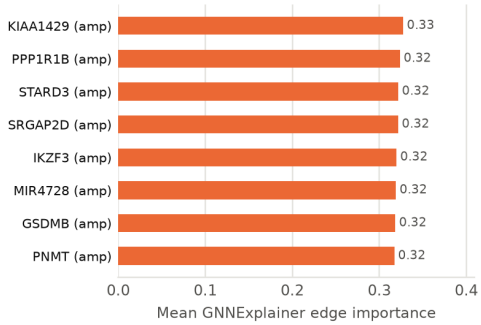
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R1 ↑ **higher risk**	N stage = N3 AND Age < 72.0	44 pts 13 events	**4.04** p 3.2e-06	44 mo
F2-R8 ↓ **lower risk**	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.32
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
GATA3 · driver	Mutation	p.S426Ifs*81	13	0.31
KIAA1429	High-level amplification	GISTIC ±2	0	0.33
PPP1R1B	High-level amplification	GISTIC ±2	1	0.32
STARD3	High-level amplification	GISTIC ±2	1	0.32
SRGAP2D	High-level amplification	GISTIC ±2	0	0.32
IKZF3	High-level amplification	GISTIC ±2	0	0.32
MIR4728	High-level amplification	GISTIC ±2	0	0.32
GSDMB	High-level amplification	GISTIC ±2	4	0.32
PNMT	High-level amplification	GISTIC ±2	16	0.32
ORMDL3	High-level amplification	GISTIC ±2	2	0.32



32 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 23 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-UU-A93S	63	IV	Her2	0.70	progressed at 3.8 mo (group A)
TCGA-B6-A0I9	62	IV	Her2	0.55	progressed at 11.9 mo (group A)
TCGA-PL-A8LX	35	IV	LumA	0.55	no progression, followed 0.2 mo
TCGA-E9-A22D	38	II	LumB	0.55	no progression, followed 41.0 mo
TCGA-A8-A08X	43	III	Her2	0.53	progressed at 12.0 mo (group B)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**51**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AR-A1AH-01** · Aneuploidy score: **12** · TMB (nonsyn., /Mb): **2.67** · MSIsensor score: **1.35** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

64th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

13% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 21%

Model-estimated progression time

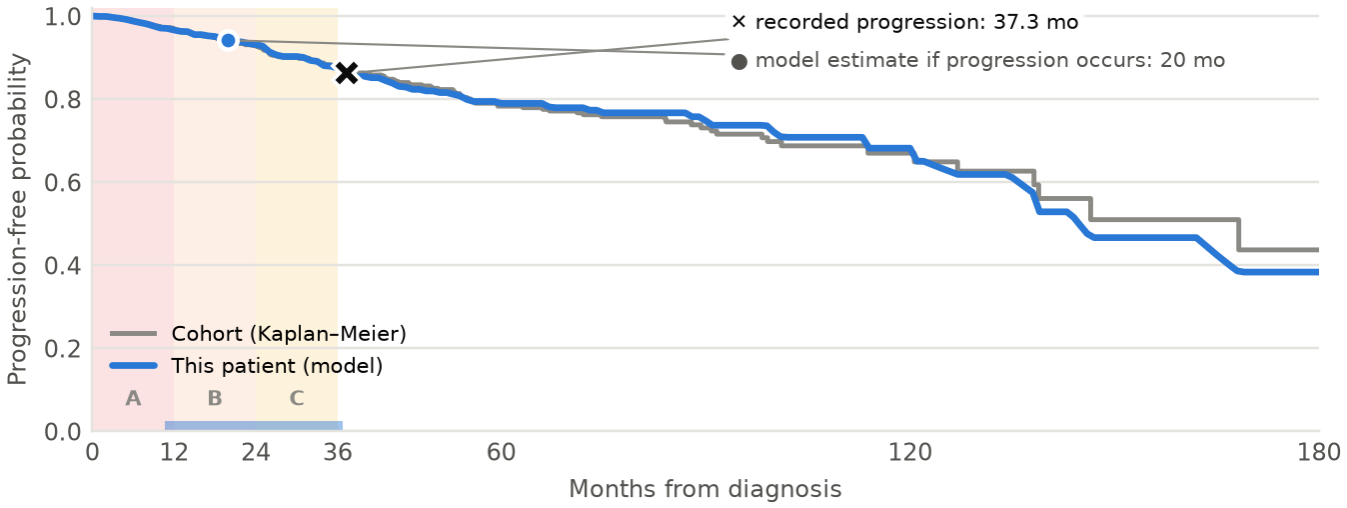
≈ 20 months

Window 12–24 months · IQR 11–37 mo
if progression occurs

Recorded progression

37.3 months

Group D · Late progression
✗ **recorded window differs** · |error| 18 mo



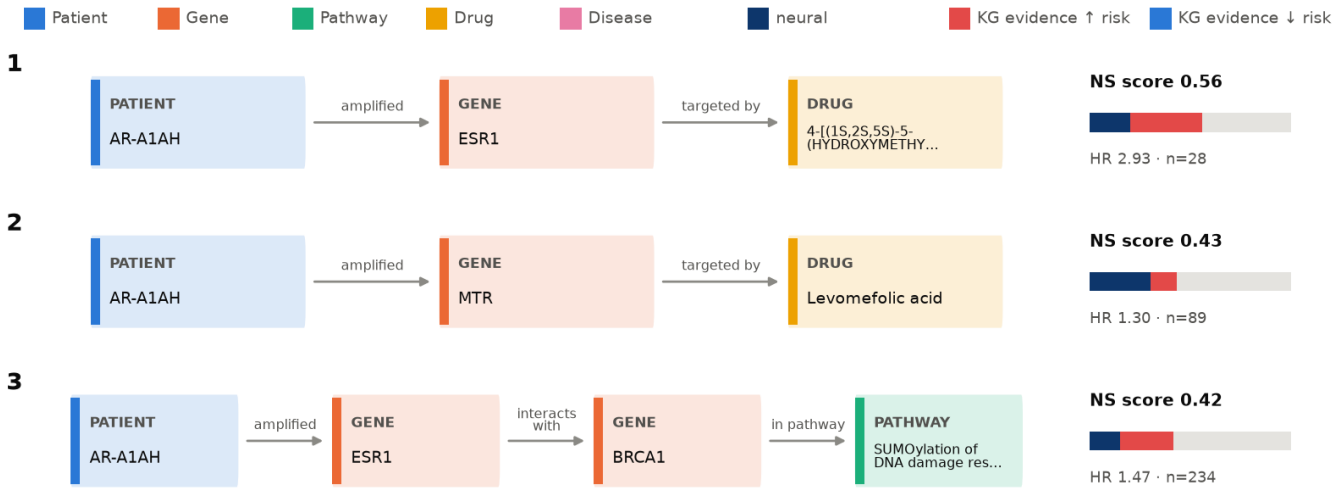
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 51-year-old patient has a Basal tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **64th risk percentile** (intermediate tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 86.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ESR1** to **4-[(1S,2S,5S)-5-(HYDROXYMETHYL)-6,8,9-TRIMETHYL-3-OXABICYCLO[3.3.1]NON-7-EN-2-YL]PHENOL**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 2.93). Progression was recorded at 37.3 months; the model's estimate given progression was 20 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

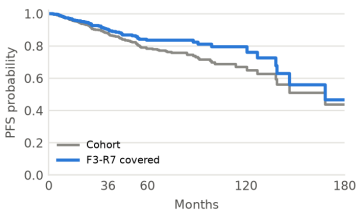


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-AR-A1AH** amplified → **ESR1** targeted by → **4-[(1S,2S,5S)-5-(HYDROXYMETHYL)-6,8,9-TRIMETHYL-3-OXABICYCLO[3.3.1]NON-7-EN-2-YL]PHENOL**	33	↑ **faster progression** HR 2.93, z +2.7, n = 28 carriers of “4-[(1S,2S,5S)-5-(HYDROXYMETHYL)-6,8,9-TRIMETHYL-3-OXABICYCLO[3.3.1]NON-7-EN-2-YL]PHENOL”	**0.56**
2	**TCGA-AR-A1AH** amplified → **MTR** targeted by → **Levomefolic acid**	51	↑ **faster progression** HR 1.30, z +0.9, n = 89 carriers of “Levomefolic acid”	**0.43**
3	**TCGA-AR-A1AH** amplified → **ESR1** interacts with → **BRCA1** in pathway → **SUMOylation of DNA damage response and repair proteins**	25	↑ **faster progression** HR 1.47, z +2.0, n = 234 carriers of “SUMOylation of DNA damage response and repair proteins”	**0.42**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

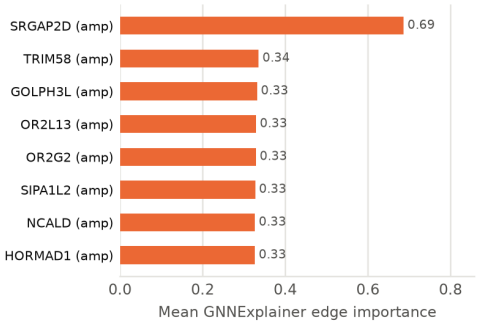
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo
F3-R8 ↓ lower risk	AJCC stage = II AND Age < 77.5 AND TMB (nonsynonymous) ≥ 0.08	433 pts 40 events	**0.50** p 3.2e-04	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ESR1 · driver	High-level amplification	GISTIC ±2	141	0.31
TP53 · driver	Mutation	p.G108Ffs*40	57	0.31
SRGAP2D	High-level amplification	GISTIC ±2	0	0.69
TRIM58	High-level amplification	GISTIC ±2	0	0.34
GOLPH3L	High-level amplification	GISTIC ±2	0	0.33
OR2L13	High-level amplification	GISTIC ±2	1	0.33
OR2G2	High-level amplification	GISTIC ±2	1	0.33
SIPA1L2	High-level amplification	GISTIC ±2	0	0.33
NCALD	High-level amplification	GISTIC ±2	1	0.33
HORMAD1	High-level amplification	GISTIC ±2	0	0.33
NBPF10	High-level amplification	GISTIC ±2	0	0.32
TPCN2	High-level amplification	GISTIC ±2	1	0.32



117 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ESR1 → Diethylstilbestrol (DB00255); ESR1 → Chlorotrianisene (DB00269); ESR1 → Conjugated estrogens (DB00286); ESR1 → Desogestrel (DB00304); ESR1 → Levonorgestrel (DB00367); ESR1 → Progesterone (DB00396); ESR1 → Lindane (DB00431); ESR1 → Raloxifene (DB00481); ESR1 → Toremifene (DB00539); ESR1 → Medroxyprogesterone acetate (DB00603) … and 164 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A7-A0CE	57	II	Basal	0.91	no progression, followed 35.3 mo
TCGA-A7-A4SE	54	II	Basal	0.88	no progression, followed 21.2 mo
TCGA-A8-A07F	65	II	LumA	0.86	no progression, followed 19.0 mo
TCGA-A8-A06Y	66	II	LumA	0.84	progressed at 26.0 mo (group C)
TCGA-B6-A0WX	40	III	Basal	0.84	progressed at 18.5 mo (group B)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group D** · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**78**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A203-01** · Aneuploidy score: **15** · TMB (nonsyn., /Mb): **2.03** · MSIsensor score: **0.31** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

66th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

12% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 21%

Model-estimated progression time

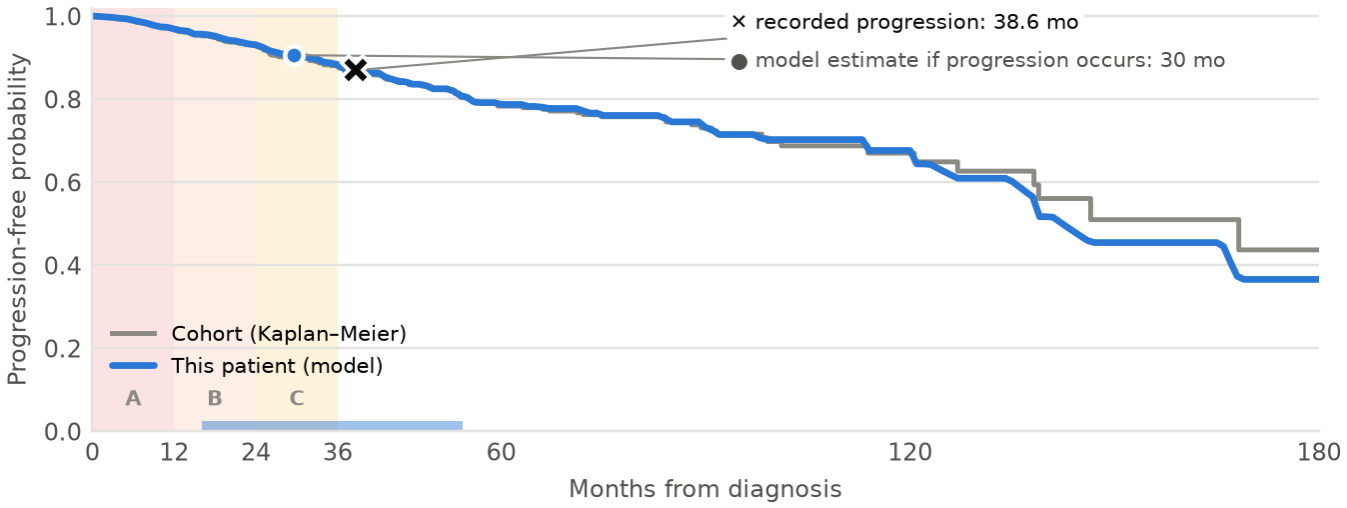
≈ 30 months

Window 24–36 months · IQR 16–54 mo
if progression occurs

Recorded progression

38.6 months

■ **Group D** · Late progression
✗ **recorded window differs** · |error| 9 mo



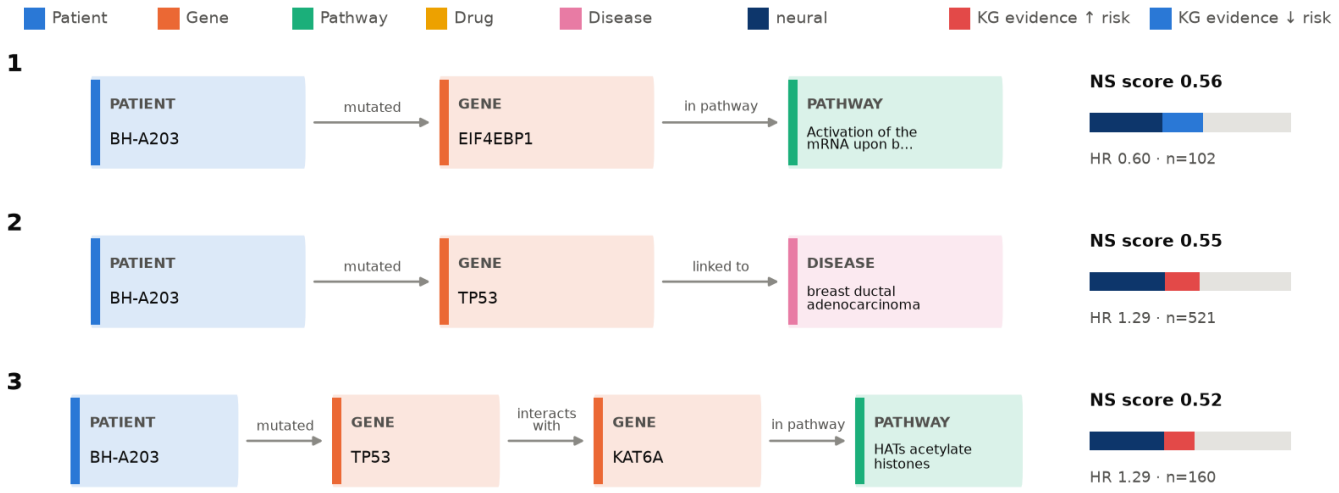
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 78-year-old patient has a Her2 tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **66th risk percentile** (intermediate tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **EIF4EBP1** to **Activation of the mRNA upon binding of the cap-binding complex and eIFs, and subsequent binding to 43S**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.60). Progression was recorded at 38.6 months; the model's estimate given progression was 30 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

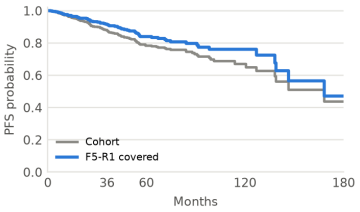


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-BH-A203** mutated → **EIF4EBP1** in pathway → **Activation of the mRNA upon binding of the cap-binding complex and eIFs, and subsequent binding to 43S**	60	↓ **slower progression** HR 0.60, z -1.5, n = 102 carriers of “Activation of the mRNA upon binding of the cap-binding complex and eIFs, and subsequent binding to 43S”	**0.56**
2	**TCGA-BH-A203** mutated → **TP53** linked to → **breast ductal adenocarcinoma**	62	↑ **faster progression** HR 1.29, z +1.3, n = 521 carriers of “breast ductal adenocarcinoma”	**0.55**
3	**TCGA-BH-A203** mutated → **TP53** interacts with → **KAT6A** in pathway → **HATs acetylate histones**	62	↑ **faster progression** HR 1.29, z +1.1, n = 160 carriers of “HATs acetylate histones”	**0.52**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

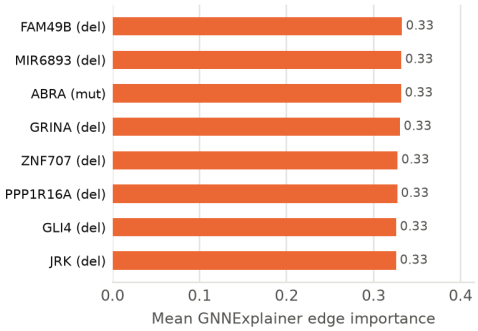
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ lower risk	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R3 ↓ lower risk	AJCC stage = II AND 39.5 ≤ Age < 78.5	422 pts 31 events	**0.36** p 4.2e-07	168 mo
F5-R4 ↓ lower risk	TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5	567 pts 64 events	**0.56** p 2.1e-03	168 mo
F5-R5 ↓ lower risk	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.H1047R	70	0.32
TP53 · driver	Mutation	p.R273H	57	0.31
FAM49B	Homozygous deletion	GISTIC ±2	0	0.33
MIR6893	Homozygous deletion	GISTIC ±2	0	0.33
ABRA	Mutation	p.E5K	0	0.33
GRINA	Homozygous deletion	GISTIC ±2	0	0.33
ZNF707	Homozygous deletion	GISTIC ±2	1	0.33
PPP1R16A	Homozygous deletion	GISTIC ±2	0	0.33
GLI4	Homozygous deletion	GISTIC ±2	0	0.33
JRK	Homozygous deletion	GISTIC ±2	0	0.33
EPPK1	Homozygous deletion	GISTIC ±2	0	0.33
LYNX1	Homozygous deletion	GISTIC ±2	0	0.33



163 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347) … and 78 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-EW-A6SD	32	II	Her2	0.80	no progression, followed 33.2 mo
TCGA-A2-A0D1	76	II	Her2	0.79	no progression, followed 34.6 mo
TCGA-C8-A12K	80	II	Basal	0.78	no progression, followed 0.0 mo
TCGA-C8-A135	64	II	Her2	0.77	no progression, followed 12.9 mo
TCGA-AR-A0TS	46	II	Basal	0.76	no progression, followed 84.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group D** · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**89**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**—**	Pathologic T	**T1C**
Prior diagnosis	**Yes**	Pathologic N	**NX**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

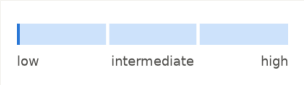
Sample: **TCGA-E2-A15O-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **0.9** · MSIsensor score: **0.19** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

1st

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 12%

Model-estimated progression time

≈ 8 months

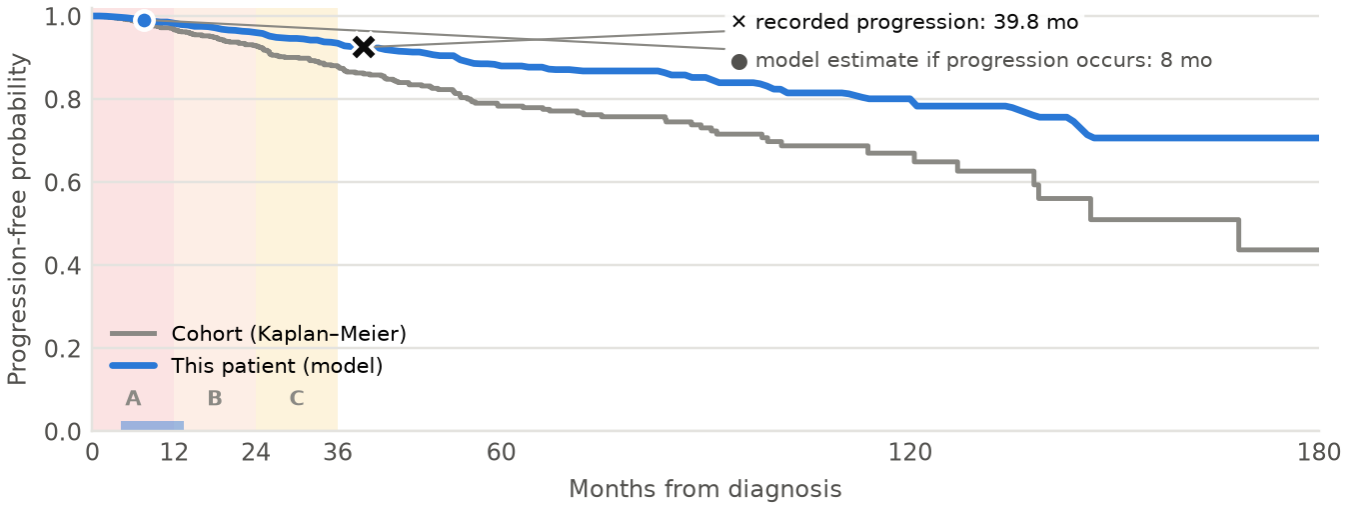
Window ≤ 12 months · IQR 4–13 mo if progression occurs

Recorded progression

39.8 months

■ **Group D** · Late progression

✗ **recorded window differs** · |error| 32 mo



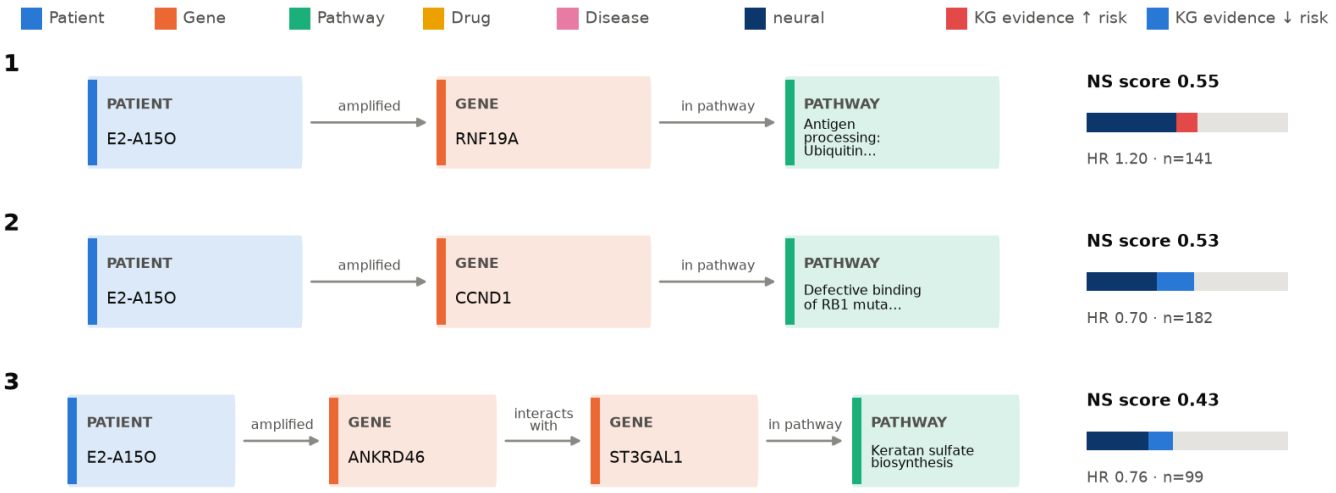
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 89-year-old patient has a LumA tumour, AJCC stage I (T1, missing, M0). The fused neuro-symbolic model places them in the **1st risk percentile** (low tier). The most significant rule covering them (*Age ≥ 83.5 AND Aneuploidy score < 23.5*) marks a group with a progression hazard 3.2× that of the remaining patients. The highest-scoring knowledge-graph path runs from **RNF19A** to **Antigen processing: Ubiquitination & Proteasome degradation**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.20). Progression was recorded at 39.8 months; the model's estimate given progression was 8 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

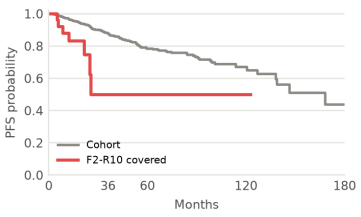


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-E2-A15O** amplified → **RNF19A** in pathway → **Antigen processing: Ubiquitination & Proteasome degradation**	74	↑ **faster progression** HR 1.20, z +0.8, n = 141 carriers of “Antigen processing: Ubiquitination & Proteasome degradation”	**0.55**
2	**TCGA-E2-A15O** amplified → **CCND1** in pathway → **Defective binding of RB1 mutants to E2F1,(E2F2, E2F3)**	58	↓ **slower progression** HR 0.70, z -1.4, n = 182 carriers of “Defective binding of RB1 mutants to E2F1,(E2F2, E2F3)”	**0.53**
3	**TCGA-E2-A15O** amplified → **ANKRD46** interacts with → **ST3GAL1** in pathway → **Keratan sulfate biosynthesis**	51	↓ **slower progression** HR 0.76, z -0.9, n = 99 carriers of “Keratan sulfate biosynthesis”	**0.43**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

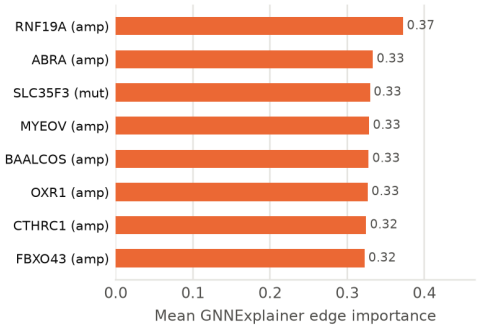
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R10 ↑ **higher risk**	Age ≥ 83.5 AND Aneuploidy score < 23.5	26 pts 7 events	**3.24** p 3.1e-03	26 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
MAP3K1 · driver	Mutation	p.P286Tfs*14	13	0.31
RNF19A	High-level amplification	GISTIC ±2	1	0.37
ABRA	High-level amplification	GISTIC ±2	0	0.33
SLC35F3	Mutation	p.D95=	0	0.33
MYEOV	High-level amplification	GISTIC ±2	0	0.33
BAALCOS	High-level amplification	GISTIC ±2	0	0.33
OXR1	High-level amplification	GISTIC ±2	0	0.33
CTHRC1	High-level amplification	GISTIC ±2	0	0.32
FBXO43	High-level amplification	GISTIC ±2	0	0.32
INTS8	High-level amplification	GISTIC ±2	1	0.32
ZNF703	High-level amplification	GISTIC ±2	1	0.32



86 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); MAP3K1 → AZD-8330 (DB06061); MAP3K1 → Binimetinib (DB11967); MAP3K1 → Fostamatinib (DB12010); FGF19 → Heparin (DB01109); SLC25A32 → Folic acid (DB00158); FGF4 → Pentosan polysulfate (DB00686) … and 48 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0ER	63	I	LumA	0.71	no progression, followed 74.4 mo
TCGA-BH-A201	64	I	LumA	0.71	no progression, followed 28.1 mo
TCGA-E2-A15C	61	I	LumA	0.68	no progression, followed 22.8 mo
TCGA-A2-A0CP	60	I	LumA	0.68	no progression, followed 92.5 mo
TCGA-E2-A15R	64	II	LumA	0.67	no progression, followed 56.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**48**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IA**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

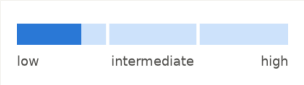
Sample: **TCGA-A2-A0SX-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **0.77** · MSIsensor score: **0.31** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

24th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 15%

Model-estimated progression time

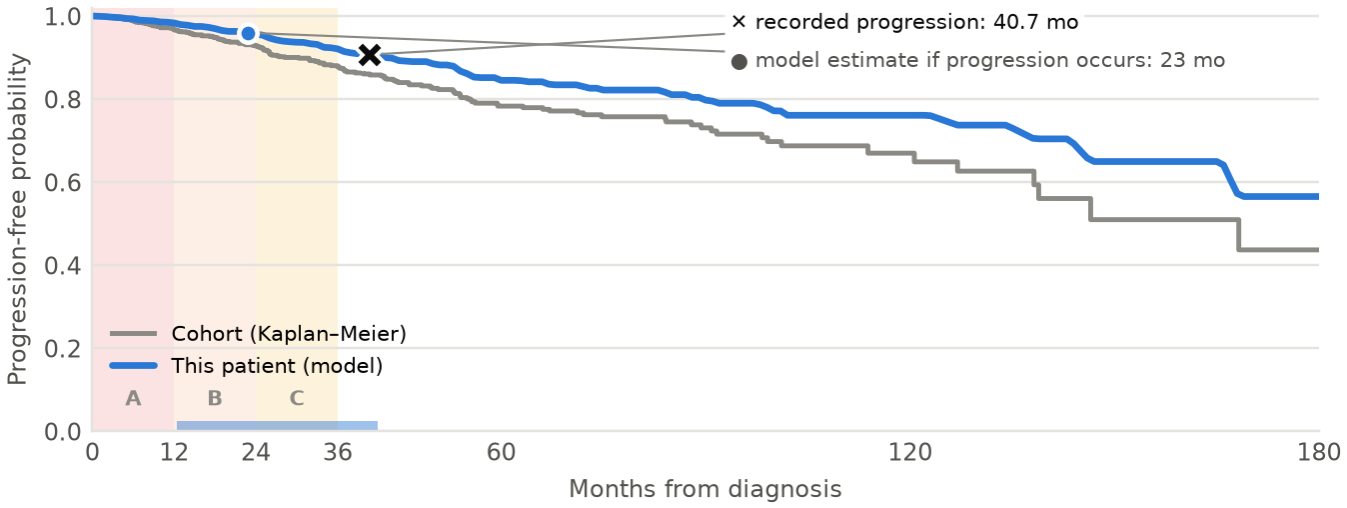
≈ 23 months

Window 12–24 months · IQR 12–42 mo
if progression occurs

Recorded progression

40.7 months

Group D · Late progression
✗ **recorded window differs** · |error| 18 mo



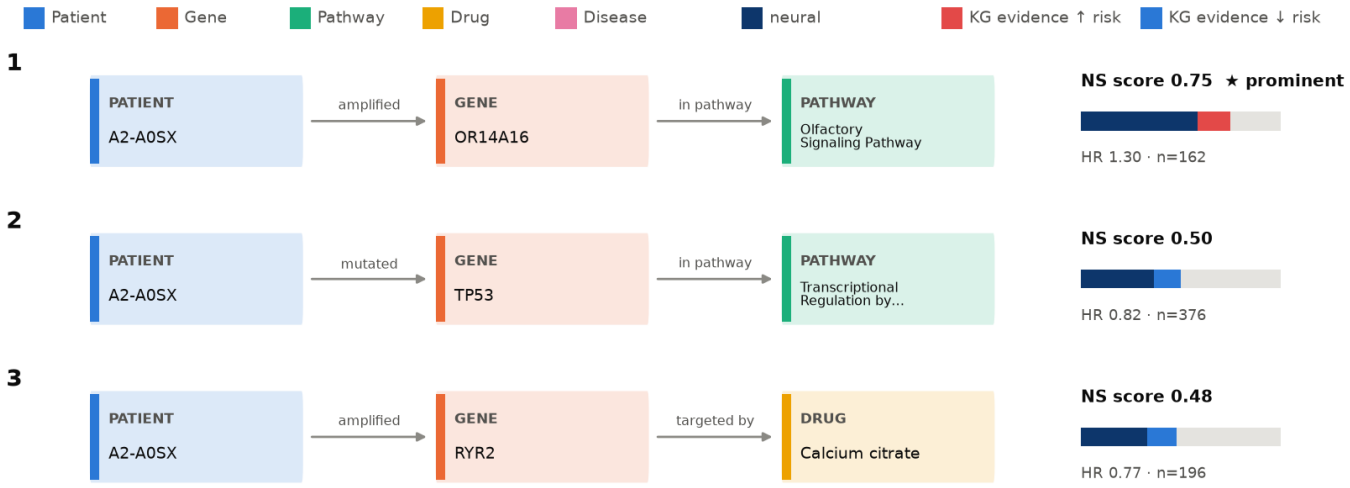
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 48-year-old patient has a Basal tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **24th risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **OR14A16** to **Olfactory Signaling Pathway**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.30). Progression was recorded at 40.7 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

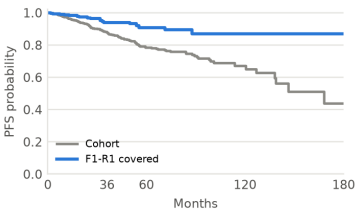


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-A2-A0SX** amplified → **OR14A16** in pathway → **Olfactory Signaling Pathway**	98	↑ **faster progression** HR 1.30, z +1.2, n = 162 carriers of “Olfactory Signaling Pathway”	**0.75**
2	**TCGA-A2-A0SX** mutated → **TP53** in pathway → **Transcriptional Regulation by VENTX**	61	↓ **slower progression** HR 0.82, z -1.0, n = 376 carriers of “Transcriptional Regulation by VENTX”	**0.50**
3	**TCGA-A2-A0SX** amplified → **RYR2** targeted by → **Calcium citrate**	55	↓ **slower progression** HR 0.77, z -1.1, n = 196 carriers of “Calcium citrate”	**0.48**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

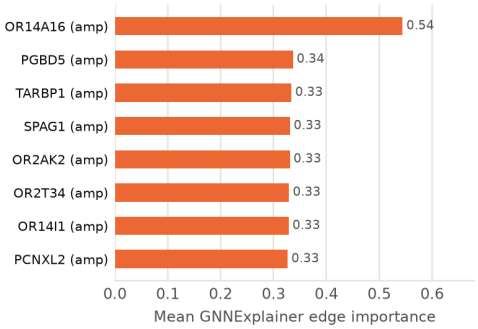
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R1 ↓ lower risk	N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45	365 pts 19 events	**0.26** p 4.0e-09	not reached
F1-R7 ↓ lower risk	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo
F1-R2 ↓ lower risk	N stage = N0 AND Age ≥ 31.5	410 pts 29 events	**0.36** p 4.5e-07	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	High-level amplification	GISTIC ±2	70	0.32
PIK3R1 · driver	Homozygous deletion	GISTIC ±2	79	0.32
TP53 · driver	Mutation	p.G199V	57	0.31
MAP3K1 · driver	Homozygous deletion	GISTIC ±2	13	0.31
OR14A16	High-level amplification	GISTIC ±2	1	0.54
PGBD5	High-level amplification	GISTIC ±2	0	0.34
TARBP1	High-level amplification	GISTIC ±2	0	0.33
SPAG1	High-level amplification	GISTIC ±2	0	0.33
OR2AK2	High-level amplification	GISTIC ±2	1	0.33
OR2T34	High-level amplification	GISTIC ±2	1	0.33
OR14I1	High-level amplification	GISTIC ±2	1	0.33
PCNXL2	High-level amplification	GISTIC ±2	0	0.33



86 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); PIK3R1 → Isoprenaline (DB01064); PIK3R1 → SF1126 (DB05210); PIK3R1 → Enzastaurin (DB06486) … and 56 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A229	37	I	LumA	0.79	no progression, followed 37.7 mo
TCGA-E9-A22G	47	II	Basal	0.78	no progression, followed 40.7 mo
TCGA-AO-A0J4	41	I	Basal	0.78	no progression, followed 52.2 mo
TCGA-B6-A409	44	III	Basal	0.76	progressed at 15.0 mo (group B)
TCGA-A2-A0D0	60	II	Basal	0.76	no progression, followed 67.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**39**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

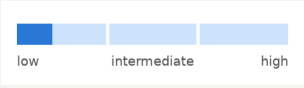
Sample: **TCGA-A2-A25B-01** · Aneuploidy score: **—** · TMB (nonsyn., /Mb): **3.27** · MSIsensor score: **0.61** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

13th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 14%

Model-estimated progression time

≈ 38 months

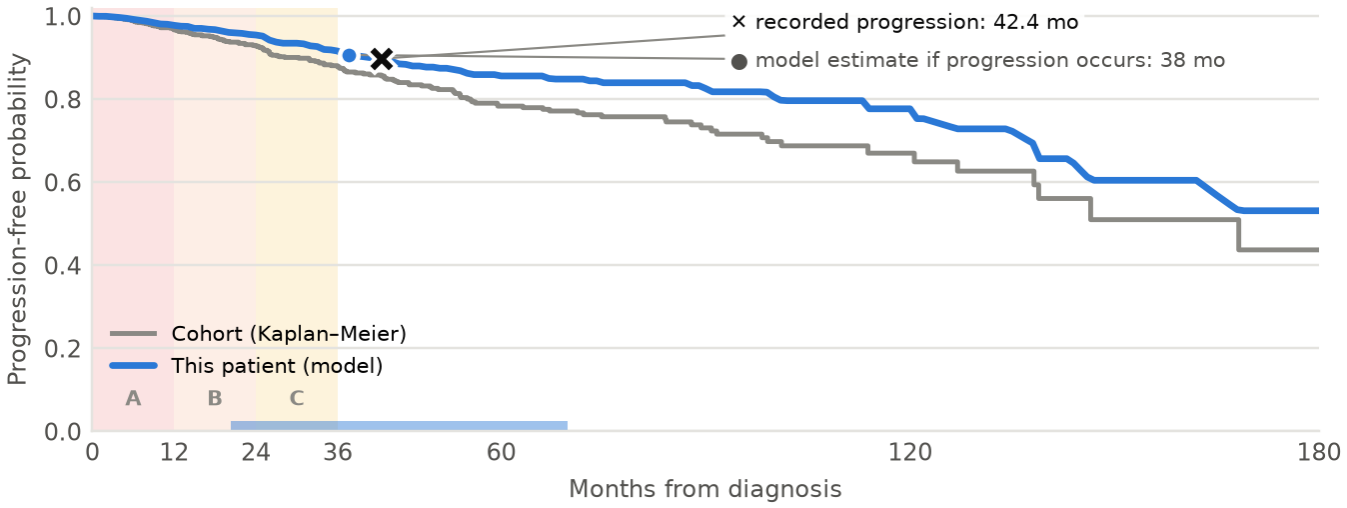
Window > 36 months · IQR 20–70 mo if progression occurs

Recorded progression

42.4 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 5 mo



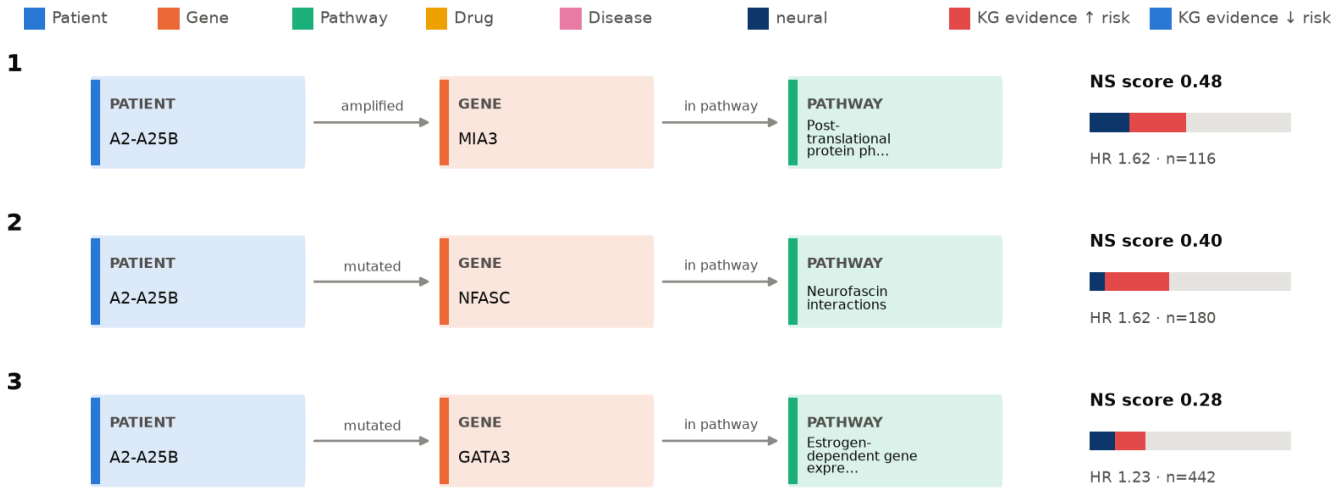
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 39-year-old patient has a LumB tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **13th risk percentile** (low tier). The most significant rule covering them (*AJCC stage = II AND Age < 77.5 AND TMB (nonsynonymous) ≥ 0.08*) marks a group with a progression hazard 0.5× that of the remaining patients. The highest-scoring knowledge-graph path runs from **MIA3** to **Post-translational protein phosphorylation**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.62). Progression was recorded at 42.4 months; the model's estimate given progression was 38 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

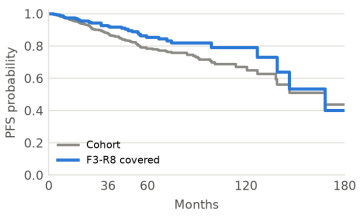


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-A2-A25B** amplified → **MIA3** in pathway → **Post-translational protein phosphorylation**	33	↑ **faster progression** HR 1.62, z +2.1, n = 116 carriers of “Post-translational protein phosphorylation”	**0.48**
2	**TCGA-A2-A25B** mutated → **NFASC** in pathway → **Neurofascin interactions**	12	↑ **faster progression** HR 1.62, z +2.4, n = 180 carriers of “Neurofascin interactions”	**0.40**
3	**TCGA-A2-A25B** mutated → **GATA3** in pathway → **Estrogen-dependent gene expression**	21	↑ **faster progression** HR 1.23, z +1.1, n = 442 carriers of “Estrogen-dependent gene expression”	**0.28**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

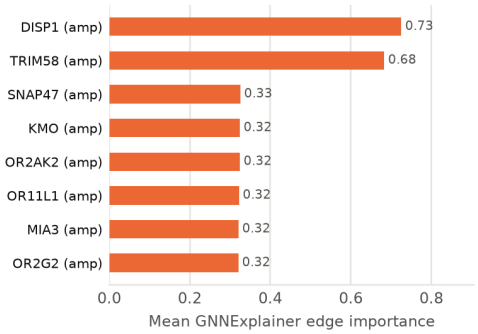
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R8 ↓ **lower risk**	AJCC stage = II AND Age < 77.5 AND TMB (nonsynonymous) ≥ 0.08	433 pts 40 events	**0.50** p 3.2e-04	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.M400Vfs*106	13	0.32
BRCA1 · driver	Mutation	p.E720*	29	0.31
FOXA1 · driver	High-level amplification	GISTIC ±2	9	0.31
DISP1	High-level amplification	GISTIC ±2	0	0.73
TRIM58	High-level amplification	GISTIC ±2	0	0.68
SNAP47	High-level amplification	GISTIC ±2	0	0.33
KMO	High-level amplification	GISTIC ±2	1	0.32
OR2AK2	High-level amplification	GISTIC ±2	1	0.32
OR11L1	High-level amplification	GISTIC ±2	1	0.32
MIA3	High-level amplification	GISTIC ±2	3	0.32
OR2G2	High-level amplification	GISTIC ±2	1	0.32
OBSCN	High-level amplification	GISTIC ±2	8	0.32



60 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

RYR2 → Flecainide (DB01195); RYR2 → Tetracaine (DB09085); RYR2 → Calcium citrate (DB11093); RYR2 → Calcium Phosphate (DB11348); RYR2 → Calcium phosphate dihydrate (DB14481); TARS2 → Threonine (DB00156); PARP1 → Theophylline (DB00277); PARP1 → Zinc (DB01593); PARP1 → Carba-nicotinamide-adenine-dinucleotide (DB02498); PARP1 → NU1025 (DB02690) … and 31 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AO-A03N	59	II	LumB	0.80	progressed at 54.1 mo (group D)
TCGA-D8-A1JB	54	II	LumA	0.78	no progression, followed 55.5 mo
TCGA-BH-A28Q	46	II	LumA	0.77	no progression, followed 36.8 mo
TCGA-A7-A6VW	48	II	Basal	0.75	no progression, followed 9.4 mo
TCGA-E9-A1NI	51	II	LumB	0.73	no progression, followed 9.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group D** · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**73**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**—**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-E2-A1LL-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **1.43** · MSIsensor score: **2.8** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

93rd

High risk tier · fused neuro-symbolic model



Probability of progression

20% by 36 mo

12 mo 6% · 24 mo 13% · 60 mo 35%

Model-estimated progression time

≈ 17 months

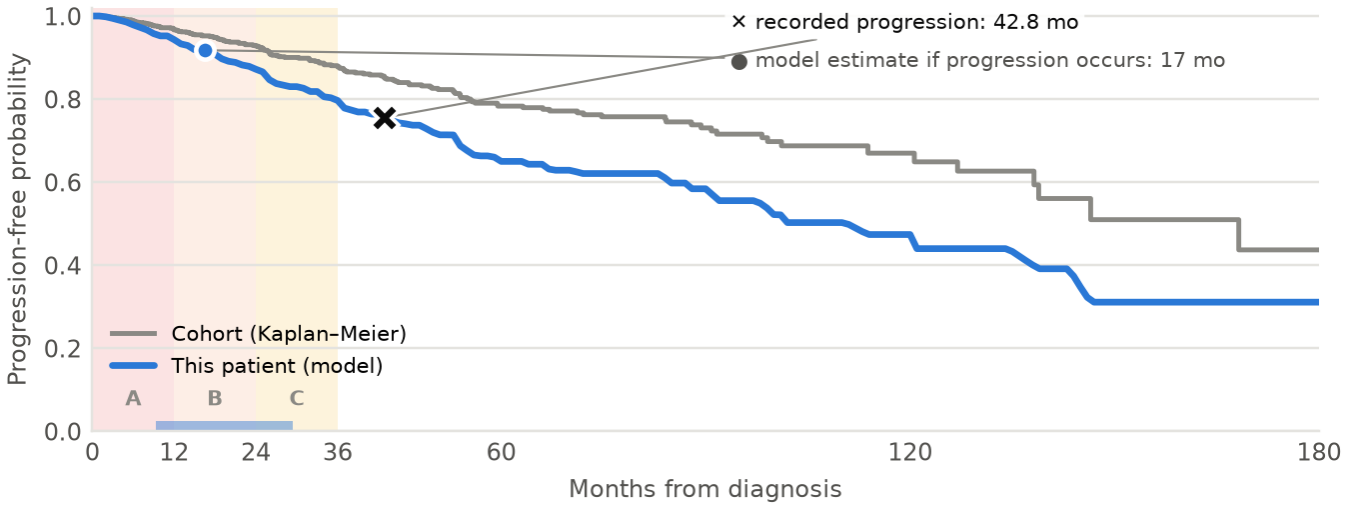
Window 12–24 months · IQR 9–29 mo if progression occurs

Recorded progression

42.8 months

■ **Group D** · Late progression

✗ **recorded window differs** · |error| 26 mo



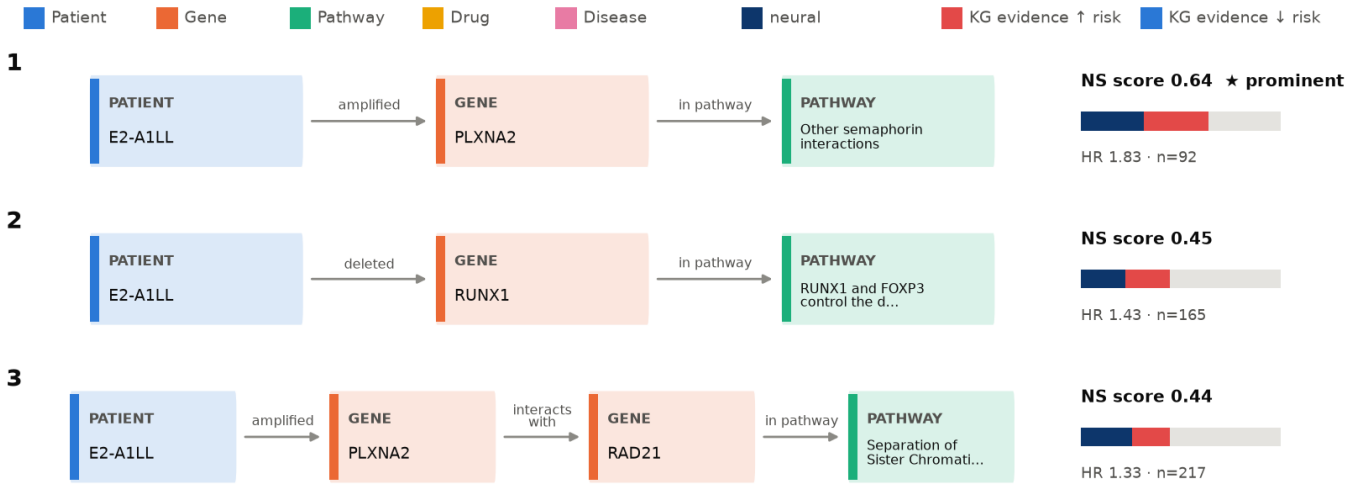
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 73-year-old patient has a Basal tumour, AJCC stage III (T3, N2, MX). The fused neuro-symbolic model places them in the **93rd risk percentile** (high tier). The most significant rule covering them (*40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.5× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PLXNA2** to **Other semaphorin interactions**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.83). Progression was recorded at 42.8 months; the model's estimate given progression was 17 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

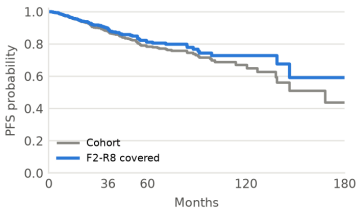


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-E2-A1LL** amplified → **PLXNA2** in pathway → **Other semaphorin interactions**	53	↑ **faster progression** HR 1.83, z +2.4, n = 92 carriers of “Other semaphorin interactions”	**0.64**
2	**TCGA-E2-A1LL** deleted → **RUNX1** in pathway → **RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)**	37	↑ **faster progression** HR 1.43, z +1.7, n = 165 carriers of “RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)”	**0.45**
3	**TCGA-E2-A1LL** amplified → **PLXNA2** interacts with → **RAD21** in pathway → **Separation of Sister Chromatids**	43	↑ **faster progression** HR 1.33, z +1.4, n = 217 carriers of “Separation of Sister Chromatids”	**0.44**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

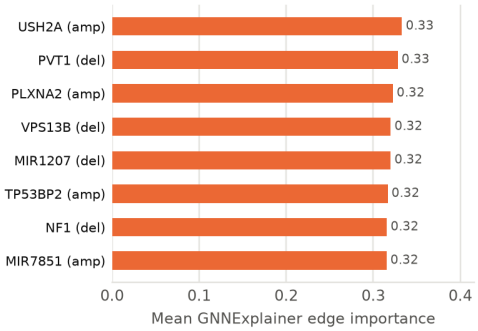
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R8 ↓ **lower risk**	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
NF1 · driver	Homozygous deletion	GISTIC ±2	6	0.32
RUNX1 · driver	Homozygous deletion	GISTIC ±2	18	0.31
USH2A	High-level amplification	GISTIC ±2	0	0.33
PVT1	Homozygous deletion	GISTIC ±2	0	0.33
PLXNA2	High-level amplification	GISTIC ±2	4	0.32
VPS13B	Homozygous deletion	GISTIC ±2	4	0.32
MIR1207	Homozygous deletion	GISTIC ±2	0	0.32
TP53BP2	High-level amplification	GISTIC ±2	12	0.32
MIR7851	High-level amplification	GISTIC ±2	0	0.32
CR1	High-level amplification	GISTIC ±2	3	0.31
SRGAP2D	High-level amplification	GISTIC ±2	0	0.31
STK3	Homozygous deletion	GISTIC ±2	2	0.31



23 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

STK3 → Fostamatinib (DB12010). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1X6	80	III	LumB	0.62	no progression, followed 17.8 mo
TCGA-AC-A2BK	78	III	Basal	0.51	no progression, followed 73.1 mo
TCGA-AR-A24R	45	III	LumB	0.51	no progression, followed 112.8 mo
TCGA-A7-A26E	71	III	LumA	0.50	no progression, followed 31.4 mo
TCGA-OL-A5RV	43	III	LumA	0.47	no progression, followed 34.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-AR-A2LK · Group D

Report date 14 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE III**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-AR-A2LK-01** · Aneuploidy score: **26** · TMB (nonsyn., /Mb): **1.07** · MSIsensor score: **1.05** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

86th

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 28%

Model-estimated progression time

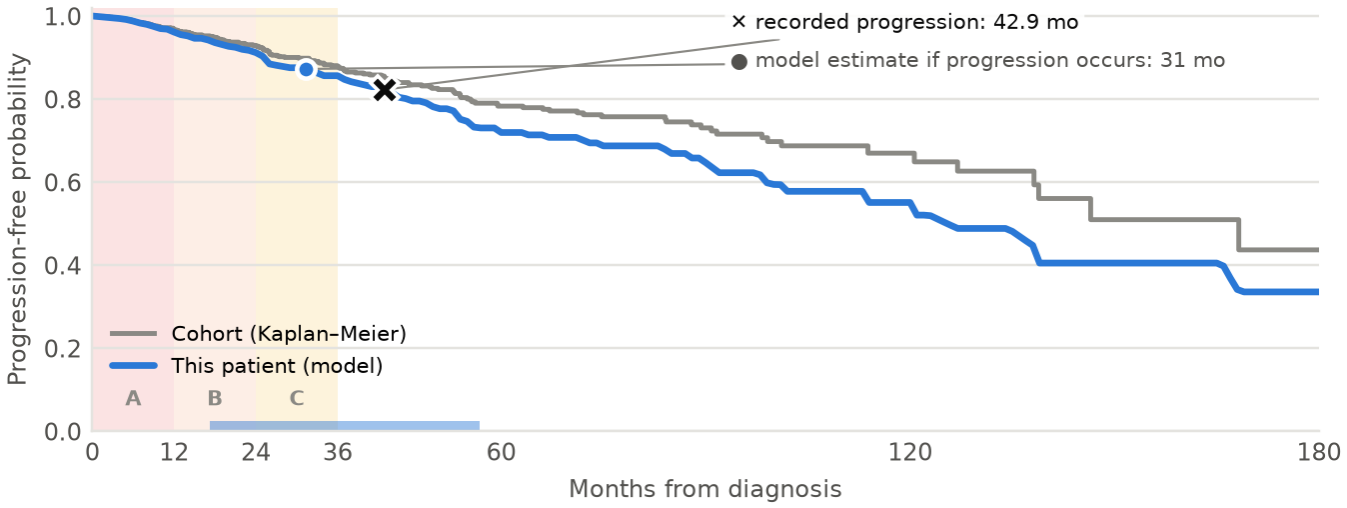
≈ 31 months

Window 24–36 months · IQR 17–57 mo
if progression occurs

Recorded progression

42.9 months

Group D · Late progression
✗ recorded window differs · |error| 12 mo



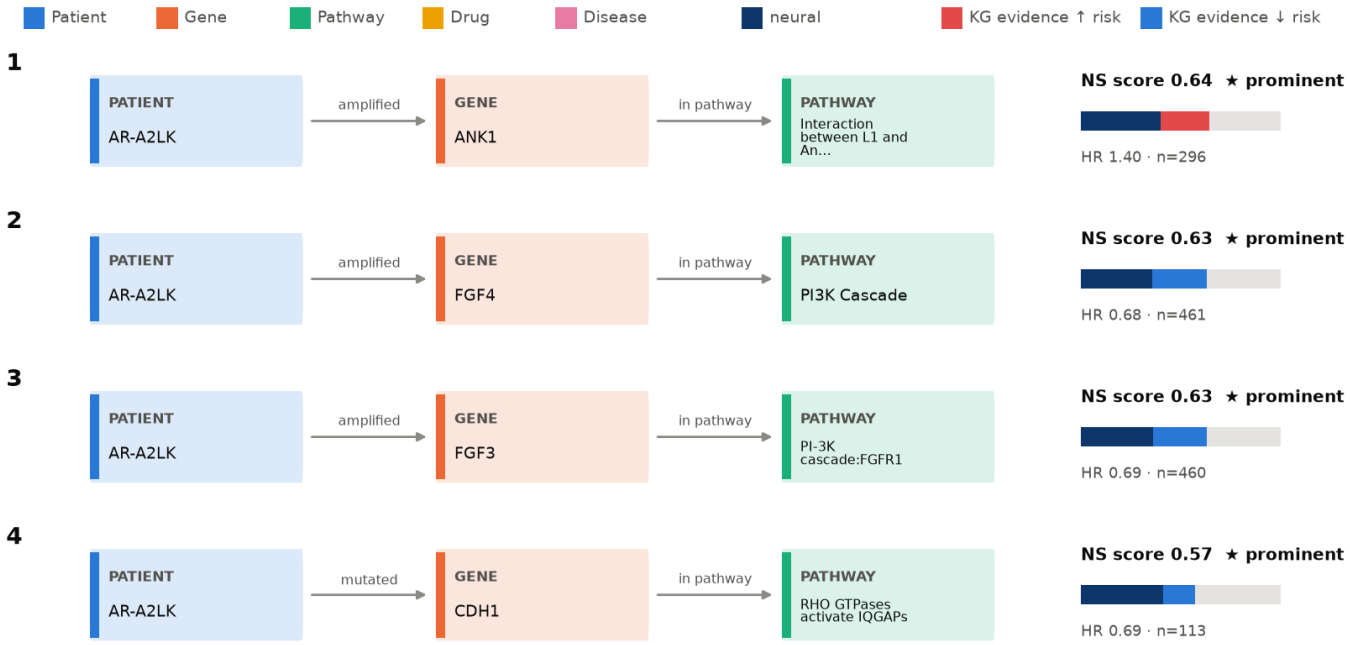
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a LumB tumour, AJCC stage III (T3, N2, M0). The fused neuro-symbolic model places them in the **86th risk percentile** (high tier). The most significant rule covering them (*Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ANK1** to **Interaction between L1 and Ankyrins**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.40). Progression was recorded at 42.9 months; the model's estimate given progression was 31 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

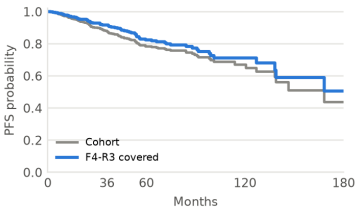


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-AR-A2LK** amplified → **ANK1** in pathway → **Interaction between L1 and Ankyrins**	67	↑ **faster progression** HR 1.40, z +1.8, n = 296 carriers of “Interaction between L1 and Ankyrins”	**0.64**
2 ★	**TCGA-AR-A2LK** amplified → **FGF4** in pathway → **PI3K Cascade**	60	↓ **slower progression** HR 0.68, z -2.0, n = 461 carriers of “PI3K Cascade”	**0.63**
3 ★	**TCGA-AR-A2LK** amplified → **FGF3** in pathway → **PI-3K cascade:FGFR1**	60	↓ **slower progression** HR 0.69, z -2.0, n = 460 carriers of “PI-3K cascade:FGFR1”	**0.63**
4 ★	**TCGA-AR-A2LK** mutated → **CDH1** in pathway → **RHO GTPases activate IQGAPs**	69	↓ **slower progression** HR 0.69, z -1.2, n = 113 carriers of “RHO GTPases activate IQGAPs”	**0.57**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

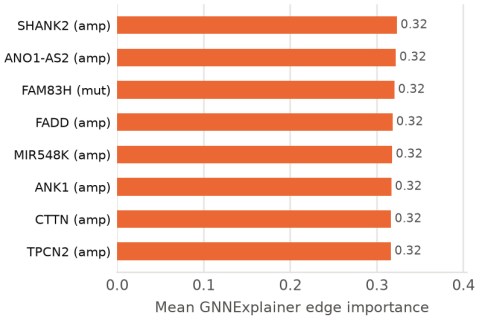
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R3 ↓ **lower risk**	Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	662 pts 77 events	**0.36** p 2.3e-07	not reached
F4-R4 ↓ **lower risk**	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo
F4-R5 ↓ **lower risk**	M stage = M0 AND Age ≥ 34.5	713 pts 89 events	**0.47** p 1.0e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.A788Efs*14	17	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
SHANK2	High-level amplification	GISTIC ±2	1	0.32
ANO1-AS2	High-level amplification	GISTIC ±2	0	0.32
FAM83H	Mutation	p.A238T	0	0.32
FADD	High-level amplification	GISTIC ±2	14	0.32
MIR548K	High-level amplification	GISTIC ±2	0	0.32
ANK1	High-level amplification	GISTIC ±2	9	0.32
CTTN	High-level amplification	GISTIC ±2	2	0.32
TPCN2	High-level amplification	GISTIC ±2	1	0.32
WISP1	Mutation	p.R171Q	0	0.32



25 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718) … and 6 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0CK	60	III	LumA	0.82	progressed at 44.8 mo (group D)
TCGA-AN-A03Y	66	II	LumB	0.74	no progression, followed 0.3 mo
TCGA-A2-A1FX	61	III	LumB	0.74	no progression, followed 60.7 mo
TCGA-PE-A5DC	72	III	LumA	0.72	no progression, followed 47.0 mo
TCGA-A8-A0A6	64	III	LumA	0.69	no progression, followed 21.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**43**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AR-A0TZ-01** · Aneuploidy score: **9** · TMB (nonsyn., /Mb): **1.87** · MSIsensor score: **0.45** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

98th

High risk tier · fused neuro-symbolic model



Probability of progression

45% by 36 mo

12 mo 12% · 24 mo 28% · 60 mo 71%

Model-estimated progression time

≈ 41 months

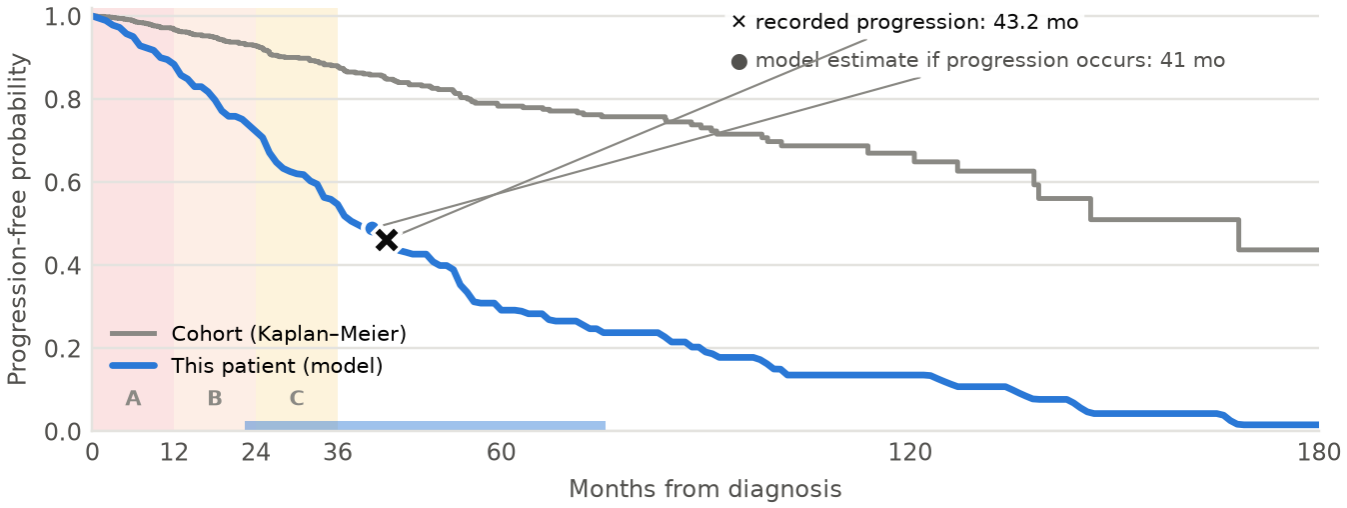
Window > 36 months · IQR 22–75 mo if progression occurs

Recorded progression

43.2 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 2 mo



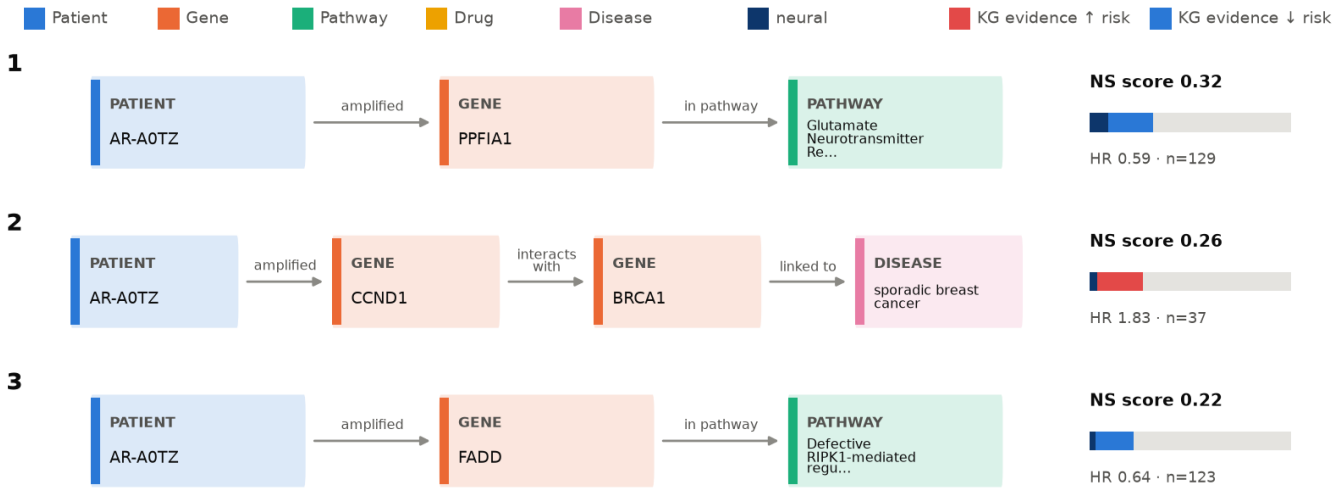
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 43-year-old patient has a LumA tumour, AJCC stage III (T2, N2, M1). The fused neuro-symbolic model places them in the **98th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **PPFIA1** to **Glutamate Neurotransmitter Release Cycle**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.59). Progression was recorded at 43.2 months; the model's estimate given progression was 41 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

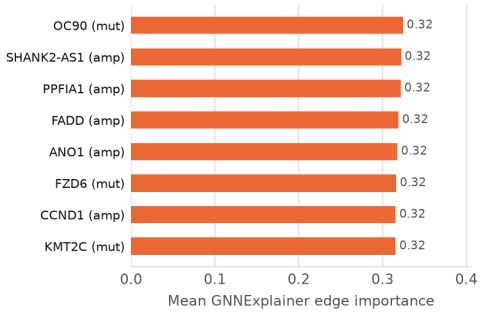
#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-AR-A0TZ** amplified → **PPFIA1** in pathway → **Glutamate Neurotransmitter Release Cycle**	15	↓ **slower progression** HR 0.59, z -1.7, n = 129 carriers of “Glutamate Neurotransmitter Release Cycle”	**0.32**
2	**TCGA-AR-A0TZ** amplified → **CCND1** interacts with → **BRCA1** linked to → **sporadic breast cancer**	6	↑ **faster progression** HR 1.83, z +1.7, n = 37 carriers of “sporadic breast cancer”	**0.26**
3	**TCGA-AR-A0TZ** amplified → **FADD** in pathway → **Defective RIPK1-mediated regulated necrosis**	5	↓ **slower progression** HR 0.64, z -1.4, n = 123 carriers of “Defective RIPK1-mediated regulated necrosis”	**0.22**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank p < 0.05, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
KMT2C · driver	Mutation	p.K4499R	4	0.32
PIK3CA · driver	Mutation	p.H1047R	70	0.31
OC90	Mutation	p.E285*	0	0.32
SHANK2-AS1	High-level amplification	GISTIC ±2	0	0.32
PPFIA1	High-level amplification	GISTIC ±2	6	0.32
FADD	High-level amplification	GISTIC ±2	14	0.32
ANO1	High-level amplification	GISTIC ±2	2	0.32
FZD6	Mutation	3'UTR	5	0.32
RNU6-83P	High-level amplification	GISTIC ±2	0	0.32
TRAPPC9	Mutation	p.Q801*	2	0.31
FGF19	High-level amplification	GISTIC ±2	13	0.31



23 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015) … and 7 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AO-A0JG	49	III	LumA	0.76	no progression, followed 26.2 mo
TCGA-AN-A0FW	67	III	LumA	0.74	no progression, followed 0.4 mo
TCGA-BH-A0BC	60	III	LumA	0.69	no progression, followed 32.0 mo
TCGA-AO-A126	39	II	LumA	0.69	progressed at 8.8 mo (group A)
TCGA-C8-A12Y	44	II	LumA	0.69	no progression, followed 48.5 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-B6-A0X5 · Group D

Report date 14 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**61**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

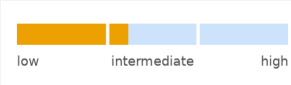
Sample: **TCGA-B6-A0X5-01** · Aneuploidy score: **3** · TMB (nonsyn., /Mb): **1.2** · MSIsensor score: **0.01** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

41st

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 6% · 60 mo 17%

Model-estimated progression time

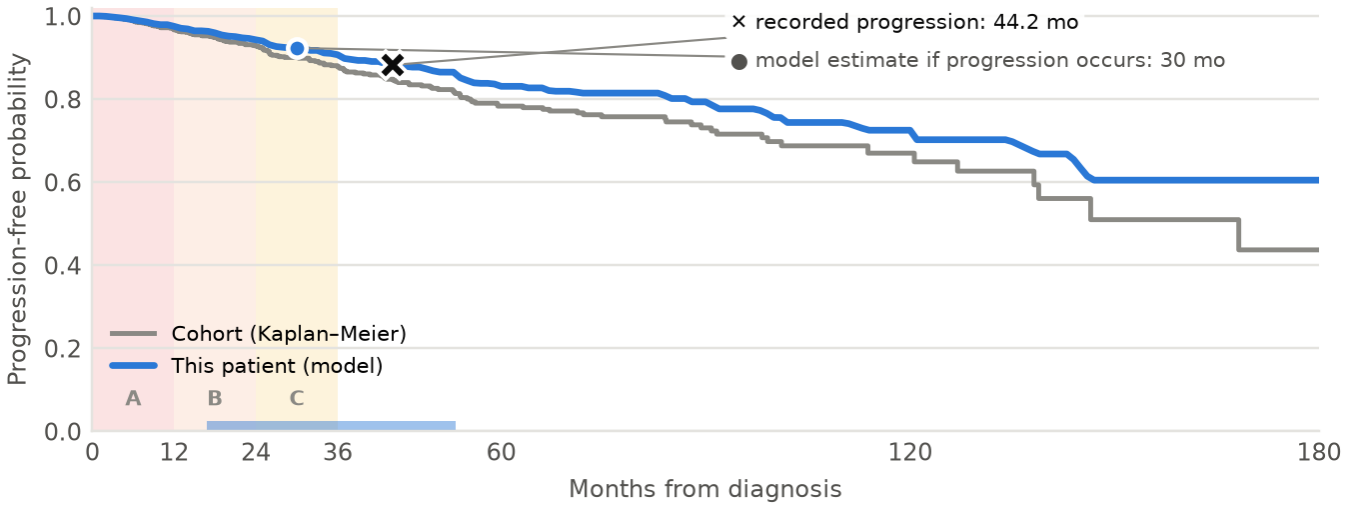
≈ 30 months

Window 24–36 months · IQR 17–53 mo
if progression occurs

Recorded progression

44.2 months

Group D · Late progression
✗ recorded window differs · |error| 14 mo



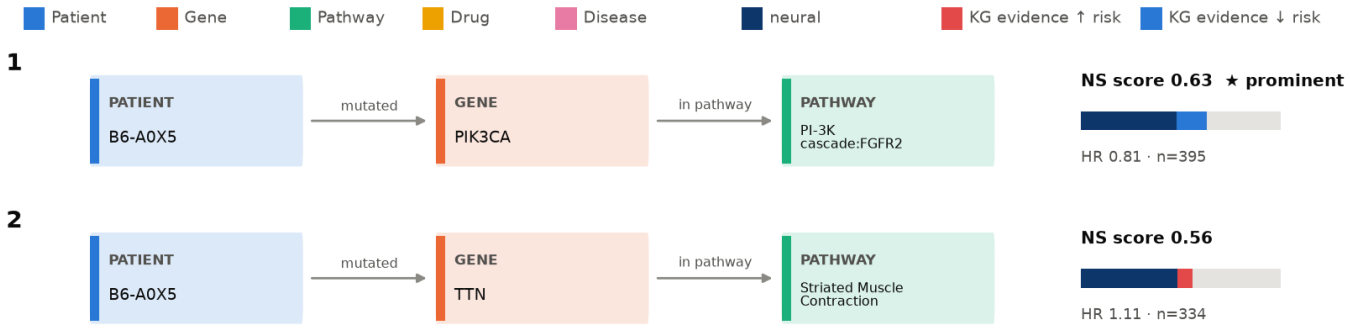
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 61-year-old patient has a LumB tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **41st risk percentile** (intermediate tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 83.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PIK3CA** to **PI-3K cascade:FGFR2**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.81). Progression was recorded at 44.2 months; the model's estimate given progression was 30 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

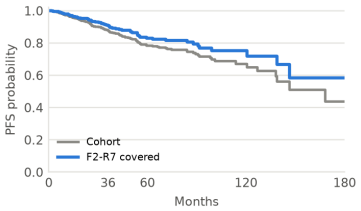


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-B6-A0X5** mutated → **PIK3CA** in pathway → **PI-3K cascade:FGFR2**	80	↓ **slower progression** HR 0.81, z -1.1, n = 395 carriers of "PI-3K cascade:FGFR2"	**0.63**
2	**TCGA-B6-A0X5** mutated → **TTN** in pathway → **Striated Muscle Contraction**	81	↑ **faster progression** HR 1.11, z +0.6, n = 334 carriers of "Striated Muscle Contraction"	**0.56**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

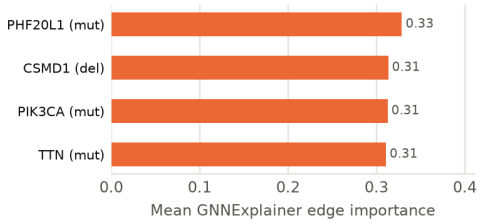
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 83.5	638 pts 66 events	**0.39** p 3.3e-07	not reached
F2-R8 ↓ lower risk	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.H1047R	70	0.31
PHF20L1	Mutation	p.A525Gfs*23	0	0.33
CSMD1	Homozygous deletion	GISTIC ±2	0	0.31
TTN	Mutation	p.N26237=	2	0.31



4 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A25C	50	II	LumB	0.97	no progression, followed 17.2 mo
TCGA-BH-A0BF	56	II	LumB	0.83	no progression, followed 43.5 mo
TCGA-A2-A0YC	59	II	LumA	0.83	progressed at 24.9 mo (group C)
TCGA-E2-A10B	67	II	LumA	0.83	progressed at 6.4 mo (group A)
TCGA-D8-A1JE	62	II	LumB	0.82	no progression, followed 18.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**53**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**EUR**	Pathologic T	**T1B**
Prior diagnosis	**No**	Pathologic N	**N3A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-A2-A0EW-01** · Aneuploidy score: **2** · TMB (nonsyn., /Mb): **0.5** · MSIsensor score: **0** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

77th

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 25%

Model-estimated progression time

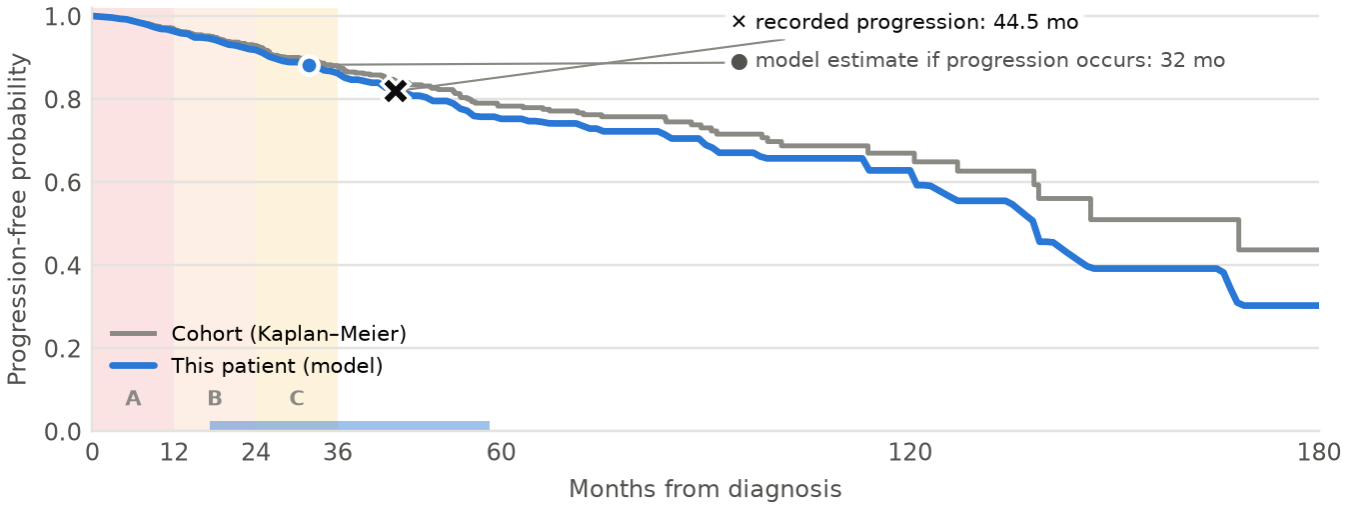
≈ 32 months

Window 24–36 months · IQR 17–58 mo
if progression occurs

Recorded progression

44.5 months

Group D · Late progression
✗ recorded window differs · |error| 13 mo



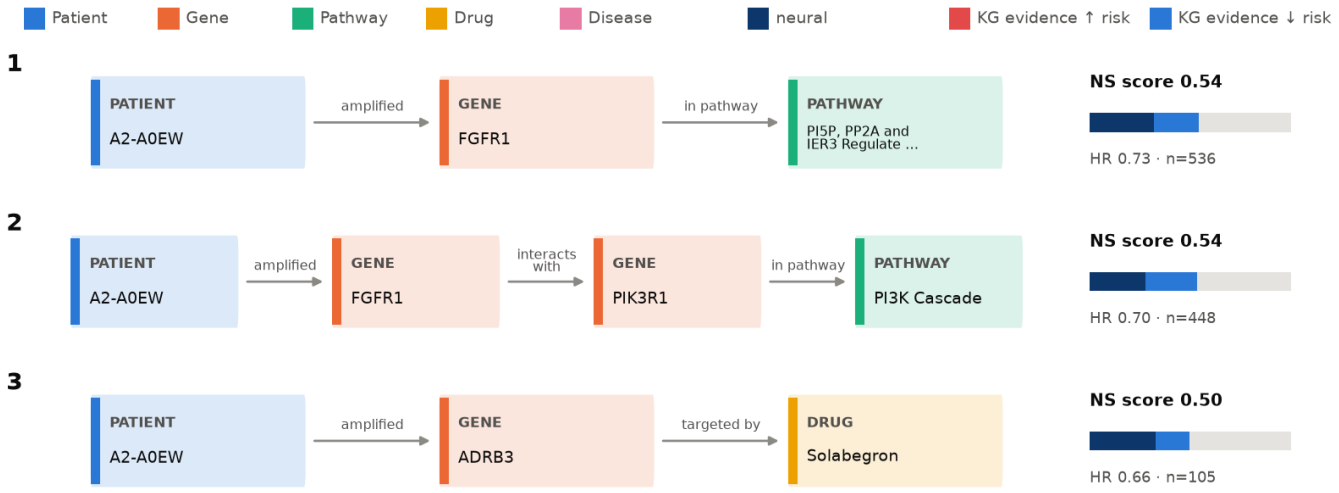
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 53-year-old patient has a LumA tumour, AJCC stage III (T1, N3, M0). The fused neuro-symbolic model places them in the **77th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **FGFR1** to **PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.73). Progression was recorded at 44.5 months; the model's estimate given progression was 32 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

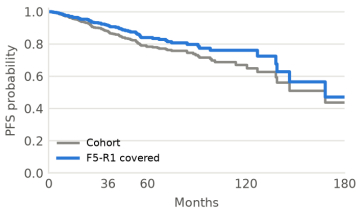


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-A2-A0EW** amplified → **FGFR1** in pathway → **PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling**	54	↓ slower progression HR 0.73, z -1.7, n = 536 carriers of “PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling”	**0.54**
2	**TCGA-A2-A0EW** amplified → **FGFR1** interacts with → **PIK3R1** in pathway → **PI3K Cascade**	46	↓ slower progression HR 0.70, z -1.9, n = 448 carriers of “PI3K Cascade”	**0.54**
3	**TCGA-A2-A0EW** amplified → **ADRB3** targeted by → **Solabegron**	55	↓ slower progression HR 0.66, z -1.3, n = 105 carriers of “Solabegron”	**0.50**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

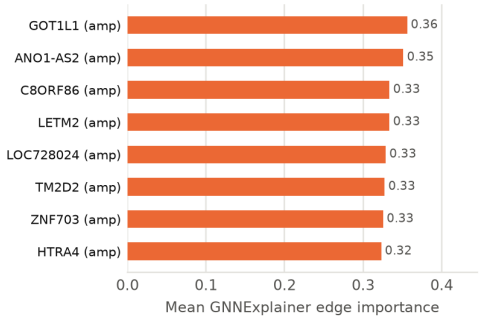
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ lower risk	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R5 ↓ lower risk	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.32
CDH1 · driver	Mutation	p.X230_splice	17	0.32
PIK3CA · driver	Mutation	p.Q546R	70	0.32
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
GOT1L1	High-level amplification	GISTIC ±2	0	0.36
ANO1-AS2	High-level amplification	GISTIC ±2	0	0.35
C8ORF86	High-level amplification	GISTIC ±2	0	0.33
LETM2	High-level amplification	GISTIC ±2	0	0.33
LOC728024	High-level amplification	GISTIC ±2	0	0.33
TM2D2	High-level amplification	GISTIC ±2	0	0.33
ZNF703	High-level amplification	GISTIC ±2	1	0.33
HTRA4	High-level amplification	GISTIC ±2	0	0.32



47 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 71 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-GM-A2DO	54	I	LumB	0.74	no progression, followed 85.3 mo
TCGA-A2-A0YK	61	II	Normal	0.72	no progression, followed 19.3 mo
TCGA-A8-A0A6	64	III	LumA	0.72	no progression, followed 21.0 mo
TCGA-B6-A0IP	74	I	LumA	0.71	no progression, followed 129.1 mo
TCGA-BH-A201	64	I	LumA	0.69	no progression, followed 28.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**60**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

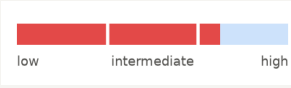
Sample: **TCGA-A2-A0CK-01** · Aneuploidy score: **4** · TMB (nonsyn., /Mb): **0.43** · MSIsensor score: **0** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

75th

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 25%

Model-estimated progression time

≈ 39 months

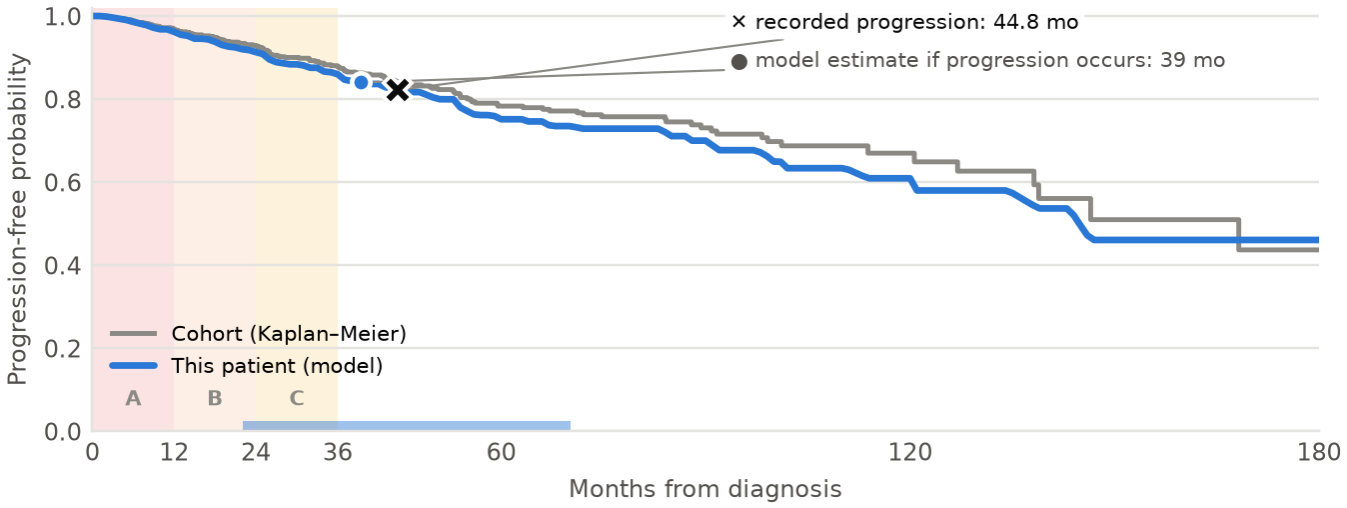
Window > 36 months · IQR 22–70 mo if progression occurs

Recorded progression

44.8 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 5 mo



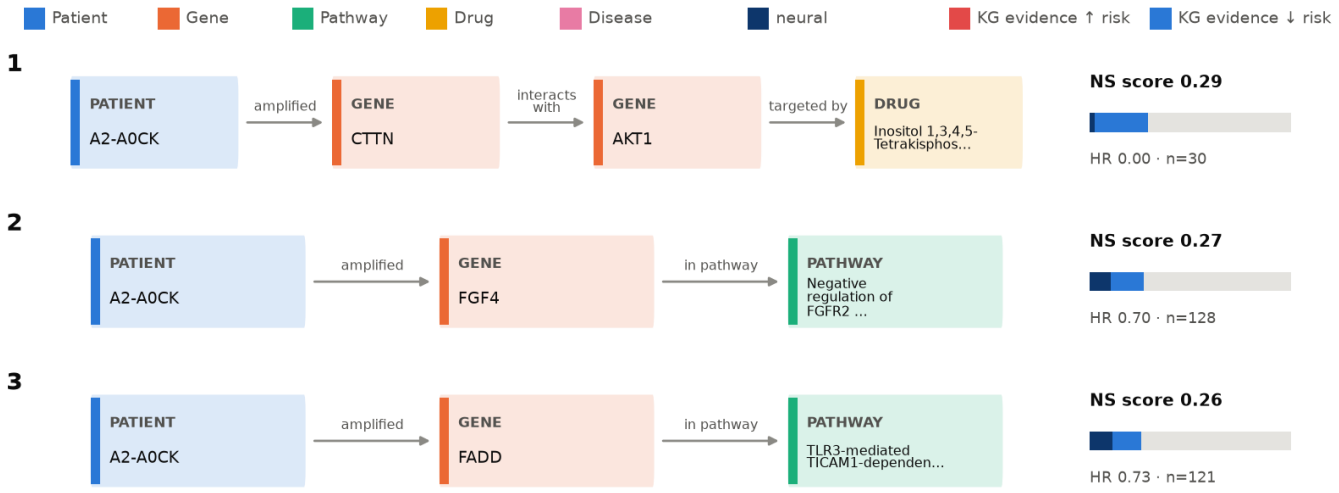
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 60-year-old patient has a LumA tumour, AJCC stage III (T3, N2, M0). The fused neuro-symbolic model places them in the **75th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 83.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **CTTN** to **Inositol 1,3,4,5-Tetrakisphosphate**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.00). Progression was recorded at 44.8 months; the model's estimate given progression was 39 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

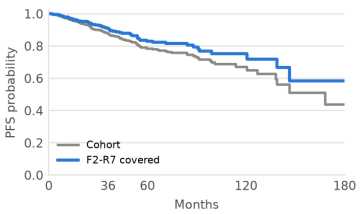


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-A2-A0CK** amplified → **CTTN** interacts with → **AKT1** targeted by → **Inositol 1,3,4,5-Tetrakisphosphate**	4	↓ **slower progression** HR 0.00, z -2.0, n = 30 carriers of “Inositol 1,3,4,5-Tetrakisphosphate”	**0.29**
2	**TCGA-A2-A0CK** amplified → **FGF4** in pathway → **Negative regulation of FGFR2 signaling**	17	↓ **slower progression** HR 0.70, z -1.2, n = 128 carriers of “Negative regulation of FGFR2 signaling”	**0.27**
3	**TCGA-A2-A0CK** amplified → **FADD** in pathway → **TLR3-mediated TICAM1-dependent programmed cell death**	19	↓ **slower progression** HR 0.73, z -1.1, n = 121 carriers of “TLR3-mediated TICAM1-dependent programmed cell death”	**0.26**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

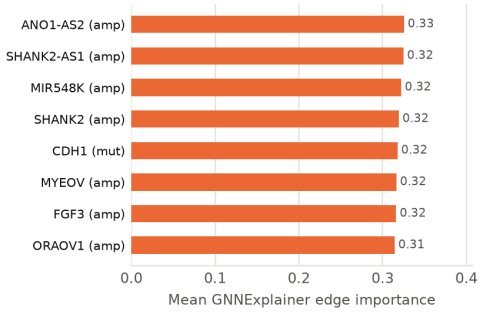
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 83.5	638 pts 66 events	**0.39** p 3.3e-07	not reached
F2-R8 ↓ lower risk	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.G278*	17	0.32
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
ANO1-AS2	High-level amplification	GISTIC ±2	0	0.33
SHANK2-AS1	High-level amplification	GISTIC ±2	0	0.32
MIR548K	High-level amplification	GISTIC ±2	0	0.32
SHANK2	High-level amplification	GISTIC ±2	1	0.32
MYEOV	High-level amplification	GISTIC ±2	0	0.32
FGF3	High-level amplification	GISTIC ±2	28	0.32
ORAOV1	High-level amplification	GISTIC ±2	0	0.31
FGF4	High-level amplification	GISTIC ±2	48	0.31
FADD	High-level amplification	GISTIC ±2	14	0.31
PPFIA1	High-level amplification	GISTIC ±2	6	0.31



15 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); FGF4 → Pentosan polysulfate (DB00686); FGF4 → Heparin (DB01109); FGF4 → TP-508 (DB05309); FGF19 → Heparin (DB01109); ANO1 → Crofelemer (DB04941). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-PE-A5DC	72	III	LumA	0.87	no progression, followed 47.0 mo
TCGA-AR-A2LK	62	III	LumB	0.82	progressed at 42.9 mo (group D)
TCGA-AC-A4ZE	63	II	LumA	0.80	no progression, followed 29.3 mo
TCGA-A7-A4SB	56	III	missing	0.75	no progression, followed 13.7 mo
TCGA-A2-A1FX	61	III	LumB	0.74	no progression, followed 60.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**45**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE X**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**No**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8523/3**

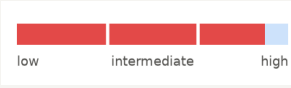
Sample: **TCGA-B6-A0IN-01** · Aneuploidy score: **7** · TMB (nonsyn., /Mb): **0.97** · MSIsensor score: **0.12** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

91st

High risk tier · fused neuro-symbolic model



Probability of progression

19% by 36 mo

12 mo 5% · 24 mo 12% · 60 mo 33%

Model-estimated progression time

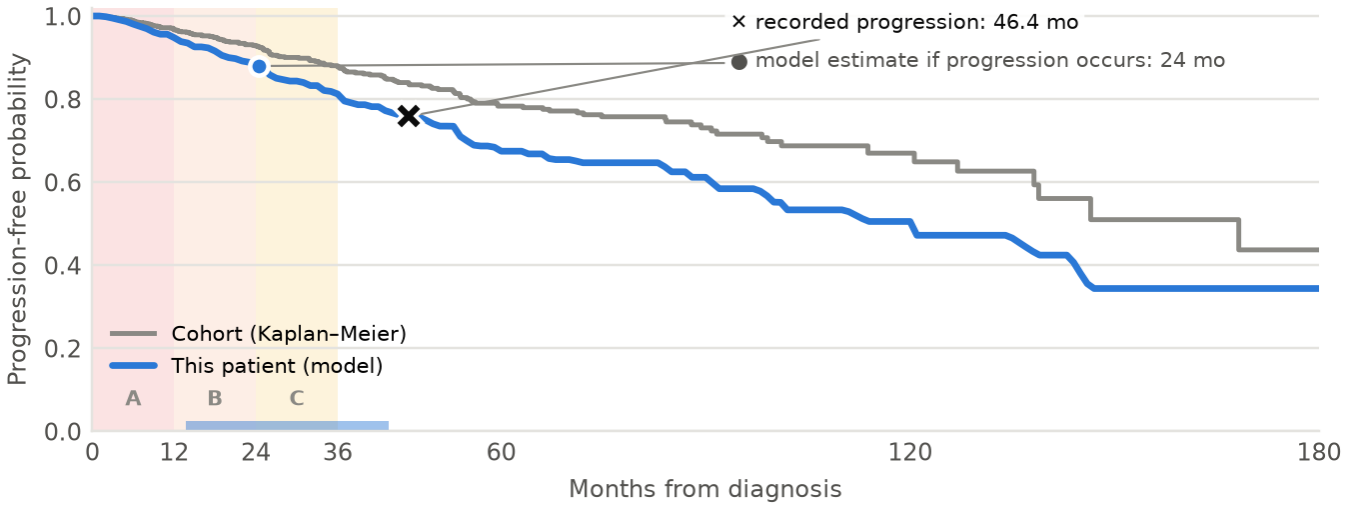
≈ 24 months

Window 24–36 months · IQR 14–44 mo
if progression occurs

Recorded progression

46.4 months

Group D · Late progression
✗ recorded window differs · |error| 22 mo



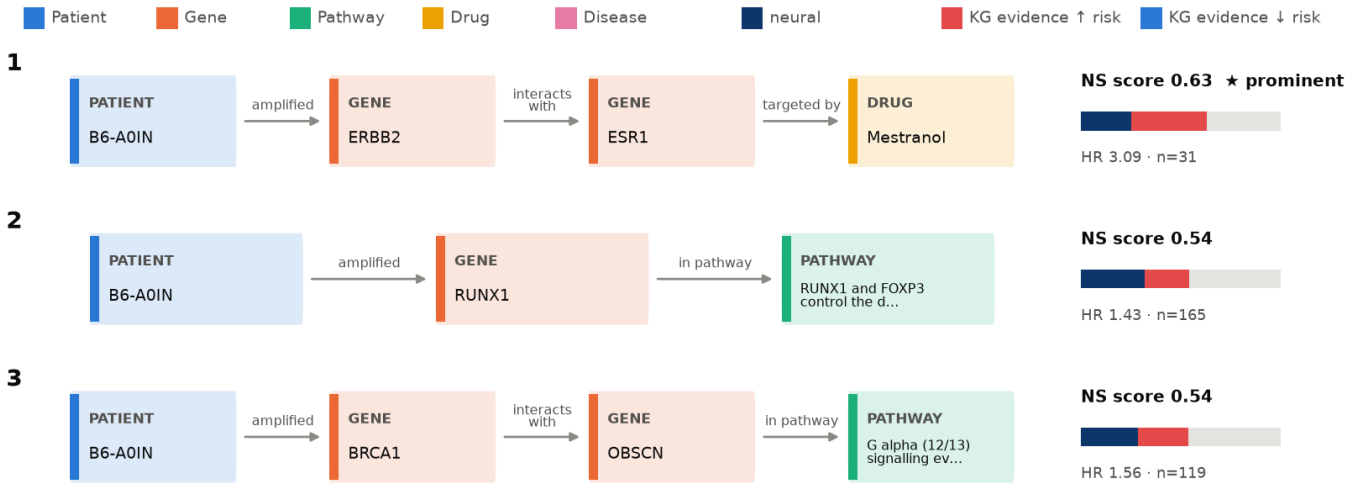
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 45-year-old patient has a LumB tumour, AJCC stage X (T1, missing, MX). The fused neuro-symbolic model places them in the **91st risk percentile** (high tier). The most significant rule covering them (*AJCC stage = III AND altered gene linked to 'breast ductal adenocarcinoma' AND Age ≥ 35.0*) marks a group with a progression hazard 2.9× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ERBB2** to **Mestranol**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 3.09). Progression was recorded at 46.4 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

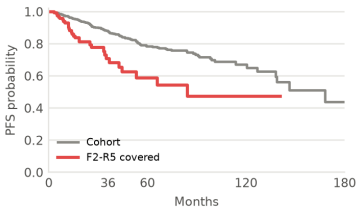


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-B6-A0IN** amplified → **ERBB2** interacts with → **ESR1** targeted by → **Mestranol**	42	↑ **faster progression** HR 3.09, z +2.8, n = 31 carriers of “Mestranol”	**0.63**
2	**TCGA-B6-A0IN** amplified → **RUNX1** in pathway → **RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)**	53	↑ **faster progression** HR 1.43, z +1.7, n = 165 carriers of “RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)”	**0.54**
3	**TCGA-B6-A0IN** amplified → **BRCA1** interacts with → **OBSCN** in pathway → **G alpha (12/13) signalling events**	48	↑ **faster progression** HR 1.56, z +1.9, n = 119 carriers of “G alpha (12/13) signalling events”	**0.54**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

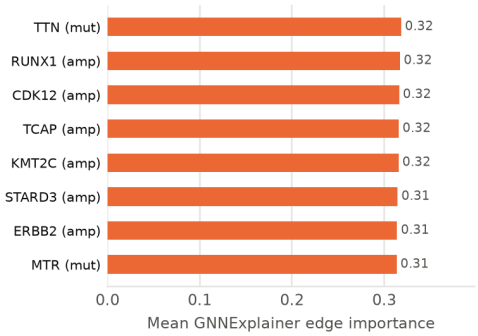
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R5 ↑ **higher risk**	AJCC stage = III AND altered gene linked to 'breast ductal adenocarcinoma' AND Age ≥ 35.0 [KG]	123 pts 30 events	**2.94** p 1.4e-07	84 mo
F2-R6 ↑ **higher risk**	AJCC stage = III AND TMB (nonsynonymous) < 7.00 AND Aneuploidy score < 26.5	176 pts 36 events	**2.19** p 6.2e-05	98 mo
F2-R8 ↓ **lower risk**	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
RUNX1 · driver	High-level amplification	GISTIC ±2	18	0.32
KMT2C · driver	High-level amplification	GISTIC ±2	4	0.32
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
BRCA1 · driver	High-level amplification	GISTIC ±2	29	0.31
GATA3 · driver	Mutation	p.S437Qfs*70	13	0.31
TTN	Mutation	p.D30372Y, p.P26895H, p.R19963*	2	0.32
CDK12	High-level amplification	GISTIC ±2	2	0.32
TCAP	High-level amplification	GISTIC ±2	1	0.32
STARD3	High-level amplification	GISTIC ±2	1	0.31
MTR	Mutation	p.A4T	18	0.31
PPP1R1B	High-level amplification	GISTIC ±2	1	0.31
PNMT	High-level amplification	GISTIC ±2	16	0.31



18 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 26 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A0WS	58	X	LumA	0.54	no progression, followed 97.5 mo
TCGA-A1-A0SN	50	II	LumB	0.53	no progression, followed 39.3 mo
TCGA-EW-A1OX	43	II	LumB	0.52	no progression, followed 30.0 mo
TCGA-EW-A1IY	38	I	LumB	0.50	no progression, followed 8.5 mo
TCGA-AC-A3QQ	54	I	LumA	0.50	no progression, followed 24.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-AR-A24L · Group D

Report date 14 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**26**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AR-A24L-01** · Aneuploidy score: **8** · TMB (nonsyn., /Mb): **1.33** · MSIsensor score: **0.09** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

38th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 17%

Model-estimated progression time

≈ 51 months

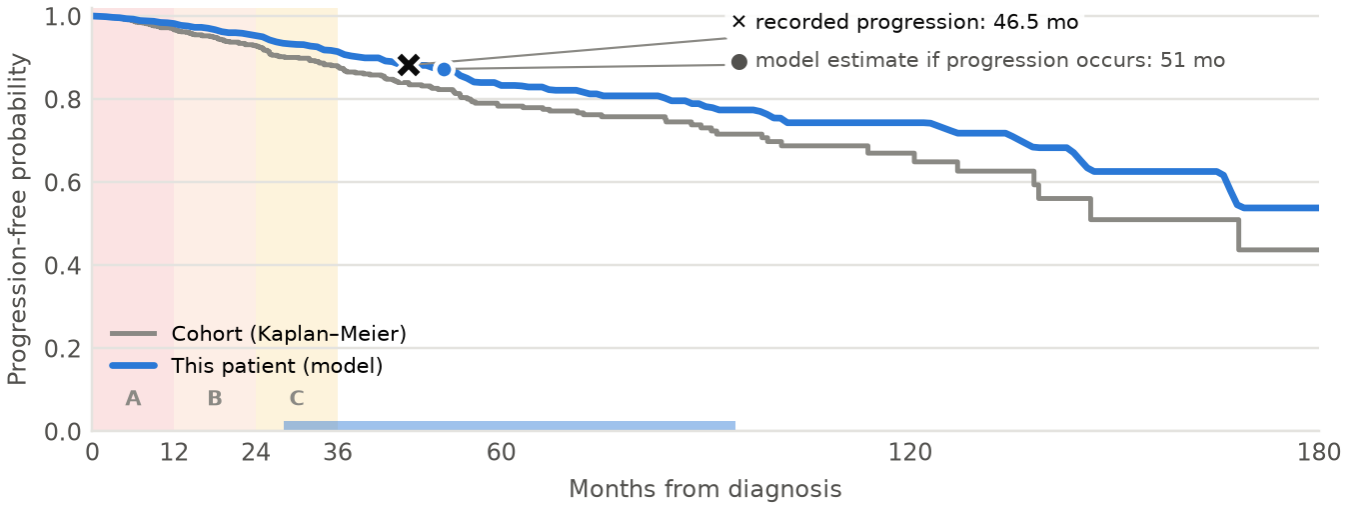
Window > 36 months · IQR 28–94 mo if progression occurs

Recorded progression

46.5 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 5 mo



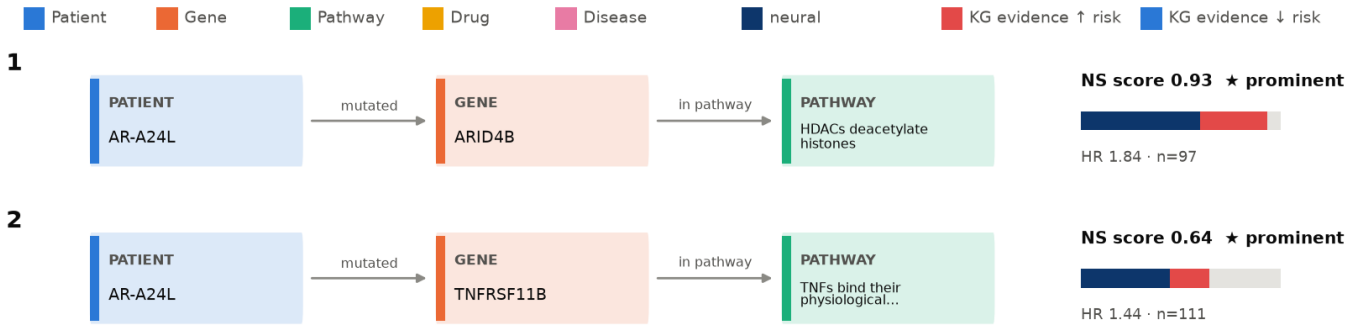
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 26-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **38th risk percentile** (intermediate tier). The highest-scoring knowledge-graph path runs from **ARID4B** to **HDACs deacetylate histones**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.84). Progression was recorded at 46.5 months; the model's estimate given progression was 51 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



2 of 4 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

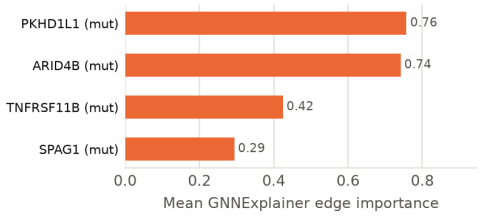
#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-AR-A24L** mutated → **ARID4B** in pathway → **HDACs deacetylate histones**	100	↑ **faster progression** HR 1.84, z +2.5, n = 97 carriers of “HDACs deacetylate histones”	**0.93**
2 ★	**TCGA-AR-A24L** mutated → **TNFRSF11B** in pathway → **TNFs bind their physiological receptors**	74	↑ **faster progression** HR 1.44, z +1.5, n = 111 carriers of “TNFs bind their physiological receptors”	**0.64**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank p < 0.05, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PKHD1L1	Mutation	p.A2130=	0	0.76
ARID4B	Mutation	p.E734Dfs*2	3	0.74
TNFRSF11B	Mutation	p.A318G	1	0.42
SPAG1	Mutation	p.*366*	0	0.29



4 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A1EW	38	II	LumA	0.96	progressed at 55.7 mo (group D)
TCGA-E9-A245	47	II	LumA	0.93	no progression, followed 0.9 mo
TCGA-BH-A0DQ	42	II	LumA	0.93	no progression, followed 3.2 mo
TCGA-BH-A0E1	52	II	LumA	0.90	no progression, followed 15.7 mo
TCGA-C8-A273	29	II	LumA	0.87	no progression, followed 16.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**39**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

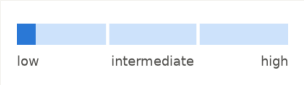
Sample: **TCGA-A2-A04V-01** · Aneuploidy score: **6** · TMB (nonsyn., /Mb): **0.9** · MSIsensor score: **0.04** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

7th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 1% · 24 mo 4% · 60 mo 14%

Model-estimated progression time

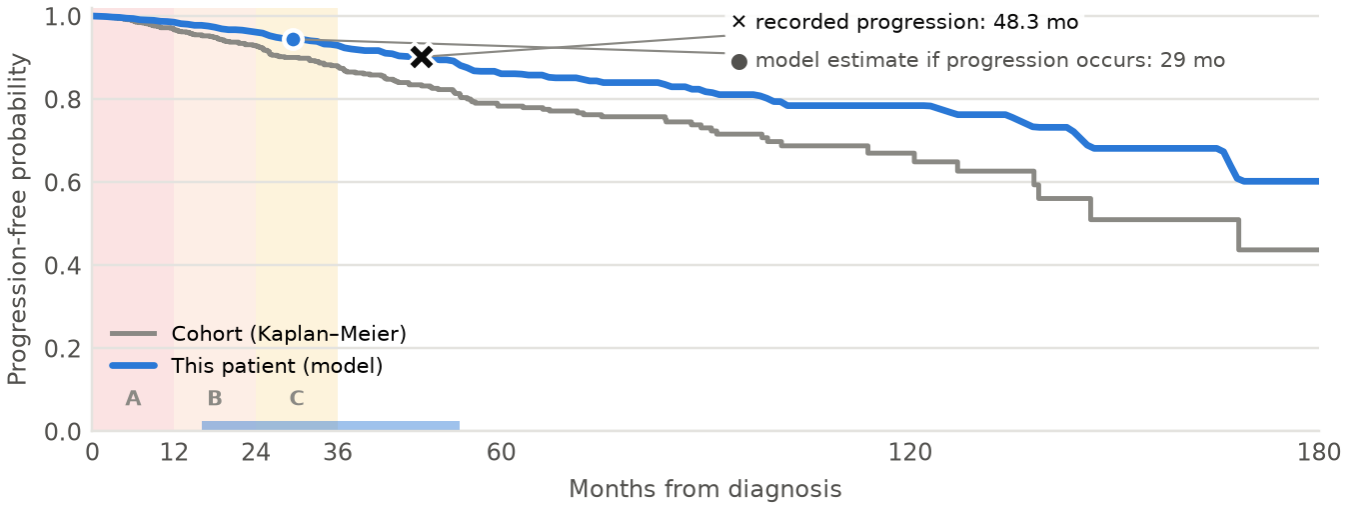
≈ 29 months

Window 24–36 months · IQR 16–54 mo
if progression occurs

Recorded progression

48.3 months

Group D · Late progression
✗ **recorded window differs** · |error| 19 mo



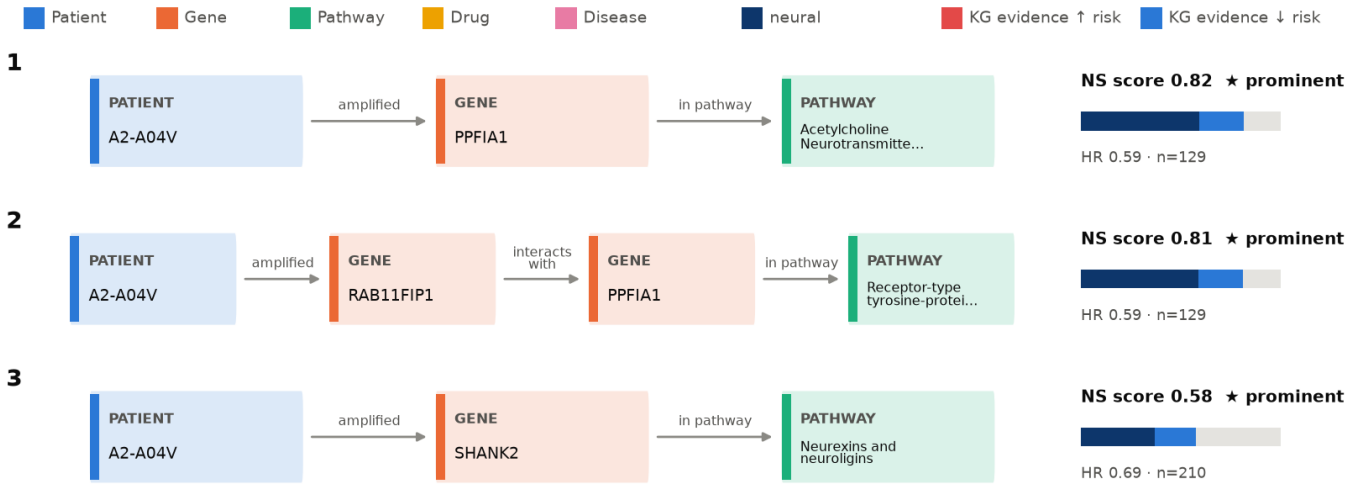
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 39-year-old patient has a LumA tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **7th risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND Age ≥ 31.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PPFIA1** to **Acetylcholine Neurotransmitter Release Cycle**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.59). Progression was recorded at 48.3 months; the model's estimate given progression was 29 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

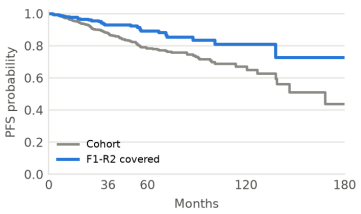


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-A2-A04V** amplified → **PPFIA1** in pathway → **Acetylcholine Neurotransmitter Release Cycle**	99	↓ **slower progression** HR 0.59, z -1.7, n = 129 carriers of "Acetylcholine Neurotransmitter Release Cycle"	**0.82**
2 ★	**TCGA-A2-A04V** amplified → **RAB11FIP1** interacts with → **PPFIA1** in pathway → **Receptor-type tyrosine-protein phosphatases**	98	↓ **slower progression** HR 0.59, z -1.7, n = 129 carriers of "Receptor-type tyrosine-protein phosphatases"	**0.81**
3 ★	**TCGA-A2-A04V** amplified → **SHANK2** in pathway → **Neurexins and neuroligins**	61	↓ **slower progression** HR 0.69, z -1.6, n = 210 carriers of "Neurexins and neuroligins"	**0.58**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

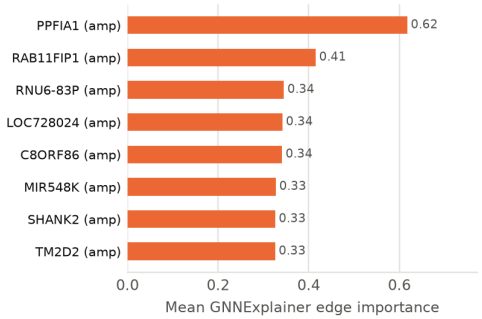
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R2 ↓ **lower risk**	N stage = N0 AND Age ≥ 31.5	410 pts 29 events	**0.36** p 4.5e-07	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
PIK3CA · driver	Mutation	p.G118D	70	0.31
PPFIA1	High-level amplification	GISTIC ±2	6	0.62
RAB11FIP1	High-level amplification	GISTIC ±2	0	0.41
RNU6-83P	High-level amplification	GISTIC ±2	0	0.34
LOC728024	High-level amplification	GISTIC ±2	0	0.34
C8ORF86	High-level amplification	GISTIC ±2	0	0.34
MIR548K	High-level amplification	GISTIC ±2	0	0.33
SHANK2	High-level amplification	GISTIC ±2	1	0.33
TM2D2	High-level amplification	GISTIC ±2	0	0.33
LETM2	High-level amplification	GISTIC ±2	0	0.32



56 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 64 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-GM-A3NW	63	II	LumA	0.87	no progression, followed 110.5 mo
TCGA-C8-A12O	50	II	LumA	0.84	no progression, followed 12.7 mo
TCGA-BH-A201	64	I	LumA	0.83	no progression, followed 28.1 mo
TCGA-E2-A15R	64	II	LumA	0.82	no progression, followed 56.9 mo
TCGA-D8-A1XF	45	II	LumB	0.82	no progression, followed 15.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group D** · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**67**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EAS**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A1EX-01** · Aneuploidy score: **12** · TMB (nonsyn., /Mb): **0.8** · MSIsensor score: **0.23** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

46th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 5% · 60 mo 17%

Model-estimated progression time

≈ 40 months

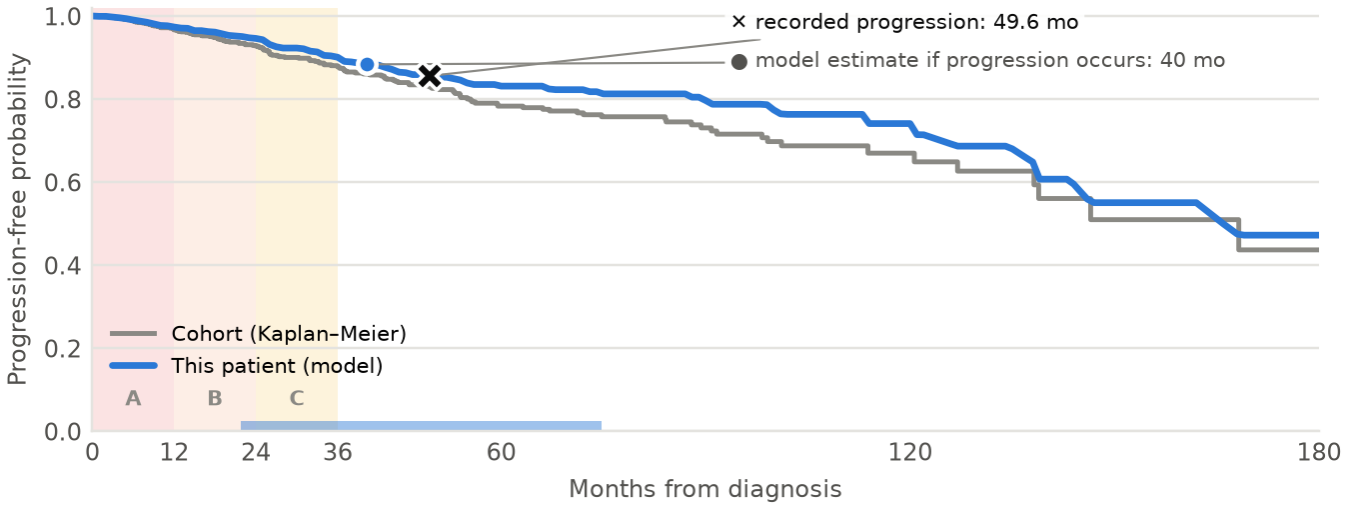
Window > 36 months · IQR 22–75 mo if progression occurs

Recorded progression

49.6 months

■ **Group D** · Late progression

✓ **estimate in recorded window** · |error| 9 mo



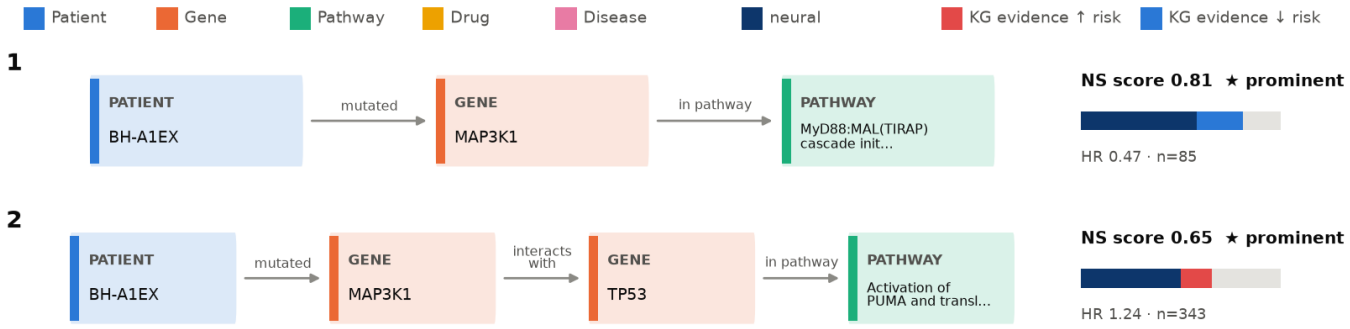
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 67-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **46th risk percentile** (intermediate tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 86.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **MAP3K1** to **MyD88:MAL(TIRAP) cascade initiated on plasma membrane**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.47). Progression was recorded at 49.6 months; the model's estimate given progression was 40 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

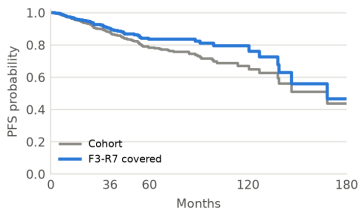


2 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-BH-A1EX** mutated → **MAP3K1** in pathway → **MyD88:MAL(TIRAP) cascade initiated on plasma membrane**	97	↓ **slower progression** HR 0.47, z -1.7, n = 85 carriers of "MyD88:MAL(TIRAP) cascade initiated on plasma membrane"	**0.81**
2 ★	**TCGA-BH-A1EX** mutated → **MAP3K1** interacts with → **TP53** in pathway → **Activation of PUMA and translocation to mitochondria**	83	↑ **faster progression** HR 1.24, z +1.2, n = 343 carriers of "Activation of PUMA and translocation to mitochondria"	**0.65**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

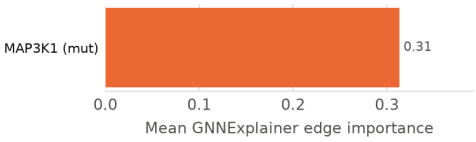
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo
F3-R8 ↓ lower risk	AJCC stage = II AND Age < 77.5 AND TMB (nonsynonymous) ≥ 0.08	433 pts 40 events	**0.50** p 3.2e-04	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MAP3K1 · driver	Mutation	p.W201R	13	0.31



1 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MAP3K1 → AZD-8330 (DB06061); MAP3K1 → Binimetinib (DB11967); MAP3K1 → Fostamatinib (DB12010). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-EW-A1P6	64	II	LumA	0.96	no progression, followed 18.5 mo
TCGA-BH-A0E1	52	II	LumA	0.94	no progression, followed 15.7 mo
TCGA-E9-A245	47	II	LumA	0.92	no progression, followed 0.9 mo
TCGA-BH-A0DH	63	II	LumA	0.90	no progression, followed 38.0 mo
TCGA-D8-A1X9	66	II	LumA	0.89	no progression, followed 23.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for

51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**47**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**AFR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

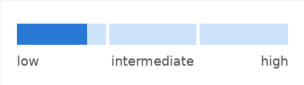
Sample: **TCGA-B6-A402-01** · Aneuploidy score: **17** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0.48** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

26th

Low risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

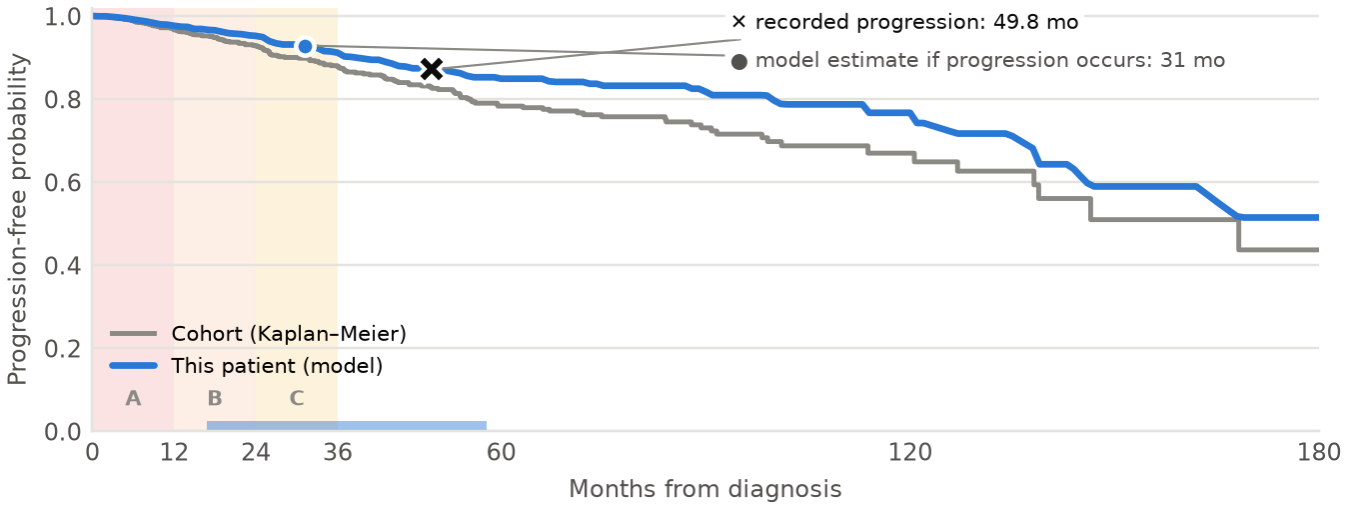
≈ 31 months

Window 24–36 months · IQR 17–58 mo
if progression occurs

Recorded progression

49.8 months

Group D · Late progression
✗ recorded window differs · |error| 19 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 47-year-old patient has a subtype-unassigned tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **26th risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND 41.5 ≤ Age < 78.5*) marks a group with a progression hazard 0.3× that of the remaining patients. Progression was recorded at 49.8 months; the model's estimate given progression was 31 months.

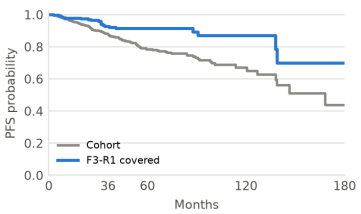
3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

No explanation path is available: none of this patient's altered panel genes has a stored PrimeKG pathway, drug, or breast-disease link (or the tumour carries no qualifying alteration). The risk estimate therefore rests on the clinical block and the symbolic rules below.

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

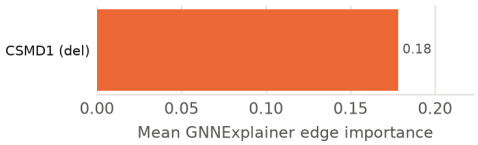
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R1 ↓ lower risk	N stage = N0 AND 41.5 ≤ Age < 78.5	351 pts 22 events	**0.30** p 5.2e-08	not reached
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo
F3-R2 ↓ lower risk	N stage = N0	418 pts 33 events	**0.37** p 4.8e-07	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CSMD1	Homozygous deletion	GISTIC ±2	0	0.18



1 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A14S	65	I	missing	0.94	no progression, followed 33.2 mo
TCGA-BH-A0B8	64	I	missing	0.91	no progression, followed 51.6 mo
TCGA-AR-A252	50	I	LumA	0.82	no progression, followed 93.3 mo
TCGA-A2-A04R	36	I	LumB	0.79	no progression, followed 121.9 mo
TCGA-A2-A0CQ	62	I	LumA	0.78	no progression, followed 88.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**45**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**—**	PAM50 subtype	**LumA**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

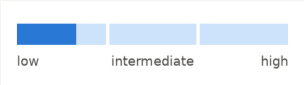
Sample: **TCGA-HN-A2OB-01** · Aneuploidy score: **2** · TMB (nonsyn., /Mb): **0.63** · MSIsensor score: **0** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

22nd

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

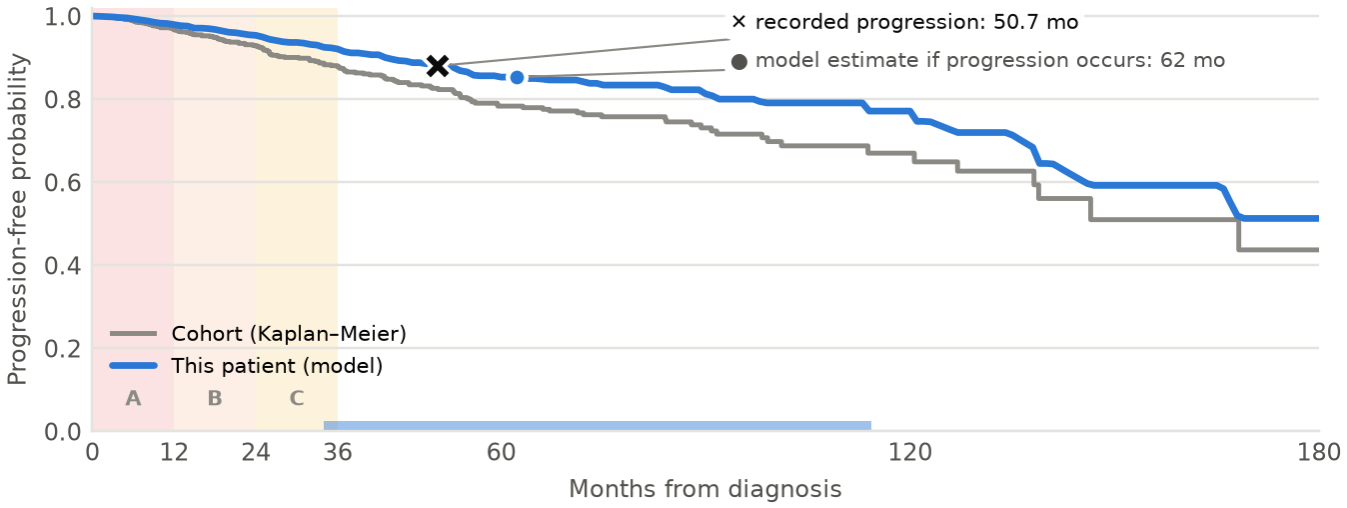
≈ 62 months

Window > 36 months · IQR 34–114 mo
if progression occurs

Recorded progression

50.7 months

Group D · Late progression
✓ **estimate in recorded window** · |error| 12 mo



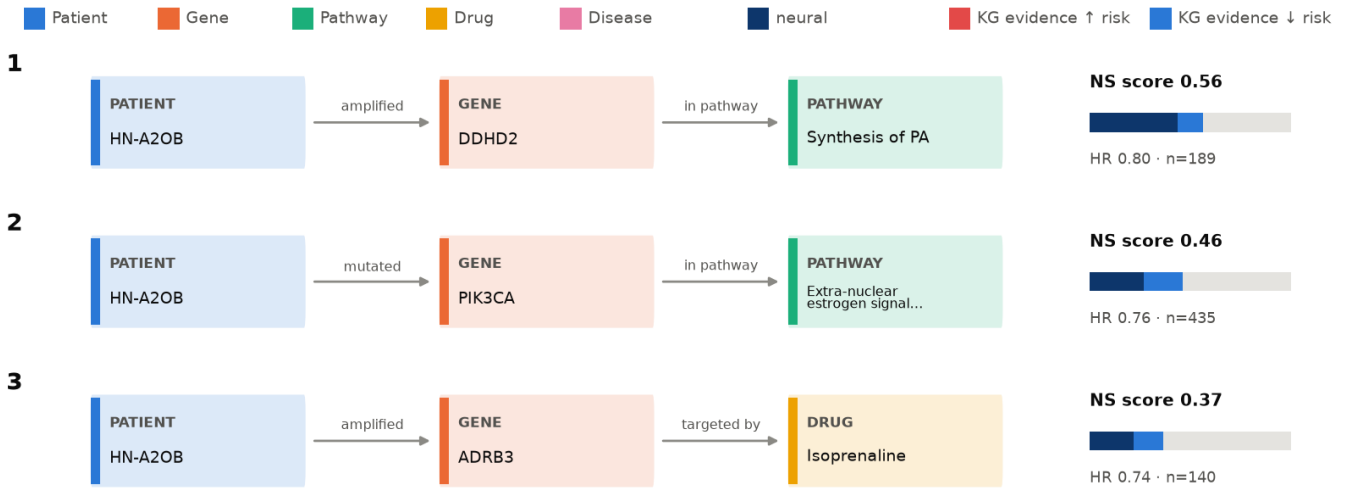
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 45-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **22nd risk percentile** (low tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **DDHD2** to **Synthesis of PA**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.80). Progression was recorded at 50.7 months; the model's estimate given progression was 62 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

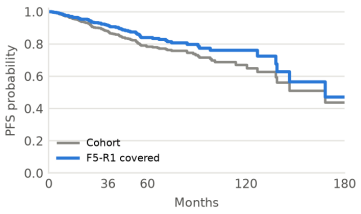


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-HN-A2OB** amplified → **DDHD2** in pathway → **Synthesis of PA**	73	↓ **slower progression** HR 0.80, z -0.9, n = 189 carriers of “Synthesis of PA”	**0.56**
2	**TCGA-HN-A2OB** mutated → **PIK3CA** in pathway → **Extra-nuclear estrogen signaling**	45	↓ **slower progression** HR 0.76, z -1.5, n = 435 carriers of “Extra-nuclear estrogen signaling”	**0.46**
3	**TCGA-HN-A2OB** amplified → **ADRB3** targeted by → **Isoprenaline**	37	↓ **slower progression** HR 0.74, z -1.1, n = 140 carriers of “Isoprenaline”	**0.37**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

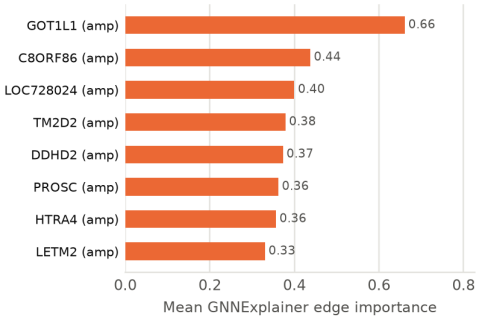
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ lower risk	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R3 ↓ lower risk	AJCC stage = II AND 39.5 ≤ Age < 78.5	422 pts 31 events	**0.36** p 4.2e-07	168 mo
F5-R5 ↓ lower risk	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
BAP1 · driver	Mutation	p.I214M	9	0.32
PIK3CA · driver	Mutation	p.E542K	70	0.31
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
GOT1L1	High-level amplification	GISTIC ±2	0	0.66
C8ORF86	High-level amplification	GISTIC ±2	0	0.44
LOC728024	High-level amplification	GISTIC ±2	0	0.40
TM2D2	High-level amplification	GISTIC ±2	0	0.38
DDHD2	High-level amplification	GISTIC ±2	1	0.37
PROSC	High-level amplification	GISTIC ±2	0	0.36
HTRA4	High-level amplification	GISTIC ±2	0	0.36
LETM2	High-level amplification	GISTIC ±2	0	0.33
RNF5P1	High-level amplification	GISTIC ±2	0	0.33



32 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109) … and 52 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0ES	52	II	LumA	0.81	no progression, followed 72.0 mo
TCGA-BH-A1ES	35	II	missing	0.80	progressed at 113.8 mo (group D)
TCGA-E2-A1IE	61	II	LumA	0.77	progressed at 59.5 mo (group D)
TCGA-WT-AB41	55	II	LumA	0.77	no progression, followed 53.0 mo
TCGA-BH-A0EI	51	II	LumA	0.74	no progression, followed 63.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**47**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

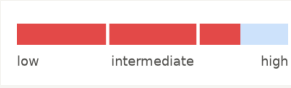
Sample: **TCGA-AR-A0U2-01** · Aneuploidy score: **22** · TMB (nonsyn., /Mb): **1.43** · MSIsensor score: **1.19** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

82nd

High risk tier · fused neuro-symbolic model



Probability of progression

16% by 36 mo

12 mo 4% · 24 mo 10% · 60 mo 27%

Model-estimated progression time

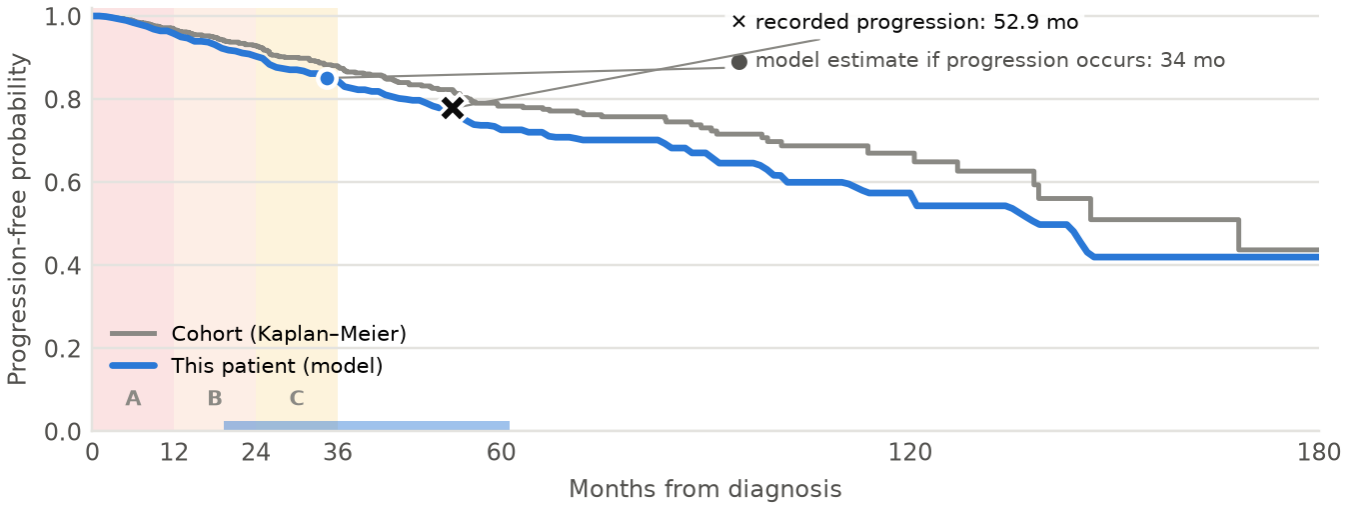
≈ 34 months

Window 24–36 months · IQR 19–61 mo
if progression occurs

Recorded progression

52.9 months

Group D · Late progression
✗ recorded window differs · |error| 19 mo



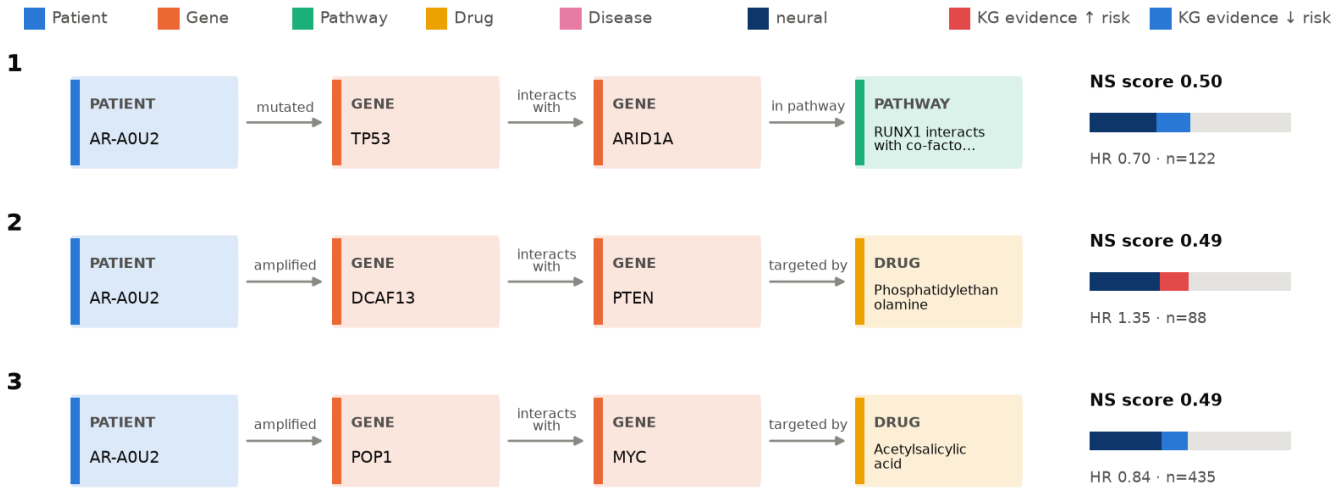
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 47-year-old patient has a LumB tumour, AJCC stage III (T2, N2, M0). The fused neuro-symbolic model places them in the **82nd risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 83.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **TP53** to **RUNX1 interacts with co-factors whose precise effect on RUNX1 targets is not known**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.70). Progression was recorded at 52.9 months; the model's estimate given progression was 34 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

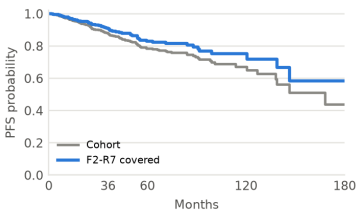


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-AR-A0U2** mutated → **TP53** interacts with → **ARID1A** in pathway → **RUNX1 interacts with co-factors whose precise effect on RUNX1 targets is not known**	56	↓ **slower progression** HR 0.70, z -1.2, n = 122 carriers of “RUNX1 interacts with co-factors whose precise effect on RUNX1 targets is not known”	**0.50**
2	**TCGA-AR-A0U2** amplified → **DCAF13** interacts with → **PTEN** targeted by → **Phosphatidylethanolamine**	58	↑ **faster progression** HR 1.35, z +1.1, n = 88 carriers of “Phosphatidylethanolamine”	**0.49**
3	**TCGA-AR-A0U2** amplified → **POP1** interacts with → **MYC** targeted by → **Acetylsalicylic acid**	60	↓ **slower progression** HR 0.84, z -1.0, n = 435 carriers of “Acetylsalicylic acid”	**0.49**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

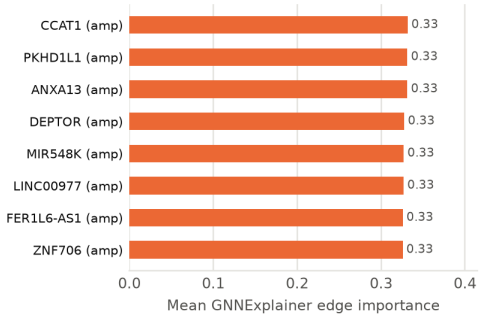
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 83.5	638 pts 66 events	**0.39** p 3.3e-07	not reached
F2-R8 ↓ lower risk	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
GATA3 · driver	Mutation	p.P408Afs*99, p.R364S	13	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
TP53 · driver	Mutation	p.C176F	57	0.31
CCAT1	High-level amplification	GISTIC ±2	0	0.33
PKHD1L1	High-level amplification	GISTIC ±2	0	0.33
ANXA13	High-level amplification	GISTIC ±2	0	0.33
DEPTOR	High-level amplification	GISTIC ±2	0	0.33
MIR548K	High-level amplification	GISTIC ±2	0	0.33
LINC00977	High-level amplification	GISTIC ±2	0	0.33
FER1L6-AS1	High-level amplification	GISTIC ±2	0	0.33
ZNF706	High-level amplification	GISTIC ±2	1	0.33



197 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404) … and 33 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-C8-A130	52	III	LumB	0.83	no progression, followed 12.2 mo
TCGA-AN-A0AS	70	III	LumA	0.81	no progression, followed 0.3 mo
TCGA-A2-A0YH	53	III	LumB	0.80	no progression, followed 21.7 mo
TCGA-BH-A0HX	54	II	LumA	0.80	no progression, followed 27.3 mo
TCGA-A8-A08P	70	III	LumB	0.79	no progression, followed 31.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**31**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**Normal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8502/3**

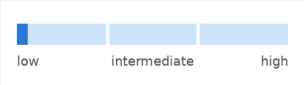
Sample: **TCGA-AO-A03U-01** · Aneuploidy score: **1** · TMB (nonsyn., /Mb): **0.17** · MSIsensor score: **0.01** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

4th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 12%

Model-estimated progression time

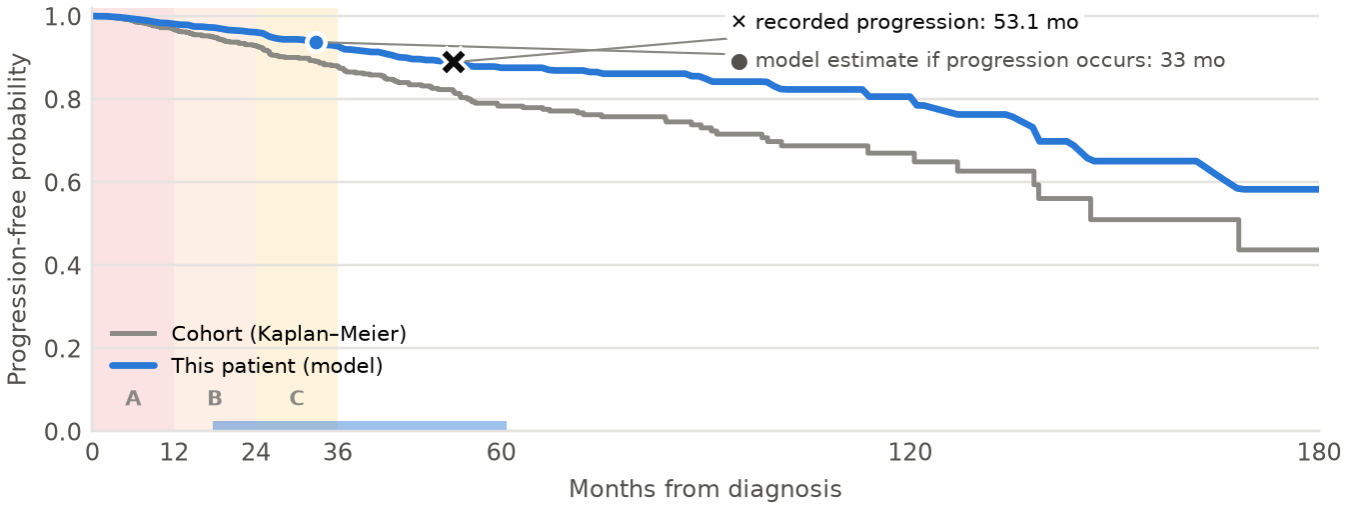
≈ 33 months

Window 24–36 months · IQR 18–61 mo
if progression occurs

Recorded progression

53.1 months

Group D · Late progression
✗ recorded window differs · |error| 20 mo



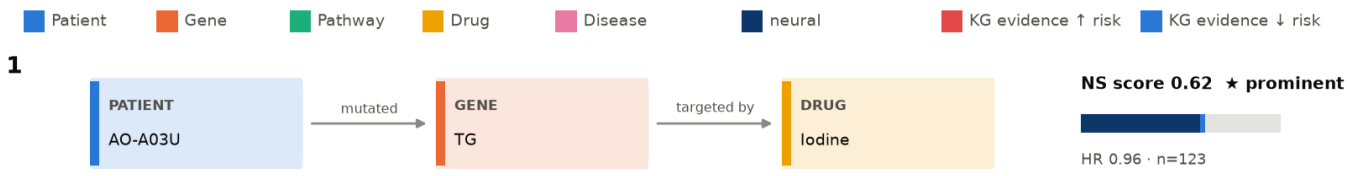
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 31-year-old patient has a Normal tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **4th risk percentile** (low tier). The most significant rule covering them (*N stage = N0*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **TG** to **Iodine**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.96). Progression was recorded at 53.1 months; the model's estimate given progression was 33 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

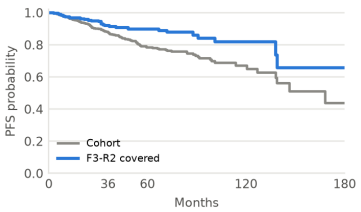


1 of 1 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-AO-A03U** mutated → **TG** targeted by → **Iodine**	100	↓ **slower progression** HR 0.96, z -0.2, n = 123 carriers of “Iodine”	**0.62**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

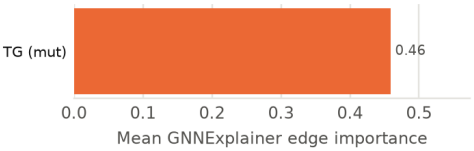
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R2 ↓ **lower risk**	N stage = N0	418 pts 33 events	**0.37** p 4.8e-07	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TG	Mutation	p.R492=	1	0.46



1 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TG → Iodine (DB05382). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A1-A0SB	70	I	Normal	0.84	no progression, followed 8.5 mo
TCGA-D8-A27E	66	I	LumA	0.72	no progression, followed 17.4 mo
TCGA-BH-A8FZ	58	I	Normal	0.71	no progression, followed 18.9 mo
TCGA-A2-A04R	36	I	LumB	0.68	no progression, followed 121.9 mo
TCGA-A7-A3IY	71	I	missing	0.68	no progression, followed 11.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**40**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-E2-A1L7-01** · Aneuploidy score: **6** · TMB (nonsyn., /Mb): **2.07** · MSIsensor score: **0.93** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

78th

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 25%

Model-estimated progression time

≈ 38 months

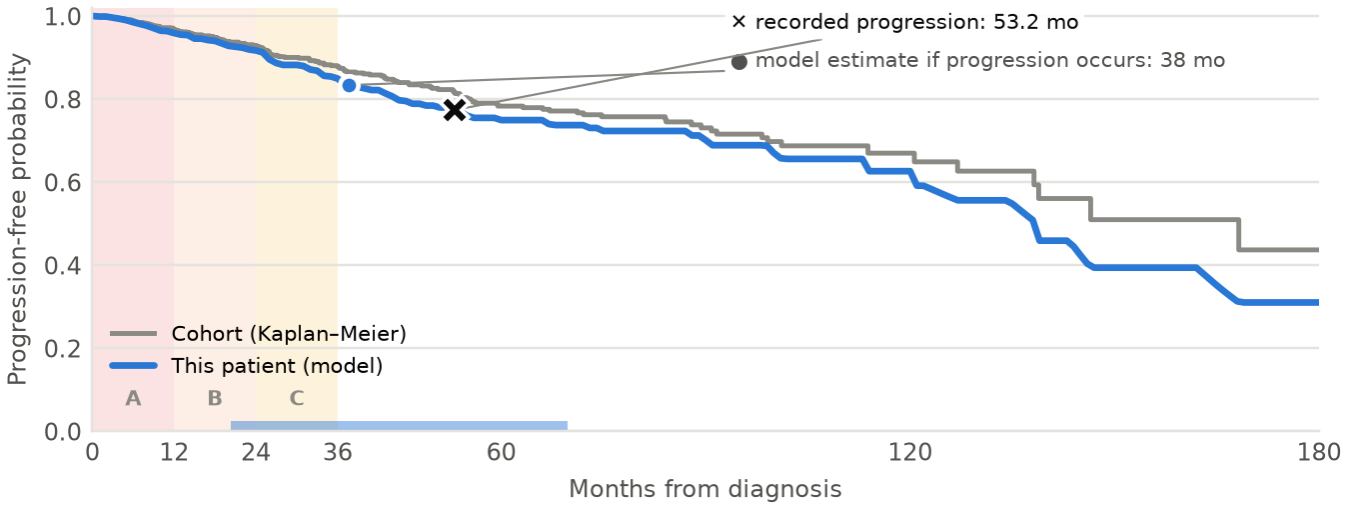
Window > 36 months · IQR 20–70 mo if progression occurs

Recorded progression

53.2 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 16 mo



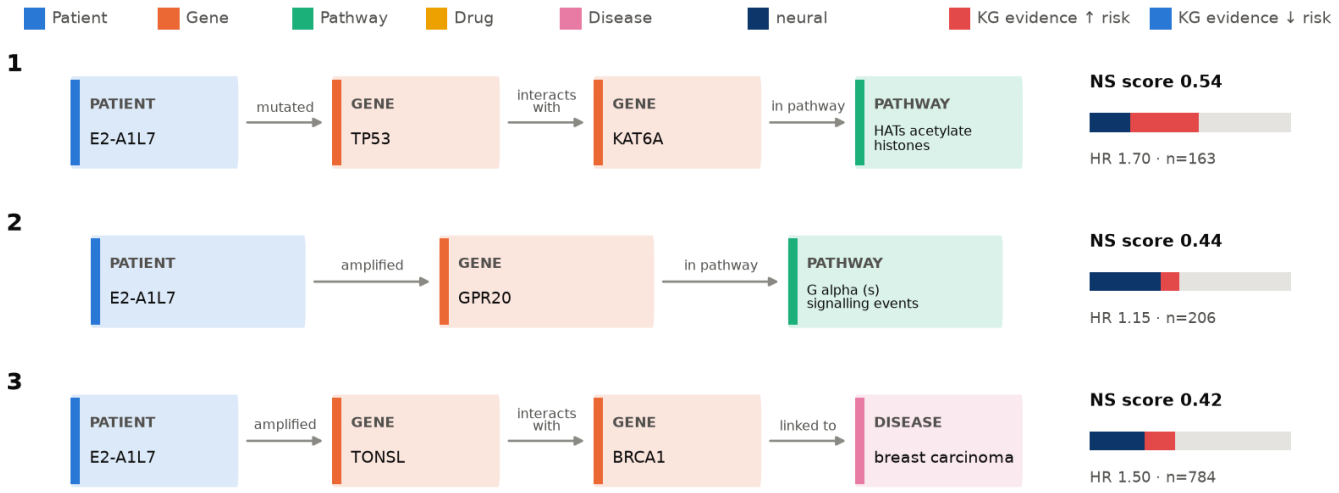
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 40-year-old patient has a LumB tumour, AJCC stage III (T2, N2, M0). The fused neuro-symbolic model places them in the **78th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **TP53** to **HATs acetylate histones**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.70). Progression was recorded at 53.2 months; the model's estimate given progression was 38 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

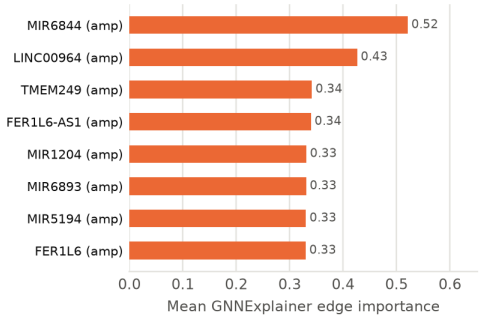
#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-E2-A1L7** mutated → **TP53** interacts with → **KAT6A** in pathway → **HATs acetylate histones**	34	↑ **faster progression** HR 1.70, z +2.6, n = 163 carriers of "HATs acetylate histones"	**0.54**
2	**TCGA-E2-A1L7** amplified → **GPR20** in pathway → **G alpha (s) signalling events**	59	↑ **faster progression** HR 1.15, z +0.7, n = 206 carriers of "G alpha (s) signalling events"	**0.44**
3	**TCGA-E2-A1L7** amplified → **TONSL** interacts with → **BRCA1** linked to → **breast carcinoma**	46	↑ **faster progression** HR 1.50, z +1.1, n = 784 carriers of "breast carcinoma"	**0.42**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank p < 0.05, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
BRCA1 · driver	Homozygous deletion	GISTIC ±2	29	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
TP53 · driver	Mutation	p.I255F	57	0.31
MIR6844	High-level amplification	GISTIC ±2	0	0.52
LINC00964	High-level amplification	GISTIC ±2	0	0.43
TMEM249	High-level amplification	GISTIC ±2	0	0.34
FER1L6-AS1	High-level amplification	GISTIC ±2	0	0.34
MIR1204	High-level amplification	GISTIC ±2	0	0.33
MIR6893	High-level amplification	GISTIC ±2	0	0.33
MIR5194	High-level amplification	GISTIC ±2	0	0.33
FER1L6	High-level amplification	GISTIC ±2	0	0.33
MIR7112-2	High-level amplification	GISTIC ±2	0	0.33



208 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548) … and 78 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A8-A09N	57	III	LumB	0.85	no progression, followed 1.0 mo
TCGA-AR-A251	51	III	Basal	0.81	no progression, followed 99.6 mo
TCGA-E9-A22E	56	III	LumB	0.79	no progression, followed 41.7 mo
TCGA-BH-A18U	72	III	LumB	0.79	no progression, followed 51.4 mo
TCGA-BH-A0DS	71	III	LumA	0.78	no progression, followed 2.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**45**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

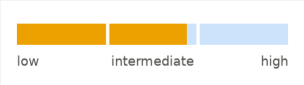
Sample: **TCGA-AR-A256-01** · Aneuploidy score: **17** · TMB (nonsyn., /Mb): **5.67** · MSIsensor score: **4.53** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

63rd

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

12% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 21%

Model-estimated progression time

≈ 16 months

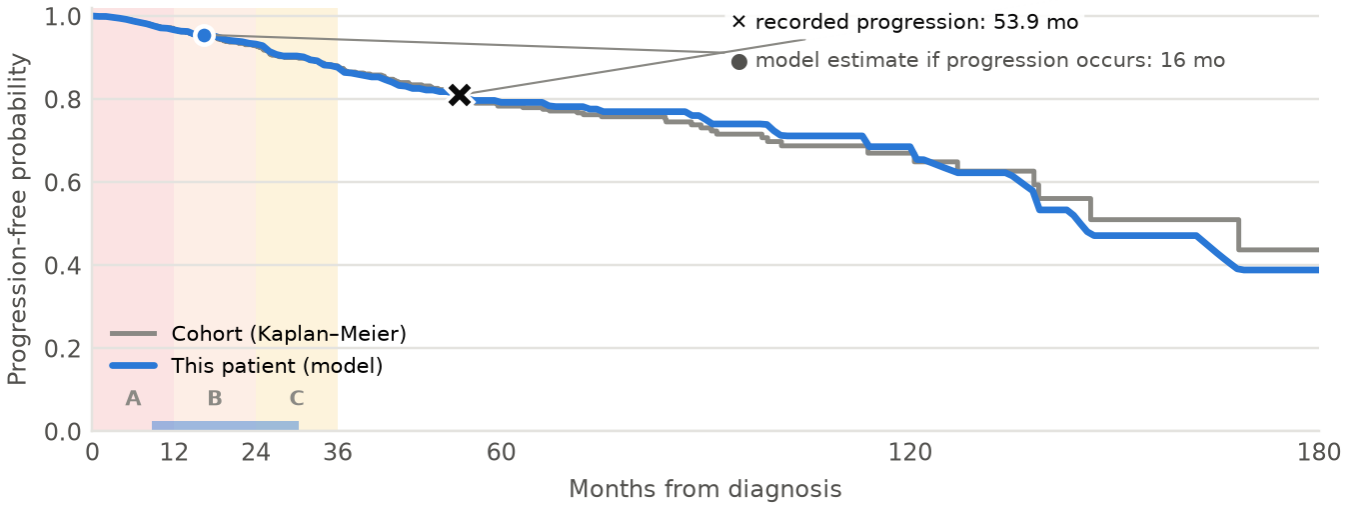
Window 12–24 months · IQR 9–30 mo if progression occurs

Recorded progression

53.9 months

Group D · Late progression

✗ recorded window differs · |error| 38 mo



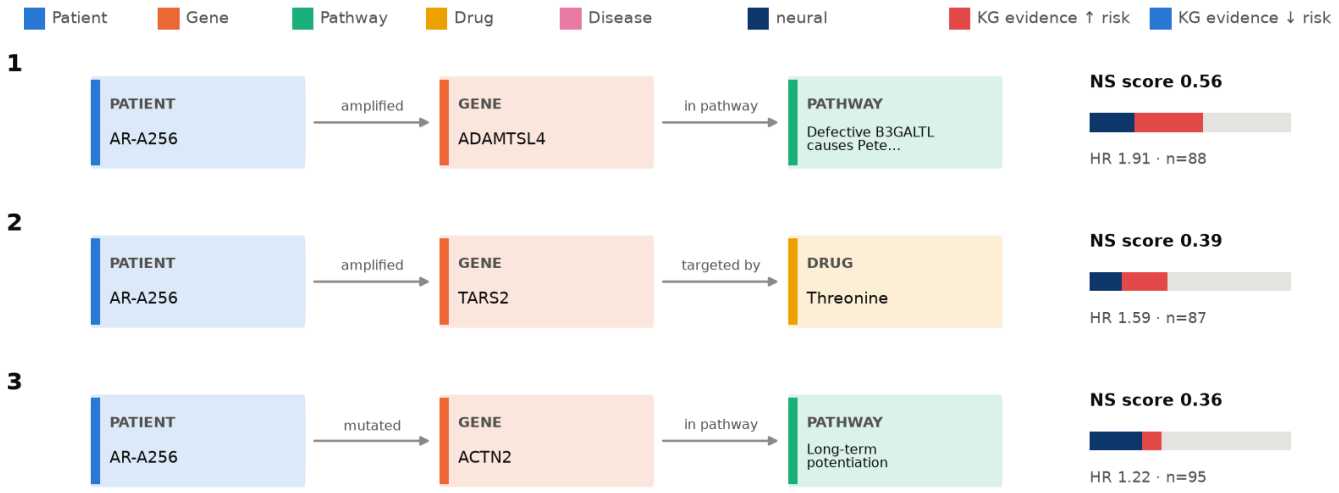
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 45-year-old patient has a Basal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **63rd risk percentile** (intermediate tier). The most significant rule covering them (*N stage = N0 AND 41.5 ≤ Age < 78.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ADAMTSL4** to **Defective B3GALTL causes Peters-plus syndrome (PpS)**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.91). Progression was recorded at 53.9 months; the model's estimate given progression was 16 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

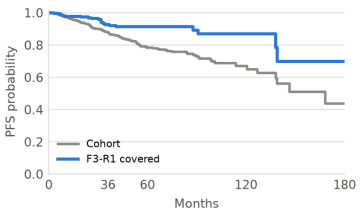


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-AR-A256** amplified → **ADAMTSL4** in pathway → **Defective B3GALTL causes Peters-plus syndrome (PpS)**	37	↑ **faster progression** HR 1.91, z +2.6, n = 88 carriers of “Defective B3GALTL causes Peters-plus syndrome (PpS)”	**0.56**
2	**TCGA-AR-A256** amplified → **TARS2** targeted by → **Threonine**	26	↑ **faster progression** HR 1.59, z +1.7, n = 87 carriers of “Threonine”	**0.39**
3	**TCGA-AR-A256** mutated → **ACTN2** in pathway → **Long-term potentiation**	43	↑ **faster progression** HR 1.22, z +0.7, n = 95 carriers of “Long-term potentiation”	**0.36**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

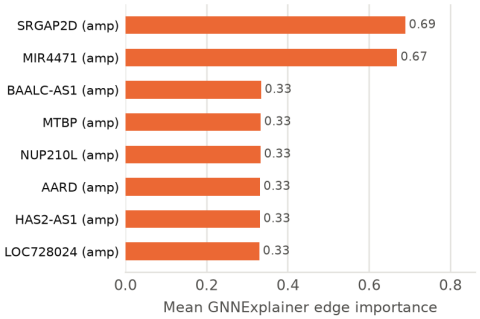
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R1 ↓ lower risk	N stage = N0 AND 41.5 ≤ Age < 78.5	351 pts 22 events	**0.30** p 5.2e-08	not reached
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo
F3-R2 ↓ lower risk	N stage = N0	418 pts 33 events	**0.37** p 4.8e-07	not reached
F3-R8 ↓ lower risk	AJCC stage = II AND Age < 77.5 AND TMB (nonsynonymous) ≥ 0.08	433 pts 40 events	**0.50** p 3.2e-04	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
TP53 · driver	Mutation	p.R110Sfs*14	57	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
SRGAP2D	High-level amplification	GISTIC ±2	0	0.69
MIR4471	High-level amplification	GISTIC ±2	0	0.67
BAALC-AS1	High-level amplification	GISTIC ±2	0	0.33
MTBP	High-level amplification	GISTIC ±2	0	0.33
NUP210L	High-level amplification	GISTIC ±2	0	0.33
AARD	High-level amplification	GISTIC ±2	0	0.33
HAS2-AS1	High-level amplification	GISTIC ±2	0	0.33
LOC728024	High-level amplification	GISTIC ±2	0	0.33
OC90	High-level amplification	GISTIC ±2	0	0.33



204 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 89 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A3QA	33	II	Basal	0.86	no progression, followed 30.2 mo
TCGA-A7-A13D	46	II	Basal	0.86	no progression, followed 31.7 mo
TCGA-E2-A14R	62	II	Basal	0.85	progressed at 31.9 mo (group C)
TCGA-AN-A0FX	52	II	Basal	0.84	no progression, followed 0.3 mo
TCGA-AC-A6IW	73	II	Basal	0.84	no progression, followed 13.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**73**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**EUR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8523/3**

Sample: **TCGA-BH-A1FR-01** · Aneuploidy score: **—** · TMB (nonsyn., /Mb): **0.47** · MSIsensor score: **0.11** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

72nd

High risk tier · fused neuro-symbolic model



Probability of progression

13% by 36 mo

12 mo 3% · 24 mo 8% · 60 mo 23%

Model-estimated progression time

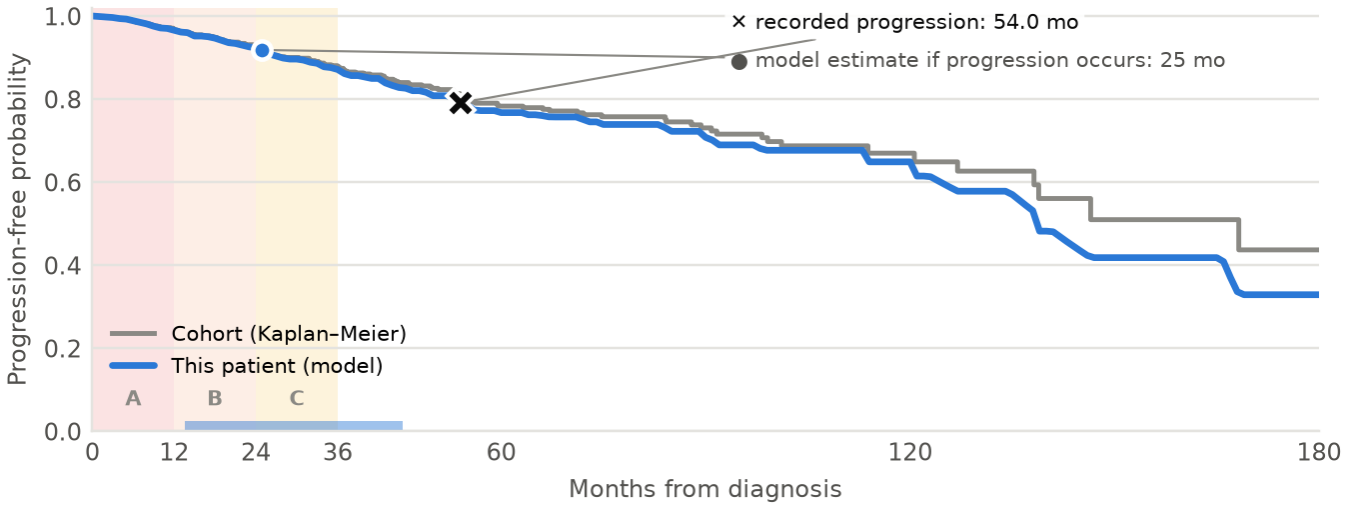
≈ 25 months

Window 24–36 months · IQR 14–46 mo
if progression occurs

Recorded progression

54.0 months

Group D · Late progression
✗ recorded window differs · |error| 29 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 73-year-old patient has a LumA tumour, AJCC stage III (T4, N1, M0). The fused neuro-symbolic model places them in the **72nd risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. Progression was recorded at 54.0 months; the model's estimate given progression was 25 months.

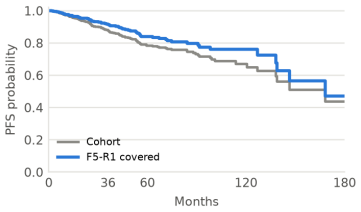
3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

No explanation path is available: none of this patient's altered panel genes has a stored PrimeKG pathway, drug, or breast-disease link (or the tumour carries no qualifying alteration). The risk estimate therefore rests on the clinical block and the symbolic rules below.

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ lower risk	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R5 ↓ lower risk	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

No mutation or high-level CNA on the 500-gene panel was recorded for this tumour.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1XC	85	III	LumA	0.94	progressed at 12.4 mo (group B)
TCGA-A2-A3KD	47	III	LumA	0.76	no progression, followed 39.6 mo
TCGA-B6-A0RI	44	III	LumA	0.70	progressed at 36.2 mo (group D)
TCGA-B6-A0RO	71	III	LumA	0.66	no progression, followed 162.0 mo
TCGA-D8-A73W	79	III	LumA	0.66	progressed at 0.7 mo (group A)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**59**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

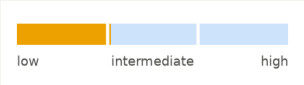
Sample: **TCGA-AO-A03N-01** · Aneuploidy score: **23** · TMB (nonsyn., /Mb): **3** · MSIsensor score: **0.23** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

35th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

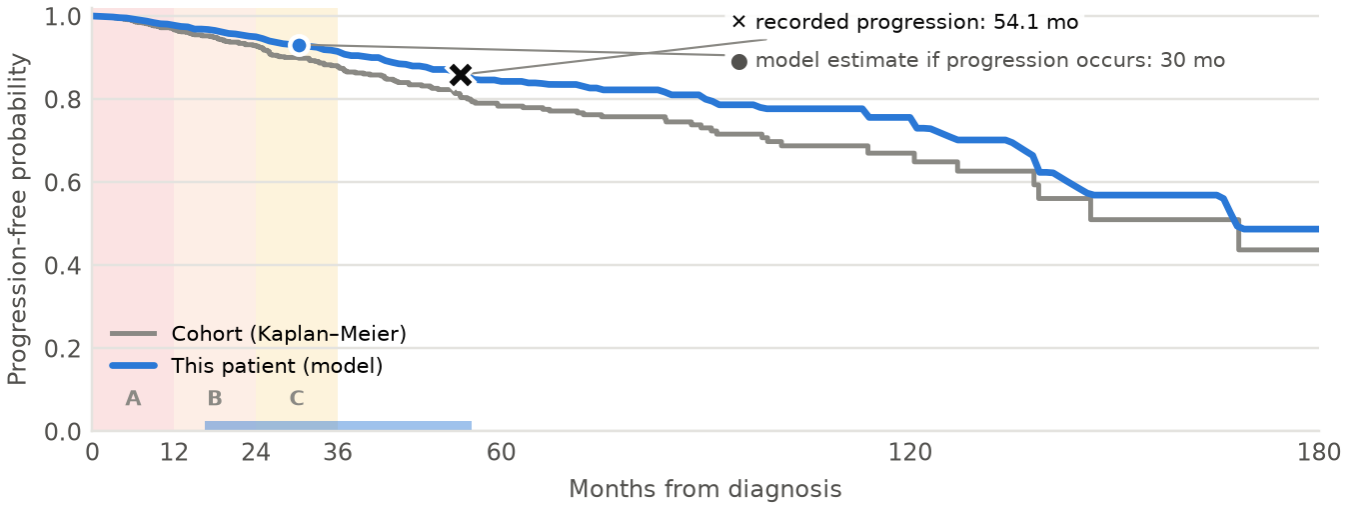
≈ 30 months

Window 24–36 months · IQR 17–56 mo
if progression occurs

Recorded progression

54.1 months

Group D · Late progression
✗ recorded window differs · |error| 24 mo



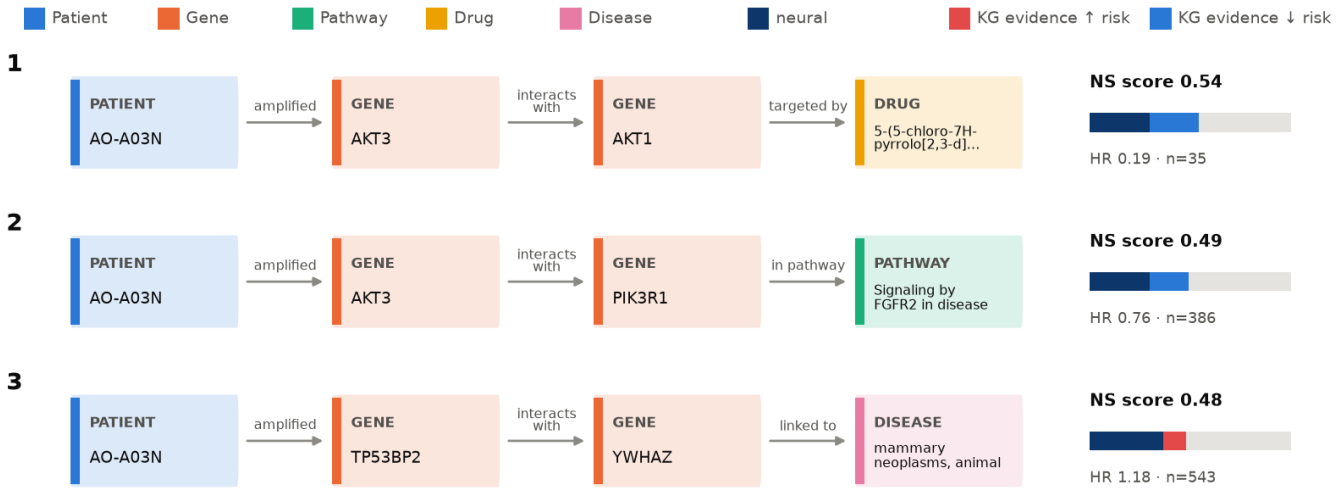
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 59-year-old patient has a LumB tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **35th risk percentile** (intermediate tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **AKT3** to **5-(5-chloro-7H-pyrrolo[2,3-d]pyrimidin-4-yl)-4,5,6,7-tetrahydro-1H-imidazo[4,5-c]pyridine**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.19). Progression was recorded at 54.1 months; the model's estimate given progression was 30 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

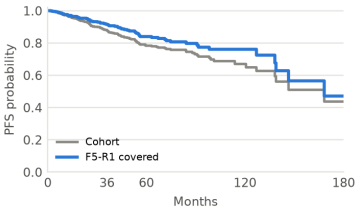


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-AO-A03N** amplified → **AKT3** interacts with → **AKT1** targeted by → **5-(5-chloro-7H-pyrrolo[2,3-d]pyrimidin-4-yl)-4,5,6,7-tetrahydro-1H-imidazo[4,5-c]pyridine**	49	↓ **slower progression** HR 0.19, z -1.9, n = 35 carriers of "5-(5-chloro-7H-pyrrolo[2,3-d]pyrimidin-4-yl)-4,5,6,7-tetrahydro-1H-imidazo[4,5-c]pyridine"	**0.54**
2	**TCGA-AO-A03N** amplified → **AKT3** interacts with → **PIK3R1** in pathway → **Signaling by FGFR2 in disease**	50	↓ **slower progression** HR 0.76, z -1.4, n = 386 carriers of "Signaling by FGFR2 in disease"	**0.49**
3	**TCGA-AO-A03N** amplified → **TP53BP2** interacts with → **YWHAZ** linked to → **mammary neoplasms, animal**	61	↑ **faster progression** HR 1.18, z +0.9, n = 543 carriers of "mammary neoplasms, animal"	**0.48**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

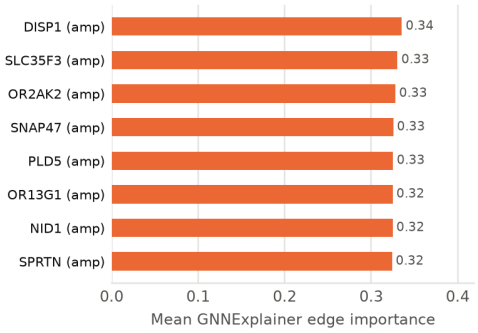
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ lower risk	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R3 ↓ lower risk	AJCC stage = II AND 39.5 ≤ Age < 78.5	422 pts 31 events	**0.36** p 4.2e-07	168 mo
F5-R4 ↓ lower risk	TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5	567 pts 64 events	**0.56** p 2.1e-03	168 mo
F5-R5 ↓ lower risk	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.31
TP53 · driver	Mutation	p.A161T	57	0.31
PIK3CA · driver	Mutation	p.E542K	70	0.31
FOXA1 · driver	High-level amplification	GISTIC ±2	9	0.31
DISP1	High-level amplification	GISTIC ±2	0	0.34
SLC35F3	High-level amplification	GISTIC ±2	0	0.33
OR2AK2	High-level amplification	GISTIC ±2	1	0.33
SNAP47	High-level amplification	GISTIC ±2	0	0.33
PLD5	High-level amplification	GISTIC ±2	0	0.33
OR13G1	High-level amplification	GISTIC ±2	1	0.32
NID1	High-level amplification	GISTIC ±2	3	0.32
SPRTN	High-level amplification	GISTIC ±2	1	0.32



83 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PTEN → Phosphatidylethanolamine (DB04327); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); PIK3CA → ATP (DB00171) … and 43 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1JB	54	II	LumA	0.85	no progression, followed 55.5 mo
TCGA-AR-A0U1	36	II	missing	0.84	no progression, followed 133.2 mo
TCGA-E9-A22G	47	II	Basal	0.84	no progression, followed 40.7 mo
TCGA-A2-A25B	39	II	LumB	0.80	progressed at 42.4 mo (group D)
TCGA-A2-A0D0	60	II	Basal	0.80	no progression, followed 67.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A1FL · Group D

Report date 14 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**69**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

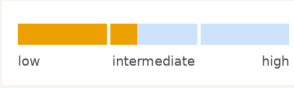
Sample: **TCGA-BH-A1FL-01** · Aneuploidy score: **4** · TMB (nonsyn., /Mb): **0.6** · MSIsensor score: **0.94** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

44th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 17%

Model-estimated progression time

≈ 49 months

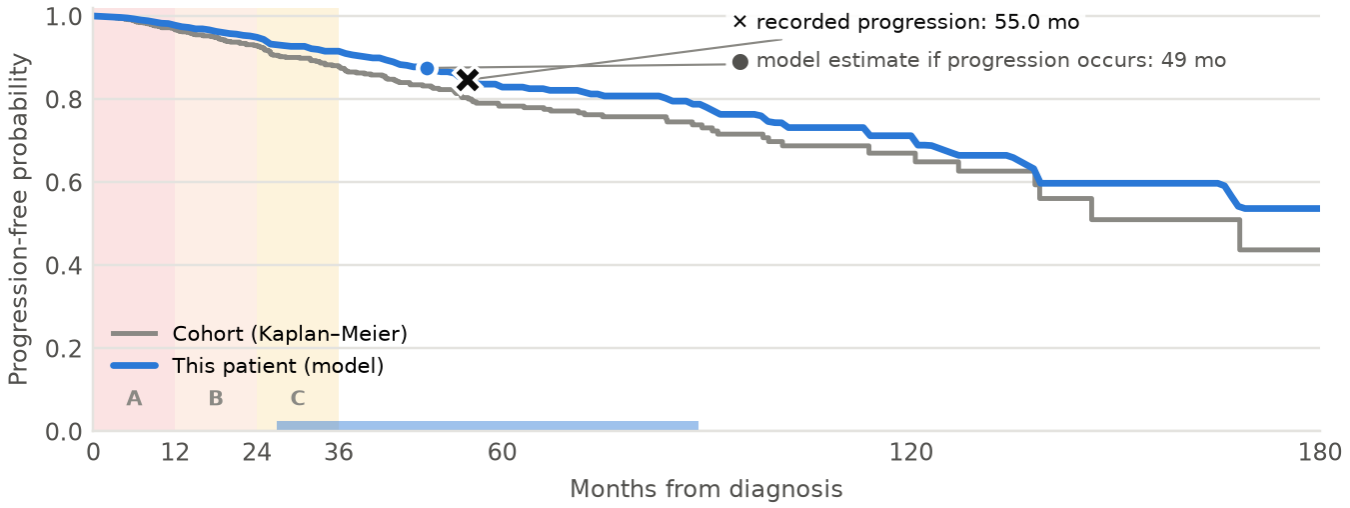
Window > 36 months · IQR 27–89 mo if progression occurs

Recorded progression

55.0 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 6 mo



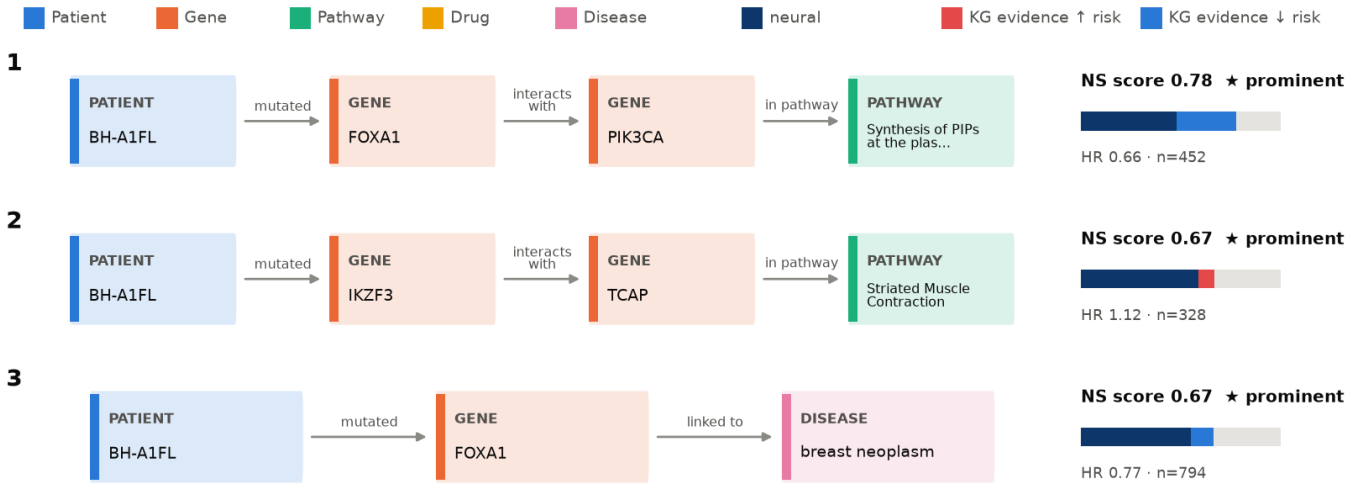
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 69-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **44th risk percentile** (intermediate tier). The most significant rule covering them (*Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **FOXA1** to **Synthesis of PIPs at the plasma membrane**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.66). Progression was recorded at 55.0 months; the model's estimate given progression was 49 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

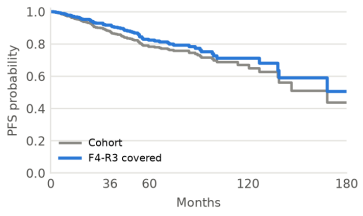


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-BH-A1FL** mutated → **FOXA1** interacts with → **PIK3CA** in pathway → **Synthesis of PIPs at the plasma membrane**	80	↓ **slower progression** HR 0.66, z -2.2, n = 452 carriers of “Synthesis of PIPs at the plasma membrane”	**0.78**
2 ★	**TCGA-BH-A1FL** mutated → **IKZF3** interacts with → **TCAP** in pathway → **Striated Muscle Contraction**	98	↑ **faster progression** HR 1.12, z +0.6, n = 328 carriers of “Striated Muscle Contraction”	**0.67**
3 ★	**TCGA-BH-A1FL** mutated → **FOXA1** linked to → **breast neoplasm**	92	↓ **slower progression** HR 0.77, z -0.8, n = 794 carriers of “breast neoplasm”	**0.67**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

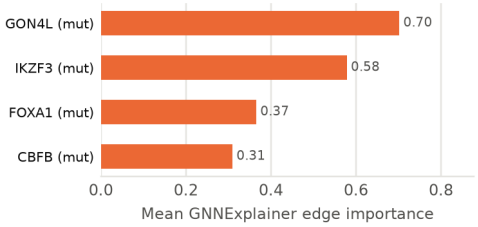
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R3 ↓ lower risk	Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	662 pts 77 events	**0.36** p 2.3e-07	not reached
F4-R4 ↓ lower risk	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo
F4-R5 ↓ lower risk	M stage = M0 AND Age ≥ 34.5	713 pts 89 events	**0.47** p 1.0e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FOXA1 · driver	Mutation	p.Y259S	9	0.37
CBFB · driver	Mutation	p.I102M	29	0.31
GON4L	Mutation	p.S1477L	0	0.70
IKZF3	Mutation	p.A20S	0	0.58



4 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A1NH	71	II	LumA	0.85	no progression, followed 18.9 mo
TCGA-BH-A0BM	54	II	LumA	0.83	no progression, followed 61.7 mo
TCGA-E2-A153	51	II	LumA	0.81	no progression, followed 23.2 mo
TCGA-BH-A0E1	52	II	LumA	0.80	no progression, followed 15.7 mo

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A1EX	67	II	LumA	0.80	progressed at 49.6 mo (group D)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**44**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Normal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IA**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

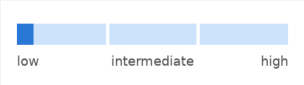
Sample: **TCGA-BH-A1FU-01** · Aneuploidy score: **26** · TMB (nonsyn., /Mb): **2.73** · MSIsensor score: **0** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

6th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 13%

Model-estimated progression time

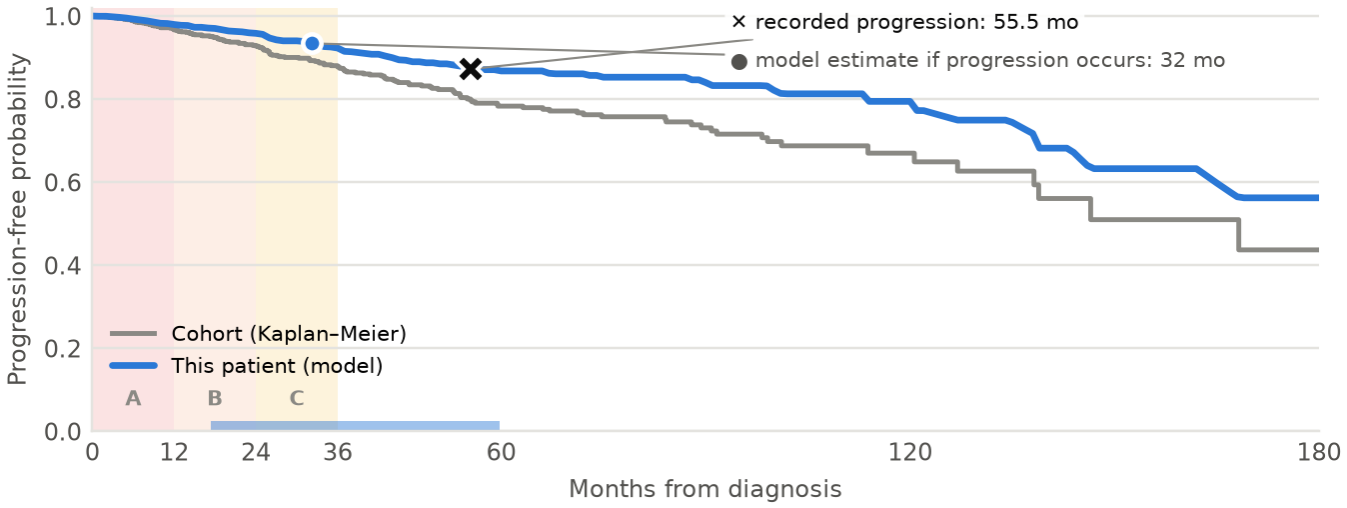
≈ 32 months

Window 24–36 months · IQR 17–60 mo
if progression occurs

Recorded progression

55.5 months

Group D · Late progression
✗ recorded window differs · |error| 23 mo



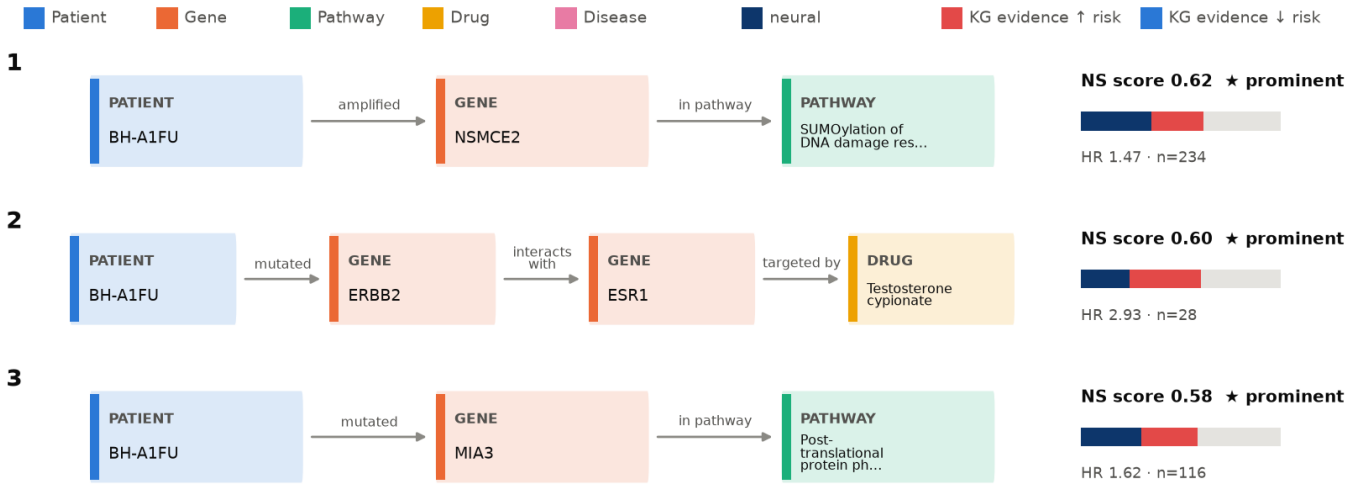
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 44-year-old patient has a Normal tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **6th risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND 41.5 ≤ Age < 78.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **NSMCE2** to **SUMOylation of DNA damage response and repair proteins**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.47). Progression was recorded at 55.5 months; the model's estimate given progression was 32 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

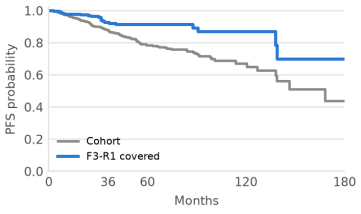


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-BH-A1FU** amplified → **NSMCE2** in pathway → **SUMOylation of DNA damage response and repair proteins**	59	↑ **faster progression** HR 1.47, z +2.0, n = 234 carriers of “SUMOylation of DNA damage response and repair proteins”	**0.62**
2 ★	**TCGA-BH-A1FU** mutated → **ERBB2** interacts with → **ESR1** targeted by → **Testosterone cypionate**	41	↑ **faster progression** HR 2.93, z +2.7, n = 28 carriers of “Testosterone cypionate”	**0.60**
3 ★	**TCGA-BH-A1FU** mutated → **MIA3** in pathway → **Post-translational protein phosphorylation**	51	↑ **faster progression** HR 1.62, z +2.1, n = 116 carriers of “Post-translational protein phosphorylation”	**0.58**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

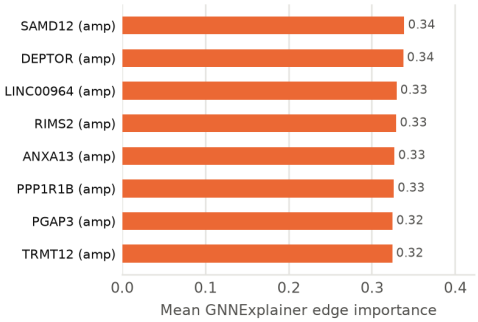
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R1 ↓ lower risk	N stage = N0 AND 41.5 ≤ Age < 78.5	351 pts 22 events	**0.30** p 5.2e-08	not reached
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo
F3-R2 ↓ lower risk	N stage = N0	418 pts 33 events	**0.37** p 4.8e-07	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.R175H	57	0.32
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.32
ERBB2 · driver	Mutation	p.R340G	55	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
SAMD12	High-level amplification	GISTIC ±2	0	0.34
DEPTOR	High-level amplification	GISTIC ±2	0	0.34
LINC00964	High-level amplification	GISTIC ±2	0	0.33
RIMS2	High-level amplification	GISTIC ±2	0	0.33
ANXA13	High-level amplification	GISTIC ±2	0	0.33
PPP1R1B	High-level amplification	GISTIC ±2	1	0.33
PGAP3	High-level amplification	GISTIC ±2	0	0.32
TRMT12	High-level amplification	GISTIC ±2	1	0.32



118 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259) … and 51 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0CX	52	II	Her2	0.75	no progression, followed 56.8 mo
TCGA-BH-A0BG	73	I	Basal	0.72	no progression, followed 61.5 mo
TCGA-D8-A1J9	48	I	LumB	0.70	no progression, followed 17.5 mo
TCGA-AO-A0J4	41	I	Basal	0.68	no progression, followed 52.2 mo
TCGA-D8-A27M	59	I	Basal	0.68	no progression, followed 13.5 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**38**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

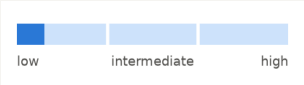
Sample: **TCGA-BH-A1EW-01** · Aneuploidy score: **15** · TMB (nonsyn., /Mb): **0.47** · MSIsensor score: **0.03** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

10th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

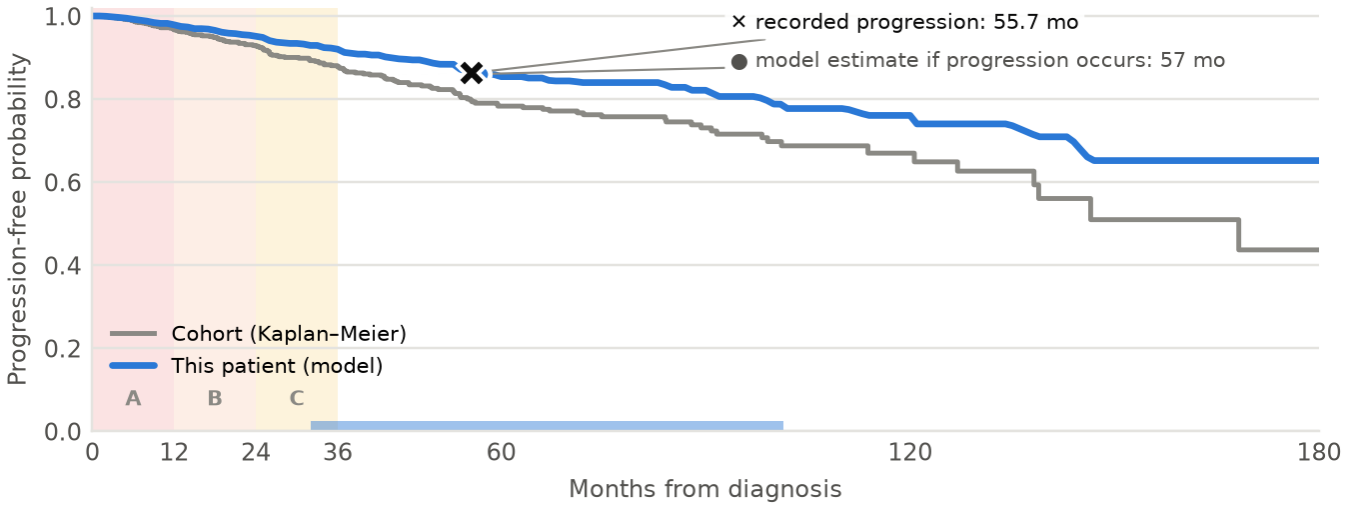
≈ 57 months

Window > 36 months · IQR 32–101 mo
if progression occurs

Recorded progression

55.7 months

Group D · Late progression
✓ **estimate in recorded window** · |error| 1 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 38-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **10th risk percentile** (low tier). Progression was recorded at 55.7 months; the model's estimate given progression was 57 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

No explanation path is available: none of this patient's altered panel genes has a stored PrimeKG pathway, drug, or breast-disease link (or the tumour carries no qualifying alteration). The risk estimate therefore rests on the clinical block and the symbolic rules below.

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank $p < 0.05$, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

No mutation or high-level CNA on the 500-gene panel was recorded for this tumour.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A245	47	II	LumA	0.99	no progression, followed 0.9 mo
TCGA-BH-A0DQ	42	II	LumA	0.98	no progression, followed 3.2 mo

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0E1	52	II	LumA	0.98	no progression, followed 15.7 mo
TCGA-AR-A24L	26	II	LumA	0.96	progressed at 46.5 mo (group D)
TCGA-C8-A273	29	II	LumA	0.89	no progression, followed 16.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A0BQ · Group D

Report date 14 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**39**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

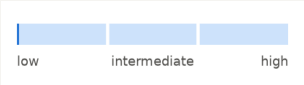
Sample: **TCGA-BH-A0BQ-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **1.5** · MSIsensor score: **0** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

1st

Low risk tier · fused neuro-symbolic model



Probability of progression

6% by 36 mo

12 mo 2% · 24 mo 3% · 60 mo 11%

Model-estimated progression time

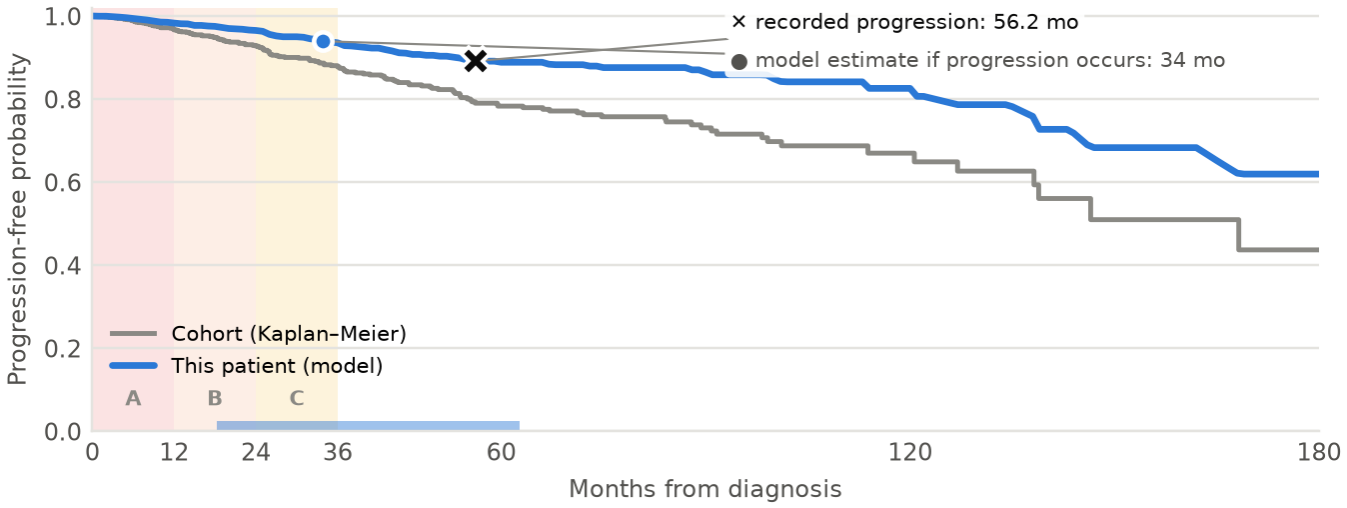
≈ 34 months

Window 24–36 months · IQR 18–63 mo
if progression occurs

Recorded progression

56.2 months

Group D · Late progression
✗ recorded window differs · |error| 22 mo



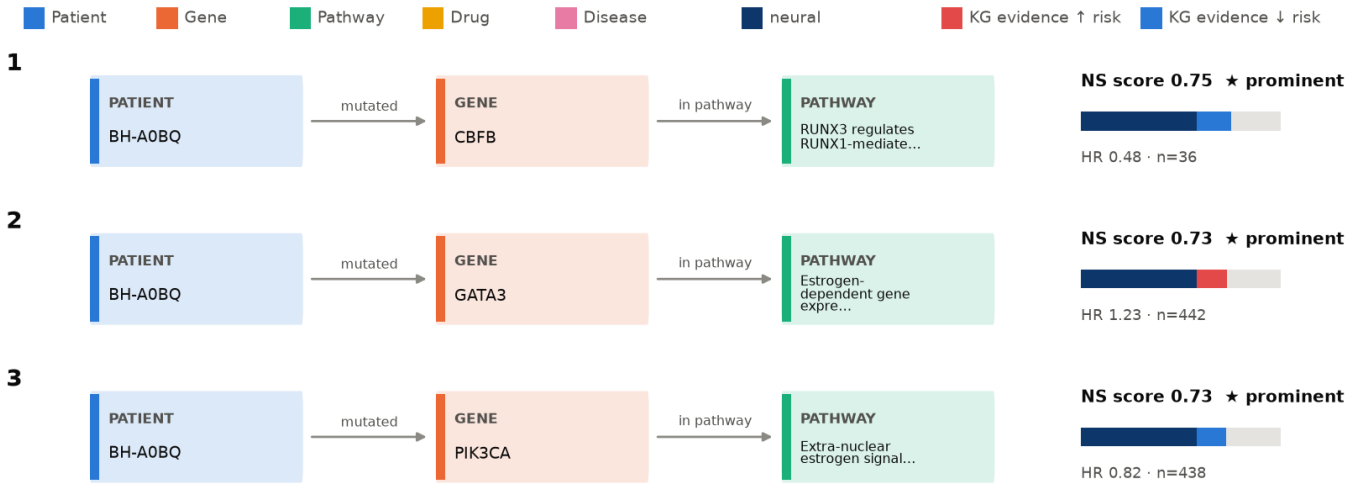
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 39-year-old patient has a LumA tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **1st risk percentile** (low tier). The most significant rule covering them (*N stage = N0*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **CBFB** to **RUNX3 regulates RUNX1-mediated transcription**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.48). Progression was recorded at 56.2 months; the model's estimate given progression was 34 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

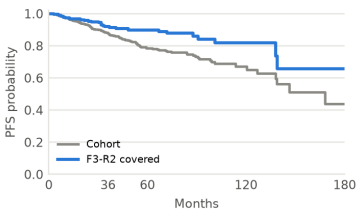


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-BH-A0BQ** mutated → **CBFB** in pathway → **RUNX3 regulates RUNX1-mediated transcription**	97	↓ **slower progression** HR 0.48, z -1.3, n = 36 carriers of "RUNX3 regulates RUNX1-mediated transcription"	**0.75**
2 ★	**TCGA-BH-A0BQ** mutated → **GATA3** in pathway → **Estrogen-dependent gene expression**	97	↑ **faster progression** HR 1.23, z +1.1, n = 442 carriers of "Estrogen-dependent gene expression"	**0.73**
3 ★	**TCGA-BH-A0BQ** mutated → **PIK3CA** in pathway → **Extra-nuclear estrogen signaling**	97	↓ **slower progression** HR 0.82, z -1.1, n = 438 carriers of "Extra-nuclear estrogen signaling"	**0.73**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

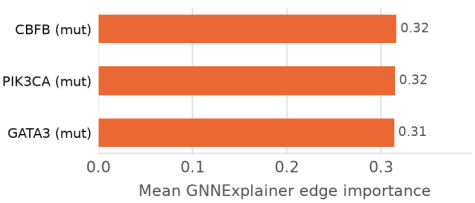
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R2 ↓ lower risk	N stage = N0	418 pts 33 events	**0.37** p 4.8e-07	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CBFB · driver	Mutation	p.X94_splice	29	0.32
PIK3CA · driver	Mutation	p.E545K	70	0.32
GATA3 · driver	Mutation	p.K357*, p.M356I	13	0.31



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0H3	46	I	LumA	0.95	no progression, followed 63.4 mo

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A0RN	60	I	LumA	0.93	no progression, followed 263.3 mo
TCGA-A2-A0EO	54	I	LumA	0.88	no progression, followed 80.3 mo
TCGA-E9-A1RA	48	I	LumA	0.87	no progression, followed 45.0 mo
TCGA-B6-A0X0	54	I	LumA	0.86	no progression, followed 129.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-3C-AAAU · Group D

Report date 14 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**55**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE X**
Genetic ancestry	**EUR**	Pathologic T	**TX**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**No**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-3C-AAAU-01** · Aneuploidy score: **19** · TMB (nonsyn., /Mb): **0.8** · MSIsensor score: **0.55** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

69th

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 7% · 60 mo 23%

Model-estimated progression time

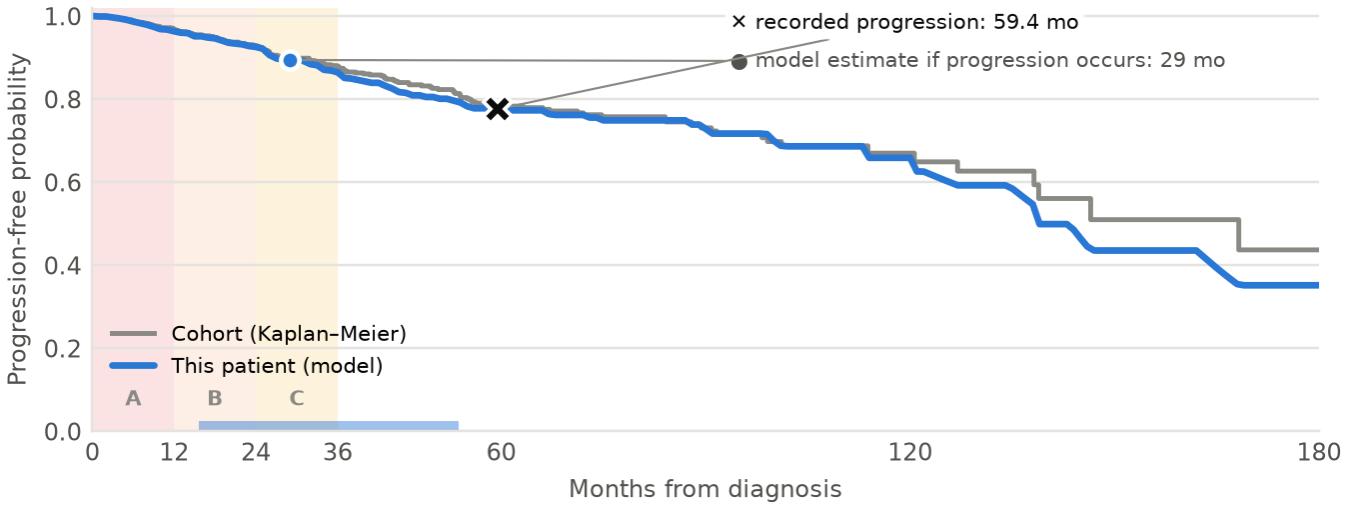
≈ 29 months

Window 24–36 months · IQR 16–54 mo
if progression occurs

Recorded progression

59.4 months

Group D · Late progression
✗ recorded window differs · |error| 30 mo



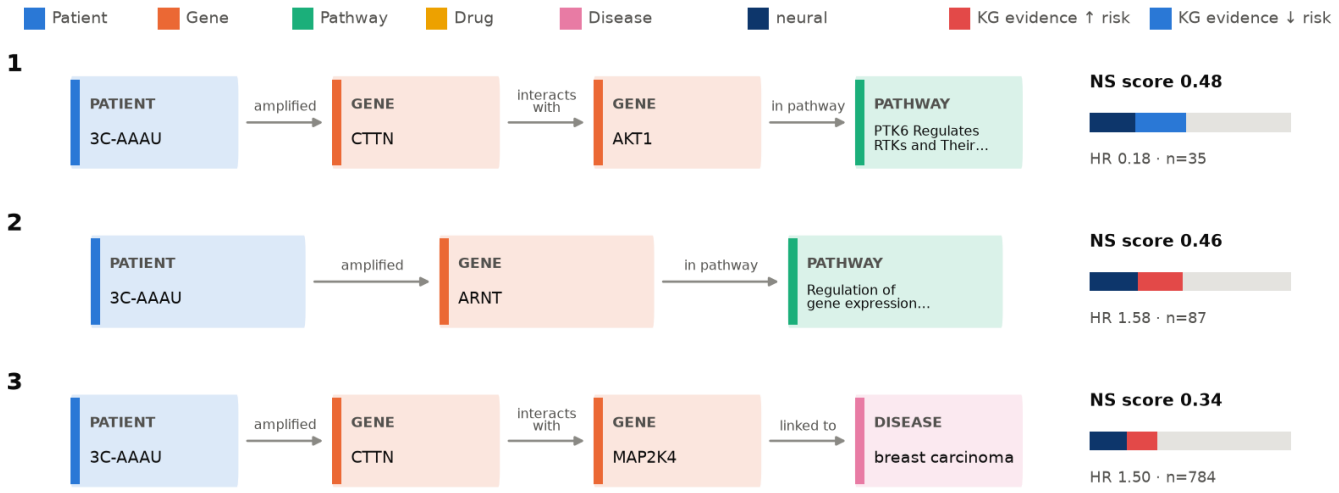
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 55-year-old patient has a LumA tumour, AJCC stage X (missing, missing, MX). The fused neuro-symbolic model places them in the **69th risk percentile** (high tier). The most significant rule covering them (*AJCC stage = III AND 6.5 ≤ Aneuploidy score < 26.5*) marks a group with a progression hazard 2.9× that of the remaining patients. The highest-scoring knowledge-graph path runs from **CTTN** to **PTK6 Regulates RTKs and Their Effectors AKT1 and DOK1**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.18). Progression was recorded at 59.4 months; the model's estimate given progression was 29 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

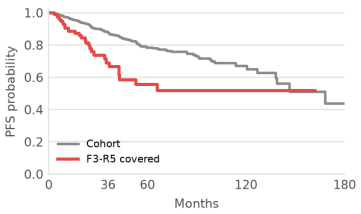


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-3C-AAAU** amplified → **CTTN** interacts with → **AKT1** in pathway → **PTK6 Regulates RTKs and Their Effectors AKT1 and DOK1**	38	↓ **slower progression** HR 0.18, z -1.9, n = 35 carriers of “PTK6 Regulates RTKs and Their Effectors AKT1 and DOK1”	**0.48**
2	**TCGA-3C-AAAU** amplified → **ARNT** in pathway → **Regulation of gene expression by Hypoxia-inducible Factor**	40	↑ **faster progression** HR 1.58, z +1.7, n = 87 carriers of “Regulation of gene expression by Hypoxia-inducible Factor”	**0.46**
3	**TCGA-3C-AAAU** amplified → **CTTN** interacts with → **MAP2K4** linked to → **breast carcinoma**	31	↑ **faster progression** HR 1.50, z +1.1, n = 784 carriers of “breast carcinoma”	**0.34**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

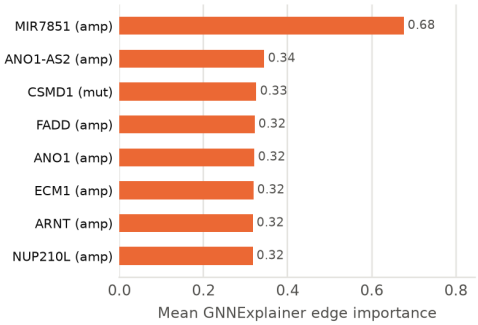
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R5 ↑ **higher risk**	AJCC stage = III AND 6.5 ≤ Aneuploidy score < 26.5	121 pts 29 events	**2.88** p 3.5e-07	not reached
F3-R6 ↑ **higher risk**	AJCC stage = III AND Aneuploidy score < 26.5 AND Age ≥ 35.0	171 pts 36 events	**2.48** p 3.8e-06	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
GATA3 · driver	Mutation	p.P408Afs*99	13	0.31
MIR7851	High-level amplification	GISTIC ±2	0	0.68
ANO1-AS2	High-level amplification	GISTIC ±2	0	0.34
CSMD1	Mutation	p.X967_splice	0	0.33
FADD	High-level amplification	GISTIC ±2	14	0.32
ANO1	High-level amplification	GISTIC ±2	2	0.32
ECM1	High-level amplification	GISTIC ±2	1	0.32
ARNT	High-level amplification	GISTIC ±2	6	0.32
NUP210L	High-level amplification	GISTIC ±2	0	0.32
TARS2	High-level amplification	GISTIC ±2	2	0.32
FGF19	High-level amplification	GISTIC ±2	13	0.32



29 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); ANO1 → Crofelemer (DB04941); TARS2 → Threonine (DB00156); FGF19 → Heparin (DB01109); FGF4 → Pentosan polysulfate (DB00686); FGF4 → Heparin (DB01109); FGF4 → TP-508 (DB05309) … and 5 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-OL-A66O	39	II	LumA	0.59	no progression, followed 17.4 mo
TCGA-A1-A0SQ	45	II	LumA	0.57	no progression, followed 18.2 mo
TCGA-A8-A07E	81	X	LumA	0.56	no progression, followed 20.0 mo
TCGA-EW-A423	75	II	LumA	0.56	no progression, followed 17.5 mo
TCGA-AR-A1AK	70	I	LumA	0.56	no progression, followed 103.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**61**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

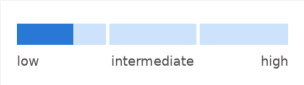
Sample: **TCGA-E2-A1IE-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **0.6** · MSIsensor score: **0.38** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

21st

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

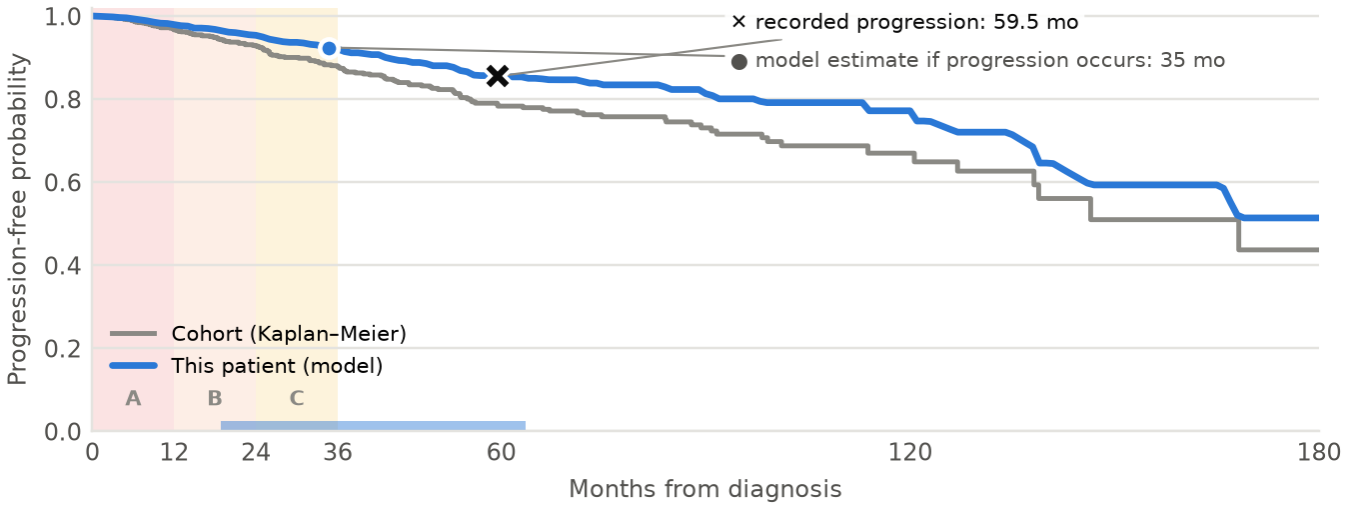
≈ 35 months

Window 24–36 months · IQR 19–64 mo
if progression occurs

Recorded progression

59.5 months

Group D · Late progression
✗ recorded window differs · |error| 25 mo



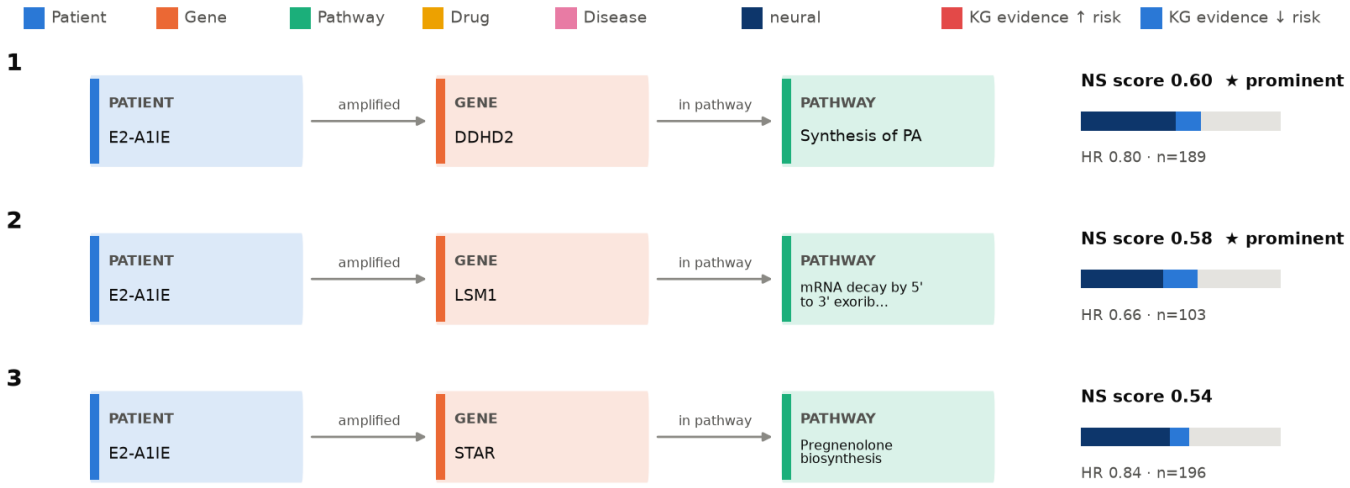
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 61-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **21st risk percentile** (low tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **DDHD2** to **Synthesis of PA**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.80). Progression was recorded at 59.5 months; the model's estimate given progression was 35 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

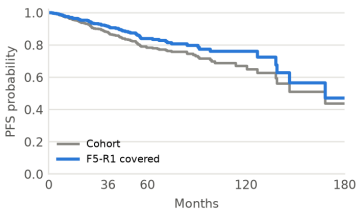


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-E2-A1IE** amplified → **DDHD2** in pathway → **Synthesis of PA**	79	↓ **slower progression** HR 0.80, z -0.9, n = 189 carriers of “Synthesis of PA”	**0.60**
2 ★	**TCGA-E2-A1IE** amplified → **LSM1** in pathway → **mRNA decay by 5' to 3' exoribonuclease**	69	↓ **slower progression** HR 0.66, z -1.3, n = 103 carriers of “mRNA decay by 5' to 3' exoribonuclease”	**0.58**
3	**TCGA-E2-A1IE** amplified → **STAR** in pathway → **Pregnenolone biosynthesis**	74	↓ **slower progression** HR 0.84, z -0.7, n = 196 carriers of “Pregnenolone biosynthesis”	**0.54**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

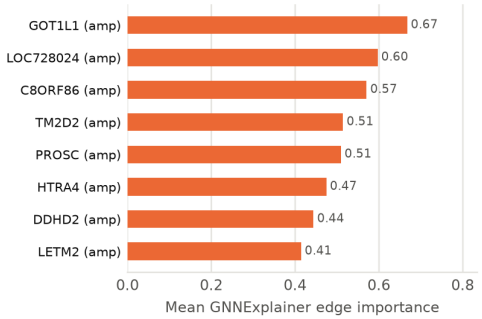
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ **lower risk**	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ **lower risk**	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R3 ↓ **lower risk**	AJCC stage = II AND 39.5 ≤ Age < 78.5	422 pts 31 events	**0.36** p 4.2e-07	168 mo
F5-R5 ↓ **lower risk**	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ATM · driver	Mutation	p.L2780R	29	0.32
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
GOT1L1	High-level amplification	GISTIC ±2	0	0.67
LOC728024	High-level amplification	GISTIC ±2	0	0.60
C8ORF86	High-level amplification	GISTIC ±2	0	0.57
TM2D2	High-level amplification	GISTIC ±2	0	0.51
PROSC	High-level amplification	GISTIC ±2	0	0.51
HTRA4	High-level amplification	GISTIC ±2	0	0.47
DDHD2	High-level amplification	GISTIC ±2	1	0.44
LETM2	High-level amplification	GISTIC ±2	0	0.41
STAR	High-level amplification	GISTIC ±2	1	0.37
LSM1	High-level amplification	GISTIC ±2	1	0.34



34 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ATM → Caffeine (DB00201); FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896) … and 47 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-EW-A1PA	59	II	LumA	0.90	no progression, followed 18.9 mo
TCGA-B6-A0WT	61	II	LumA	0.88	progressed at 127.0 mo (group D)
TCGA-BH-A204	80	II	LumB	0.83	no progression, followed 83.3 mo
TCGA-WT-AB41	55	II	LumA	0.81	no progression, followed 53.0 mo
TCGA-A2-A0ES	52	II	LumA	0.80	no progression, followed 72.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group D** · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**54**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

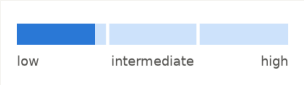
Sample: **TCGA-AO-A03P-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **1.33** · MSIsensor score: **1.02** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

29th

Low risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

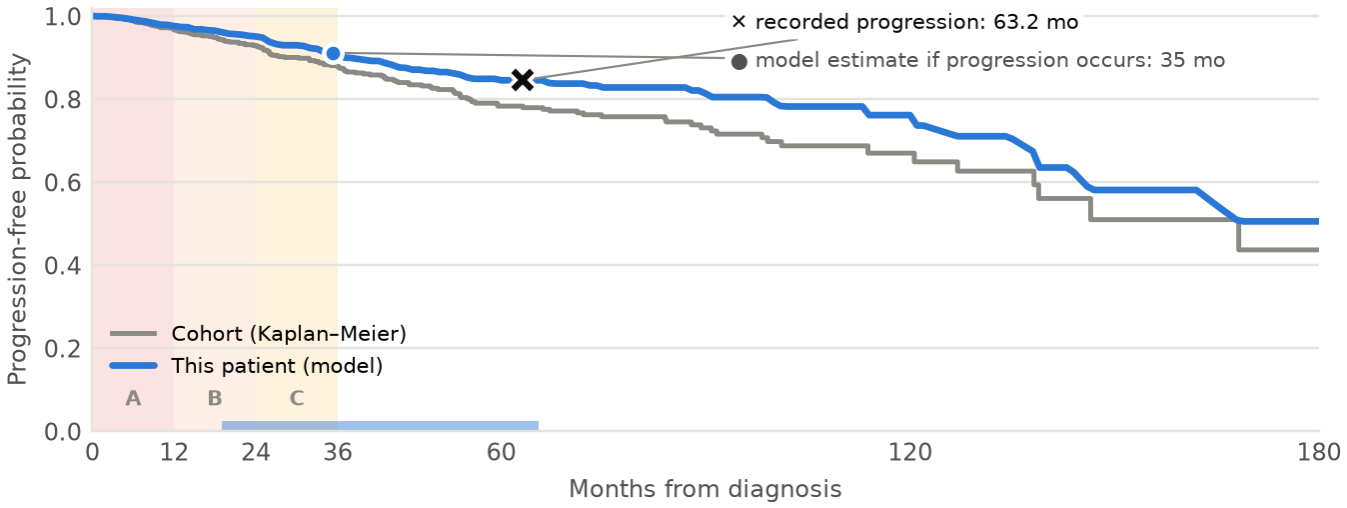
≈ 35 months

Window 24–36 months · IQR 19–65 mo
if progression occurs

Recorded progression

63.2 months

■ **Group D** · Late progression
✗ **recorded window differs** · |error| 28 mo



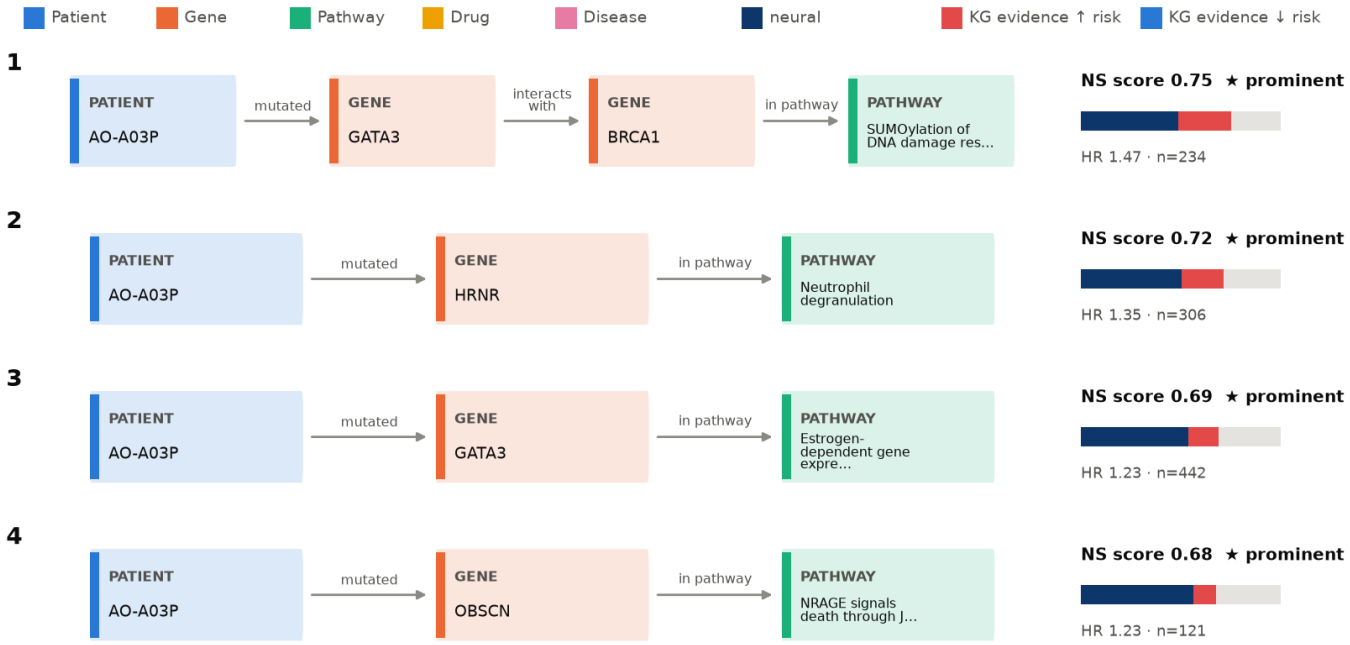
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 54-year-old patient has a LumB tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **29th risk percentile** (low tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 86.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **GATA3** to **SUMOylation of DNA damage response and repair proteins**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.47). Progression was recorded at 63.2 months; the model's estimate given progression was 35 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

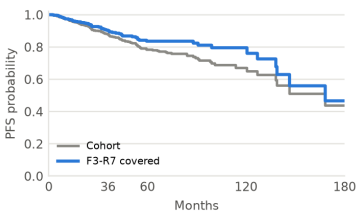


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-AO-A03P** mutated → **GATA3** interacts with → **BRCA1** in pathway → **SUMOylation of DNA damage response and repair proteins**	82	↑ **faster progression** HR 1.47, z +2.0, n = 234 carriers of “SUMOylation of DNA damage response and repair proteins”	**0.75**
2 ★	**TCGA-AO-A03P** mutated → **HRNR** in pathway → **Neutrophil degranulation**	84	↑ **faster progression** HR 1.35, z +1.6, n = 306 carriers of “Neutrophil degranulation”	**0.72**
3 ★	**TCGA-AO-A03P** mutated → **GATA3** in pathway → **Estrogen-dependent gene expression**	90	↑ **faster progression** HR 1.23, z +1.1, n = 442 carriers of “Estrogen-dependent gene expression”	**0.69**
4 ★	**TCGA-AO-A03P** mutated → **OBSCN** in pathway → **NRAGE signals death through JNK**	94	↑ **faster progression** HR 1.23, z +0.8, n = 121 carriers of “NRAGE signals death through JNK”	**0.68**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

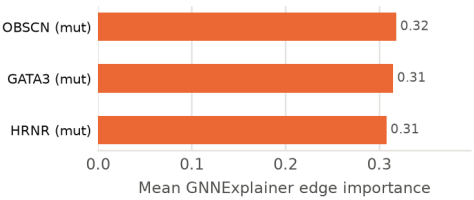
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo
F3-R8 ↓ lower risk	AJCC stage = II AND Age < 77.5 AND TMB (nonsynonymous) ≥ 0.08	433 pts 40 events	**0.50** p 3.2e-04	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.M442Ifs*65	13	0.31
OBSCN	Mutation	p.G3509V	8	0.32
HRNR	Mutation	p.S2027=	3	0.31



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

HRNR → Zinc (DB01593); HRNR → Zinc acetate (DB14487). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0C7	48	II	LumB	0.86	no progression, followed 91.0 mo
TCGA-D8-A1XR	56	II	LumB	0.85	no progression, followed 15.8 mo
TCGA-A8-A082	58	II	LumB	0.82	no progression, followed 18.0 mo
TCGA-B6-A1KC	67	II	LumB	0.81	no progression, followed 43.6 mo
TCGA-BH-A0HL	56	II	missing	0.80	no progression, followed 2.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**40**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0WY-01** · Aneuploidy score: **25** · TMB (nonsyn., /Mb): **1.03** · MSIsensor score: **0.08** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

77th

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 25%

Model-estimated progression time

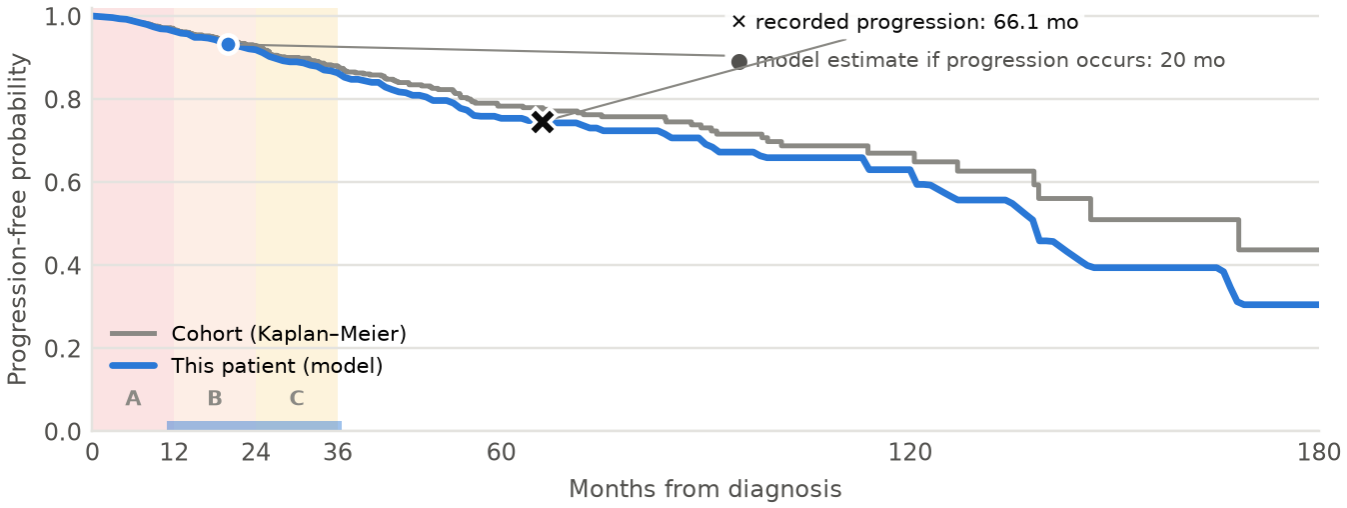
≈ 20 months

Window 12–24 months · IQR 11–37 mo
if progression occurs

Recorded progression

66.1 months

Group D · Late progression
✗ recorded window differs · |error| 46 mo



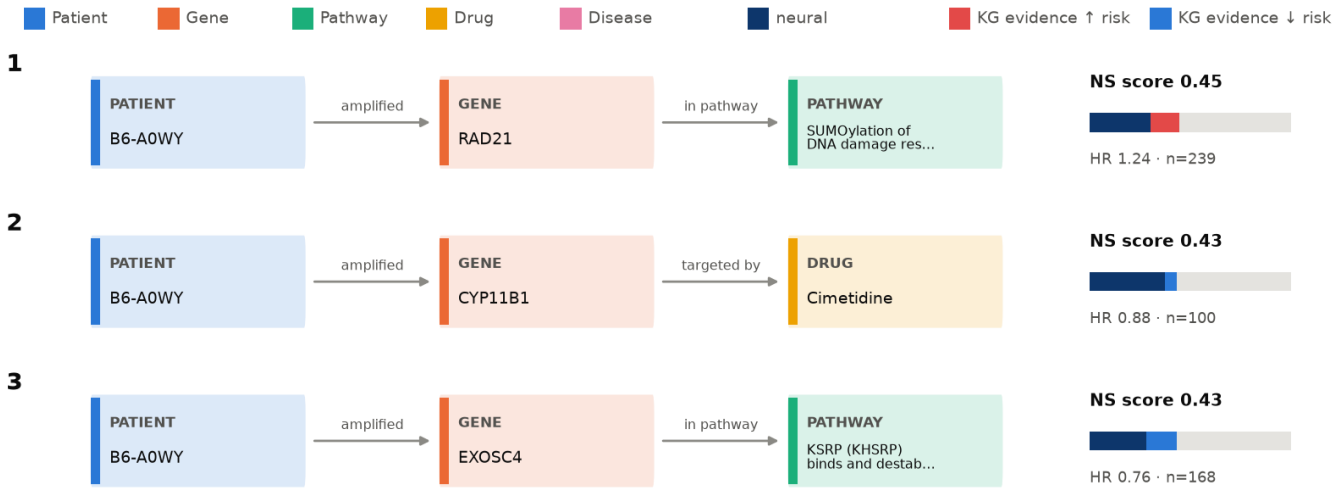
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 40-year-old patient has a LumA tumour, AJCC stage III (T3, N1, M0). The fused neuro-symbolic model places them in the **77th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **RAD21** to **SUMOylation of DNA damage response and repair proteins**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.24). Progression was recorded at 66.1 months; the model's estimate given progression was 20 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

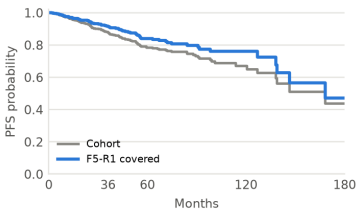


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-B6-A0WY** amplified → **RAD21** in pathway → **SUMOylation of DNA damage response and repair proteins**	50	↑ **faster progression** HR 1.24, z +1.1, n = 239 carriers of “SUMOylation of DNA damage response and repair proteins”	**0.45**
2	**TCGA-B6-A0WY** amplified → **CYP11B1** targeted by → **Cimetidine**	63	↓ **slower progression** HR 0.88, z -0.4, n = 100 carriers of “Cimetidine”	**0.43**
3	**TCGA-B6-A0WY** amplified → **EXOSC4** in pathway → **KSRP (KHSRP) binds and destabilizes mRNA**	47	↓ **slower progression** HR 0.76, z -1.1, n = 168 carriers of “KSRP (KHSRP) binds and destabilizes mRNA”	**0.43**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

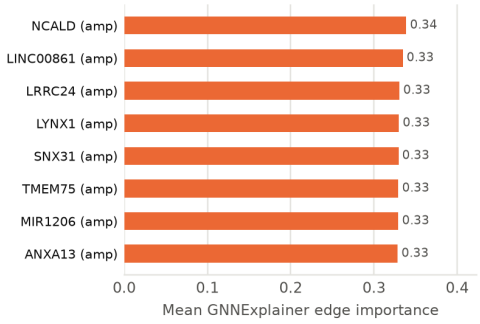
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ **lower risk**	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ **lower risk**	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R4 ↓ **lower risk**	TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5	567 pts 64 events	**0.56** p 2.1e-03	168 mo
F5-R5 ↓ **lower risk**	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
PIK3CA · driver	Mutation	p.E109_I112delinsD	70	0.32
PIK3R1 · driver	Mutation	p.I571_L573del	79	0.31
NCALD	High-level amplification	GISTIC ±2	1	0.34
LINC00861	High-level amplification	GISTIC ±2	0	0.33
LRRC24	High-level amplification	GISTIC ±2	0	0.33
LYNX1	High-level amplification	GISTIC ±2	0	0.33
SNX31	High-level amplification	GISTIC ±2	0	0.33
TMEM75	High-level amplification	GISTIC ±2	0	0.33
MIR1206	High-level amplification	GISTIC ±2	0	0.33
ANXA13	High-level amplification	GISTIC ±2	0	0.33
SCRT1	High-level amplification	GISTIC ±2	0	0.33



314 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); PIK3R1 → Isoprenaline (DB01064) … and 93 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AN-A0XS	63	III	LumA	0.89	no progression, followed 0.3 mo
TCGA-A8-A06T	75	III	LumA	0.89	no progression, followed 53.1 mo
TCGA-AR-A24V	52	II	LumA	0.85	no progression, followed 105.3 mo
TCGA-C8-A12N	58	II	LumA	0.85	no progression, followed 11.8 mo
TCGA-A2-A04N	66	I	LumA	0.85	no progression, followed 143.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**38**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0RS-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **4.63** · MSIsensor score: **0.22** · AJCC edition: **3RD**

2 · Progression assessment

Model risk (cohort percentile)

64th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 22%

Model-estimated progression time

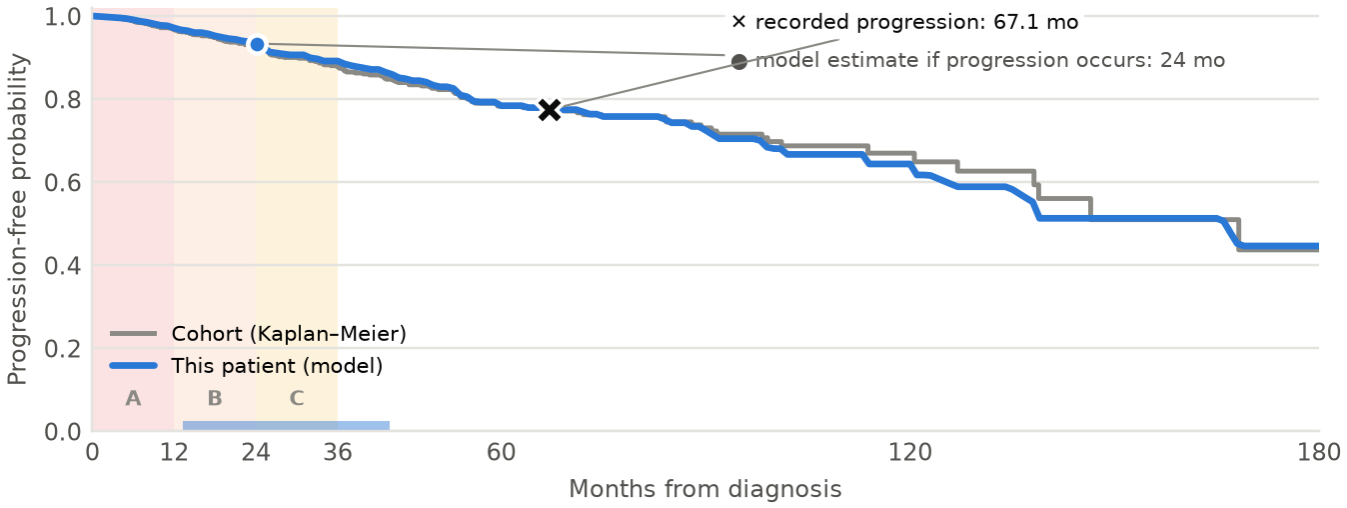
≈ 24 months

Window 24–36 months · IQR 13–44 mo
if progression occurs

Recorded progression

67.1 months

Group D · Late progression
✗ recorded window differs · |error| 43 mo



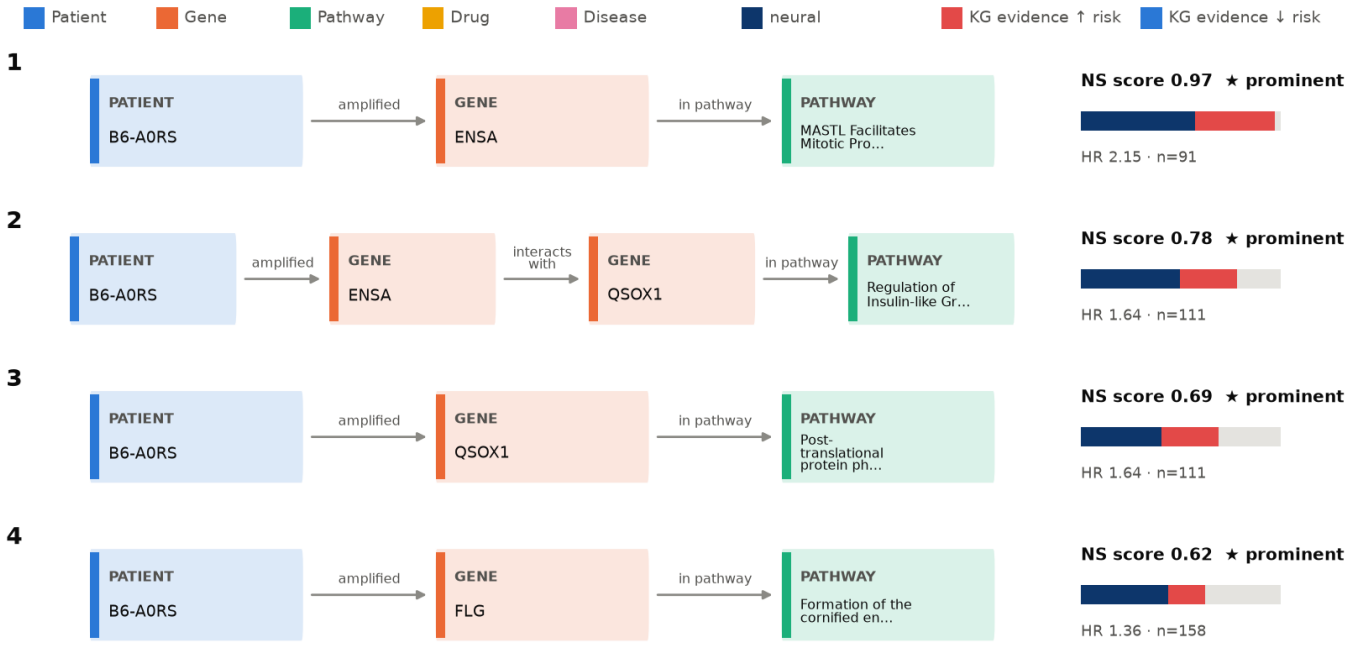
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 38-year-old patient has a Her2 tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **64th risk percentile** (intermediate tier). The most significant rule covering them (*Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ENSA** to **MASTL Facilitates Mitotic Progression**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 2.15). Progression was recorded at 67.1 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

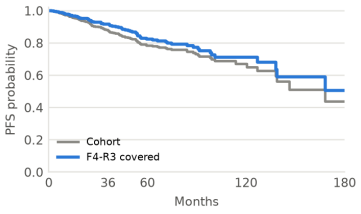


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-B6-A0RS** amplified → **ENSA** in pathway → **MASTL Facilitates Mitotic Progression**	95	↑ **faster progression** HR 2.15, z +3.3, n = 91 carriers of “MASTL Facilitates Mitotic Progression”	**0.97**
2 ★	**TCGA-B6-A0RS** amplified → **ENSA** interacts with → **QSOX1** in pathway → **Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)**	83	↑ **faster progression** HR 1.64, z +2.1, n = 111 carriers of “Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)”	**0.78**
3 ★	**TCGA-B6-A0RS** amplified → **QSOX1** in pathway → **Post-translational protein phosphorylation**	68	↑ **faster progression** HR 1.64, z +2.1, n = 111 carriers of “Post-translational protein phosphorylation”	**0.69**
4 ★	**TCGA-B6-A0RS** amplified → **FLG** in pathway → **Formation of the cornified envelope**	73	↑ **faster progression** HR 1.36, z +1.4, n = 158 carriers of “Formation of the cornified envelope”	**0.62**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

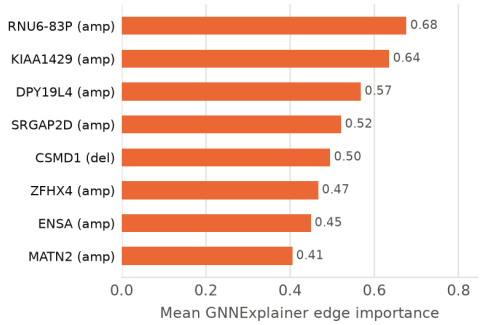
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R3 ↓ lower risk	Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	662 pts 77 events	**0.36** p 2.3e-07	not reached
F4-R2 ↓ lower risk	N stage = N0 AND 34.5 ≤ Age < 83.5	386 pts 32 events	**0.43** p 2.5e-05	not reached
F4-R4 ↓ lower risk	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo
F4-R5 ↓ lower risk	M stage = M0 AND Age ≥ 34.5	713 pts 89 events	**0.47** p 1.0e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	High-level amplification	GISTIC ±2	70	0.31
TP53 · driver	Mutation	p.R273C	57	0.31
RNU6-83P	High-level amplification	GISTIC ±2	0	0.68
KIAA1429	High-level amplification	GISTIC ±2	0	0.64
DPY19L4	High-level amplification	GISTIC ±2	0	0.57
SRGAP2D	High-level amplification	GISTIC ±2	0	0.52
CSMD1	Homozygous deletion	GISTIC ±2	0	0.50
ZFHX4	High-level amplification	GISTIC ±2	0	0.47
ENSA	High-level amplification	GISTIC ±2	1	0.45
MATN2	High-level amplification	GISTIC ±2	0	0.41
HMCN1	High-level amplification	GISTIC ±2	0	0.37
FLG	High-level amplification	GISTIC ±2	1	0.37



54 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347) … and 30 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A248	51	II	Her2	0.71	no progression, followed 1.9 mo
TCGA-B6-A0RH	51	II	Her2	0.69	progressed at 71.2 mo (group D)
TCGA-AO-A0J6	61	II	Basal	0.67	no progression, followed 37.5 mo
TCGA-E9-A244	54	II	Basal	0.66	no progression, followed 0.7 mo
TCGA-AR-A24K	46	II	LumB	0.66	no progression, followed 50.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**51**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0RH-01** · Aneuploidy score: **10** · TMB (nonsyn., /Mb): **1.27** · MSIsensor score: **0.22** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

71st

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 23%

Model-estimated progression time

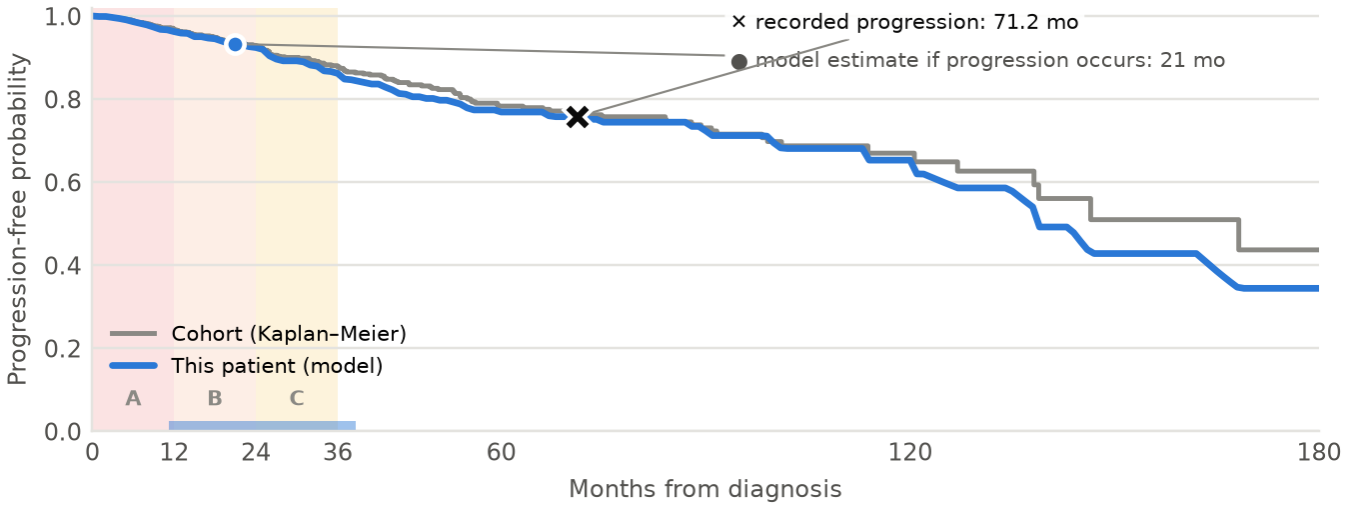
≈ 21 months

Window 12–24 months · IQR 11–39 mo
if progression occurs

Recorded progression

71.2 months

Group D · Late progression
✗ recorded window differs · |error| 50 mo



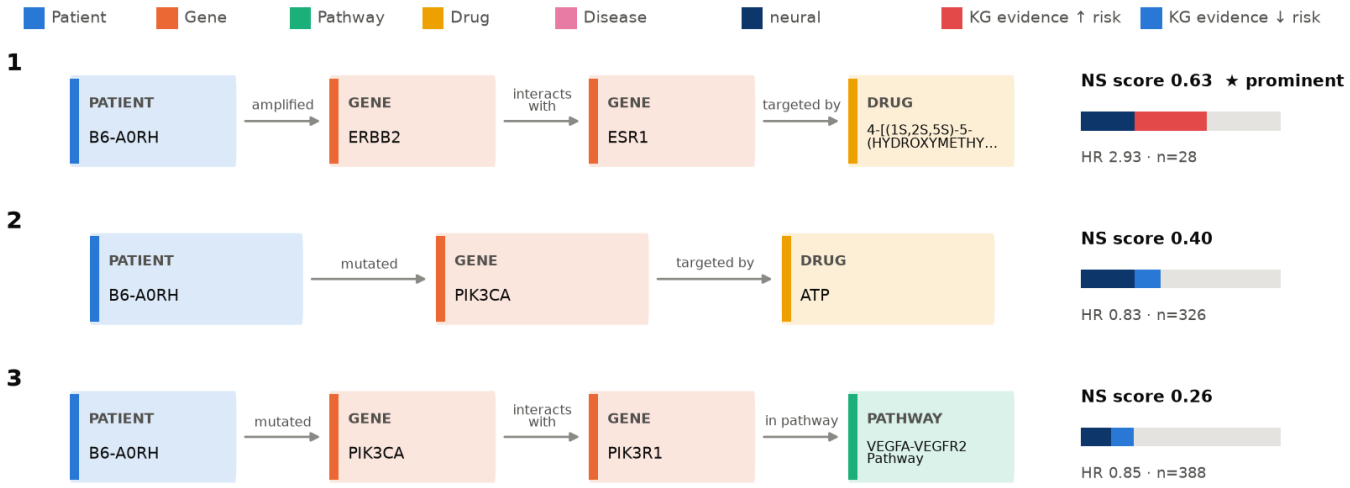
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 51-year-old patient has a Her2 tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **71st risk percentile** (high tier). The most significant rule covering them (*N stage = N0 AND 41.5 ≤ Age < 78.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ERBB2** to **4-[(1S,2S,5S)-5-(HYDROXYMETHYL)-6,8,9-TRIMETHYL-3-OXABICYCLO[3.3.1]NON-7-EN-2-YL]PHENOL**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 2.93). Progression was recorded at 71.2 months; the model's estimate given progression was 21 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

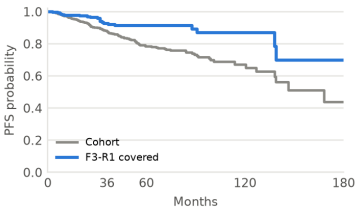


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-B6-A0RH** amplified → **ERBB2** interacts with → **ESR1** targeted by → **4-[(1S,2S,5S)-5-(HYDROXYMETHYL)-6,8,9-TRIMETHYL-3-OXABICYCLO[3.3.1]NON-7-EN-2-YL]PHENOL**	45	↑ **faster progression** HR 2.93, z +2.7, n = 28 carriers of “4-[(1S,2S,5S)-5-(HYDROXYMETHYL)-6,8,9-TRIMETHYL-3-OXABICYCLO[3.3.1]NON-7-EN-2-YL]PHENOL”	**0.63**
2	**TCGA-B6-A0RH** mutated → **PIK3CA** targeted by → **ATP**	45	↓ **slower progression** HR 0.83, z -1.0, n = 326 carriers of “ATP”	**0.40**
3	**TCGA-B6-A0RH** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **VEGFA-VEGFR2 Pathway**	25	↓ **slower progression** HR 0.85, z -0.8, n = 388 carriers of “VEGFA-VEGFR2 Pathway”	**0.26**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

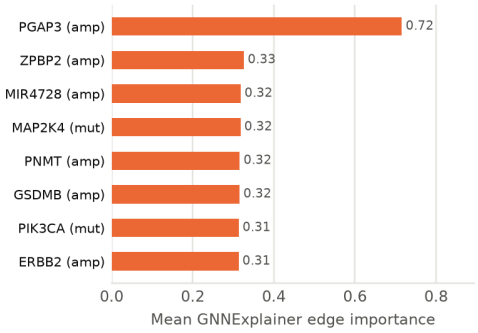
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R1 ↓ lower risk	N stage = N0 AND 41.5 ≤ Age < 78.5	351 pts 22 events	**0.30** p 5.2e-08	not reached
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo
F3-R2 ↓ lower risk	N stage = N0	418 pts 33 events	**0.37** p 4.8e-07	not reached
F3-R8 ↓ lower risk	AJCC stage = II AND Age < 77.5 AND TMB (nonsynonymous) ≥ 0.08	433 pts 40 events	**0.50** p 3.2e-04	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MAP2K4 · driver	Mutation	p.X211_splice	12	0.32
PIK3CA · driver	High-level amplification	GISTIC ±2	70	0.31
PIK3CA · driver	Mutation	p.E545K	70	0.31
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
TP53 · driver	Mutation	p.Y220S	57	0.31
PGAP3	High-level amplification	GISTIC ±2	0	0.72
ZPBP2	High-level amplification	GISTIC ±2	0	0.33
MIR4728	High-level amplification	GISTIC ±2	0	0.32
PNMT	High-level amplification	GISTIC ±2	16	0.32
GSDMB	High-level amplification	GISTIC ±2	4	0.32
GRB7	High-level amplification	GISTIC ±2	12	0.31
ECM1	Mutation	p.*15*	1	0.31



22 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241) … and 42 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A3XV	46	II	Her2	0.89	no progression, followed 32.7 mo
TCGA-D8-A1JG	62	II	Her2	0.89	no progression, followed 53.0 mo
TCGA-C8-A1HF	48	II	Her2	0.88	no progression, followed 10.9 mo
TCGA-AN-A0FV	58	II	Her2	0.88	no progression, followed 0.3 mo
TCGA-EW-A1OZ	56	II	LumB	0.83	no progression, followed 40.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**34**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A1FN-01** · Aneuploidy score: **12** · TMB (nonsyn., /Mb): **3.27** · MSIsensor score: **0.72** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

36th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 6% · 60 mo 17%

Model-estimated progression time

≈ 53 months

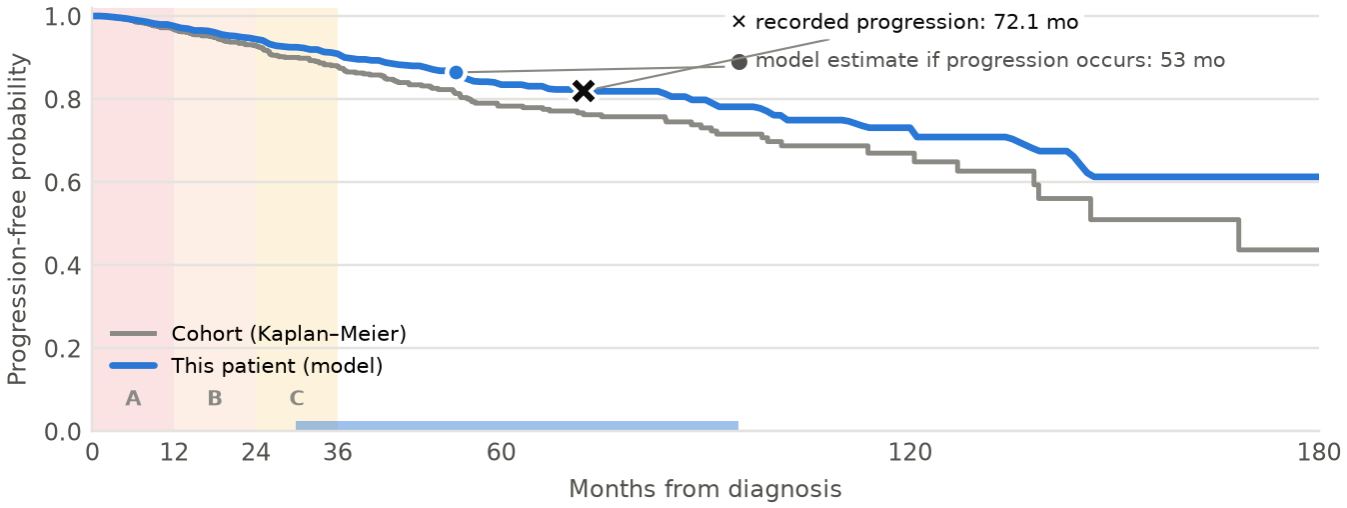
Window > 36 months · IQR 30–95 mo if progression occurs

Recorded progression

72.1 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 19 mo



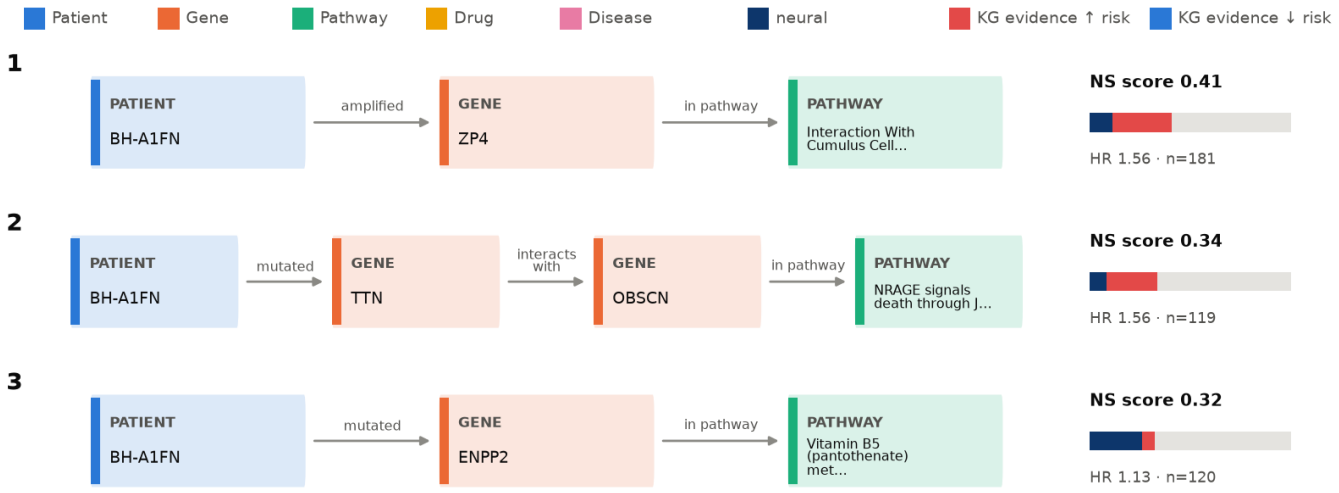
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 34-year-old patient has a LumB tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **36th risk percentile** (intermediate tier). The most significant rule covering them (*N stage = N0 AND TMB (nonsynonymous) ≥ 0.78 AND Age < 83.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ZP4** to **Interaction With Cumulus Cells And The Zona Pellucida**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.56). Progression was recorded at 72.1 months; the model's estimate given progression was 53 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

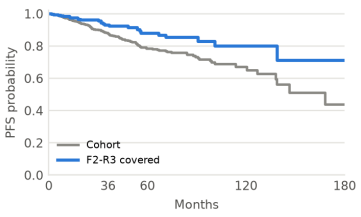


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-BH-A1FN** amplified → **ZP4** in pathway → **Interaction With Cumulus Cells And The Zona Pellucida**	19	↑ **faster progression** HR 1.56, z +2.2, n = 181 carriers of “Interaction With Cumulus Cells And The Zona Pellucida”	**0.41**
2	**TCGA-BH-A1FN** mutated → **TTN** interacts with → **OBSCN** in pathway → **NRAGE signals death through JNK**	14	↑ **faster progression** HR 1.56, z +1.9, n = 119 carriers of “NRAGE signals death through JNK”	**0.34**
3	**TCGA-BH-A1FN** mutated → **ENPP2** in pathway → **Vitamin B5 (pantothenate) metabolism**	44	↑ **faster progression** HR 1.13, z +0.5, n = 120 carriers of “Vitamin B5 (pantothenate) metabolism”	**0.32**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

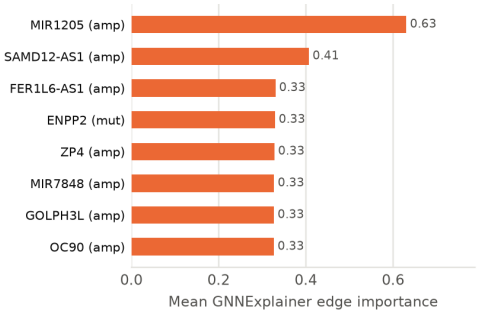
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R3 ↓ lower risk	N stage = N0 AND TMB (nonsynonymous) ≥ 0.78 AND Age < 83.5	317 pts 25 events	**0.41** p 3.7e-05	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PTEN · driver	High-level amplification	GISTIC ±2	22	0.32
PTEN · driver	Mutation	p.X154_splice	22	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
RB1 · driver	Mutation	p.Q93*	25	0.31
TP53 · driver	Mutation	p.R280T	57	0.31
MIR1205	High-level amplification	GISTIC ±2	0	0.63
SAMD12-AS1	High-level amplification	GISTIC ±2	0	0.41
FER1L6-AS1	High-level amplification	GISTIC ±2	0	0.33
ENPP2	High-level amplification	GISTIC ±2	1	0.33
ENPP2	Mutation	p.E371Q	1	0.33
ZP4	High-level amplification	GISTIC ±2	1	0.33
MIR7848	High-level amplification	GISTIC ±2	0	0.33



203 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PTEN → Phosphatidylethanolamine (DB04327); PTEN → Phosphatidylethanolamine (DB04327); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); RB1 → Insulin human (DB00030); RB1 → Insulin pork (DB00071); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404) … and 43 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A7-A13D	46	II	Basal	0.87	no progression, followed 31.7 mo
TCGA-AR-A0TP	43	II	Basal	0.84	no progression, followed 140.5 mo
TCGA-A2-A04T	62	II	Basal	0.80	no progression, followed 73.8 mo
TCGA-B6-A0I1	73	II	Basal	0.79	progressed at 36.8 mo (group D)
TCGA-A7-A4SE	54	II	Basal	0.78	no progression, followed 21.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**31**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

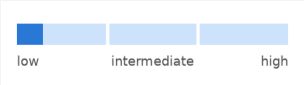
Sample: **TCGA-BH-A1FE-01** · Aneuploidy score: **6** · TMB (nonsyn., /Mb): **0.63** · MSIsensor score: **0.03** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

10th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

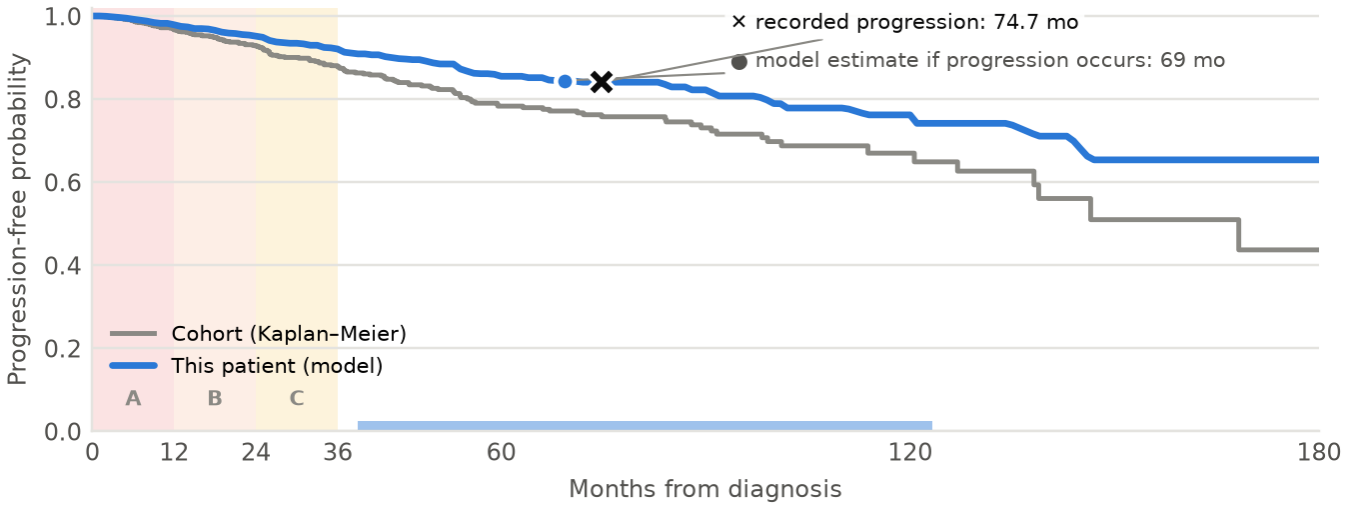
≈ 69 months

Window > 36 months · IQR 39–123 mo
if progression occurs

Recorded progression

74.7 months

Group D · Late progression
✓ **estimate in recorded window** · |error| 5 mo



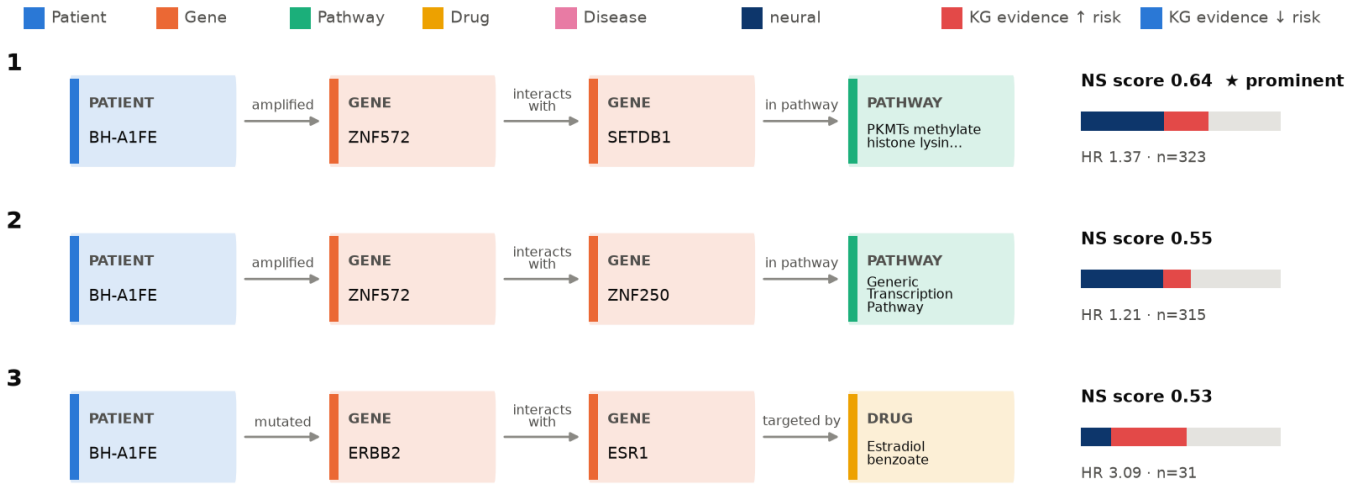
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 31-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **10th risk percentile** (low tier). The highest-scoring knowledge-graph path runs from **ZNF572** to **PKMTs methylate histone lysines**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.37). Progression was recorded at 74.7 months; the model's estimate given progression was 69 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

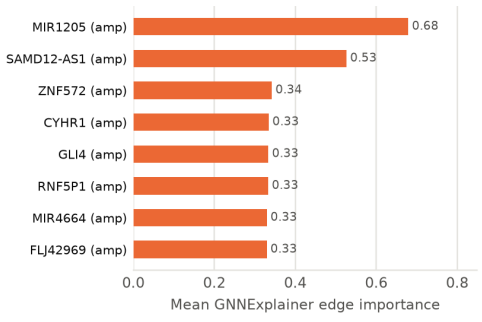
#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-BH-A1FE** amplified → **ZNF572** interacts with → **SETDB1** in pathway → **PKMTs methylate histone lysines**	69	↑ **faster progression** HR 1.37, z +1.7, n = 323 carriers of “PKMTs methylate histone lysines”	**0.64**
2	**TCGA-BH-A1FE** amplified → **ZNF572** interacts with → **ZNF250** in pathway → **Generic Transcription Pathway**	69	↑ **faster progression** HR 1.21, z +1.0, n = 315 carriers of “Generic Transcription Pathway”	**0.55**
3	**TCGA-BH-A1FE** mutated → **ERBB2** interacts with → **ESR1** targeted by → **Estradiol benzoate**	26	↑ **faster progression** HR 3.09, z +2.8, n = 31 carriers of “Estradiol benzoate”	**0.53**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank p < 0.05, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	Mutation	p.D769Y	55	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
PIK3CA · driver	Mutation	p.K111E	70	0.31
RUNX1 · driver	High-level amplification	GISTIC ±2	18	0.31
TP53 · driver	Mutation	p.N247I	57	0.31
MIR1205	High-level amplification	GISTIC ±2	0	0.68
SAMD12-AS1	High-level amplification	GISTIC ±2	0	0.53
ZNF572	High-level amplification	GISTIC ±2	0	0.34
CYHR1	High-level amplification	GISTIC ±2	0	0.33
GLI4	High-level amplification	GISTIC ±2	0	0.33
RNF5P1	High-level amplification	GISTIC ±2	0	0.33
MIR4664	High-level amplification	GISTIC ±2	0	0.33



295 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 106 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A15E	40	II	LumA	0.88	no progression, followed 20.7 mo
TCGA-C8-A135	64	II	Her2	0.87	no progression, followed 12.9 mo
TCGA-AR-A24U	47	II	Her2	0.86	no progression, followed 102.8 mo
TCGA-LL-A5YN	46	II	LumA	0.86	no progression, followed 14.7 mo
TCGA-E2-A14Y	35	II	Basal	0.86	no progression, followed 69.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-A8-A08T · Group D

Report date 14 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**64**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**—**	PAM50 subtype	**LumA**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A8-A08T-01** · Aneuploidy score: **8** · TMB (nonsyn., /Mb): **1.3** · MSIsensor score: **0.24** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

68% by 36 mo

12 mo 25% · 24 mo 46% · 60 mo 87%

Model-estimated progression time

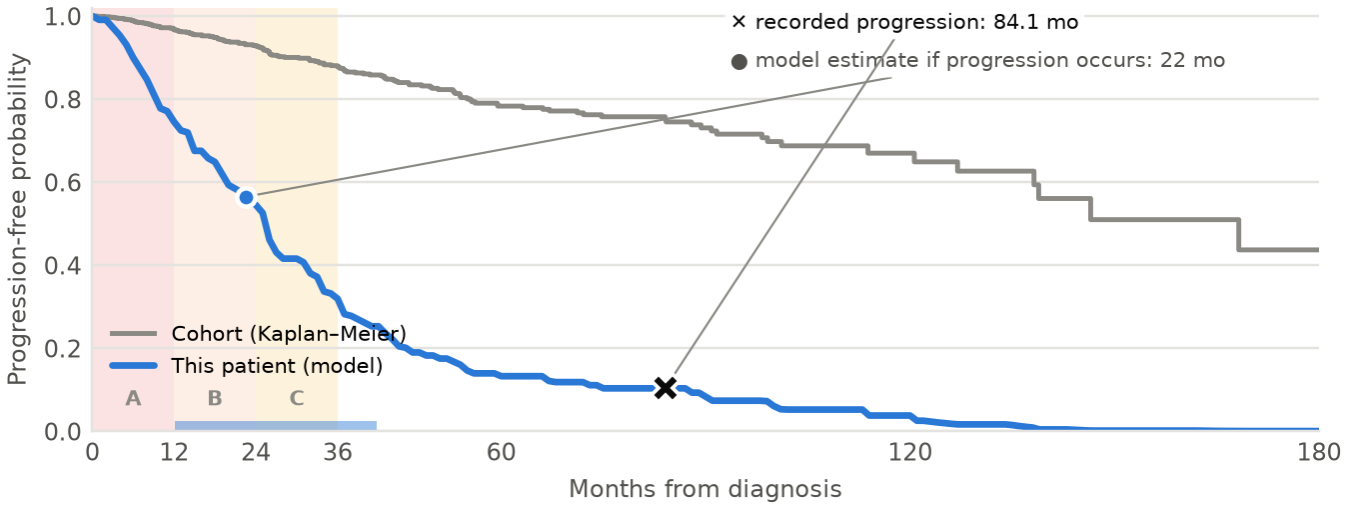
≈ 22 months

Window 12–24 months · IQR 12–42 mo
if progression occurs

Recorded progression

84.1 months

Group D · Late progression
✗ recorded window differs · |error| 62 mo



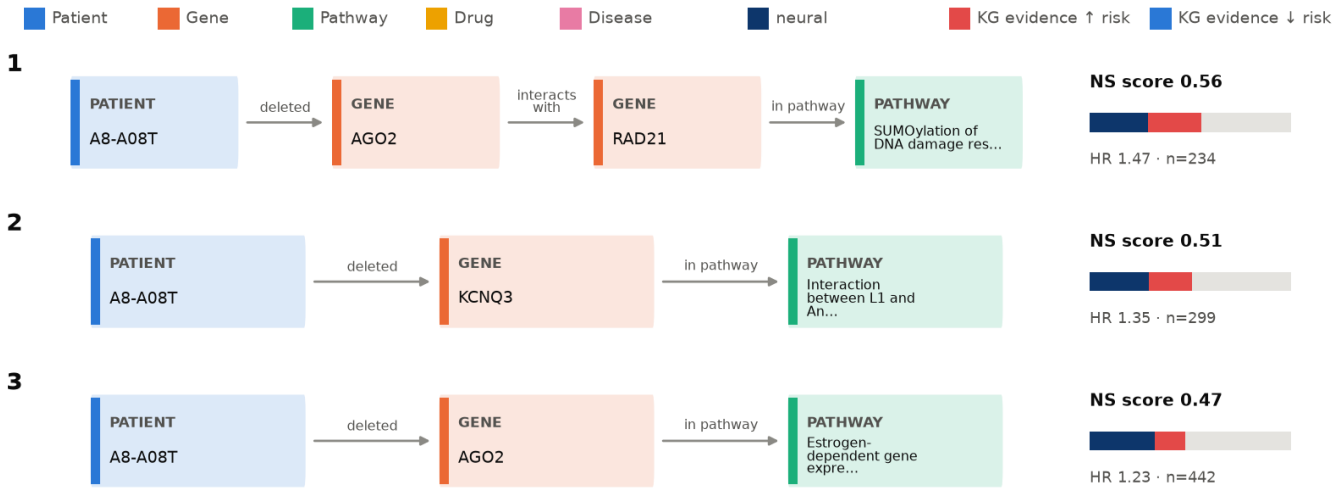
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 64-year-old patient has a LumA tumour, AJCC stage IV (T2, N1, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **AGO2** to **SUMOylation of DNA damage response and repair proteins**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.47). Progression was recorded at 84.1 months; the model's estimate given progression was 22 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

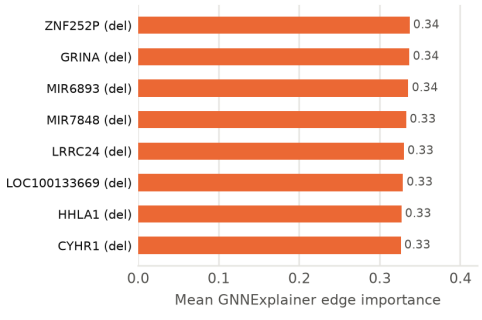
#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-A8-A08T** deleted → **AGO2** interacts with → **RAD21** in pathway → **SUMOylation of DNA damage response and repair proteins**	49	↑ **faster progression** HR 1.47, z +2.0, n = 234 carriers of "SUMOylation of DNA damage response and repair proteins"	**0.56**
2	**TCGA-A8-A08T** deleted → **KCNQ3** in pathway → **Interaction between L1 and Ankyrins**	49	↑ **faster progression** HR 1.35, z +1.6, n = 299 carriers of "Interaction between L1 and Ankyrins"	**0.51**
3	**TCGA-A8-A08T** deleted → **AGO2** in pathway → **Estrogen-dependent gene expression**	54	↑ **faster progression** HR 1.23, z +1.1, n = 442 carriers of "Estrogen-dependent gene expression"	**0.47**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank p < 0.05, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3R1 · driver	Homozygous deletion	GISTIC ±2	79	0.31
PIK3CA · driver	Mutation	p.E545K	70	0.31
CBFB · driver	Mutation	p.Q74K, p.W73*	29	0.31
MAP3K1 · driver	Homozygous deletion	GISTIC ±2	13	0.31
ZNF252P	Homozygous deletion	GISTIC ±2	0	0.34
GRINA	Homozygous deletion	GISTIC ±2	0	0.34
MIR6893	Homozygous deletion	GISTIC ±2	0	0.34
MIR7848	Homozygous deletion	GISTIC ±2	0	0.33
LRRC24	Homozygous deletion	GISTIC ±2	0	0.33
LOC100133669	Homozygous deletion	GISTIC ±2	0	0.33
HHLA1	Homozygous deletion	GISTIC ±2	0	0.33
CYHR1	Homozygous deletion	GISTIC ±2	0	0.33



155 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3R1 → Isoprenaline (DB01064); PIK3R1 → SF1126 (DB05210); PIK3R1 → Enzastaurin (DB06486); PIK3R1 → Wortmannin (DB08059); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015) … and 77 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A203	78	II	Her2	0.68	progressed at 38.6 mo (group D)
TCGA-B6-A0RM	57	X	LumA	0.65	no progression, followed 78.0 mo
TCGA-BH-A1FE	31	II	LumA	0.65	progressed at 74.7 mo (group D)
TCGA-BH-A0E7	79	II	LumA	0.64	no progression, followed 44.8 mo
TCGA-AN-A0FZ	45	III	LumA	0.63	no progression, followed 0.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**43**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8522/3**

Sample: **TCGA-AR-A24O-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **0.67** · MSIsensor score: **0.08** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

82nd

High risk tier · fused neuro-symbolic model



Probability of progression

16% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 27%

Model-estimated progression time

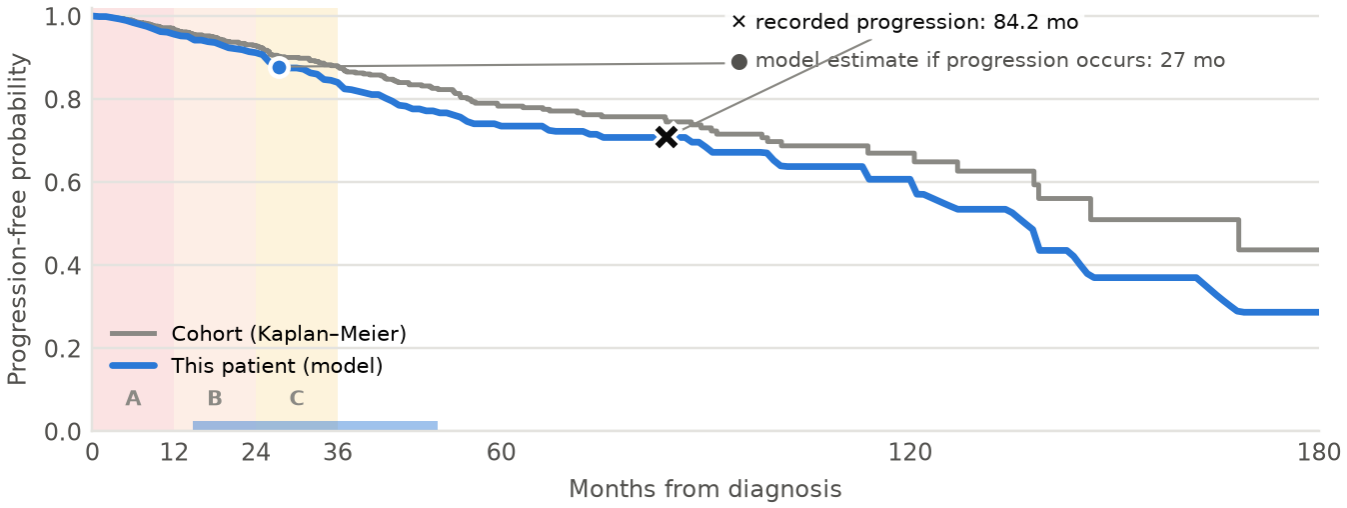
≈ 27 months

Window 24–36 months · IQR 15–51 mo
if progression occurs

Recorded progression

84.2 months

Group D · Late progression
✗ recorded window differs · |error| 57 mo



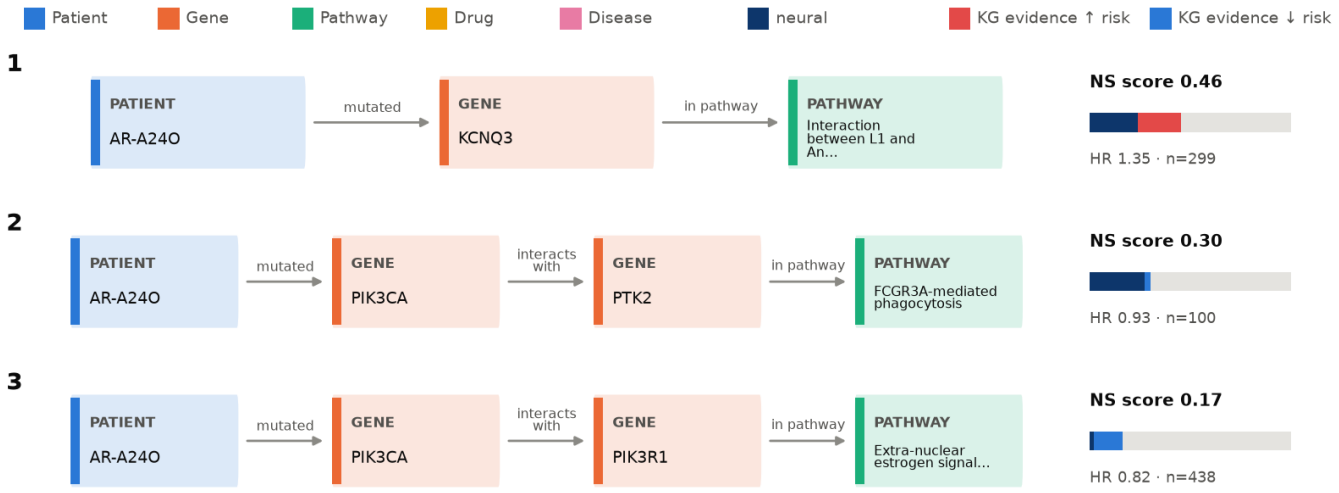
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 43-year-old patient has a LumA tumour, AJCC stage III (T3, N2, M0). The fused neuro-symbolic model places them in the **82nd risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 86.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **KCNQ3** to **Interaction between L1 and Ankyrins**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.35). Progression was recorded at 84.2 months; the model's estimate given progression was 27 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

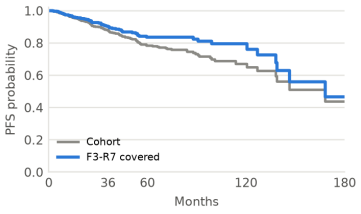


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-AR-A24O** mutated → **KCNQ3** in pathway → **Interaction between L1 and Ankyrins**	40	↑ faster progression HR 1.35, z +1.6, n = 299 carriers of “Interaction between L1 and Ankyrins”	**0.46**
2	**TCGA-AR-A24O** mutated → **PIK3CA** interacts with → **PTK2** in pathway → **FCGR3A-mediated phagocytosis**	45	↓ slower progression HR 0.93, z -0.2, n = 100 carriers of “FCGR3A-mediated phagocytosis”	**0.30**
3	**TCGA-AR-A24O** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **Extra-nuclear estrogen signaling**	3	↓ slower progression HR 0.82, z -1.1, n = 438 carriers of “Extra-nuclear estrogen signaling”	**0.17**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

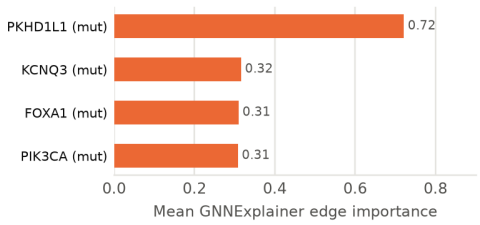
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FOXA1 · driver	Mutation	p.I176V	9	0.31
PIK3CA · driver	Mutation	p.H1047L	70	0.31
PKHD1L1	Mutation	p.Y1437=	0	0.72
KCNQ3	Mutation	p.T655M	8	0.32



4 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); KCNQ3 → Amitriptyline (DB00321); KCNQ3 → Diclofenac (DB00586); KCNQ3 → Meclofenamic acid (DB00939) … and 3 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0YL	48	III	LumA	0.74	no progression, followed 48.5 mo
TCGA-AR-A24M	38	III	LumA	0.74	no progression, followed 120.3 mo
TCGA-B6-A0RV	42	III	LumA	0.73	no progression, followed 169.5 mo
TCGA-C8-A133	65	III	LumA	0.73	no progression, followed 0.0 mo
TCGA-B6-A0IE	38	III	LumA	0.72	progressed at 27.4 mo (group C)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**47**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**—**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8522/3**

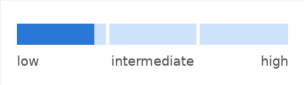
Sample: **TCGA-BH-A0C3-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **0.37** · MSIsensor score: **0.05** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

29th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

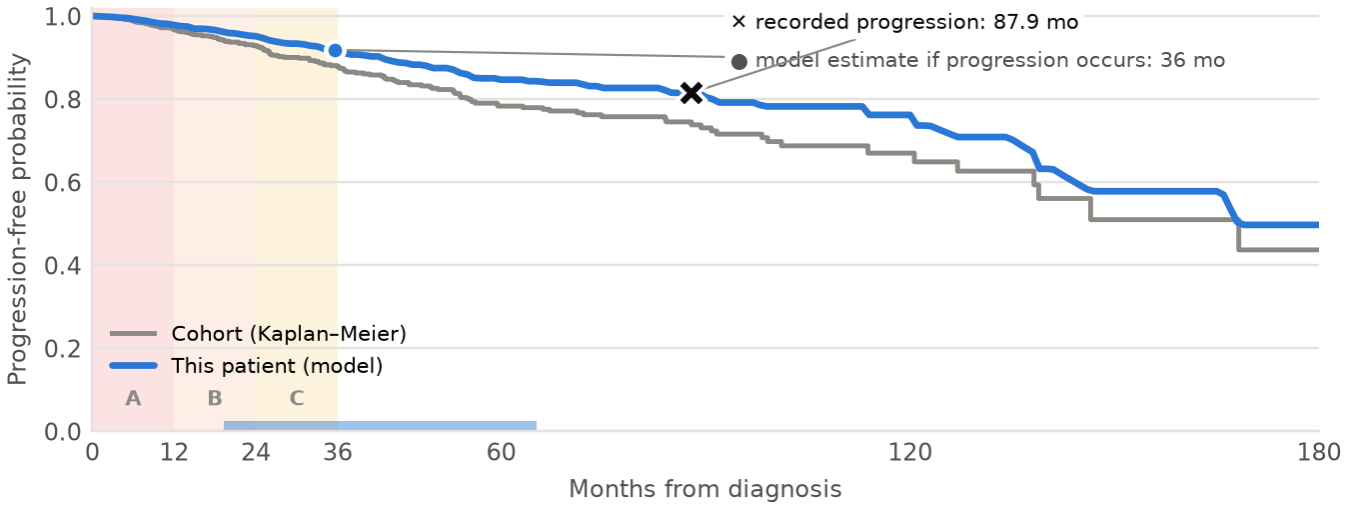
≈ 36 months

Window 24–36 months · IQR 19–65 mo
if progression occurs

Recorded progression

87.9 months

Group D · Late progression
✗ recorded window differs · |error| 52 mo



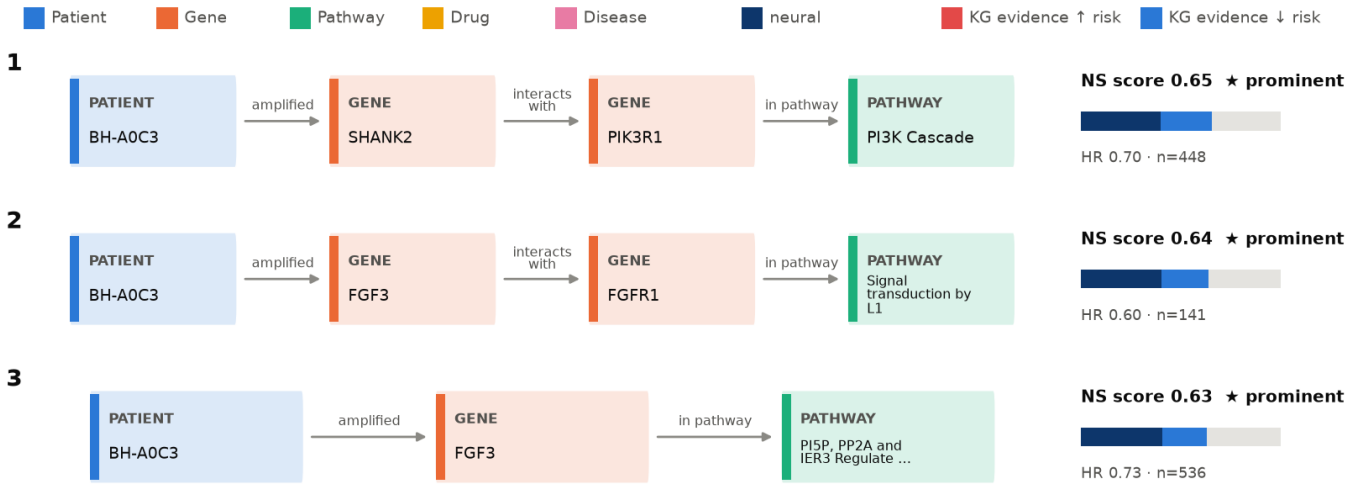
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 47-year-old patient has a LumB tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **29th risk percentile** (low tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **SHANK2** to **PI3K Cascade**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.70). Progression was recorded at 87.9 months; the model's estimate given progression was 36 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

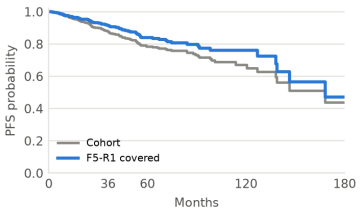


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-BH-A0C3** amplified → **SHANK2** interacts with → **PIK3R1** in pathway → **PI3K Cascade**	66	↓ slower progression HR 0.70, z -1.9, n = 448 carriers of “PI3K Cascade”	**0.65**
2 ★	**TCGA-BH-A0C3** amplified → **FGF3** interacts with → **FGFR1** in pathway → **Signal transduction by L1**	67	↓ slower progression HR 0.60, z -1.8, n = 141 carriers of “Signal transduction by L1”	**0.64**
3 ★	**TCGA-BH-A0C3** amplified → **FGF3** in pathway → **PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling**	68	↓ slower progression HR 0.73, z -1.7, n = 536 carriers of “PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling”	**0.63**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

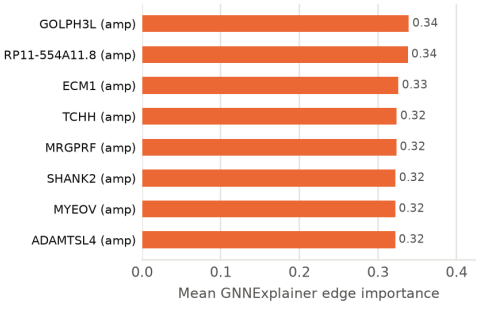
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ lower risk	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R5 ↓ lower risk	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.G187S	57	0.31
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
RUNX1 · driver	Mutation	p.D198G	18	0.31
GOLPH3L	High-level amplification	GISTIC ±2	0	0.34
RP11-554A11.8	High-level amplification	GISTIC ±2	0	0.34
ECM1	High-level amplification	GISTIC ±2	1	0.33
TCHH	High-level amplification	GISTIC ±2	1	0.32
MRGPRF	High-level amplification	GISTIC ±2	0	0.32
SHANK2	High-level amplification	GISTIC ±2	1	0.32
MYEOV	High-level amplification	GISTIC ±2	0	0.32
ADAMTSL4	High-level amplification	GISTIC ±2	2	0.32
CA14	High-level amplification	GISTIC ±2	6	0.32



35 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169) … and 15 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-Z7-A8R6	46	I	LumB	0.77	no progression, followed 107.0 mo
TCGA-AR-A1AP	80	I	LumA	0.74	no progression, followed 93.9 mo
TCGA-GM-A2DL	50	I	LumA	0.69	no progression, followed 115.7 mo
TCGA-A2-A0YF	67	I	LumA	0.68	no progression, followed 50.5 mo
TCGA-A1-A0SE	56	I	LumA	0.68	no progression, followed 43.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**64**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N3**
Radiation therapy	**No**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

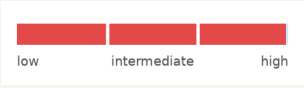
Sample: **TCGA-B6-A0IB-01** · Aneuploidy score: **21** · TMB (nonsyn., /Mb): **2.2** · MSIsensor score: **0.25** · AJCC edition: **3RD**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

68% by 36 mo

12 mo 25% · 24 mo 49% · 60 mo 88%

Model-estimated progression time

≈ 7 months

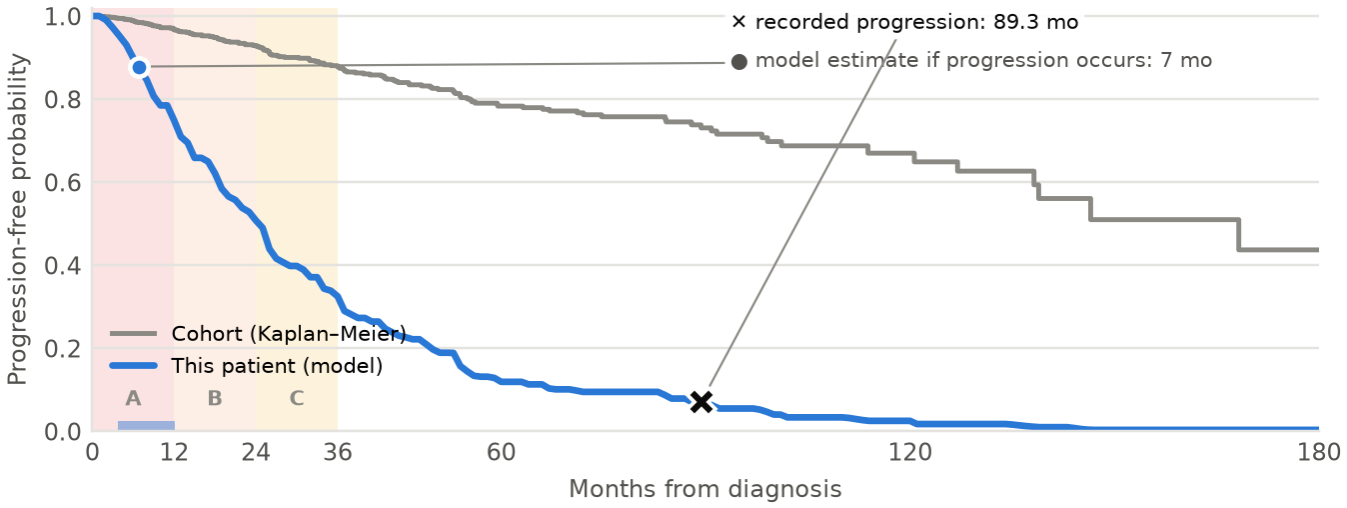
Window ≤ 12 months · IQR 4–12 mo if progression occurs

Recorded progression

89.3 months

Group D · Late progression

✗ recorded window differs · |error| 83 mo



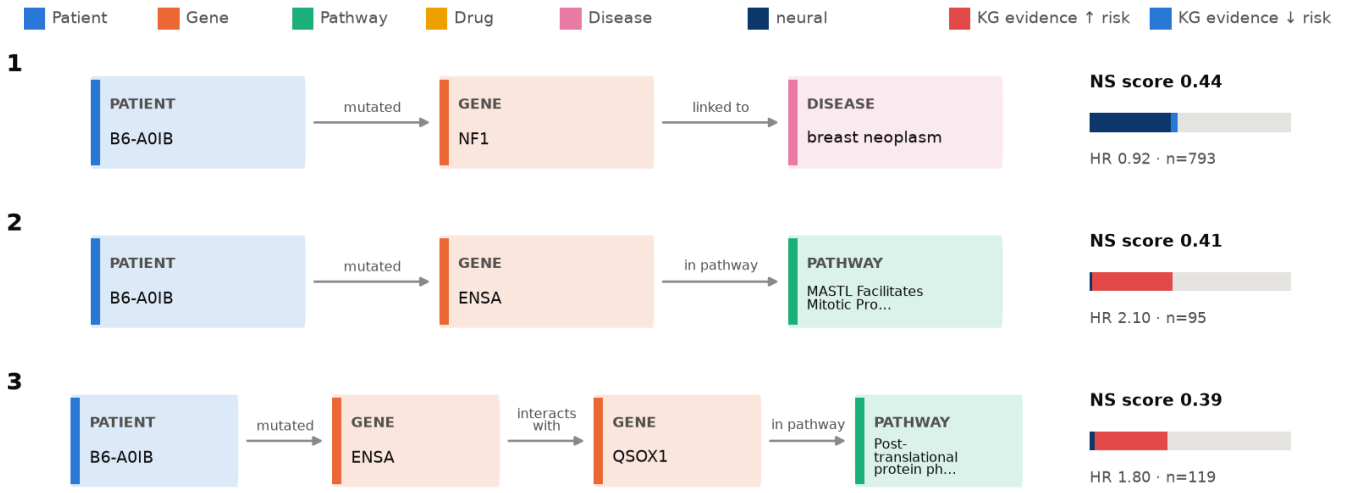
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 64-year-old patient has a LumB tumour, AJCC stage IV (T3, N3, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The most significant rule covering them (*N stage = N3 AND Age < 72.0*) marks a group with a progression hazard 4.0× that of the remaining patients. The highest-scoring knowledge-graph path runs from **NF1** to **breast neoplasm**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.92). Progression was recorded at 89.3 months; the model's estimate given progression was 7 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

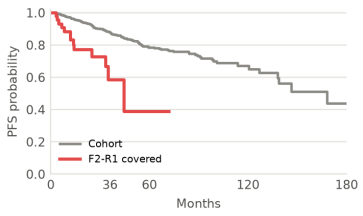


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-B6-A0IB** mutated → **NF1** linked to → **breast neoplasm**	67	↓ **slower progression** HR 0.92, z -0.3, n = 793 carriers of “breast neoplasm”	**0.44**
2	**TCGA-B6-A0IB** mutated → **ENSA** in pathway → **MASTL Facilitates Mitotic Progression**	2	↑ **faster progression** HR 2.10, z +3.1, n = 95 carriers of “MASTL Facilitates Mitotic Progression”	**0.41**
3	**TCGA-B6-A0IB** mutated → **ENSA** interacts with → **QSOX1** in pathway → **Post-translational protein phosphorylation**	4	↑ **faster progression** HR 1.80, z +2.7, n = 119 carriers of “Post-translational protein phosphorylation”	**0.39**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

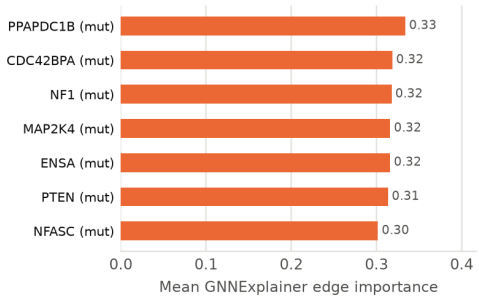
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R1 ↑ higher risk	N stage = N3 AND Age < 72.0	44 pts 13 events	**4.04** p 3.2e-06	44 mo
F2-R8 ↓ lower risk	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
NF1 · driver	Mutation	p.E1894Q	6	0.32
MAP2K4 · driver	Mutation	p.X73_splice	12	0.32
PTEN · driver	Mutation	p.X165_splice	22	0.31
PPAPDC1B	Mutation	p.D82=	0	0.33
CDC42BPA	Mutation	p.R1708W	4	0.32
ENSA	Mutation	p.L37V	1	0.32
NFASC	Mutation	p.K321R	3	0.30



7 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PTEN → Phosphatidylethanolamine (DB04327). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0CS	73	IV	LumA	0.59	no progression, followed 77.2 mo
TCGA-A2-A0T2	66	IV	Basal	0.58	progressed at 8.4 mo (group A)
TCGA-A8-A07W	76	IV	LumB	0.54	no progression, followed 10.0 mo
TCGA-A2-A0SV	63	IV	LumB	0.50	progressed at 27.1 mo (group C)
TCGA-A2-A0SW	82	IV	LumB	0.50	progressed at 21.5 mo (group B)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**46**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

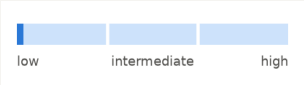
Sample: **TCGA-BH-A18K-01** · Aneuploidy score: **18** · TMB (nonsyn., /Mb): **1.67** · MSIsensor score: **0.3** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

2nd

Low risk tier · fused neuro-symbolic model



Probability of progression

6% by 36 mo

12 mo 1% · 24 mo 4% · 60 mo 13%

Model-estimated progression time

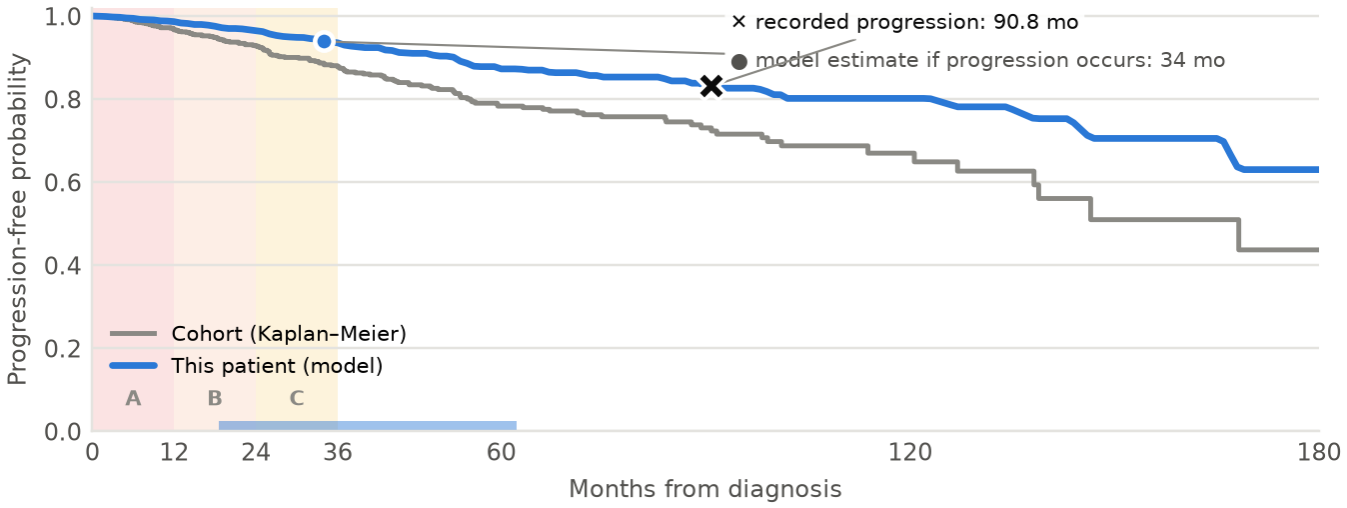
≈ 34 months

Window 24–36 months · IQR 19–62 mo
if progression occurs

Recorded progression

90.8 months

Group D · Late progression
✗ **recorded window differs** · |error| 57 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 46-year-old patient has a LumA tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **2nd risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ITGA10** to **Platelet Adhesion to exposed collagen**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.49). Progression was recorded at 90.8 months; the model's estimate given progression was 34 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

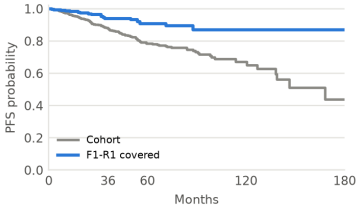


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-BH-A18K** mutated → **ITGA10** in pathway → **Platelet Adhesion to exposed collagen**	98	↑ **faster progression** HR 1.49, z +1.5, n = 93 carriers of “Platelet Adhesion to exposed collagen”	**0.79**
2 ★	**TCGA-BH-A18K** mutated → **SPTA1** in pathway → **Interaction between L1 and Ankyrins**	85	↑ **faster progression** HR 1.32, z +1.5, n = 301 carriers of “Interaction between L1 and Ankyrins”	**0.71**
3 ★	**TCGA-BH-A18K** mutated → **PIK3R1** in pathway → **RND1 GTPase cycle**	84	↑ **faster progression** HR 1.41, z +1.5, n = 131 carriers of “RND1 GTPase cycle”	**0.70**
4 ★	**TCGA-BH-A18K** mutated → **FGFR1** in pathway → **Signal transduction by L1**	85	↓ **slower progression** HR 0.75, z -1.1, n = 143 carriers of “Signal transduction by L1”	**0.65**
5 ★	**TCGA-BH-A18K** mutated → **KMT2C** linked to → **cribriform carcinoma of breast**	89	↑ **faster progression** HR 1.12, z +0.5, n = 662 carriers of “cribriform carcinoma of breast”	**0.60**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

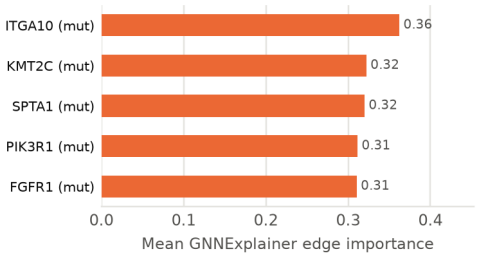
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R1 ↓ lower risk	N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45	365 pts 19 events	**0.26** p 4.0e-09	not reached
F1-R7 ↓ lower risk	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo
F1-R2 ↓ lower risk	N stage = N0 AND Age ≥ 31.5	410 pts 29 events	**0.36** p 4.5e-07	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.V3284F	4	0.32
PIK3R1 · driver	Mutation	p.D569dup	79	0.31
FGFR1 · driver	Mutation	p.I788=	43	0.31
ITGA10	Mutation	p.L990=	3	0.36
SPTA1	Mutation	p.D420=	4	0.32



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3R1 → Isoprenaline (DB01064); PIK3R1 → SF1126 (DB05210); PIK3R1 → Enzastaurin (DB06486); PIK3R1 → Wortmannin (DB08059); FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525) … and 10 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0EM	73	I	LumA	0.77	no progression, followed 101.7 mo
TCGA-BH-A0BO	54	I	LumA	0.77	no progression, followed 72.2 mo
TCGA-E2-A1IO	37	I	LumA	0.76	no progression, followed 61.0 mo
TCGA-BH-A1ET	55	I	LumA	0.76	no progression, followed 82.8 mo
TCGA-BH-A0EB	69	I	LumA	0.76	no progression, followed 24.5 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-B6-A0RP · Group D

Report date 14 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**73**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8522/3**

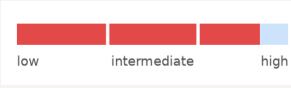
Sample: **TCGA-B6-A0RP-01** · Aneuploidy score: **20** · TMB (nonsyn., /Mb): **0.7** · MSIsensor score: **0.01** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

90th

High risk tier · fused neuro-symbolic model



Probability of progression

18% by 36 mo

12 mo 5% · 24 mo 10% · 60 mo 30%

Model-estimated progression time

≈ 15 months

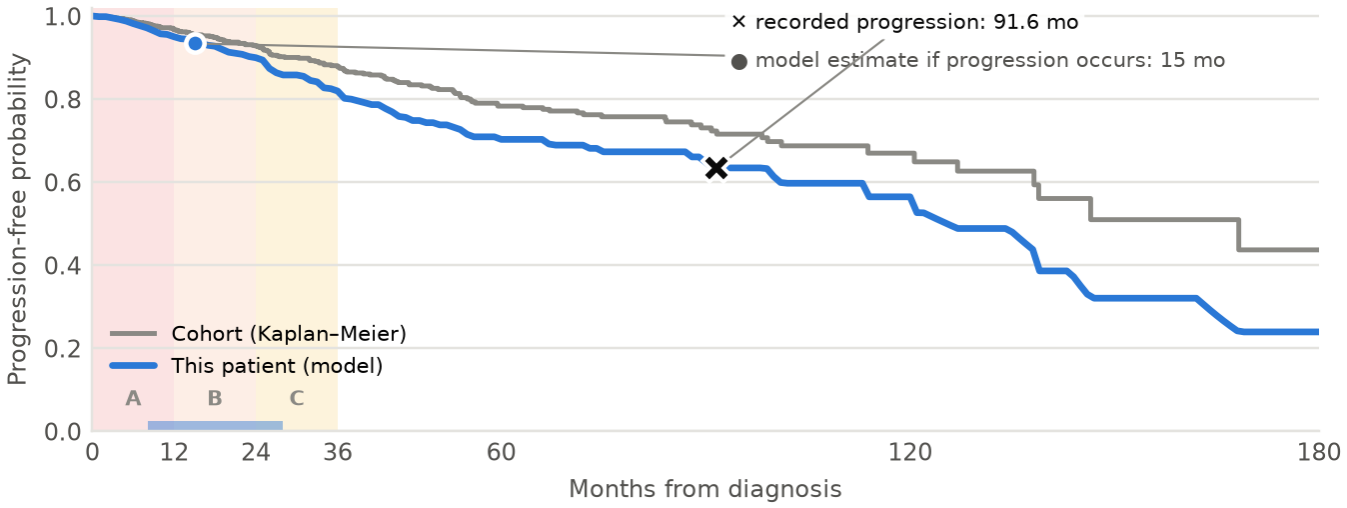
Window 12–24 months · IQR 8–28 mo if progression occurs

Recorded progression

91.6 months

Group D · Late progression

✗ recorded window differs · |error| 77 mo



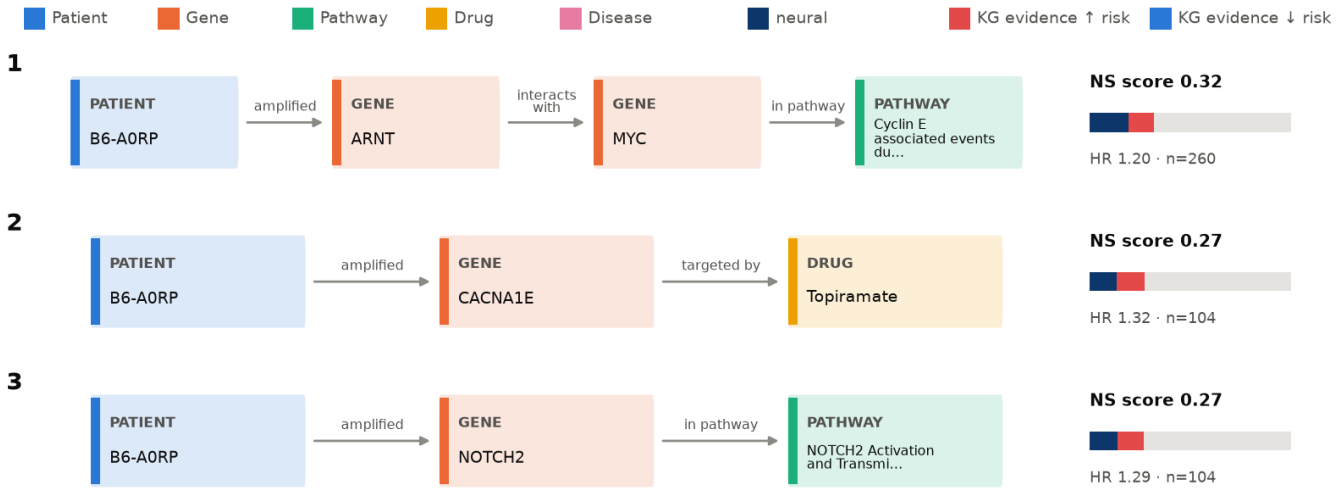
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 73-year-old patient has a LumA tumour, AJCC stage III (T2, N1, M0). The fused neuro-symbolic model places them in the **90th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 86.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ARNT** to **Cyclin E associated events during G1/S transition** ; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.20). Progression was recorded at 91.6 months; the model's estimate given progression was 15 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

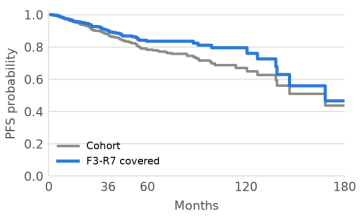


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-B6-A0RP** amplified → **ARNT** interacts with → **MYC** in pathway → **Cyclin E associated events during G1/S transition**	32	↑ **faster progression** HR 1.20, z +0.9, n = 260 carriers of “Cyclin E associated events during G1/S transition ”	**0.32**
2	**TCGA-B6-A0RP** amplified → **CACNA1E** targeted by → **Topiramate**	23	↑ **faster progression** HR 1.32, z +1.0, n = 104 carriers of “Topiramate”	**0.27**
3	**TCGA-B6-A0RP** amplified → **NOTCH2** in pathway → **NOTCH2 Activation and Transmission of Signal to the Nucleus**	23	↑ **faster progression** HR 1.29, z +1.0, n = 104 carriers of “NOTCH2 Activation and Transmission of Signal to the Nucleus”	**0.27**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

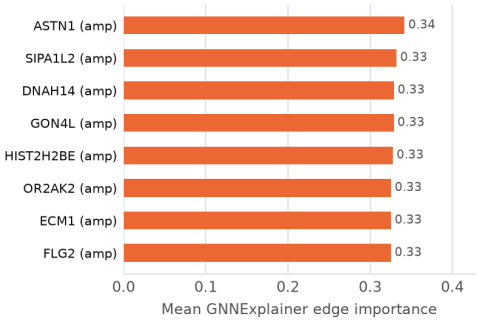
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.D4425Y	4	0.32
PIK3CA · driver	Mutation	p.N345K	70	0.32
ASTN1	High-level amplification	GISTIC ±2	0	0.34
SIPA1L2	High-level amplification	GISTIC ±2	0	0.33
DNAH14	High-level amplification	GISTIC ±2	0	0.33
GON4L	High-level amplification	GISTIC ±2	0	0.33
HIST2H2BE	High-level amplification	GISTIC ±2	0	0.33
OR2AK2	High-level amplification	GISTIC ±2	1	0.33
ECM1	High-level amplification	GISTIC ±2	1	0.33
FLG2	High-level amplification	GISTIC ±2	1	0.33
FAM72C	High-level amplification	GISTIC ±2	0	0.32
SRGAP2D	High-level amplification	GISTIC ±2	0	0.32



111 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); CACNA1E → Ergocalciferol (DB00153); CACNA1E → Enflurane (DB00228); CACNA1E → Topiramate (DB00273) … and 55 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AN-A0XP	69	III	LumA	0.91	no progression, followed 0.3 mo
TCGA-A8-A09B	58	III	LumA	0.87	no progression, followed 12.0 mo
TCGA-BH-A0BA	51	III	LumA	0.87	no progression, followed 37.2 mo
TCGA-D8-A1XG	86	III	LumA	0.86	no progression, followed 14.7 mo
TCGA-A8-A07G	65	II	LumA	0.85	no progression, followed 19.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**34**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

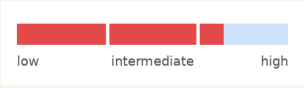
Sample: **TCGA-A2-A25E-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **1.7** · MSIsensor score: **1.75** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

76th

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 25%

Model-estimated progression time

≈ 41 months

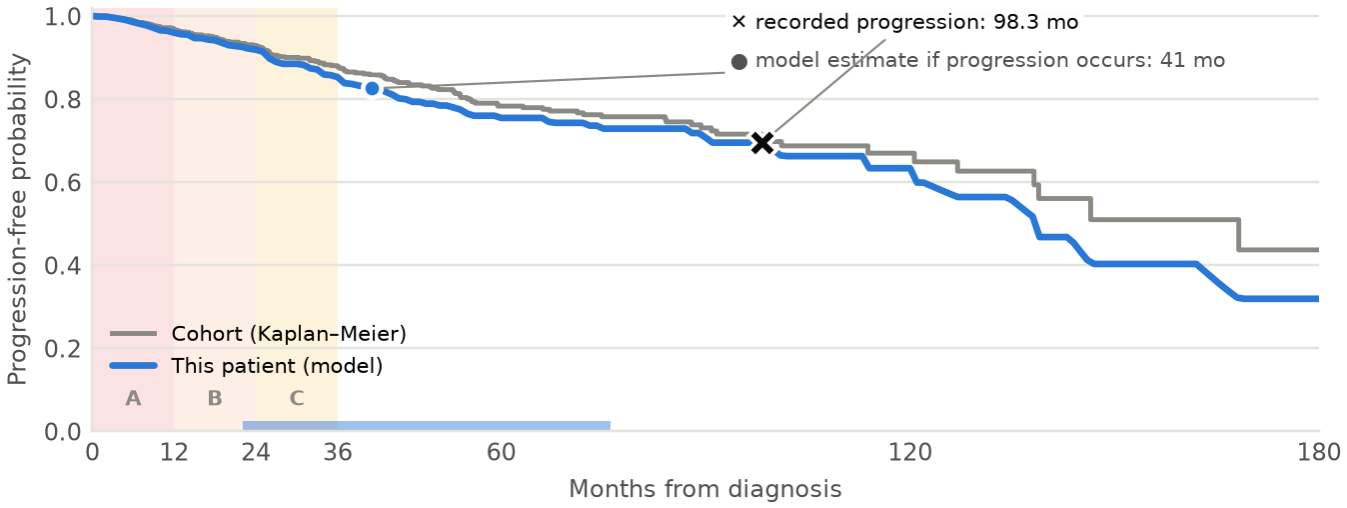
Window > 36 months · IQR 22–76 mo if progression occurs

Recorded progression

98.3 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 57 mo



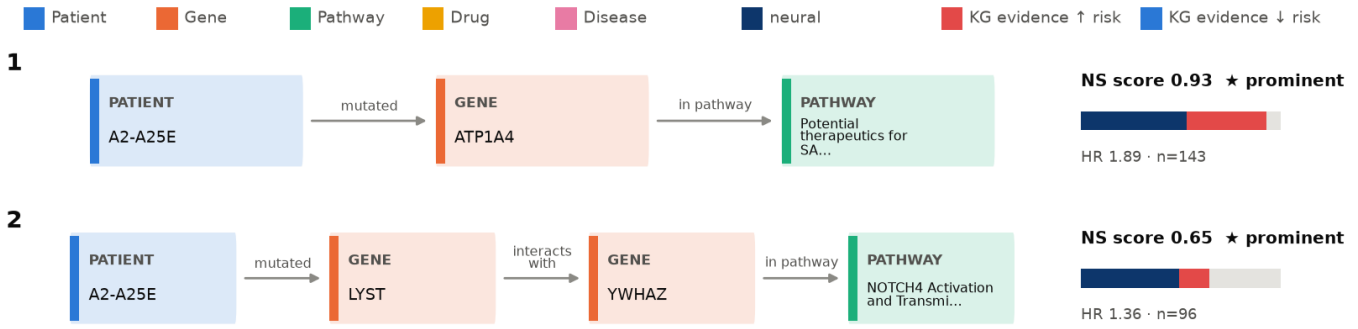
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 34-year-old patient has a LumB tumour, AJCC stage III (T2, N2, M0). The fused neuro-symbolic model places them in the **76th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **ATP1A4** to **Potential therapeutics for SARS**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.89). Progression was recorded at 98.3 months; the model's estimate given progression was 41 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



2 of 19 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

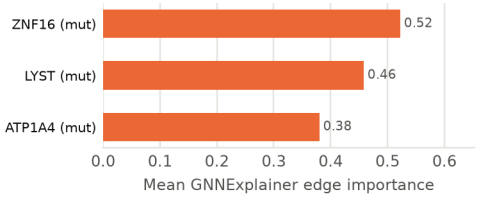
#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-A2-A25E** mutated → **ATP1A4** in pathway → **Potential therapeutics for SARS**	88	↑ **faster progression** HR 1.89, z +3.0, n = 143 carriers of “Potential therapeutics for SARS”	**0.93**
2 ★	**TCGA-A2-A25E** mutated → **LYST** interacts with → **YWHAZ** in pathway → **NOTCH4 Activation and Transmission of Signal to the Nucleus**	82	↑ **faster progression** HR 1.36, z +1.1, n = 96 carriers of “NOTCH4 Activation and Transmission of Signal to the Nucleus”	**0.65**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank p < 0.05, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ZNF16	Mutation	p.D184N	0	0.52
LYST	Mutation	p.N3046D	0	0.46
ATP1A4	Mutation	p.A328G	4	0.38



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ATP1A4 → Rubidium Rb-82 (DB09479). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A0TT	53	III	LumB	0.88	no progression, followed 109.0 mo
TCGA-D8-A27N	36	III	LumB	0.79	no progression, followed 17.1 mo
TCGA-A2-A1G6	50	III	Normal	0.79	no progression, followed 16.5 mo
TCGA-AR-A1AU	39	III	LumA	0.76	no progression, followed 94.3 mo
TCGA-C8-A1HI	40	III	LumA	0.74	no progression, followed 11.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group D** · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**46**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

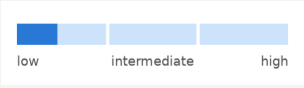
Sample: **TCGA-GM-A2DA-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **1.07** · MSIsensor score: **0.08** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

15th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 14%

Model-estimated progression time

≈ 39 months

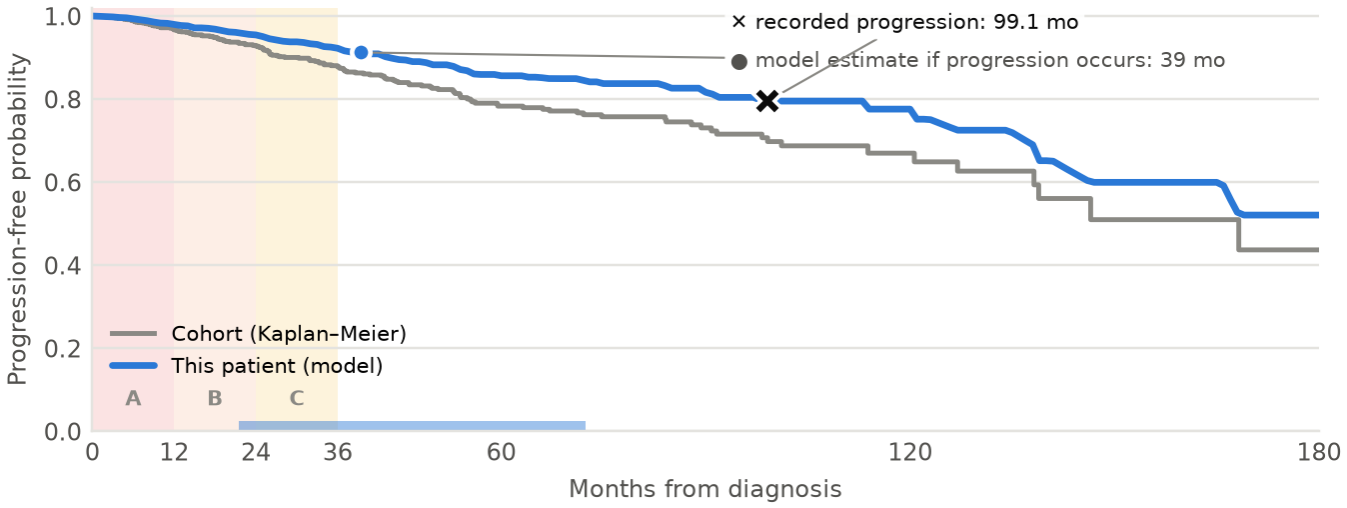
Window > 36 months · IQR 21–72 mo if progression occurs

Recorded progression

99.1 months

■ **Group D** · Late progression

✓ **estimate in recorded window** · |error| 60 mo



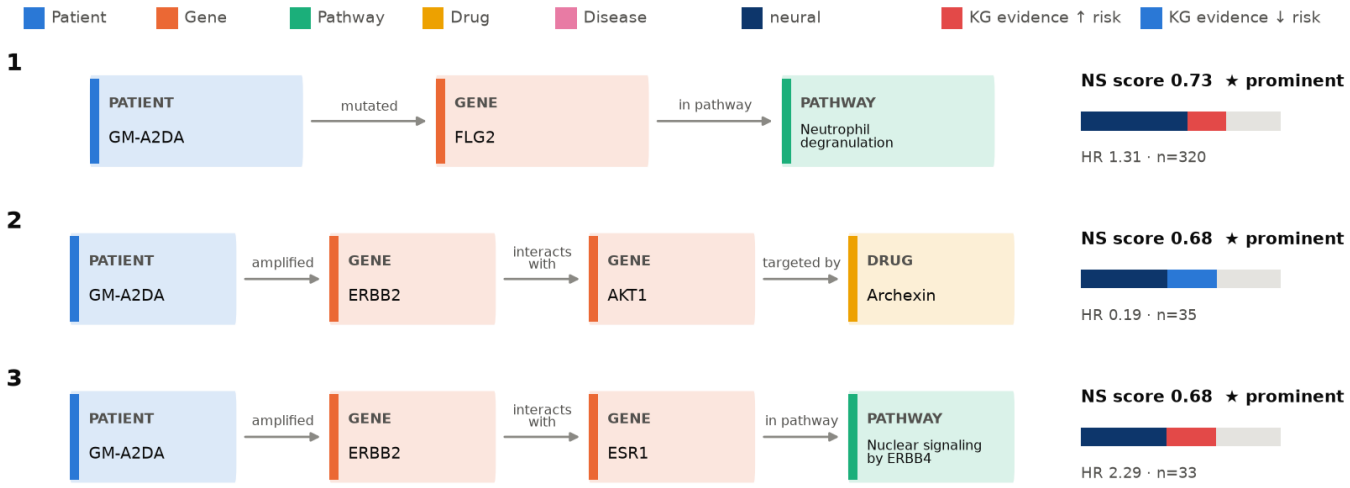
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 46-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **15th risk percentile** (low tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **FLG2** to **Neutrophil degranulation**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.31). Progression was recorded at 99.1 months; the model's estimate given progression was 39 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

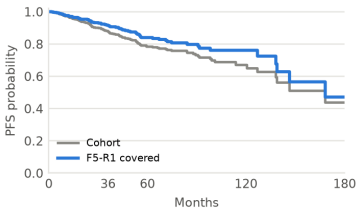


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-GM-A2DA** mutated → **FLG2** in pathway → **Neutrophil degranulation**	89	↑ **faster progression** HR 1.31, z +1.5, n = 320 carriers of “Neutrophil degranulation”	**0.73**
2 ★	**TCGA-GM-A2DA** amplified → **ERBB2** interacts with → **AKT1** targeted by → **Archexin**	72	↓ **slower progression** HR 0.19, z -1.9, n = 35 carriers of “Archexin”	**0.68**
3 ★	**TCGA-GM-A2DA** amplified → **ERBB2** interacts with → **ESR1** in pathway → **Nuclear signaling by ERBB4**	72	↑ **faster progression** HR 2.29, z +1.9, n = 33 carriers of “Nuclear signaling by ERBB4”	**0.68**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

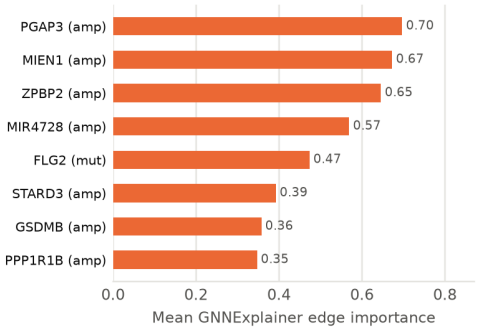
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ lower risk	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R3 ↓ lower risk	AJCC stage = II AND 39.5 ≤ Age < 78.5	422 pts 31 events	**0.36** p 4.2e-07	168 mo
F5-R4 ↓ lower risk	TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5	567 pts 64 events	**0.56** p 2.1e-03	168 mo
F5-R5 ↓ lower risk	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
PGAP3	High-level amplification	GISTIC ±2	0	0.70
MIEN1	High-level amplification	GISTIC ±2	0	0.67
ZPBP2	High-level amplification	GISTIC ±2	0	0.65
MIR4728	High-level amplification	GISTIC ±2	0	0.57
FLG2	Mutation	p.G1757A	1	0.47
STARD3	High-level amplification	GISTIC ±2	1	0.39
GSDMB	High-level amplification	GISTIC ±2	4	0.36
PPP1R1B	High-level amplification	GISTIC ±2	1	0.35
MUC16	Mutation	p.G14473S	8	0.32
GRB7	High-level amplification	GISTIC ±2	12	0.32
CDH17	Mutation	p.V294=	1	0.32



17 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 20 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-C8-A132	56	II	LumA	0.94	no progression, followed 12.6 mo
TCGA-E2-A1LB	41	II	Her2	0.87	no progression, followed 75.8 mo
TCGA-A8-A07P	68	II	LumA	0.83	no progression, followed 11.0 mo
TCGA-E2-A15H	38	II	LumA	0.78	no progression, followed 12.9 mo
TCGA-E9-A245	47	II	LumA	0.78	no progression, followed 0.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**40**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IA**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8523/3**

Sample: **TCGA-B6-A0RU-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **1.53** · MSIsensor score: **0.25** · AJCC edition: **3RD**

2 · Progression assessment

Model risk (cohort percentile)

52nd

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 18%

Model-estimated progression time

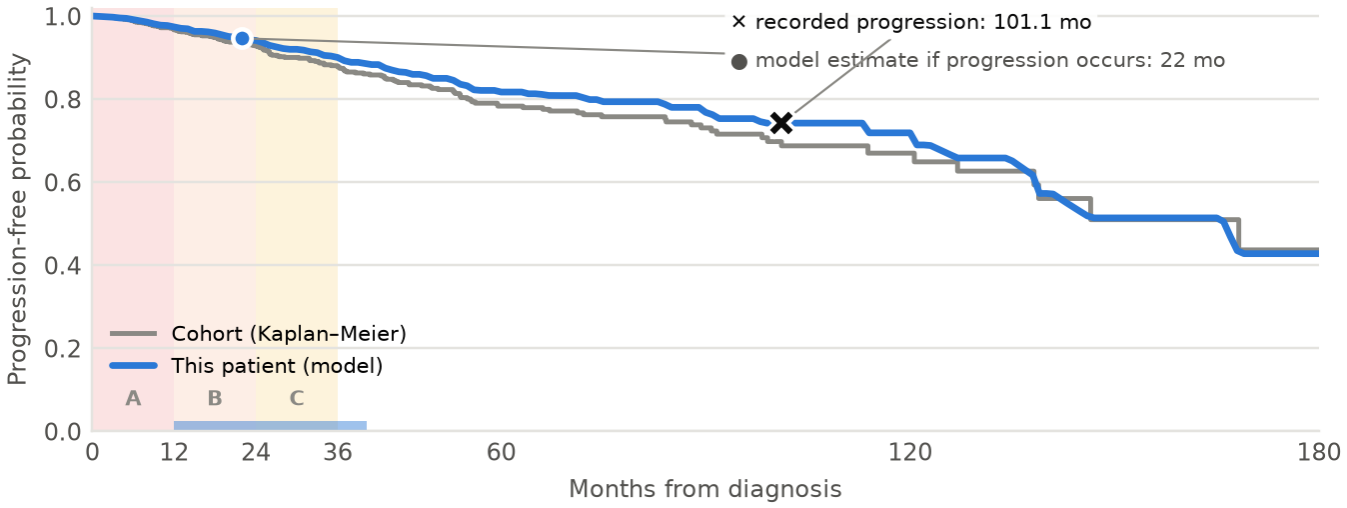
≈ 22 months

Window 12–24 months · IQR 12–40 mo
if progression occurs

Recorded progression

101.1 months

Group D · Late progression
✗ **recorded window differs** · |error| 79 mo



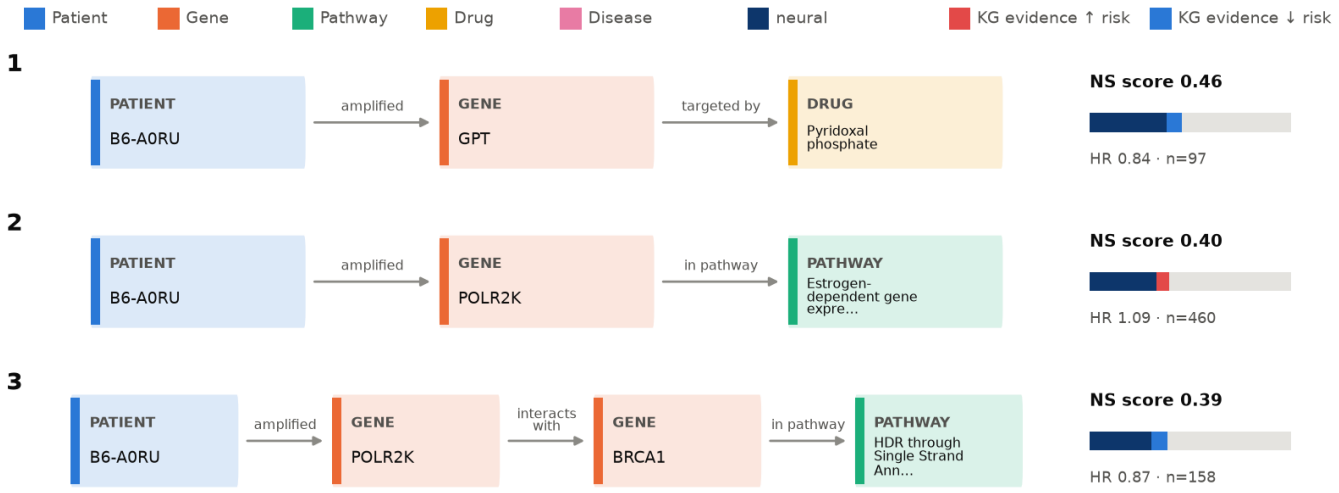
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 40-year-old patient has a Basal tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **52nd risk percentile** (intermediate tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **GPT** to **Pyridoxal phosphate**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.84). Progression was recorded at 101.1 months; the model's estimate given progression was 22 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

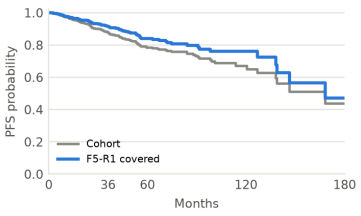


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-B6-A0RU** amplified → **GPT** targeted by → **Pyridoxal phosphate**	64	↓ **slower progression** HR 0.84, z -0.6, n = 97 carriers of “Pyridoxal phosphate”	**0.46**
2	**TCGA-B6-A0RU** amplified → **POLR2K** in pathway → **Estrogen-dependent gene expression**	55	↑ **faster progression** HR 1.09, z +0.5, n = 460 carriers of “Estrogen-dependent gene expression”	**0.40**
3	**TCGA-B6-A0RU** amplified → **POLR2K** interacts with → **BRCA1** in pathway → **HDR through Single Strand Annealing (SSA)**	52	↓ **slower progression** HR 0.87, z -0.6, n = 158 carriers of “HDR through Single Strand Annealing (SSA)”	**0.39**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

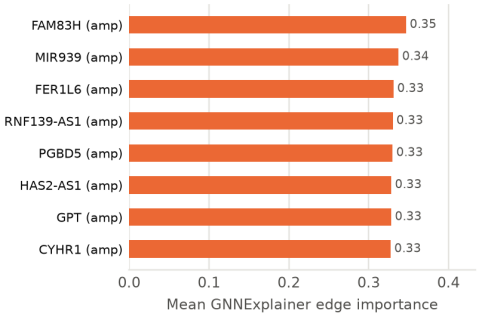
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ **lower risk**	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ **lower risk**	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R4 ↓ **lower risk**	TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5	567 pts 64 events	**0.56** p 2.1e-03	168 mo
F5-R5 ↓ **lower risk**	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.R196*	57	0.32
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
FAM83H	High-level amplification	GISTIC ±2	0	0.35
MIR939	High-level amplification	GISTIC ±2	0	0.34
FER1L6	High-level amplification	GISTIC ±2	0	0.33
RNF139-AS1	High-level amplification	GISTIC ±2	0	0.33
PGBD5	High-level amplification	GISTIC ±2	0	0.33
HAS2-AS1	High-level amplification	GISTIC ±2	0	0.33
GPT	High-level amplification	GISTIC ±2	5	0.33
CYHR1	High-level amplification	GISTIC ±2	0	0.33
RNU6-83P	High-level amplification	GISTIC ±2	0	0.33



328 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); PTEN → Phosphatidylethanolamine (DB04327); MYC → Acetylsalicylic acid (DB00945) … and 108 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AO-A0J4	41	I	Basal	0.87	no progression, followed 52.2 mo
TCGA-C8-A27B	48	II	Basal	0.82	no progression, followed 14.4 mo
TCGA-B6-A0I1	73	II	Basal	0.81	progressed at 36.8 mo (group D)
TCGA-BH-A0BW	71	I	Basal	0.81	no progression, followed 77.9 mo
TCGA-D8-A27M	59	I	Basal	0.80	no progression, followed 13.5 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**35**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A1ES-01** · Aneuploidy score: **7** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0.25** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

63rd

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 2% · 24 mo 6% · 60 mo 22%

Model-estimated progression time

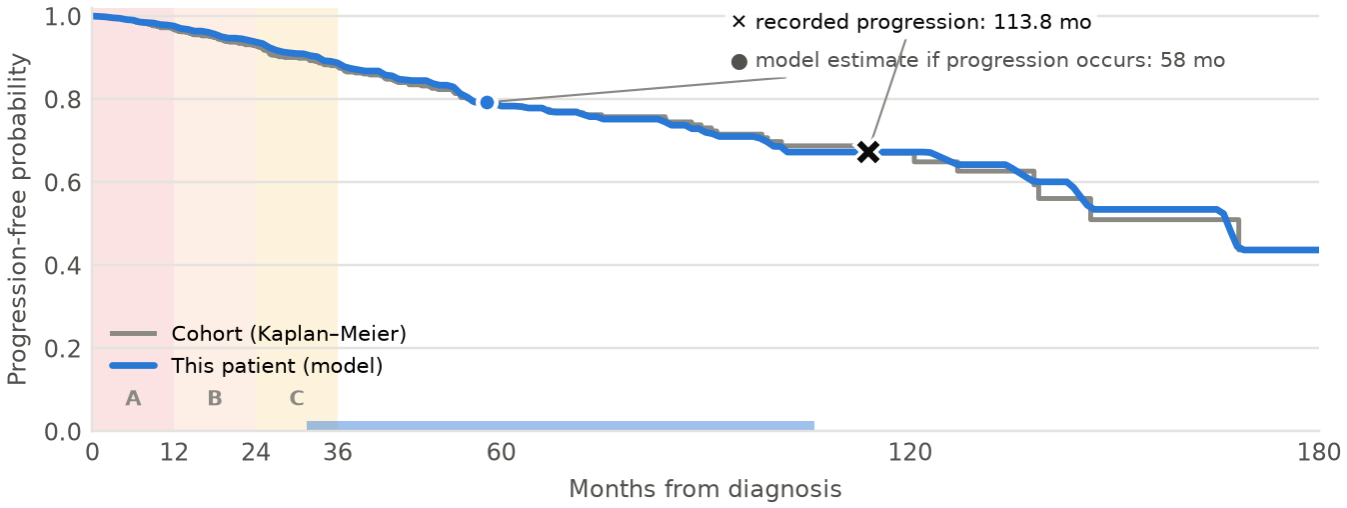
≈ 58 months

Window > 36 months · IQR 32–106 mo
if progression occurs

Recorded progression

113.8 months

Group D · Late progression
✓ **estimate in recorded window** · |error| 56 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 35-year-old patient has a subtype-unassigned tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **63rd risk percentile** (intermediate tier). The highest-scoring knowledge-graph path runs from **ADAM2** to **Interaction With Cumulus Cells And The Zona Pellucida**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.28). Progression was recorded at 113.8 months; the model's estimate given progression was 58 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

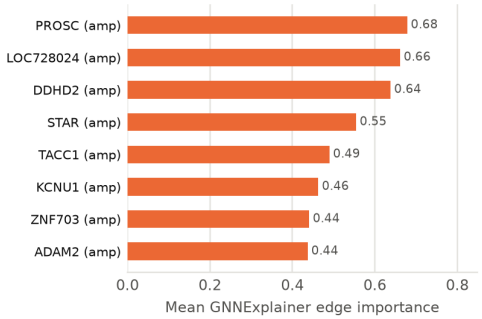
#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-BH-A1ES** amplified → **ADAM2** in pathway → **Interaction With Cumulus Cells And The Zona Pellucida**	97	↑ **faster progression** HR 1.28, z +1.2, n = 183 carriers of “Interaction With Cumulus Cells And The Zona Pellucida”	**0.74**
2 ★	**TCGA-BH-A1ES** amplified → **STAR** in pathway → **Pregnenolone biosynthesis**	98	↑ **faster progression** HR 1.15, z +0.7, n = 196 carriers of “Pregnenolone biosynthesis”	**0.68**
3 ★	**TCGA-BH-A1ES** amplified → **DDHD2** in pathway → **Synthesis of PA**	99	↑ **faster progression** HR 1.12, z +0.5, n = 192 carriers of “Synthesis of PA”	**0.67**
4 ★	**TCGA-BH-A1ES** amplified → **FGFR1** targeted by → **Lenvatinib**	98	↓ **slower progression** HR 0.88, z -0.5, n = 114 carriers of “Lenvatinib”	**0.65**
5 ★	**TCGA-BH-A1ES** amplified → **KCNU1** in pathway → **Sperm Motility And Taxes**	87	↓ **slower progression** HR 0.82, z -0.7, n = 108 carriers of “Sperm Motility And Taxes”	**0.62**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank p < 0.05, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.42
PROSC	High-level amplification	GISTIC ±2	0	0.68
LOC728024	High-level amplification	GISTIC ±2	0	0.66
DDHD2	High-level amplification	GISTIC ±2	1	0.64
STAR	High-level amplification	GISTIC ±2	1	0.55
TACC1	High-level amplification	GISTIC ±2	0	0.49
KCNU1	High-level amplification	GISTIC ±2	1	0.46
ZNF703	High-level amplification	GISTIC ±2	1	0.44
ADAM2	High-level amplification	GISTIC ±2	1	0.44
RAB11FIP1	High-level amplification	GISTIC ±2	0	0.41
CSMD1	High-level amplification	GISTIC ±2	0	0.40
HTRA4	High-level amplification	GISTIC ±2	0	0.37



31 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 45 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A3XY	49	II	missing	0.80	progressed at 26.4 mo (group C)
TCGA-HN-A2OB	45	II	LumA	0.80	progressed at 50.7 mo (group D)
TCGA-E2-A1IE	61	II	LumA	0.78	progressed at 59.5 mo (group D)
TCGA-BH-A0EI	51	II	LumA	0.77	no progression, followed 63.3 mo
TCGA-B6-A0WT	61	II	LumA	0.75	progressed at 127.0 mo (group D)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**60**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

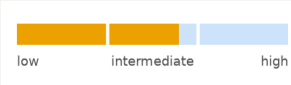
Sample: **TCGA-BH-A1FB-01** · Aneuploidy score: **—** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **—** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

60th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 2% · 24 mo 6% · 60 mo 20%

Model-estimated progression time

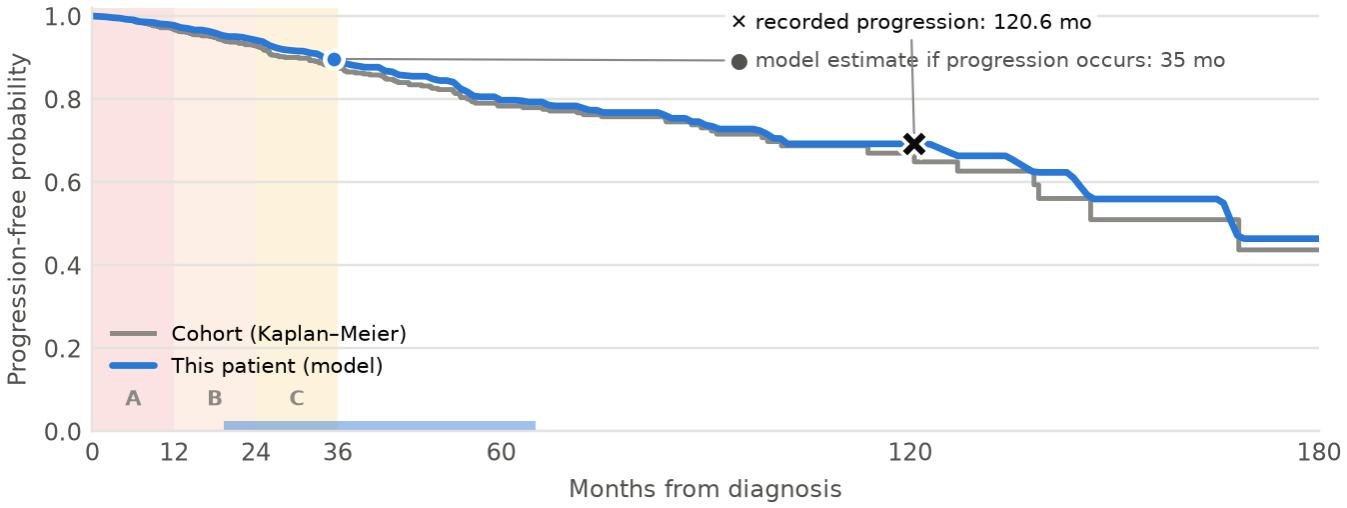
≈ 35 months

Window 24–36 months · IQR 19–65 mo
if progression occurs

Recorded progression

120.6 months

Group D · Late progression
✗ recorded window differs · |error| 85 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 60-year-old patient has a subtype-unassigned tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **60th risk percentile** (intermediate tier). The most significant rule covering them (*AJCC stage = II AND Age ≥ 39.5 AND TMB (nonsynonymous) < 141.95*) marks a group with a progression hazard 0.5× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ZNF703** to **Regulation of HSF1-mediated heat shock response**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.83). Progression was recorded at 120.6 months; the model's estimate given progression was 35 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

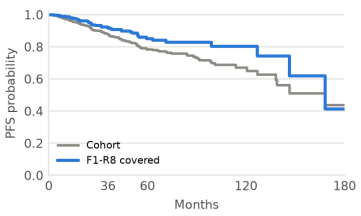


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-BH-A1FB** deleted → **ZNF703** interacts with → **BAG4** in pathway → **Regulation of HSF1-mediated heat shock response**	96	↓ **slower progression** HR 0.83, z -0.9, n = 217 carriers of “Regulation of HSF1-mediated heat shock response”	**0.70**
2 ★	**TCGA-BH-A1FB** deleted → **STAR** in pathway → **Pregnenolone biosynthesis**	98	↑ **faster progression** HR 1.15, z +0.7, n = 196 carriers of “Pregnenolone biosynthesis”	**0.68**
3 ★	**TCGA-BH-A1FB** deleted → **DDHD2** in pathway → **Synthesis of PA**	99	↑ **faster progression** HR 1.12, z +0.5, n = 192 carriers of “Synthesis of PA”	**0.67**
4 ★	**TCGA-BH-A1FB** deleted → **RAB11FIP1** interacts with → **PPFIA1** in pathway → **Acetylcholine Neurotransmitter Release Cycle**	65	↓ **slower progression** HR 0.59, z -1.7, n = 129 carriers of “Acetylcholine Neurotransmitter Release Cycle”	**0.62**
5 ★	**TCGA-BH-A1FB** deleted → **KCNU1** in pathway → **Sperm Motility And Taxes**	85	↓ **slower progression** HR 0.82, z -0.7, n = 108 carriers of “Sperm Motility And Taxes”	**0.60**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

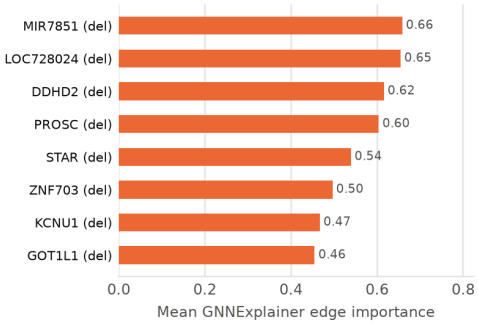
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R8 ↓ lower risk	AJCC stage = II AND Age ≥ 39.5 AND TMB (nonsynonymous) < 141.95	452 pts 39 events	**0.47** p 7.9e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CBFB · driver	Homozygous deletion	GISTIC ±2	29	0.32
CDH1 · driver	Homozygous deletion	GISTIC ±2	17	0.31
MIR7851	Homozygous deletion	GISTIC ±2	0	0.66
LOC728024	Homozygous deletion	GISTIC ±2	0	0.65
DDHD2	Homozygous deletion	GISTIC ±2	1	0.62
PROSC	Homozygous deletion	GISTIC ±2	0	0.60
STAR	Homozygous deletion	GISTIC ±2	1	0.54
ZNF703	Homozygous deletion	GISTIC ±2	1	0.50
KCNU1	Homozygous deletion	GISTIC ±2	1	0.47
GOT1L1	Homozygous deletion	GISTIC ±2	0	0.46
BAG4	Homozygous deletion	GISTIC ±2	4	0.42
RAB11FIP1	Homozygous deletion	GISTIC ±2	0	0.38



22 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ADRB3 → Amphetamine (DB00182); ADRB3 → Bethanidine (DB00217); ADRB3 → Olanzapine (DB00334); ADRB3 → Norepinephrine (DB00368); ADRB3 → Nortriptyline (DB00540); ADRB3 → Propranolol (DB00571); ADRB3 → Paroxetine (DB00715); ADRB3 → Trimipramine (DB00726); ADRB3 → Cryptenamine (DB00785); ADRB3 → Alprenolol (DB00866) … and 28 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A1FL	69	II	LumA	0.70	progressed at 55.0 mo (group D)
TCGA-BH-A1ES	35	II	missing	0.70	progressed at 113.8 mo (group D)
TCGA-LL-A442	56	II	missing	0.69	no progression, followed 29.2 mo
TCGA-A2-A3XY	49	II	missing	0.68	progressed at 26.4 mo (group C)
TCGA-A1-A0SG	61	II	LumA	0.67	no progression, followed 14.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**61**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

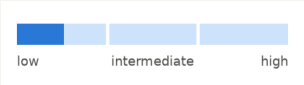
Sample: **TCGA-B6-A0WT-01** · Aneuploidy score: **3** · TMB (nonsyn., /Mb): **0.47** · MSIsensor score: **0.05** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

17th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

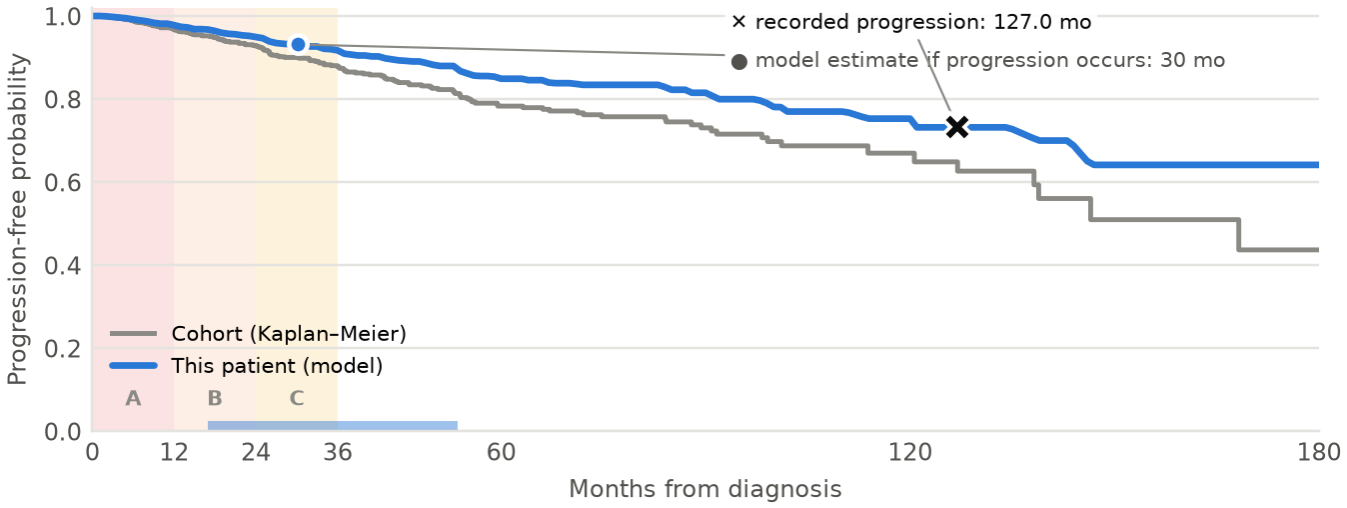
≈ 30 months

Window 24–36 months · IQR 17–54 mo
if progression occurs

Recorded progression

127.0 months

Group D · Late progression
✗ recorded window differs · |error| 97 mo



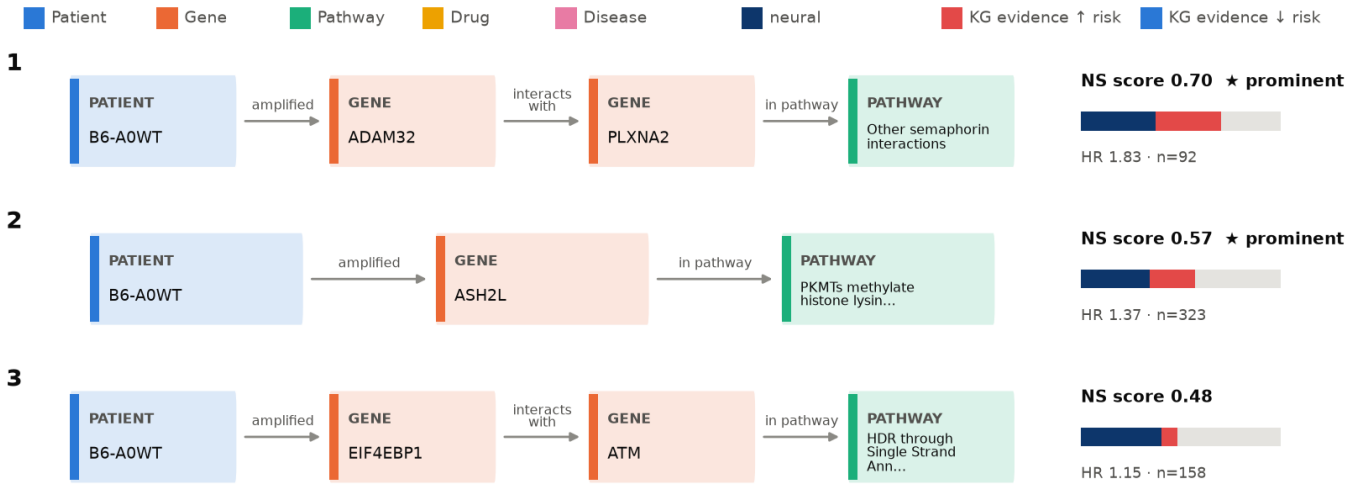
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 61-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **17th risk percentile** (low tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 83.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ADAM32** to **Other semaphorin interactions**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.83). Progression was recorded at 127.0 months; the model's estimate given progression was 30 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

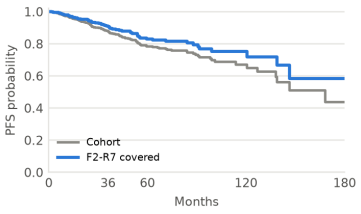


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-B6-A0WT** amplified → **ADAM32** interacts with → **PLXNA2** in pathway → **Other semaphorin interactions**	63	↑ **faster progression** HR 1.83, z +2.4, n = 92 carriers of “Other semaphorin interactions”	**0.70**
2 ★	**TCGA-B6-A0WT** amplified → **ASH2L** in pathway → **PKMTs methylate histone lysines**	57	↑ **faster progression** HR 1.37, z +1.7, n = 323 carriers of “PKMTs methylate histone lysines”	**0.57**
3	**TCGA-B6-A0WT** amplified → **EIF4EBP1** interacts with → **ATM** in pathway → **HDR through Single Strand Annealing (SSA)**	67	↑ **faster progression** HR 1.15, z +0.6, n = 158 carriers of “HDR through Single Strand Annealing (SSA)”	**0.48**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

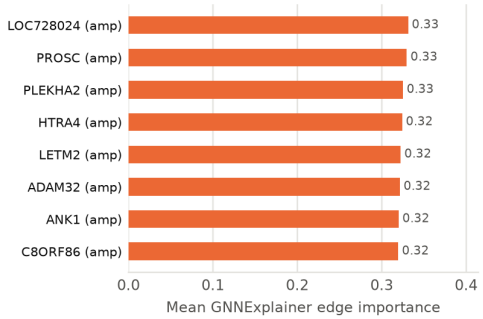
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 83.5	638 pts 66 events	**0.39** p 3.3e-07	not reached
F2-R8 ↓ lower risk	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
SF3B1 · driver	Mutation	p.K700E	4	0.31
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
PTEN · driver	Mutation	p.R130Q	22	0.31
LOC728024	High-level amplification	GISTIC ±2	0	0.33
PROSC	High-level amplification	GISTIC ±2	0	0.33
PLEKHA2	High-level amplification	GISTIC ±2	1	0.33
HTRA4	High-level amplification	GISTIC ±2	0	0.32
LETM2	High-level amplification	GISTIC ±2	0	0.32
ADAM32	High-level amplification	GISTIC ±2	0	0.32
ANK1	High-level amplification	GISTIC ±2	9	0.32
C8ORF86	High-level amplification	GISTIC ±2	0	0.32
ADAM2	High-level amplification	GISTIC ±2	1	0.32



34 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

SF3B1 → H3B-8800 (DB14017); FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896) … and 47 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-EW-A1PA	59	II	LumA	0.88	no progression, followed 18.9 mo
TCGA-E2-A1IE	61	II	LumA	0.88	progressed at 59.5 mo (group D)
TCGA-BH-A204	80	II	LumB	0.81	no progression, followed 83.3 mo
TCGA-WT-AB41	55	II	LumA	0.79	no progression, followed 53.0 mo
TCGA-A2-A0ES	52	II	LumA	0.77	no progression, followed 72.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**69**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

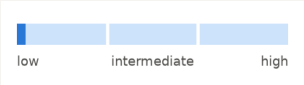
Sample: **TCGA-AR-A2LE-01** · Aneuploidy score: **4** · TMB (nonsyn., /Mb): **8.63** · MSIsensor score: **0.03** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

3rd

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 13%

Model-estimated progression time

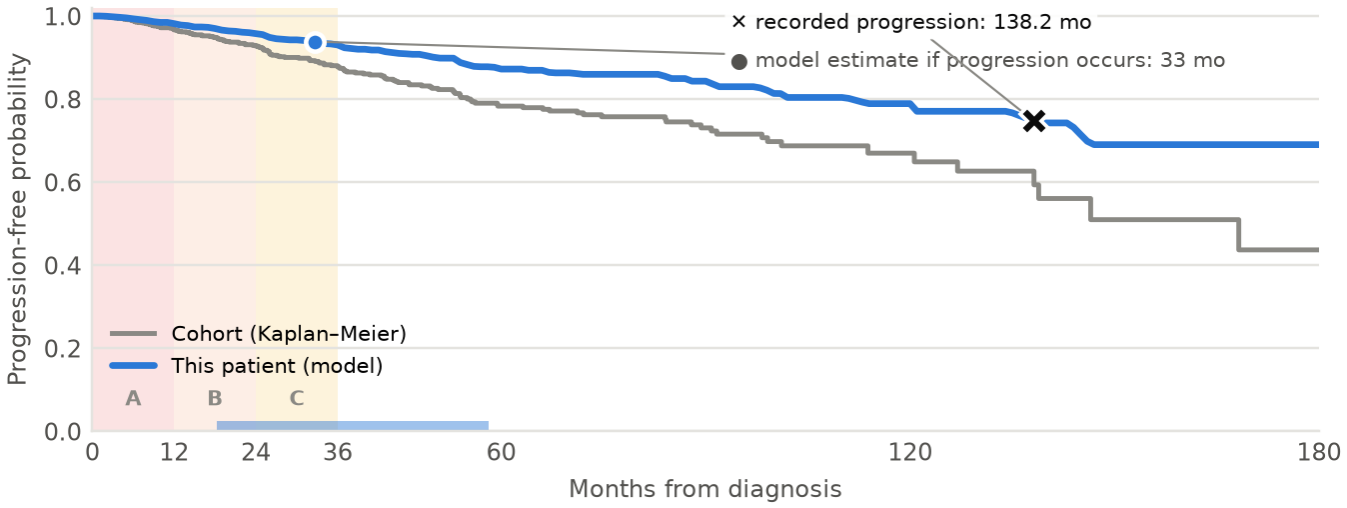
≈ 33 months

Window 24–36 months · IQR 18–58 mo
if progression occurs

Recorded progression

138.2 months

Group D · Late progression
✗ recorded window differs · |error| 106 mo



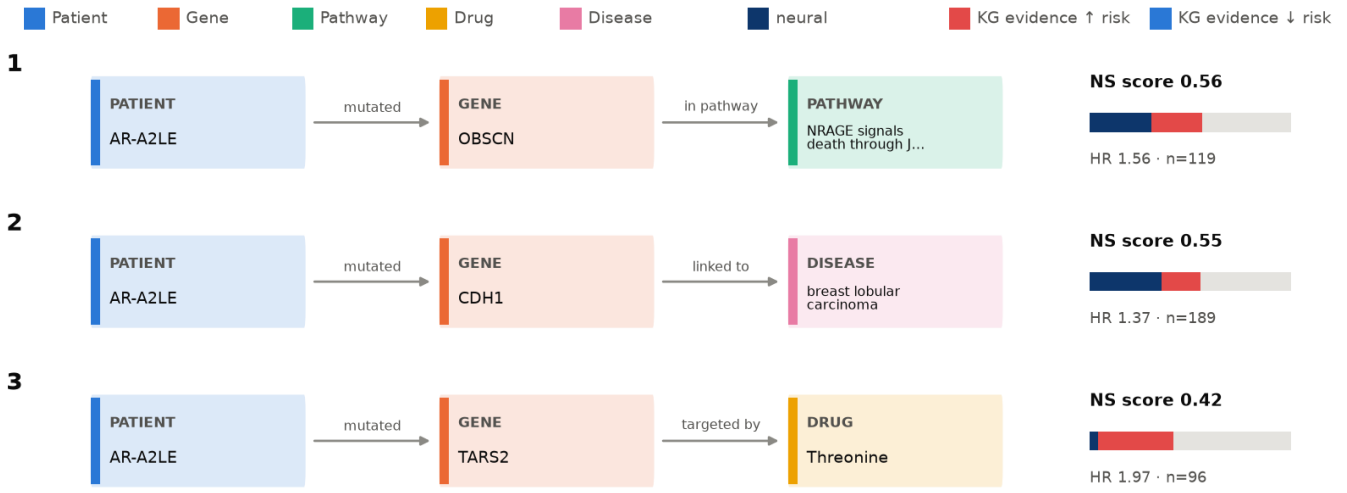
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 69-year-old patient has a LumA tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **3rd risk percentile** (low tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 83.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **OBSCN** to **NRAGE signals death through JNK**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.56). Progression was recorded at 138.2 months; the model's estimate given progression was 33 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

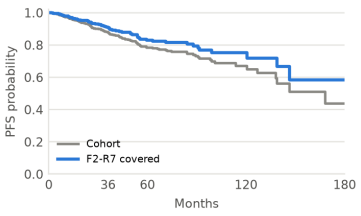


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-AR-A2LE** mutated → **OBSCN** in pathway → **NRAGE signals death through JNK**	51	↑ **faster progression** HR 1.56, z +1.9, n = 119 carriers of “NRAGE signals death through JNK”	**0.56**
2	**TCGA-AR-A2LE** mutated → **CDH1** linked to → **breast lobular carcinoma**	59	↑ **faster progression** HR 1.37, z +1.5, n = 189 carriers of “breast lobular carcinoma”	**0.55**
3	**TCGA-AR-A2LE** mutated → **TARS2** targeted by → **Threonine**	7	↑ **faster progression** HR 1.97, z +2.8, n = 96 carriers of “Threonine”	**0.42**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

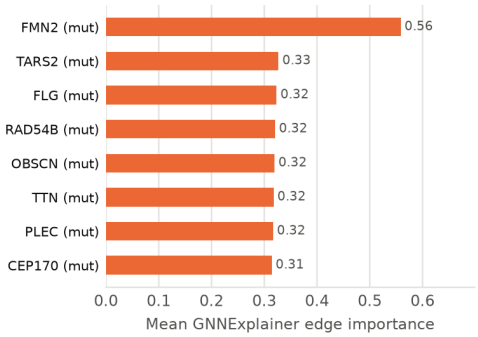
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 83.5	638 pts 66 events	**0.39** p 3.3e-07	not reached
F2-R2 ↓ lower risk	N stage = N0 AND 41.5 ≤ Age < 78.5	343 pts 24 events	**0.37** p 5.3e-06	not reached
F2-R3 ↓ lower risk	N stage = N0 AND TMB (nonsynonymous) ≥ 0.78 AND Age < 83.5	317 pts 25 events	**0.41** p 3.7e-05	not reached
F2-R8 ↓ lower risk	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.Q255*	17	0.31
ARID1A · driver	Mutation	p.S711*	10	0.30
FMN2	Mutation	p.E134K	0	0.56
TARS2	Mutation	p.S649C	2	0.33
FLG	Mutation	p.R903T	1	0.32
RAD54B	Mutation	p.R658*	0	0.32
OBSCN	Mutation	p.Q1151H	8	0.32
TTN	Mutation	p.N26888Kfs*9, p.S13089C	2	0.32
PLEC	Mutation	p.X610_splice	3	0.32
CEP170	Mutation	p.E558K	0	0.31
GNPAT	Mutation	p.R163Q	3	0.31
MUC16	Mutation	p.Q14194H	8	0.31



14 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TARS2 → Threonine (DB00156); MUC16 → Oregovomab (DB04964); ATP6V1C1 → Tiludronic acid (DB01133); ATP6V1C1 → D-tartaric acid (DB01694); ATP6V1C1 → Thonzonium (DB09552). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A1IJ	57	I	LumA	0.77	no progression, followed 28.4 mo
TCGA-A2-A0EP	56	I	LumA	0.74	no progression, followed 118.5 mo
TCGA-BH-A8FY	87	I	LumA	0.69	no progression, followed 9.7 mo
TCGA-E2-A576	69	I	LumA	0.67	no progression, followed 34.3 mo
TCGA-B6-A0IP	74	I	LumA	0.66	no progression, followed 129.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

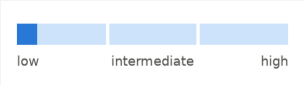
Sample: **TCGA-B6-A2IU-01** · Aneuploidy score: **9** · TMB (nonsyn., /Mb): **1.23** · MSIsensor score: **0.03** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

7th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 14%

Model-estimated progression time

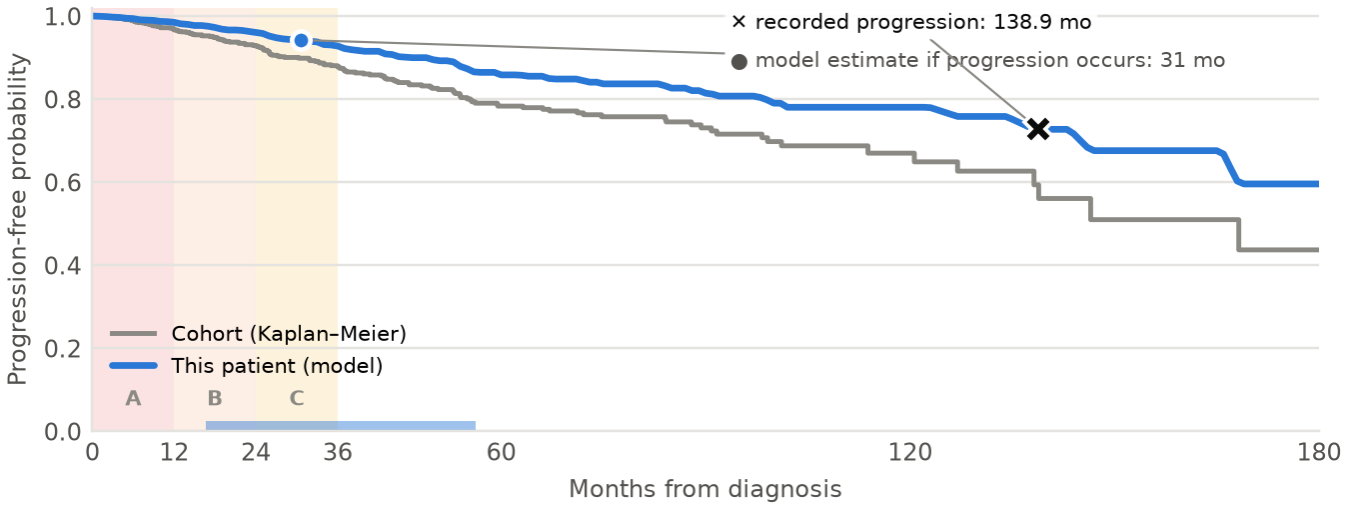
≈ 31 months

Window 24–36 months · IQR 17–56 mo
if progression occurs

Recorded progression

138.9 months

Group D · Late progression
✗ **recorded window differs** · |error| 108 mo



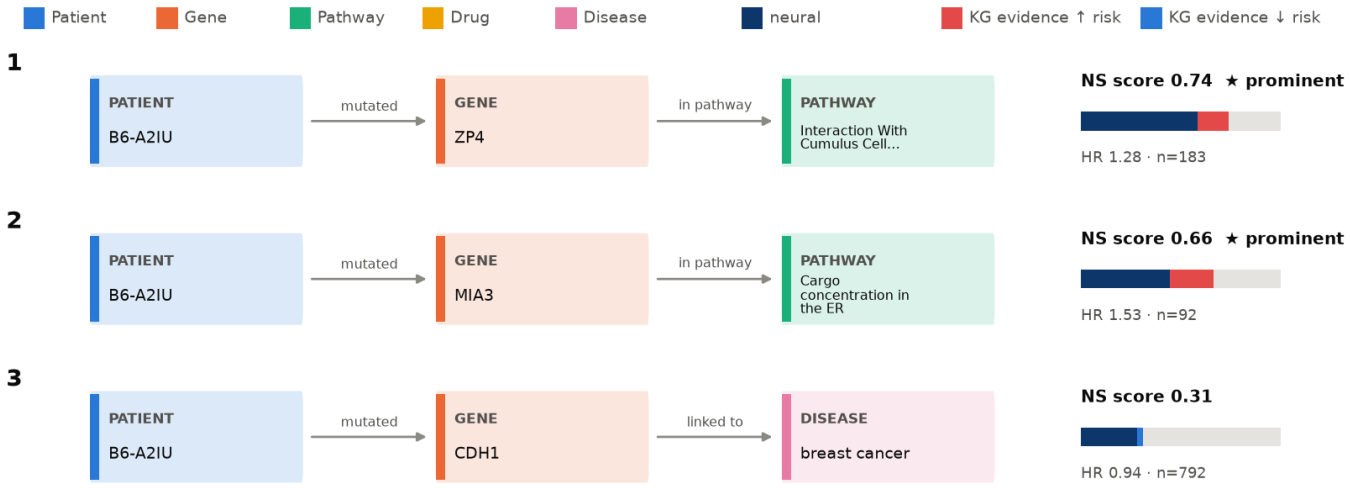
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a LumA tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **7th risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ZP4** to **Interaction With Cumulus Cells And The Zona Pellucida**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.28). Progression was recorded at 138.9 months; the model's estimate given progression was 31 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

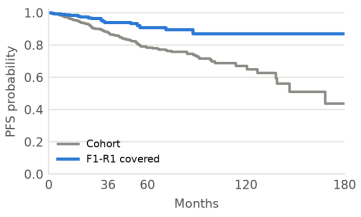


Only 2 of the 21 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-B6-A2IU** mutated → **ZP4** in pathway → **Interaction With Cumulus Cells And The Zona Pellucida**	98	↑ faster progression HR 1.28, z +1.2, n = 183 carriers of “Interaction With Cumulus Cells And The Zona Pellucida”	**0.74**
2 ★	**TCGA-B6-A2IU** mutated → **MIA3** in pathway → **Cargo concentration in the ER**	74	↑ faster progression HR 1.53, z +1.6, n = 92 carriers of “Cargo concentration in the ER”	**0.66**
3	**TCGA-B6-A2IU** mutated → **CDH1** linked to → **breast cancer**	47	↓ slower progression HR 0.94, z -0.2, n = 792 carriers of “breast cancer”	**0.31**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

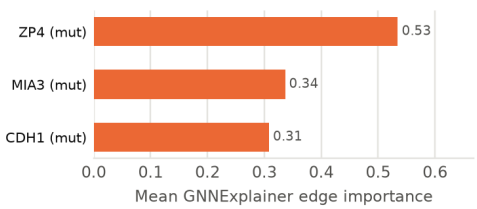
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R1 ↓ lower risk	N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45	365 pts 19 events	**0.26** p 4.0e-09	not reached
F1-R7 ↓ lower risk	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo
F1-R2 ↓ lower risk	N stage = N0 AND Age ≥ 31.5	410 pts 29 events	**0.36** p 4.5e-07	not reached
F1-R8 ↓ lower risk	AJCC stage = II AND Age ≥ 39.5 AND TMB (nonsynonymous) < 141.95	452 pts 39 events	**0.47** p 7.9e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.Q195*	17	0.31
ZP4	Mutation	p.P484=	1	0.53
MIA3	Mutation	p.I387V	3	0.34



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A2LN	65	II	LumA	0.90	no progression, followed 38.2 mo
TCGA-AR-A5QM	62	II	LumA	0.89	progressed at 31.8 mo (group C)

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-PE-A5DD	64	II	LumA	0.89	no progression, followed 64.2 mo
TCGA-E2-A10F	47	II	LumA	0.87	no progression, followed 28.9 mo
TCGA-A8-A091	61	II	LumA	0.86	progressed at 8.0 mo (group A)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**50**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

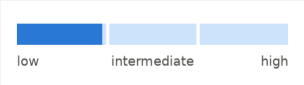
Sample: **TCGA-B6-A0IG-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **1.23** · MSIsensor score: **0.24** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

32nd

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

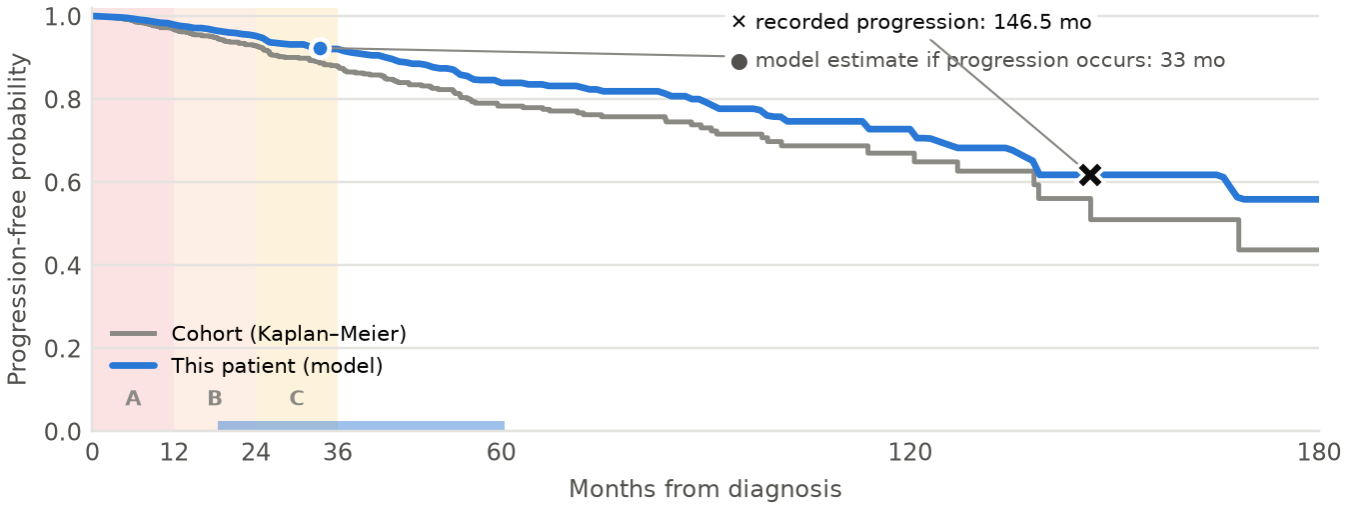
≈ 33 months

Window 24–36 months · IQR 18–60 mo
if progression occurs

Recorded progression

146.5 months

Group D · Late progression
✗ **recorded window differs** · |error| 113 mo



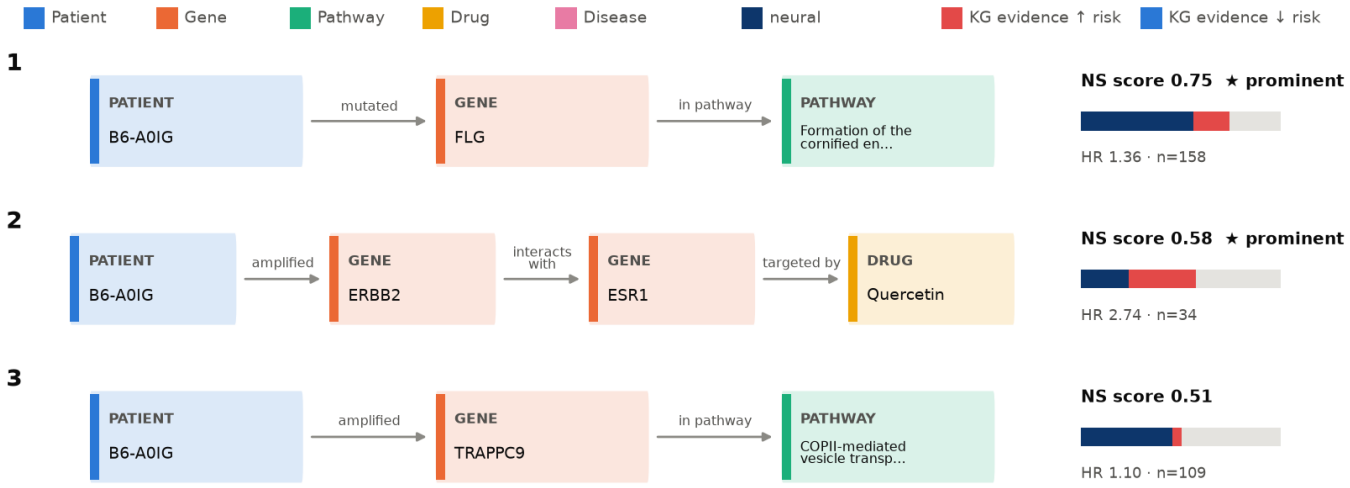
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 50-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **32nd risk percentile** (low tier). The most significant rule covering them (*Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **FLG** to **Formation of the cornified envelope**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.36). Progression was recorded at 146.5 months; the model's estimate given progression was 33 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

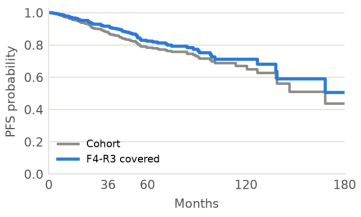


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-B6-A0IG** mutated → **FLG** in pathway → **Formation of the cornified envelope**	94	↑ **faster progression** HR 1.36, z +1.4, n = 158 carriers of “Formation of the cornified envelope”	**0.75**
2 ★	**TCGA-B6-A0IG** amplified → **ERBB2** interacts with → **ESR1** targeted by → **Quercetin**	40	↑ **faster progression** HR 2.74, z +2.5, n = 34 carriers of “Quercetin”	**0.58**
3	**TCGA-B6-A0IG** amplified → **TRAPPC9** in pathway → **COPII-mediated vesicle transport**	77	↑ **faster progression** HR 1.10, z +0.4, n = 109 carriers of “COPII-mediated vesicle transport”	**0.51**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

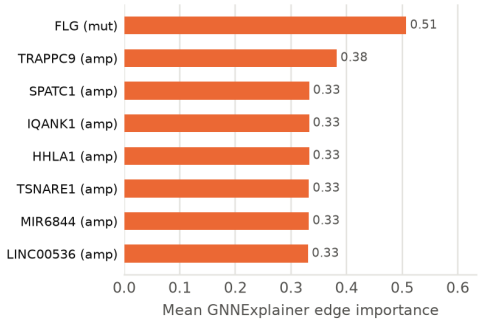
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R3 ↓ lower risk	Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	662 pts 77 events	**0.36** p 2.3e-07	not reached
F4-R4 ↓ lower risk	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo
F4-R5 ↓ lower risk	M stage = M0 AND Age ≥ 34.5	713 pts 89 events	**0.47** p 1.0e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
BRCA1 · driver	High-level amplification	GISTIC ±2	29	0.32
RB1 · driver	Mutation	p.K873E	25	0.31
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
FLG	Mutation	p.R461W	1	0.51
TRAPPC9	High-level amplification	GISTIC ±2	2	0.38
SPATC1	High-level amplification	GISTIC ±2	0	0.33
IQANK1	High-level amplification	GISTIC ±2	0	0.33
HHLA1	High-level amplification	GISTIC ±2	0	0.33
TSNARE1	High-level amplification	GISTIC ±2	0	0.33
MIR6844	High-level amplification	GISTIC ±2	0	0.33



274 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 115 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A24V	52	II	LumA	0.84	no progression, followed 105.3 mo
TCGA-E2-A150	48	II	Basal	0.81	no progression, followed 63.6 mo
TCGA-C8-A12L	67	II	Her2	0.79	no progression, followed 11.9 mo
TCGA-A8-A097	65	II	LumA	0.79	no progression, followed 12.0 mo
TCGA-LL-A5YN	46	II	LumA	0.78	no progression, followed 14.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group D** · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**50**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE II**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

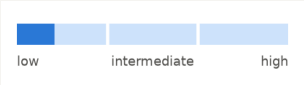
Sample: **TCGA-B6-A0WZ-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **1.87** · MSIsensor score: **0.04** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

14th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

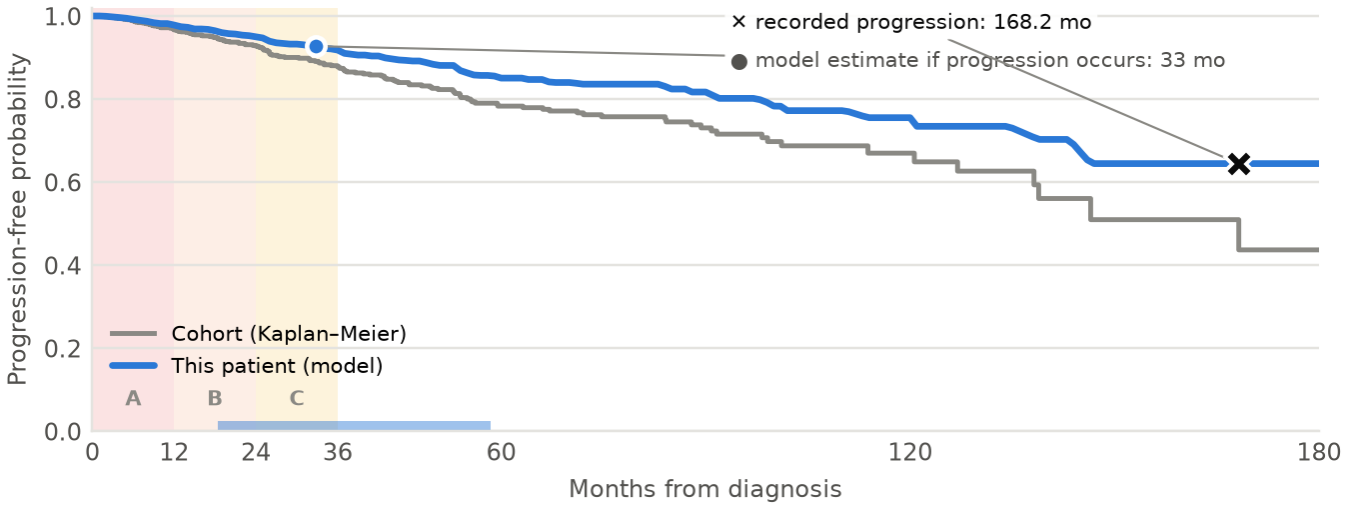
≈ 33 months

Window 24–36 months · IQR 18–58 mo
if progression occurs

Recorded progression

168.2 months

■ **Group D** · Late progression
✗ **recorded window differs** · |error| 135 mo



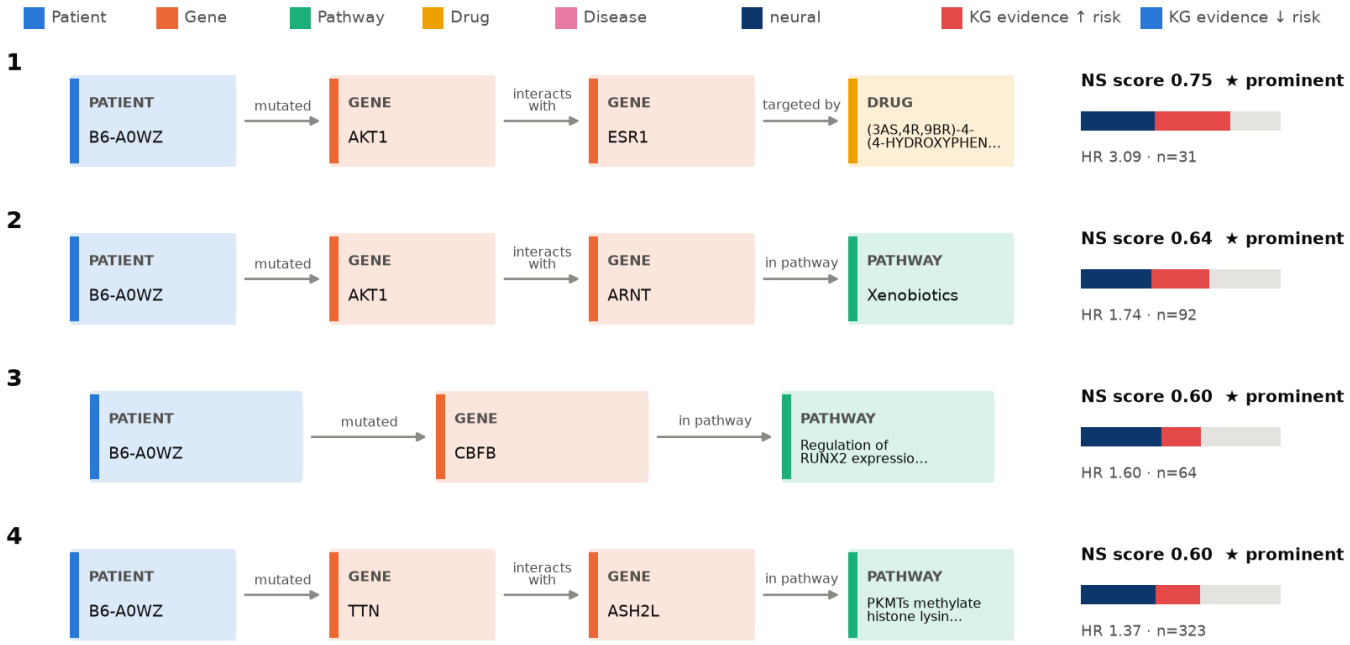
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 50-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **14th risk percentile** (low tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 83.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **AKT1** to **(3AS,4R,9BR)-4-(4-HYDROXYPHENYL)-6-(METHOXYMETHYL)-1,2,3,3A,4,9B-HEXAHYDROCYCLOPENTA[C]CHROMEN-8-OL**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 3.09). Progression was recorded at 168.2 months; the model's estimate given progression was 33 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

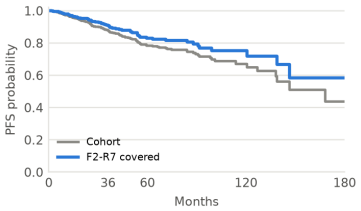


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-B6-A0WZ** mutated → **AKT1** interacts with → **ESR1** targeted by → **(3AS,4R,9BR)-4-(4-HYDROXYPHENYL)-6-(METHOXYMETHYL)-1,2,3,3A,4,9B-HEXAHYDROCYCLOPENTA[C]CHROMEN-8-OL**	62	↑ **faster progression** HR 3.09, z +2.8, n = 31 carriers of "(3AS,4R,9BR)-4-(4-HYDROXYPHENYL)-6-(METHOXYMETHYL)-1,2,3,3A,4,9B-HEXAHYDROCYCLOPENTA[C]CHROMEN-8-OL"	**0.75**
2 ★	**TCGA-B6-A0WZ** mutated → **AKT1** interacts with → **ARNT** in pathway → **Xenobiotics**	59	↑ **faster progression** HR 1.74, z +2.2, n = 92 carriers of "Xenobiotics"	**0.64**
3 ★	**TCGA-B6-A0WZ** mutated → **CBFB** in pathway → **Regulation of RUNX2 expression and activity**	67	↑ **faster progression** HR 1.60, z +1.5, n = 64 carriers of "Regulation of RUNX2 expression and activity"	**0.60**
4 ★	**TCGA-B6-A0WZ** mutated → **TTN** interacts with → **ASH2L** in pathway → **PKMTs methylate histone lysines**	62	↑ **faster progression** HR 1.37, z +1.7, n = 323 carriers of "PKMTs methylate histone lysines"	**0.60**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

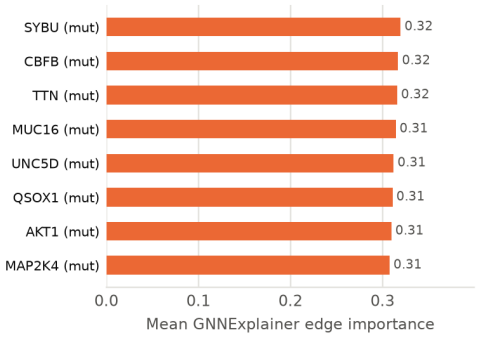
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 83.5	638 pts 66 events	**0.39** p 3.3e-07	not reached
F2-R8 ↓ lower risk	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CBFB · driver	Mutation	p.V86=	29	0.32
AKT1 · driver	Mutation	p.E17K	54	0.31
MAP2K4 · driver	Mutation	p.S184L	12	0.31
SYBU	Mutation	p.E444K	0	0.32
TTN	Mutation	p.L19843=	2	0.32
MUC16	Mutation	p.I8478=	8	0.31
UNC5D	Mutation	p.E736K	1	0.31
QSOX1	Mutation	p.L190=	4	0.31



8 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

AKT1 → ATP (DB00171); AKT1 → Arsenic trioxide (DB01169); AKT1 → Genistein (DB01645); AKT1 → Inositol 1,3,4,5-Tetrakisphosphate (DB01863); AKT1 → Resveratrol (DB02709); AKT1 → Archexin (DB05971); AKT1 → Enzastaurin (DB06486); AKT1 → Perifosine (DB06641); AKT1 → N-[2-(5-methyl-4H-1,2,4-triazol-3-yl)phenyl]-7H-pyrrolo[2,3-d]pyrimidin-4-amine (DB07584); AKT1 → 5-(5-chloro-7H-pyrrolo[2,3-d]pyrimidin-4-yl)-4,5,6,7-tetrahydro-1H-imidazo[4,5-c]pyridine (DB07585) … and 1 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A1L9	40	II	LumA	0.77	no progression, followed 19.7 mo
TCGA-E9-A22H	42	II	LumB	0.76	no progression, followed 40.5 mo
TCGA-BH-A0W4	46	II	LumA	0.76	no progression, followed 25.0 mo
TCGA-E2-A15I	44	II	LumA	0.76	no progression, followed 55.6 mo
TCGA-D8-A141	40	missing	LumA	0.75	no progression, followed 20.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.